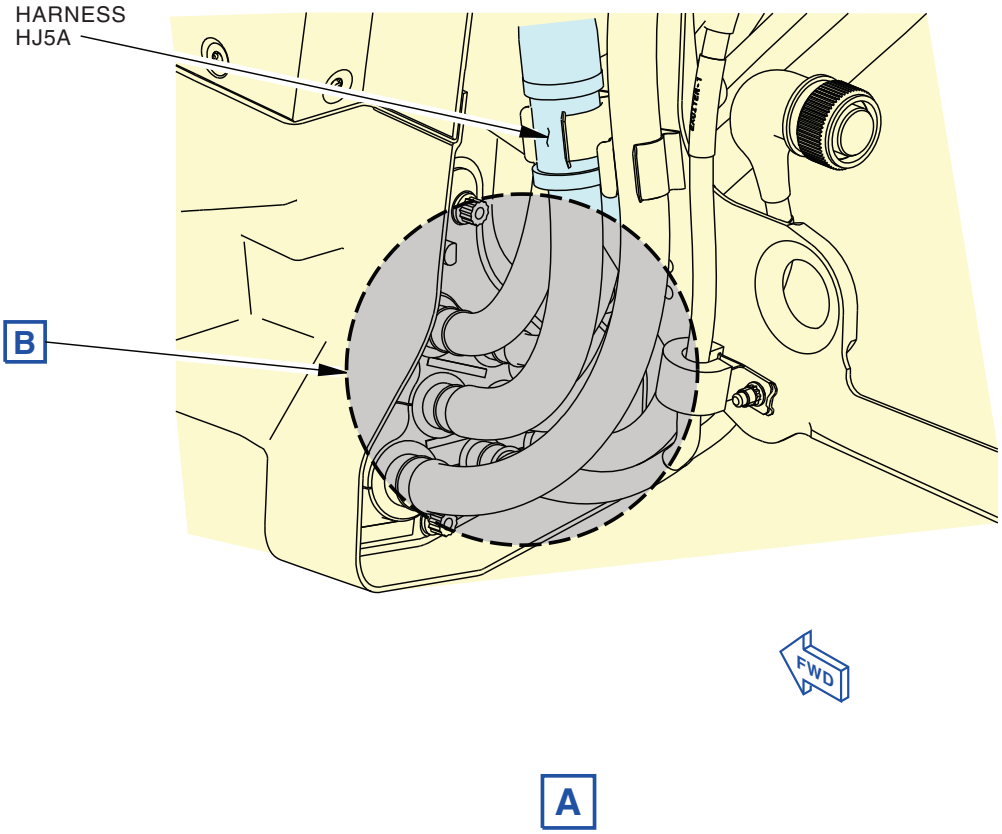
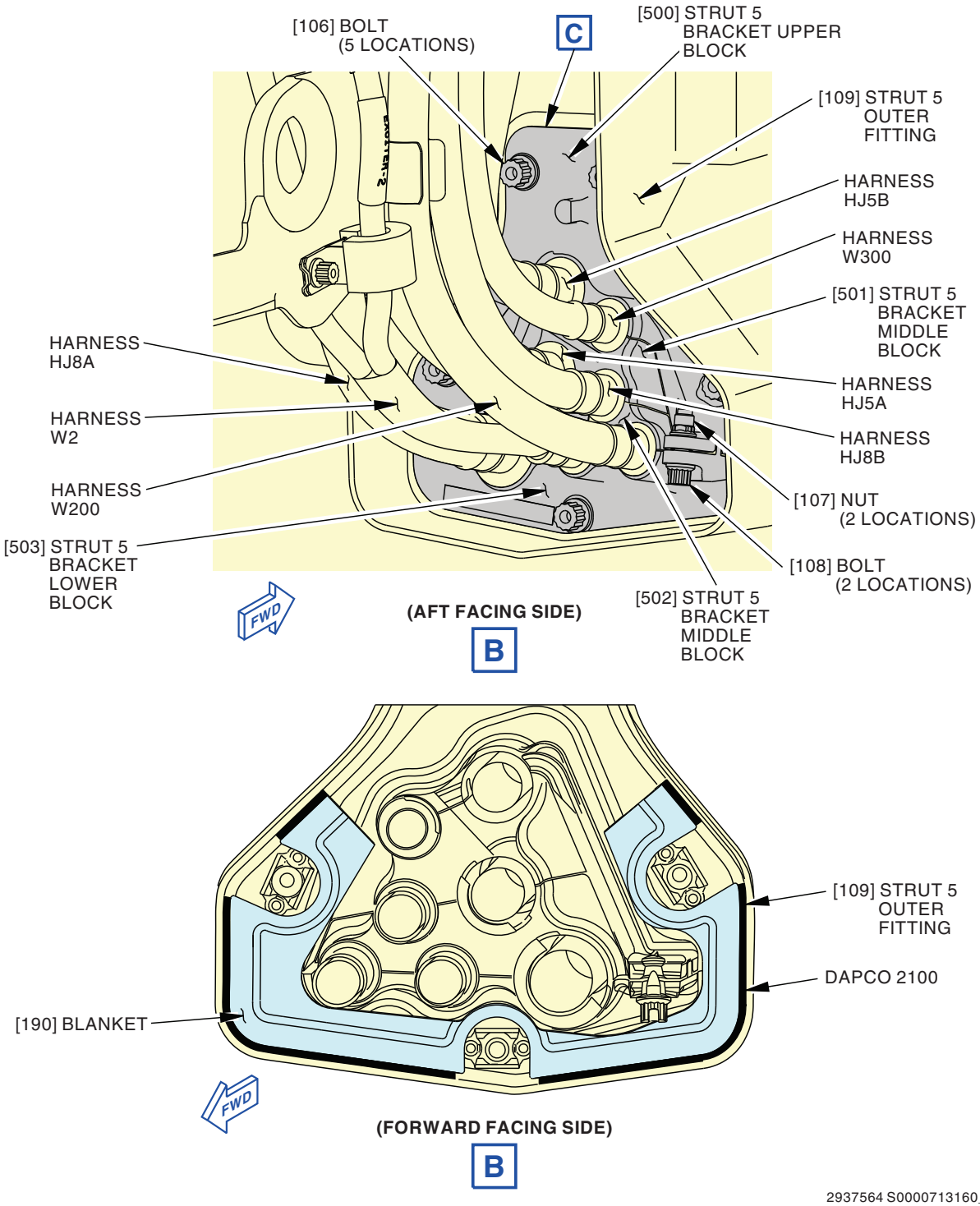



2937795 S0000713384_V2

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 2 of 9)**



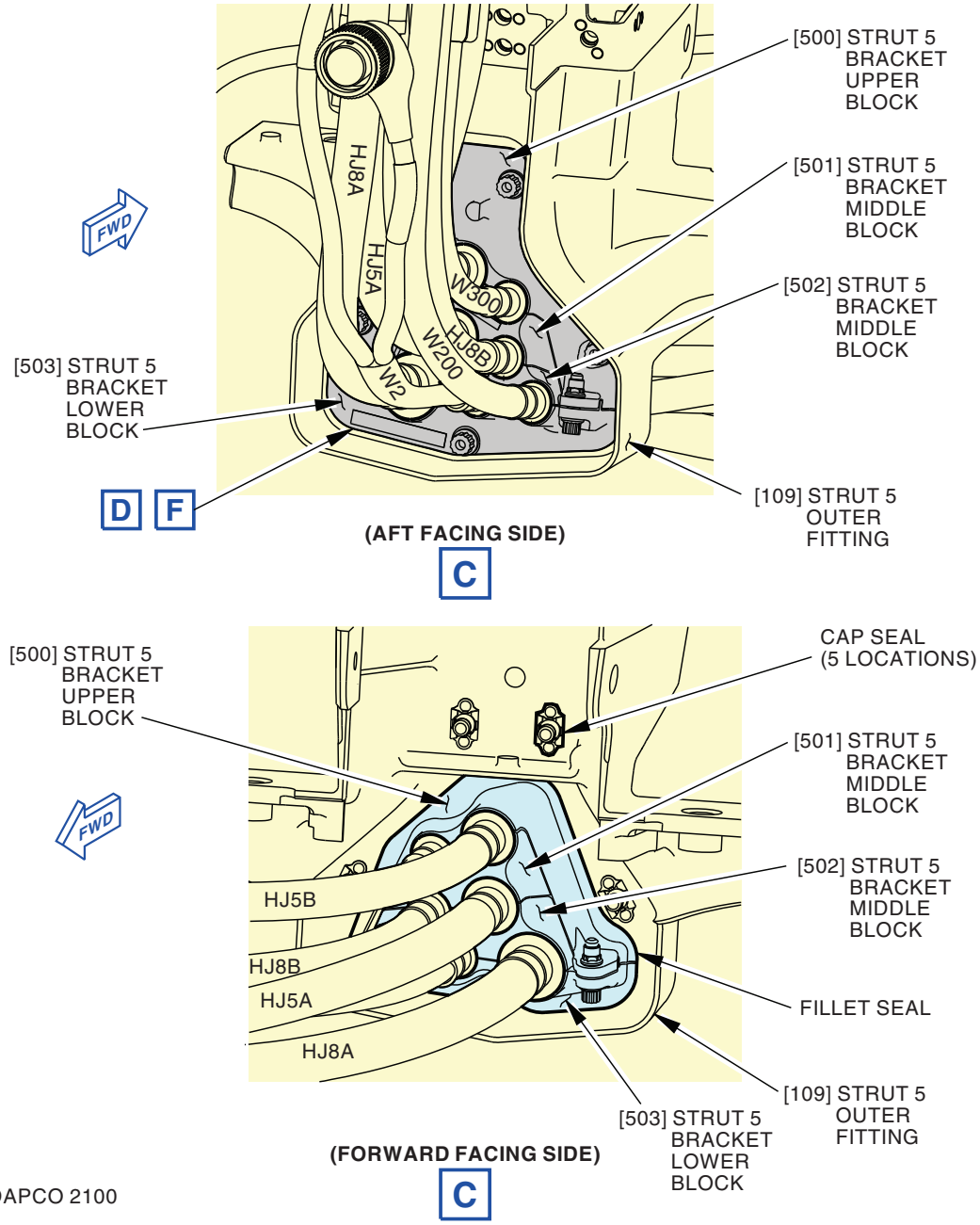
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 3 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

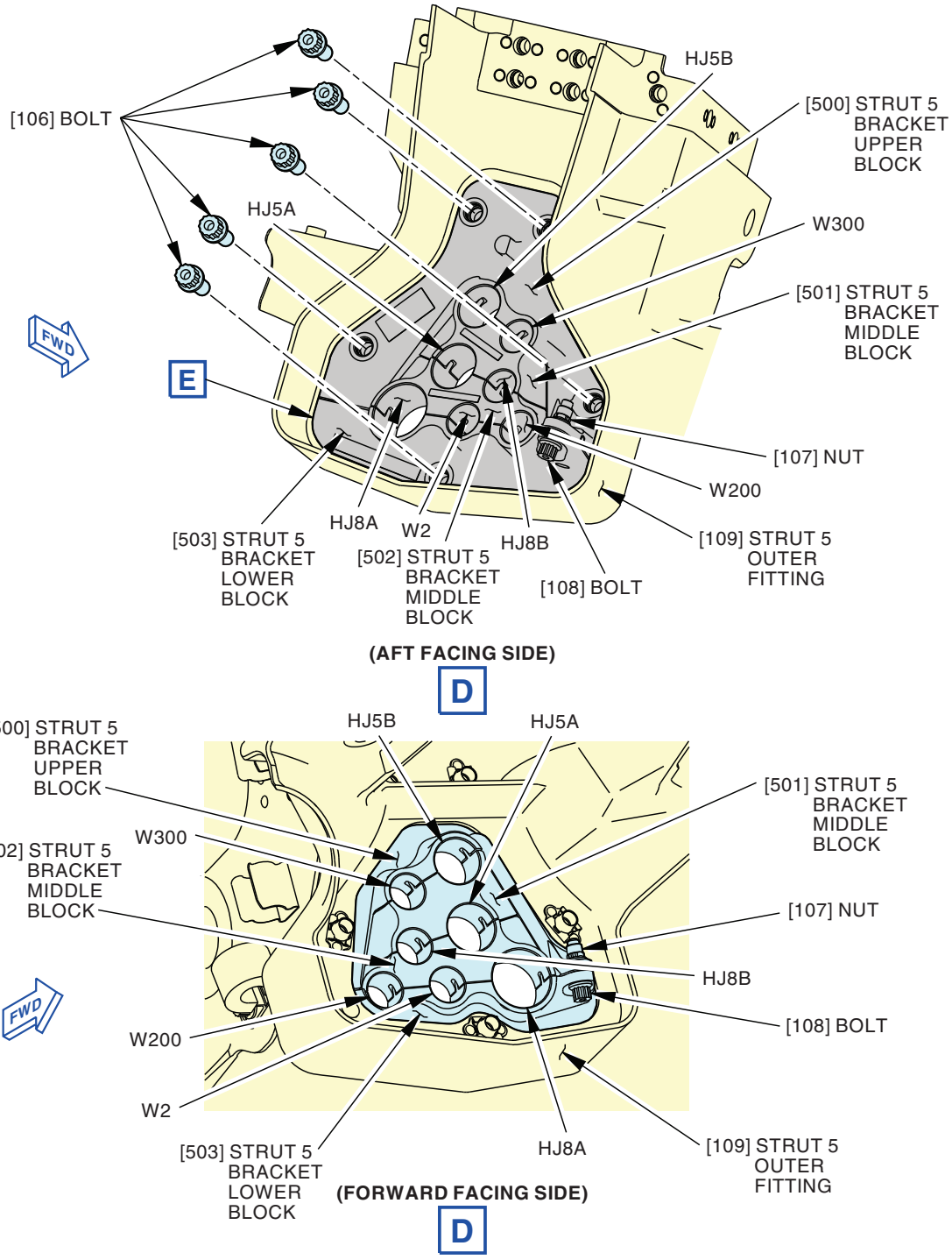
IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 4 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

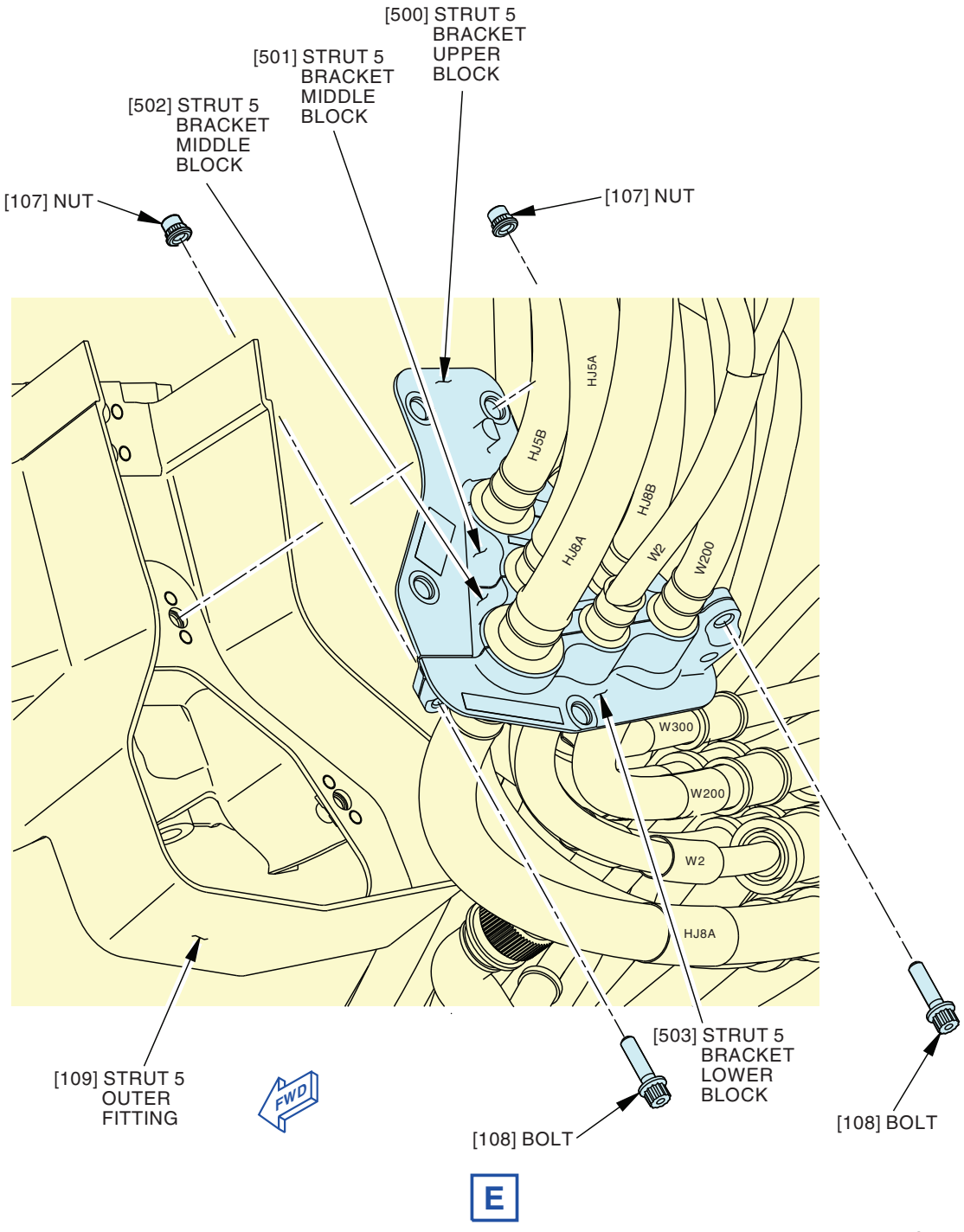
HARNESSES NOT SHOWN

2943980 S0000720369_V2

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 5 of 9)**



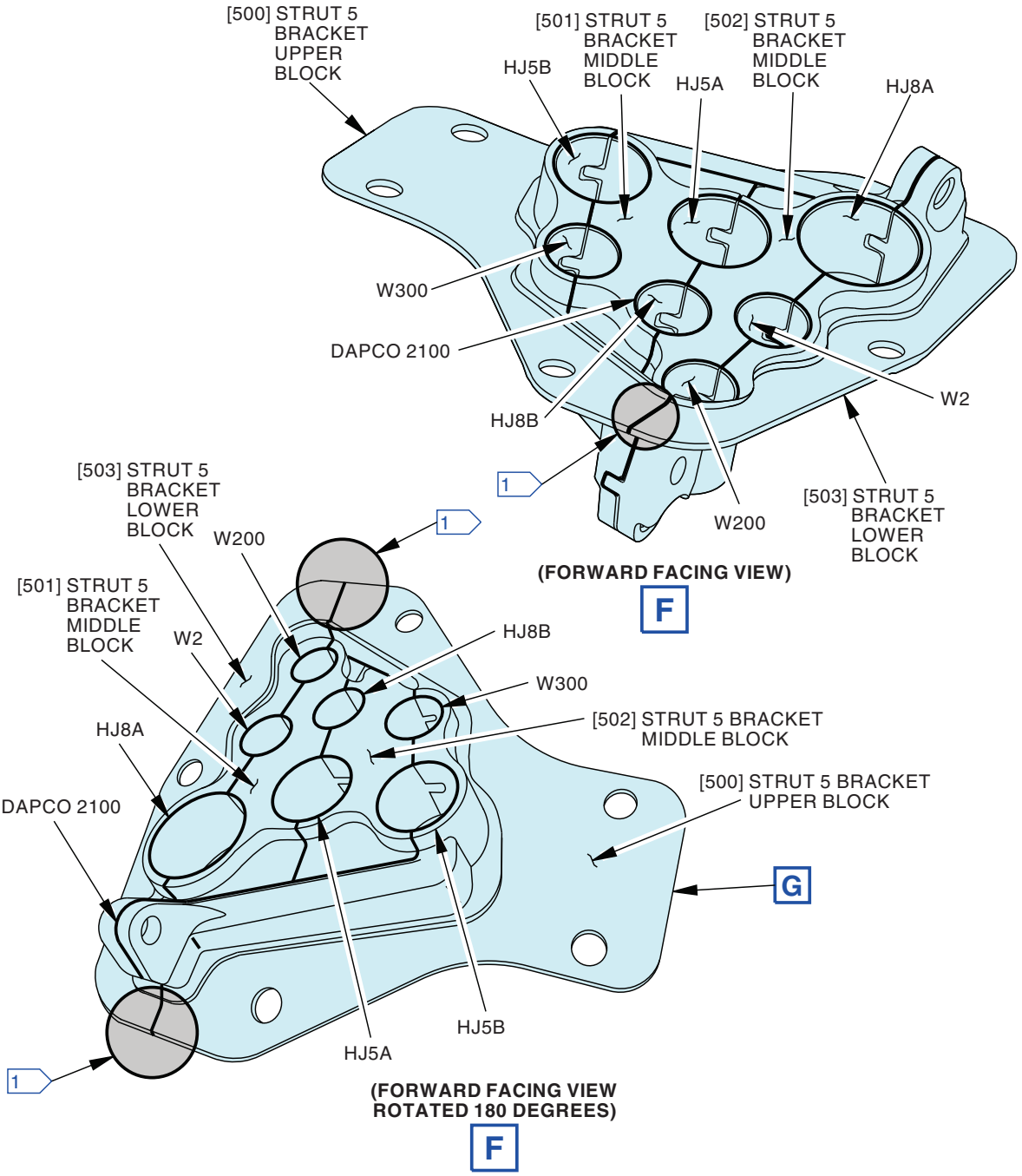
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 6 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

HARNESSES NOT SHOWN.

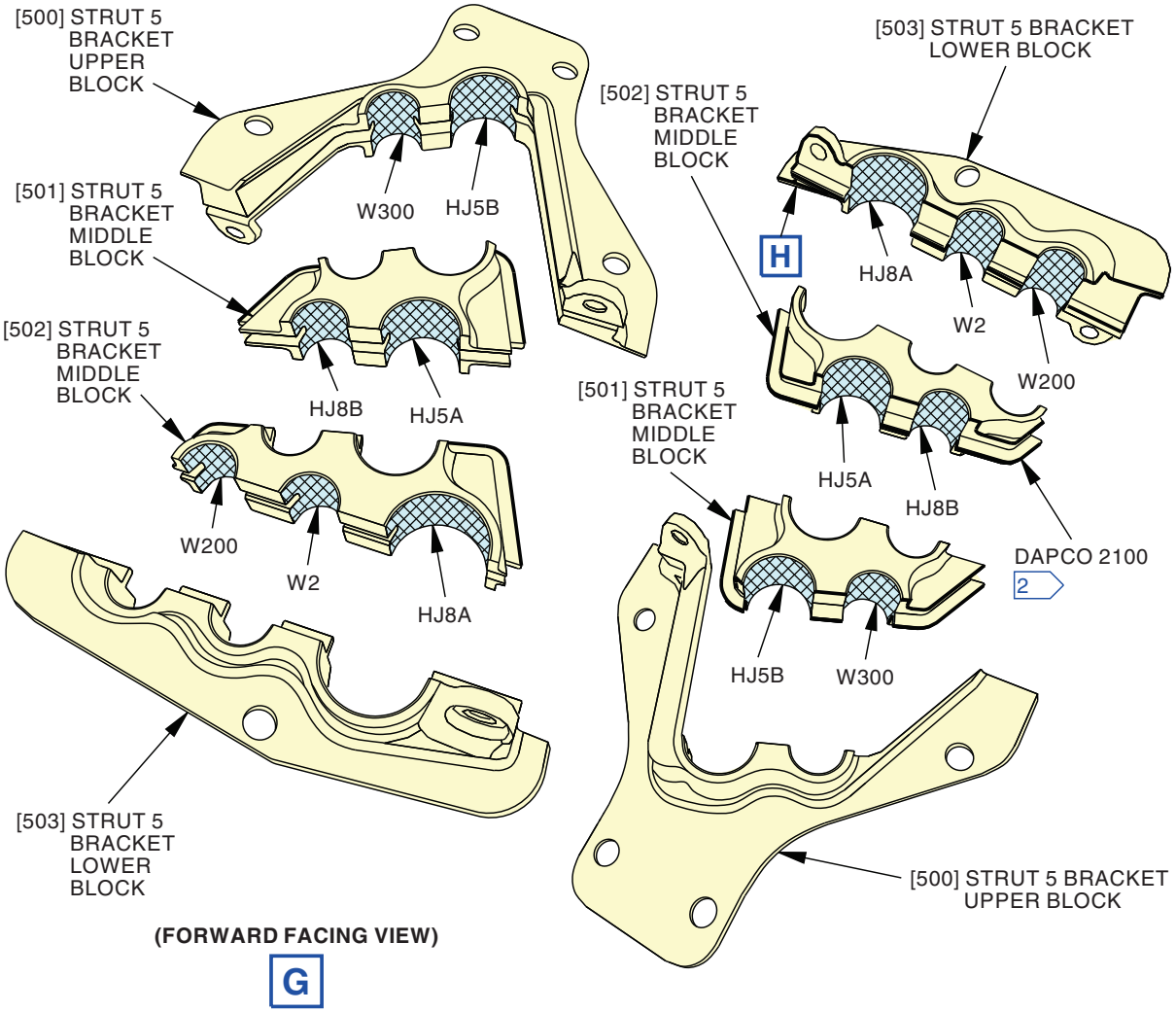
1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 7 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

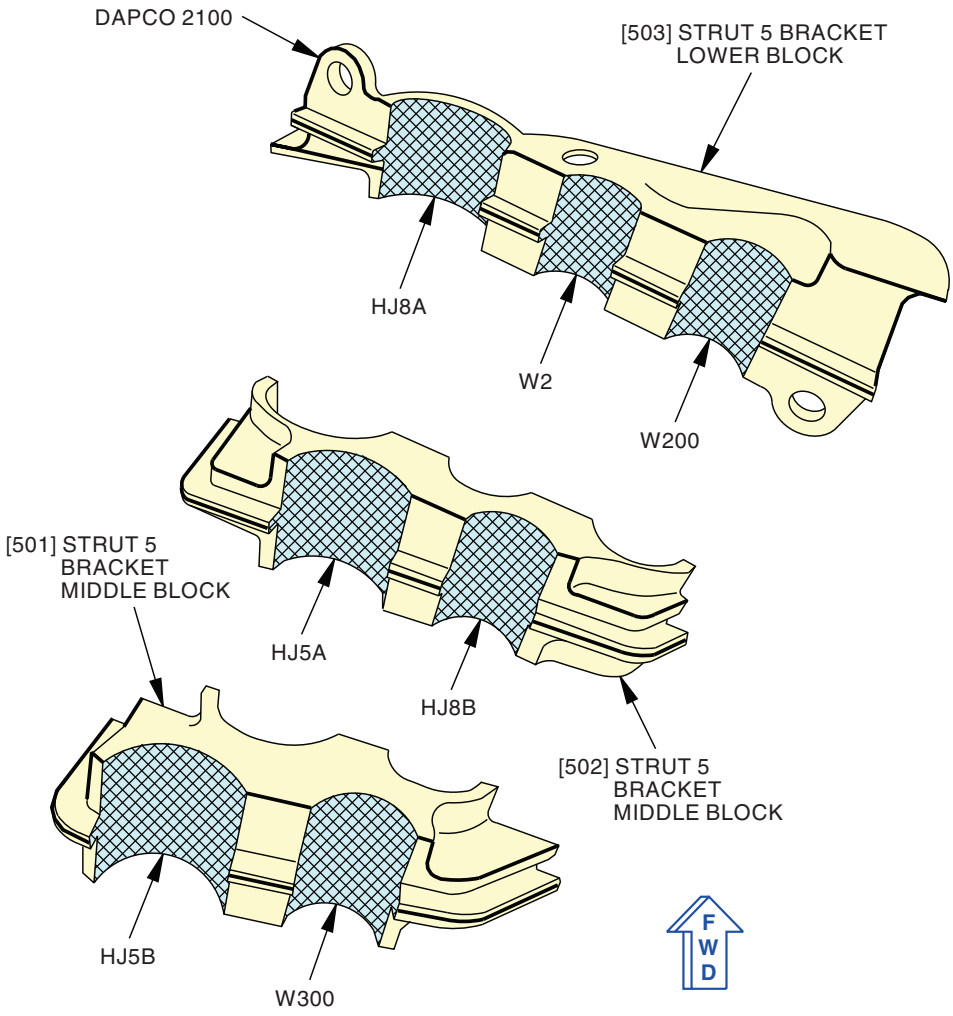
2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 8 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

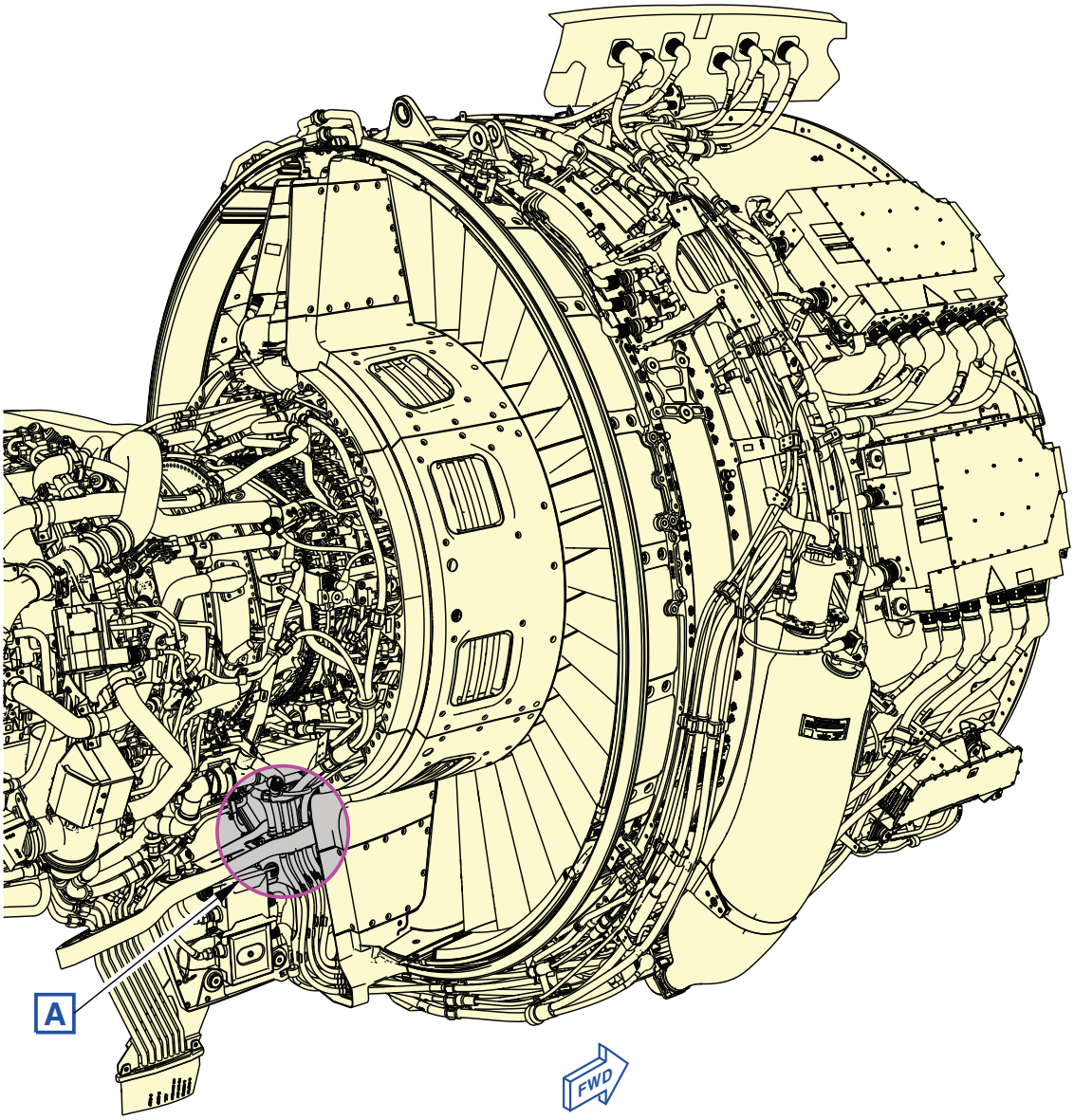



2949941 S0000726813_V1

**Strut Bracket Block Installation**
**Figure 416/73-22-01-990-878-G00 (Sheet 9 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

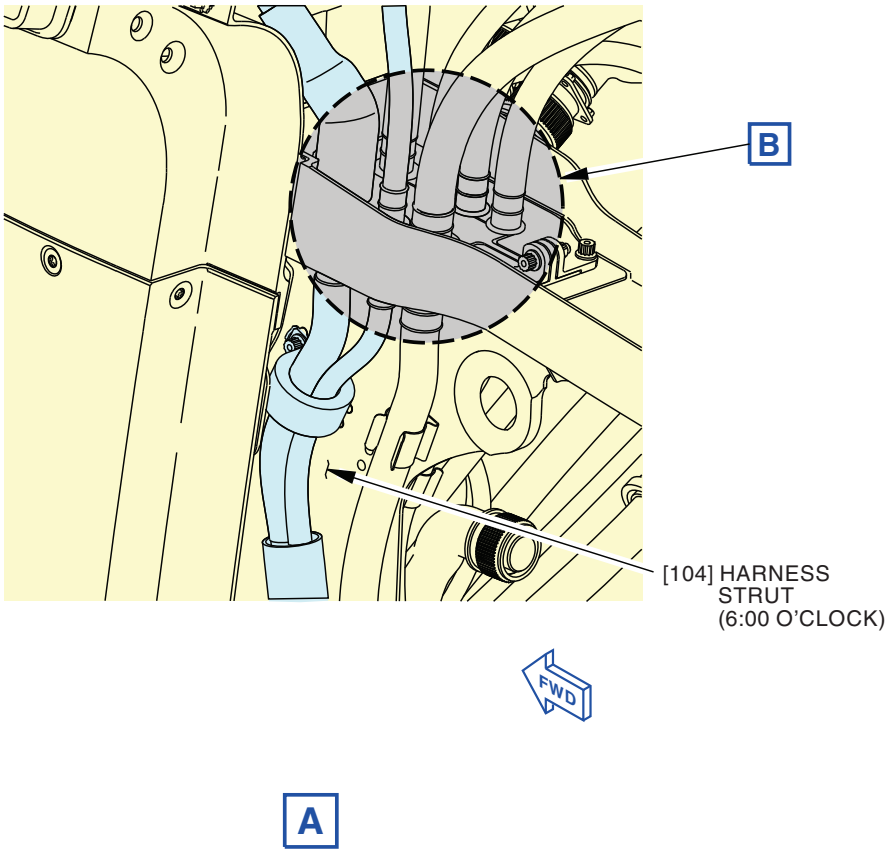



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
2938818 S0000714836_V3

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 1 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

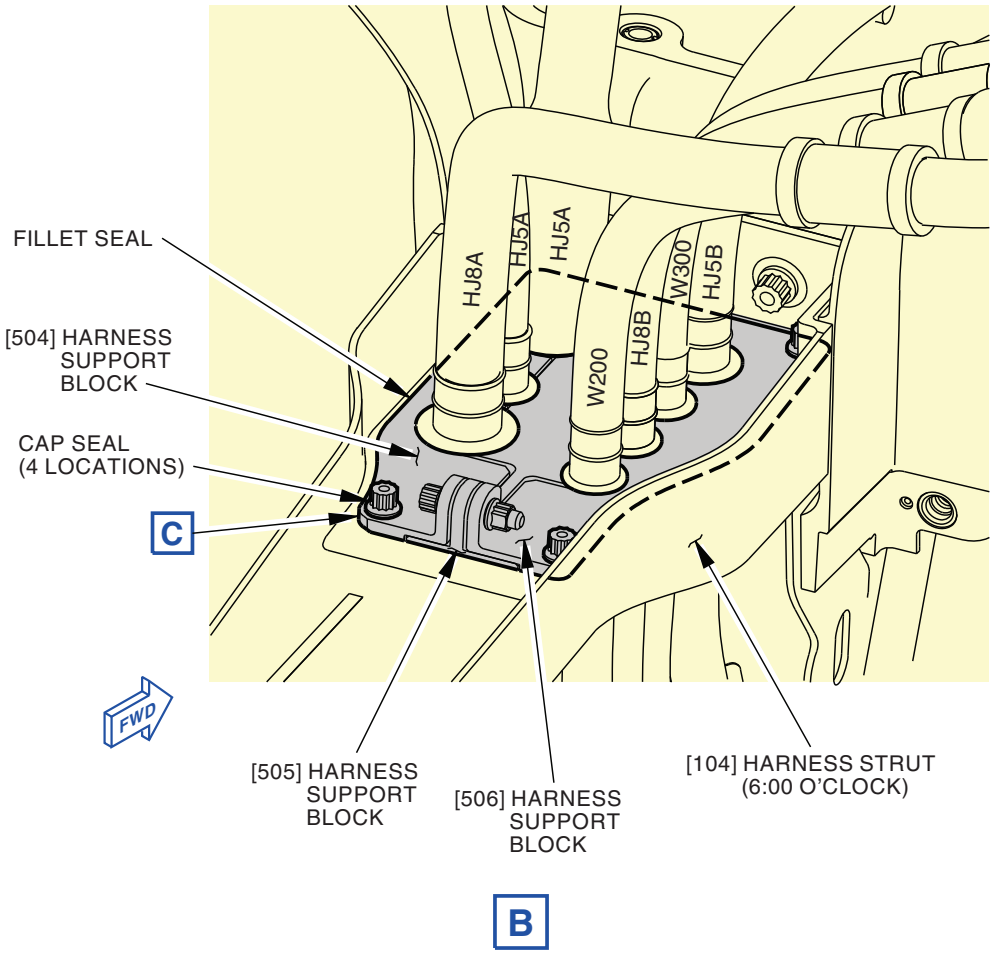



2952925 S0000730238_V1

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 2 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**LEGEND:**

——— DAPCO 2100

**NOTE:**

THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE HARNESS SUPPORT BLOCK. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

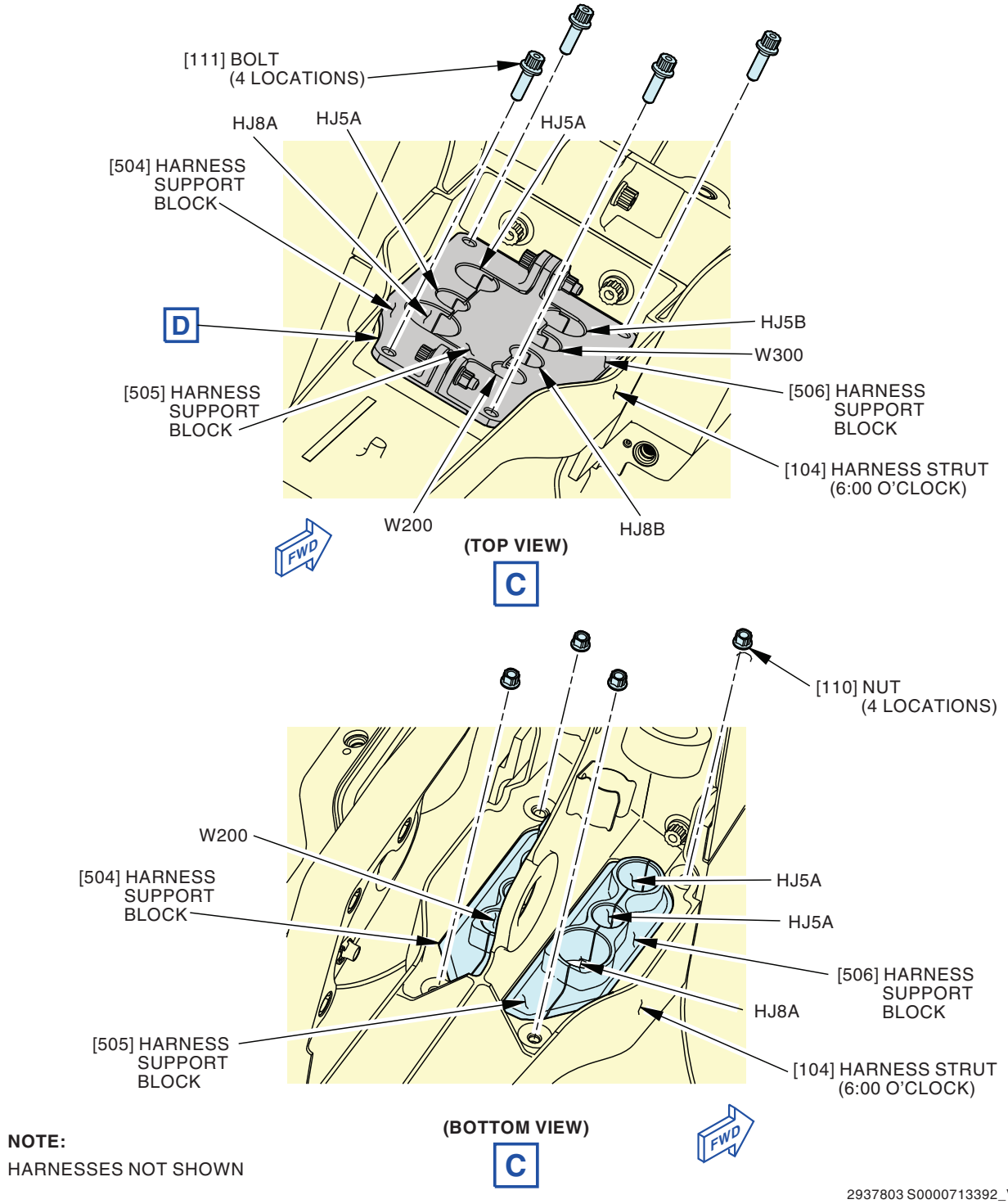
THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937807 S0000713396_V4

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 3 of 11)**



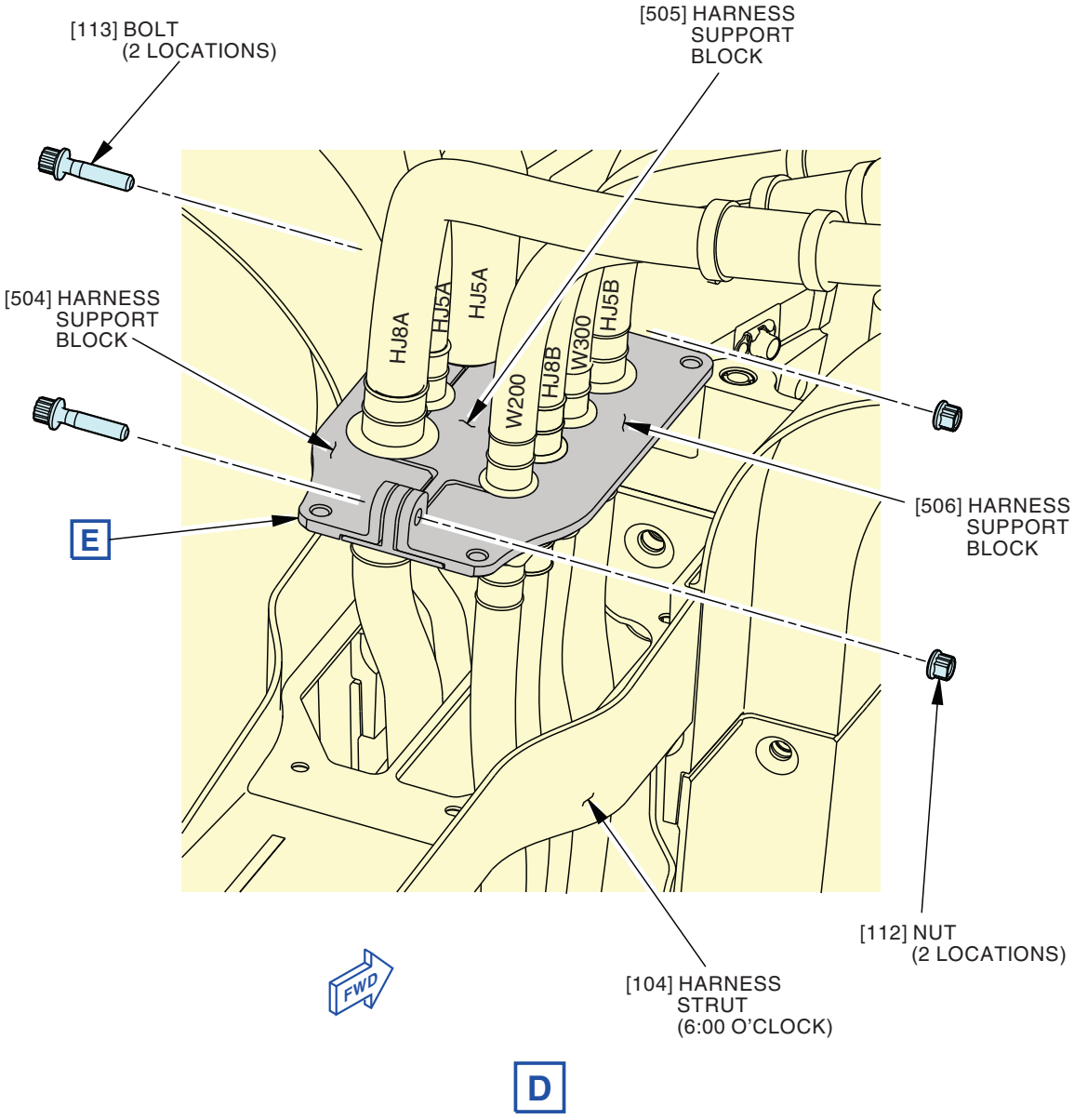
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 4 of 11)**



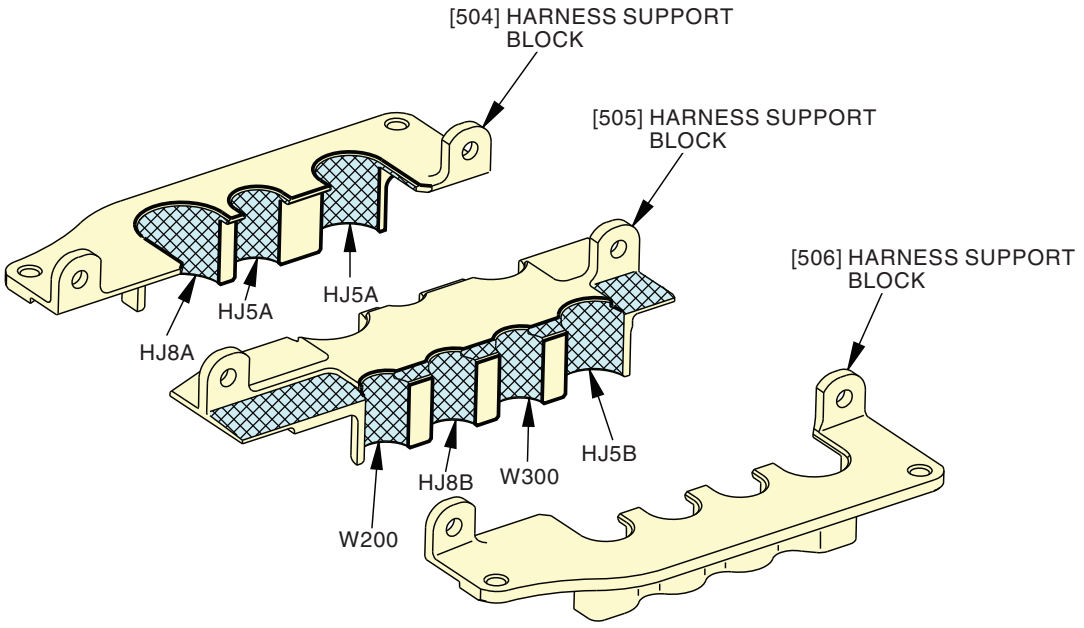
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 5 of 11)**

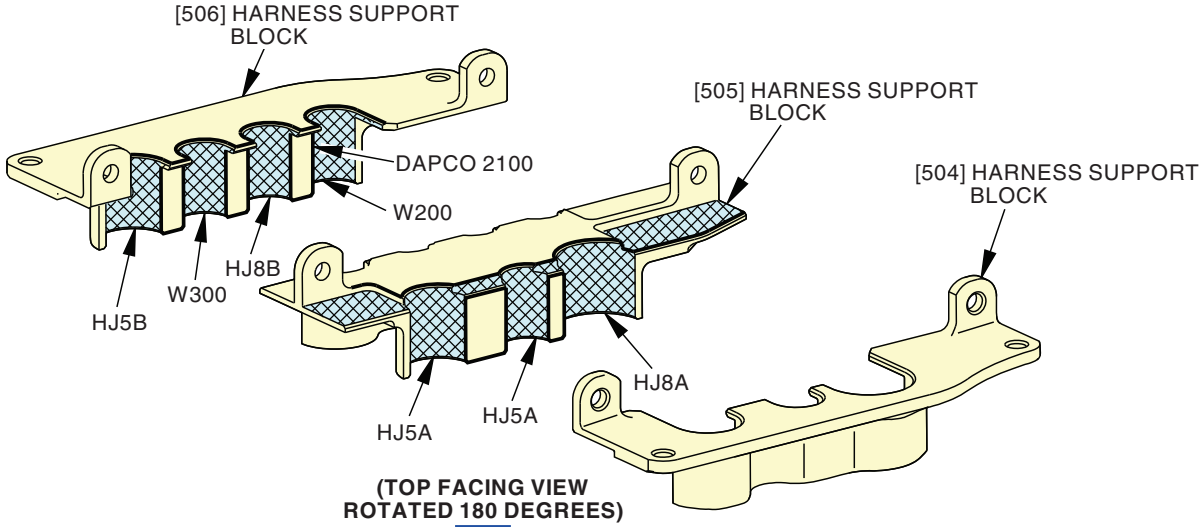


ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(TOP FACING VIEW)**

E




**(TOP FACING VIEW ROTATED 180 DEGREES)**

E

**LEGEND:**

DAPCO 2100

**NOTE:**

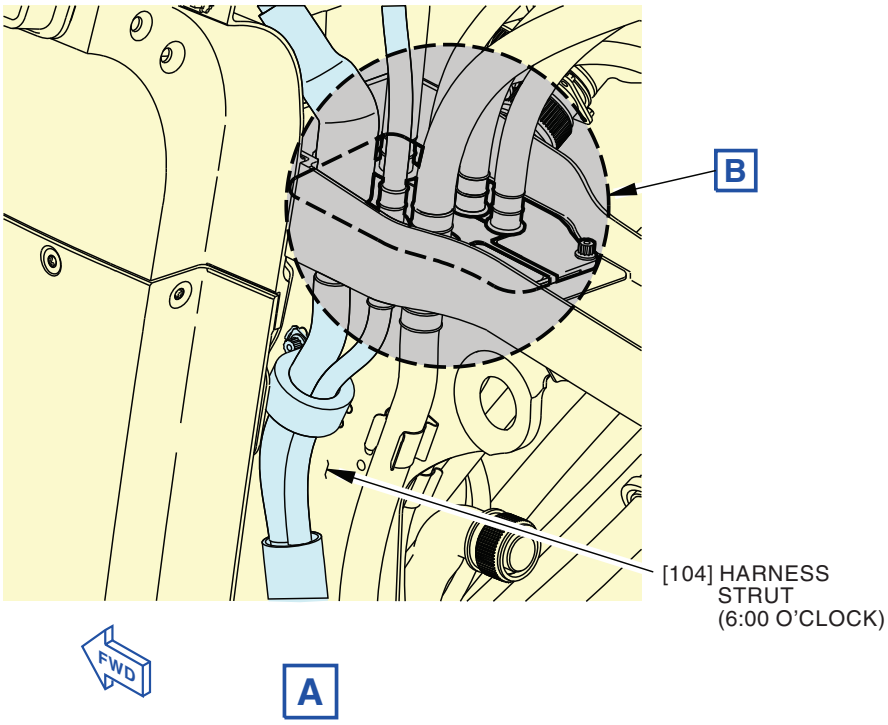
THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937805 S0000713394_V4

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 6 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

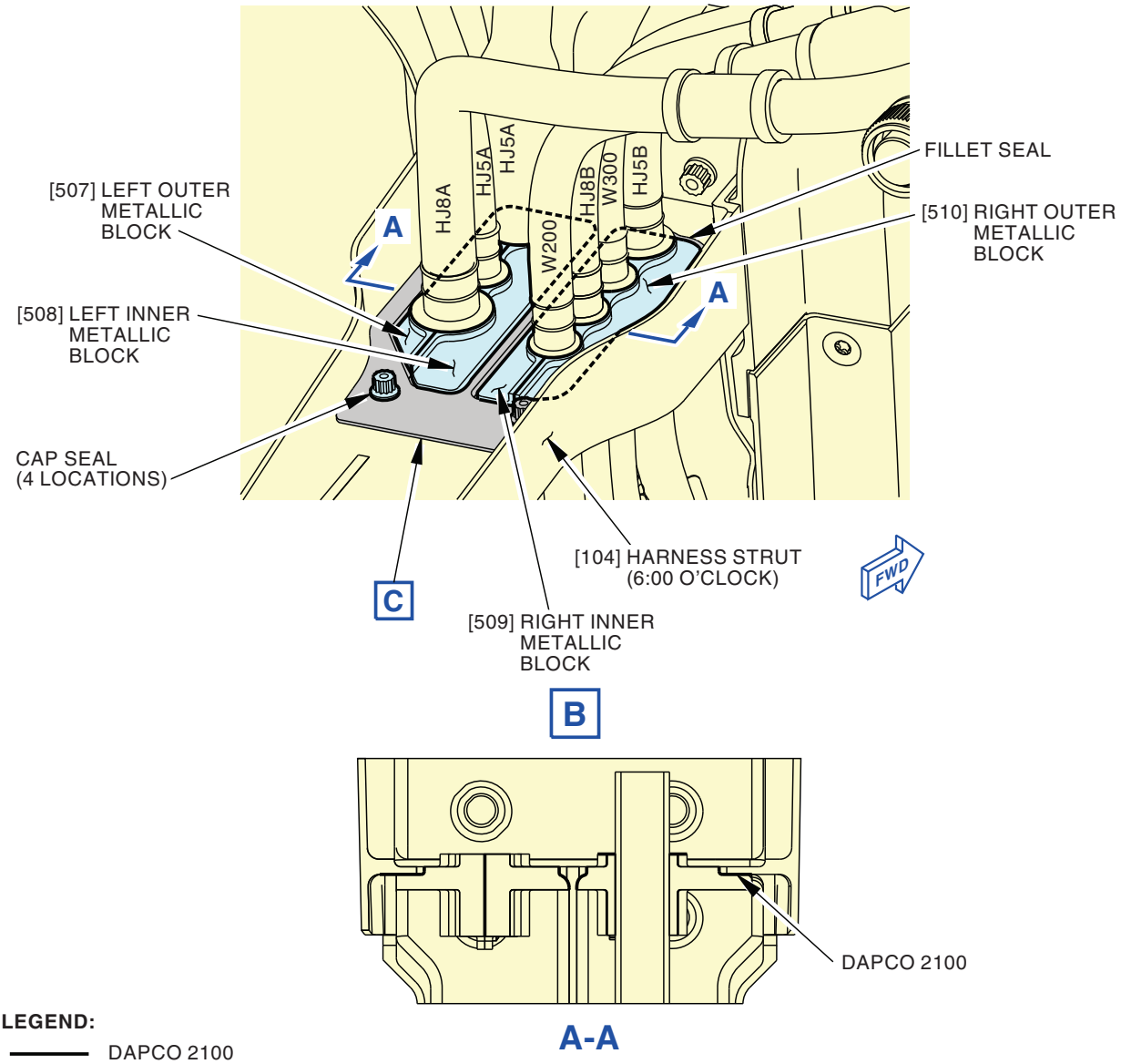



2962032 S0000739201_V1

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 7 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**LEGEND:**

DAPCO 2100

**NOTE:**

THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE METALLIC BLOCKS. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

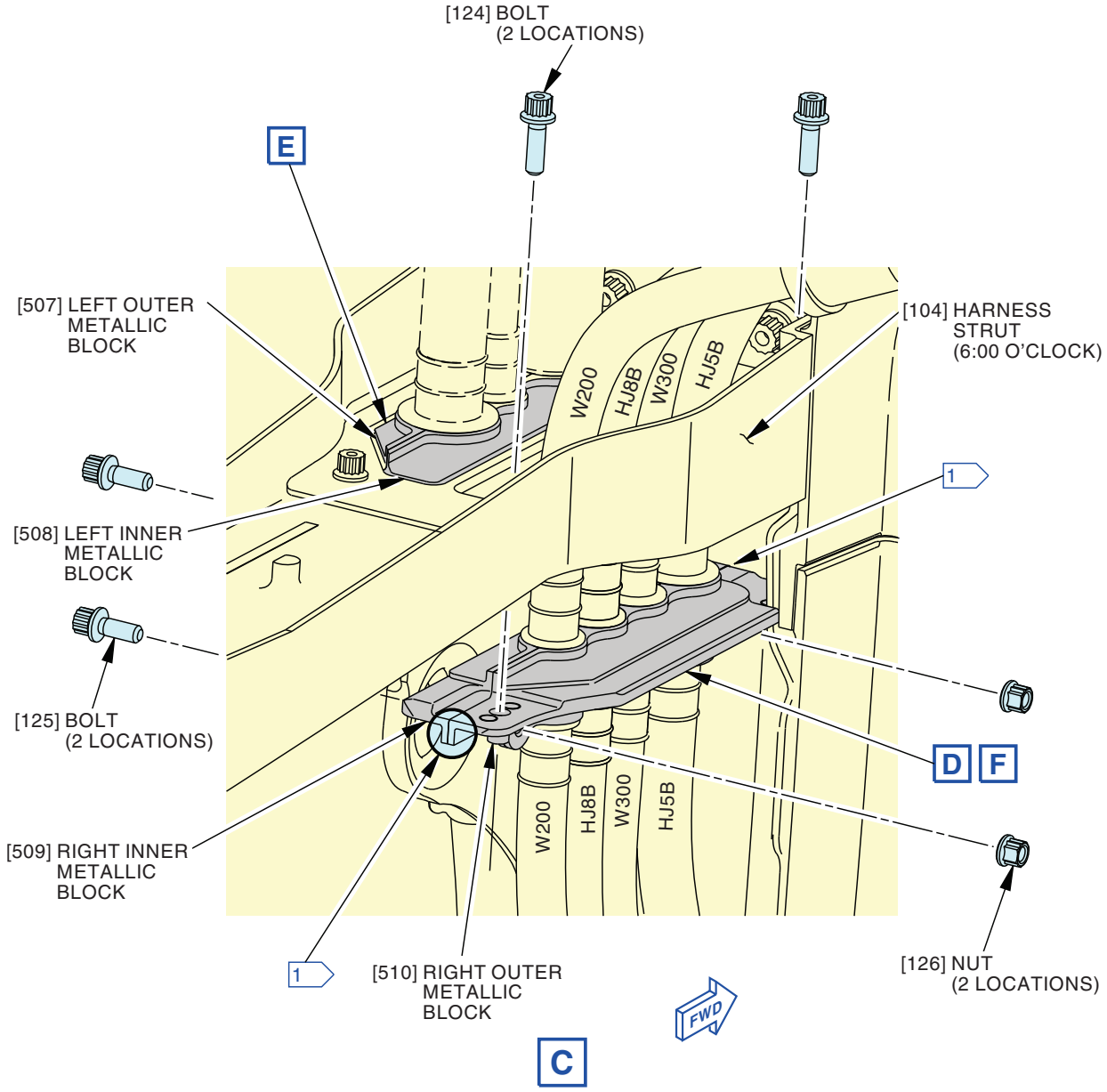
THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN BOTH RIGHT AND LEFT METALLIC BLOCKS HAVE BEEN REMOVED.

ICNLEAP1B732201D5882800059A00101
2962033 S0000739202_V2

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 8 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

THIS VIEW IS SPECIFIC FOR RIGHT INNER AND OUTER METALLIC BLOCKS. SAME LOGIC IS APPLIED FOR LEFT INNER AND OUTER METALLIC BLOCKS.

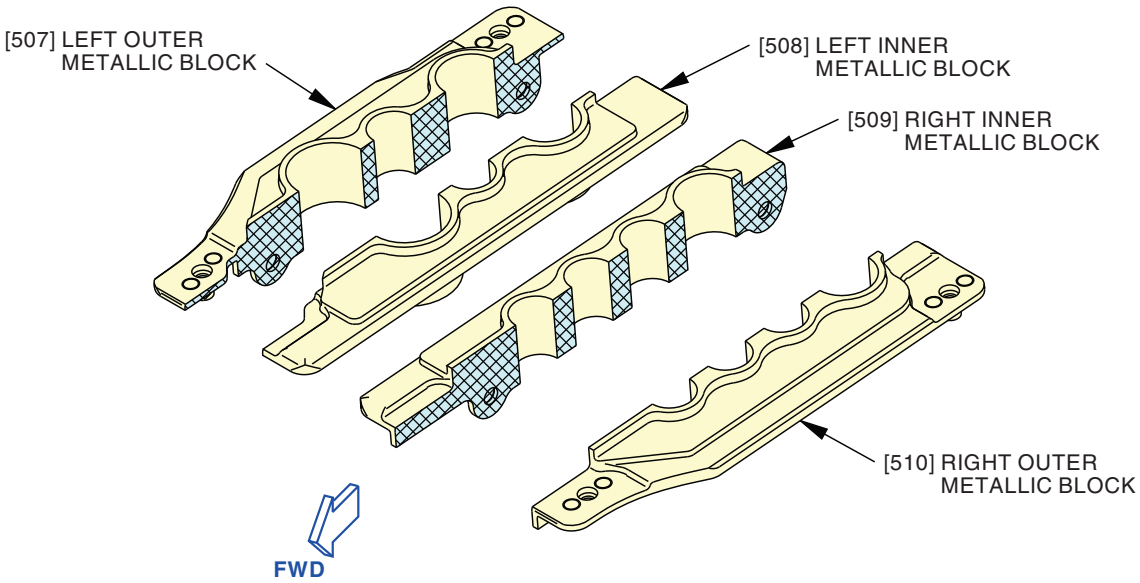
1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

ICNLEAP1B732201D5882800063A00101
2962034 S0000739203_V2

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 9 of 11)**

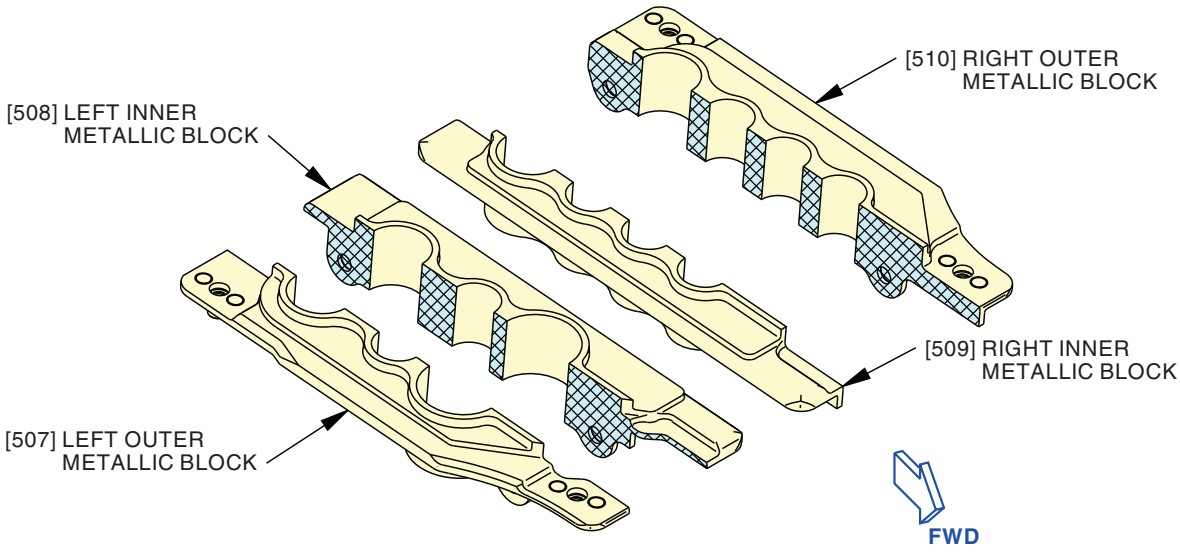


ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(TOP FACING VIEW)**

D




**(TOP FACING VIEW ROTATED 90 DEGREES)**

D

**LEGEND:**

DAPCO 2100

**NOTE:**

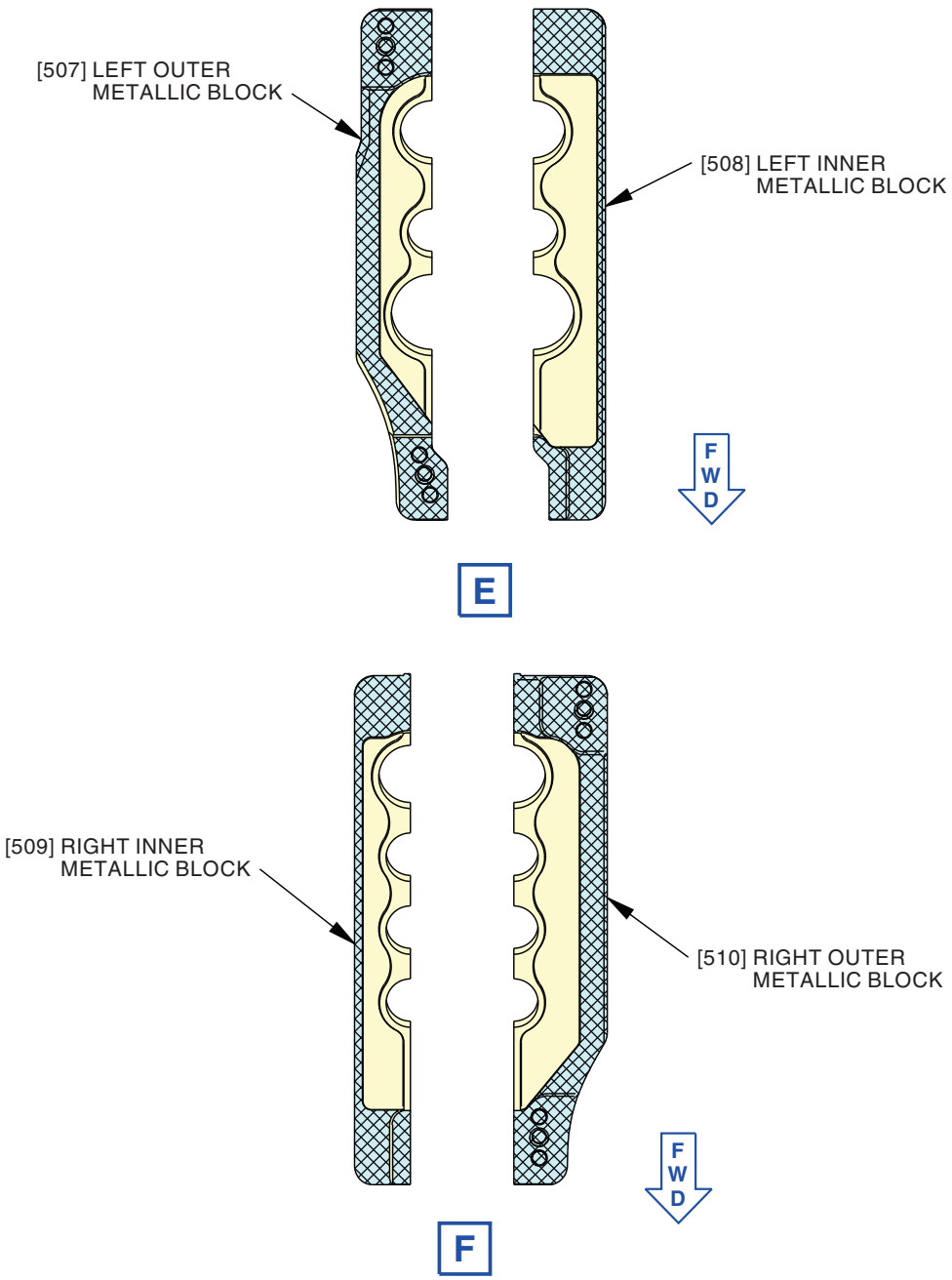
THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994662 S0000771309_V1

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 10 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**LEGEND:**

DAPCO 2100

**NOTE:**

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994663 S0000771310_V1

**Harness Support Block Installation**
**Figure 417/73-22-01-990-879-G00 (Sheet 11 of 11)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-400-809-G00**

## 19. Electronic Engine Control (EEC) Electrical Harness (HJ8B) Installation

(Figure 415, Figure 416, and Figure 417)

### A. General

(1) This task gives the instructions on how to install the electronic engine control (EEC) electrical harness (HJ8B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-00-00-800-804-G00 | Power Plant Test Reference Table (P/B 501) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 72-23-04-400-801-G00 | Fan Frame Module Segment Shrouds Installation (P/B 401) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

### D. Consumable Materials

| Reference | Description | Specification |
|---|---|---|
| A50396 | Sealant - Dapco 2200 Primerless Silicone Firewall Sealant | BMS5-63 Type II Class B-1/2 |
| A50483 [CP2927] | Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant) | |
| D00599 [CP2442] | Oil - Engine | |
| D00601 [CP2101] | Vaseline - Graphite Mineral | |
| G51328 [CP2674] | Cloth - Lint-Free | |

### E. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### F. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**G. Electronic Engine Control (EEC) Electrical Harness (HJ8B) Installation**

SUBTASK 73-22-01-210-007-G00

(1) During the installation, make sure that EEC electrical harness (HJ8B) [378] is not stressed or damaged.

SUBTASK 73-22-01-210-008-G00

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

SUBTASK 73-22-01-420-019-G00

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(3) Install the EEC electrical harness (HJ8B) [378] as follows (TASK 70-00-01-910-803-G00):

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Make sure to apply silicone sealant during an assembly of harnesses and all related parts.

1) Apply sealant on edges that are the interfaces of parts.

(d) Install the EEC electrical harness (HJ8B) [378] in a correct position with the clips [377] along its routing.

(e) Install the EEC electrical harness (HJ8B) [378] on the harness strut (6:00 o'clock) [104] as follows:

**PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288**

1) Install the harness support block [504], harness support block [505], and harness support block [506] on the harness strut (6:00 o'clock) [104] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the four nuts [110] and two nuts [112].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses (View E, Figure 417).

c) Make sure that the EEC electrical harness (HJ8B) [378] and harnesses are in the correct position on the harness support block [504], harness support block [505], and harness support block [506] (View D, Figure 417).



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)**

d) Install the two bolts [113] and two nuts [112] to assemble the harness support block [504], harness support block [505], and harness support block [506] in the correct position with the harnesses.

e) Tighten the nuts [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [112] with a lint-free cloth, G51328 [CP2674].

**CAUTION** MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

g) Make sure that there is minimum amount of the squeeze-out on the lower side.

h) Install the harness support block [504], harness support block [505], and harness support block [506] in the correct position on the harness strut (6:00 o'clock) [104].

**CAUTION** MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

i) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104] (View B, Figure 417).

j) Install the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

k) Tighten the nuts [110] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

l) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [110] with a lint-free cloth, G51328 [CP2674].

**CAUTION** MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

m) Make sure that there is minimum amount of the squeeze-out on the lower side.

n) Apply more sealant to form a correct fillet seal around the contour of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

**CAUTION** MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

o) Clean the four nuts [110] and four bolts [111] that will have sealant applied to them.



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)**

p) Apply Dapco 2100 sealant, A50483 [CP2927], to the heads of the four bolts [111].

**PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288**

2) Remove any remaining of old sealant residue left on the harnesses or harness strut (6:00 o'clock) [104].

3) Install the left outer metallic block [507] and left inner metallic block [508] on the harness strut (6:00 o'clock) [104] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the two bolts [124] and two nuts [126].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507] and left inner metallic block [508].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507], left inner metallic block [508], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the left outer metallic block [507] and left inner metallic block [508] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the two nuts [126] with a lint-free cloth, G51328 [CP2674].

g) Install the left outer metallic block [507] and left inner metallic block [508] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

h) Install the two bolts [124] that attach the left outer metallic block [507] and left inner metallic block [508] to the harness strut (6:00 o'clock) [104].

i) Tighten the two bolts [124] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

j) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the two bolts [124] with a lint-free cloth, G51328 [CP2674].

k) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

**CAUTION** MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].



ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION: MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

m) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

n) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of left outer metallic block [507] and left inner metallic block [508].

4) Install the new right inner metallic block [509] and right outer metallic block [510] on the harness strut (6:00 o'clock) [104] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the two bolts [124] and two nuts [126].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509] and right outer metallic block [510].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the right inner metallic block [509] and right outer metallic block [510] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the two nuts [126] with a lint-free cloth, G51328 [CP2674].

g) Install the right inner metallic block [509] and right outer metallic block [510] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

h) Install the two bolts [124] that attach the right inner metallic block [509] and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

i) Tighten the two bolts [124] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

j) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the two bolts [124] with a lint-free cloth, G51328 [CP2674].

k) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.



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CAUTION: MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the visible faces of the two bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

CAUTION: MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

m) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

n) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of right inner metallic block [509] and right outer metallic block [510].

**PTF ALL**

5) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the five bolts [106] and two nuts [107].

b) Apply Dapco 2100 sealant, A50483 [CP2927] to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

c) Make sure that the EEC electrical harness (HJ8B) [378] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] (View E, Figure 416).

d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [107] with a lint-free cloth, G51328 [CP2674].

g) Make sure that the fillet seal of the squeeze-out is on the forward facing side.



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CAUTION: MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

h) Make the sealant squeeze-out flush with the surface on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 416).

i) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

j) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 outer fitting [109] (View C, Figure 417).

k) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109].

l) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

m) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the bolts [106] with a lint-free cloth, G51328 [CP2674].

CAUTION: MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

n) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 417).

o) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

CAUTION: MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

p) Clean the nut plate side of the bolts [106] and nut plates.

q) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

(f) Attach blanket [190] to the strut 5 outer fitting [109].

1) Apply Dapco 2100 sealant, A50483 [CP2927], between the blanket [190] and strut 5 outer fitting [109] (View B, Figure 416).

NOTE: Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

**PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410**

(g) Apply engine oil, D00599 [CP2442], on the pressure faces and threads of the bolt [384].



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**PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410 (Continued)**

(h) Install the bolt [384] and clamp [385] that attach the EEC electrical harness (HJ8B) [378] to the forward stator case bracket.

(i) Tighten the bolt [384] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

**PTF ALL**

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION** USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(j) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ8B) [378] as follows:

1) Connect the EEC electrical harness (HJ8B) [378] connector to the T3 sensor channel B receptacle [387].

2) Connect the EEC electrical harness (HJ8B) [378] connector to the EGT harness 2 (right) receptacle [386].

3) Connect the EEC electrical harness (HJ8B) [378] connector to the No. 1 bearing acclerometer receptacle [382].

(k) Install the bolt [380] and clamp [381] that attach the EEC electrical harness (HJ8B) [378] to the fan case bracket.

(l) Tighten the bolt [380] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION** USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(m) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ8B) [378] as follows:

(n) Connect the EEC electrical harness (HJ8B) [378] to the EEC HJ8A harness receptacle [145].

(o) Connect the EEC electrical harness (HJ8B) [378] connector to the EEC channel B J8 receptacle [379].

(p) Make sure that the work area is clean of all tools and other items.

**SUBTASK 73-22-01-420-020-G00**

(4) Install the segment shroud 5 [305] (TASK 72-23-04-400-801-G00).



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SUBTASK 73-22-01-420-021-G00

(5) Install the segment shroud 2 [68] (TASK 72-23-04-400-801-G00).

SUBTASK 73-22-01-420-022-G00

(6) Install the segment shroud 1 [304] (TASK 72-23-04-400-801-G00).

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-030-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-069-G00

(2) Remove the safety tags and close these circuit breakers:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |



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(Continued)

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-070-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |

SUBTASK 73-22-01-860-071-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-860-072-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**I. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ8B) Installation Test**

SUBTASK 73-22-01-730-009-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

**——— END OF TASK ———**

**TASK 73-22-01-000-810-G00**

## 20. Electronic Engine Control (EEC) Electrical Harness (W4A) Removal

(Figure 418)

**A. General**

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (W4A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

**B. References**

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |



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(Continued)

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

**D. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 424 | Right Fan Cowl, Engine 2 |

**F. Prepare for the Removal**

SUBTASK 73-22-01-860-084-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-085-G00

(2) Open these applicable circuit breakers and install safety tags:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-086-G00

(3) For the right engine, open this circuit breaker and install safety tag:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |



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SUBTASK 73-22-01-860-087-G00

(4) For the left engine, open this circuit breaker and install safety tag:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-010-035-G00

(5) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 424 | Right Fan Cowl, Engine 2 |

## G. Electronic Engine Control (EEC) Electrical Harness (W4A) Removal

SUBTASK 73-22-01-020-025-G00

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (W4A) [400] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (W4A) [400] connector from the EEC channel A J4 receptacle [401].

(b) Disconnect the EEC electrical harness (W4A) [400] connector from the EEC channel A J1 receptacle [402].

(c) Disconnect the EEC electrical harness (W4A) [400] connector from the fan cowl support beam.

(d) Remove the nut [403], bolt [404], and clamp [405] that attach the EEC electrical harness (W4A) [400] to the engine.

(e) Remove the EEC electrical harness (W4A) [400] from the four clips [406] along its routing.

(f) Remove the bolt [407], clip nut [409], and clamp [408] that attach the EEC electrical harness (W4A) [400] to the engine.

(g) Carefully remove the EEC electrical harness (W4A) [400] from the engine.

(h) Install the protective covers on all of the harness connectors and the component receptacles.

**END OF TASK**



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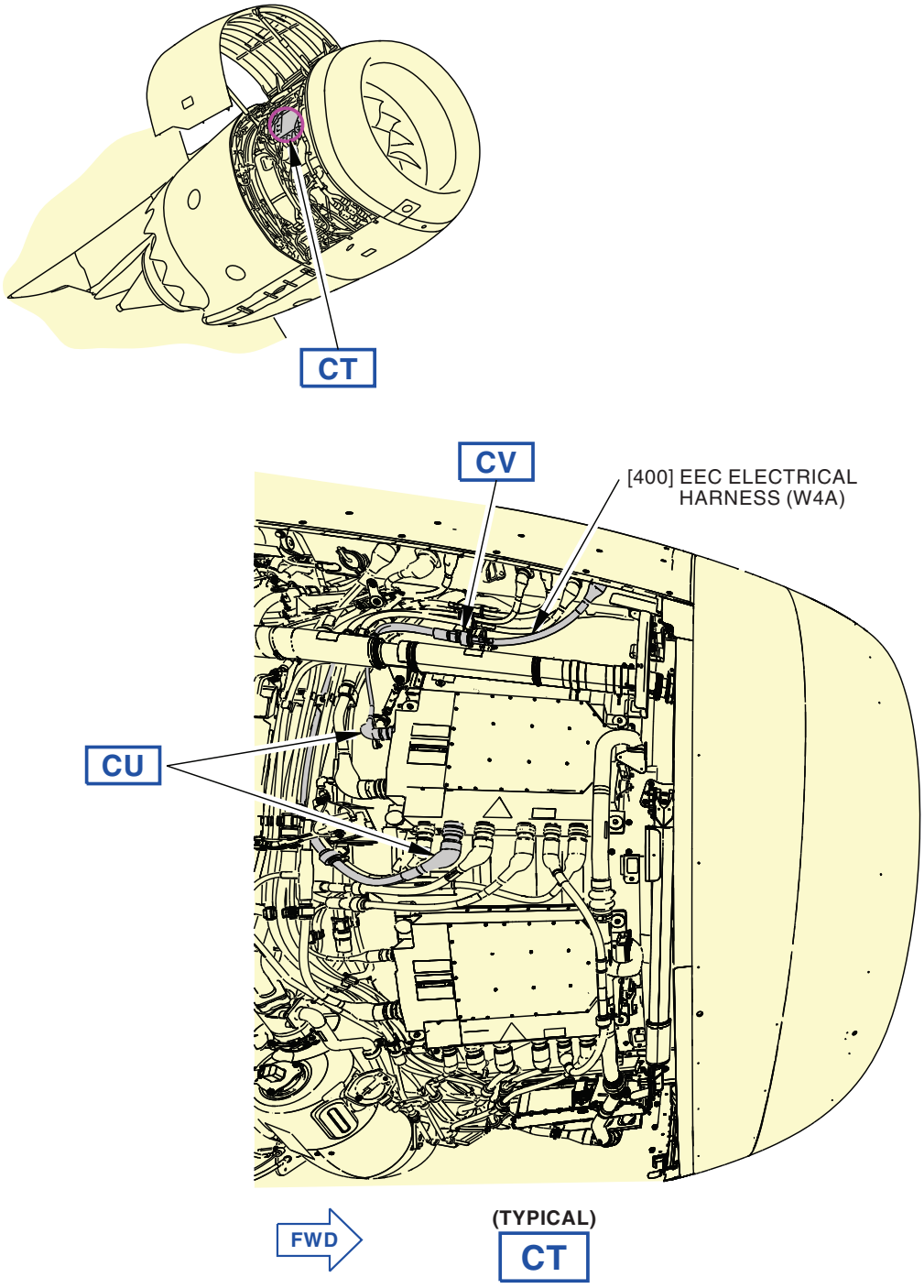


**EEC Electrical Harness (W4A) Installation**
**Figure 418/73-22-01-990-862-G00 (Sheet 1 of 2)**



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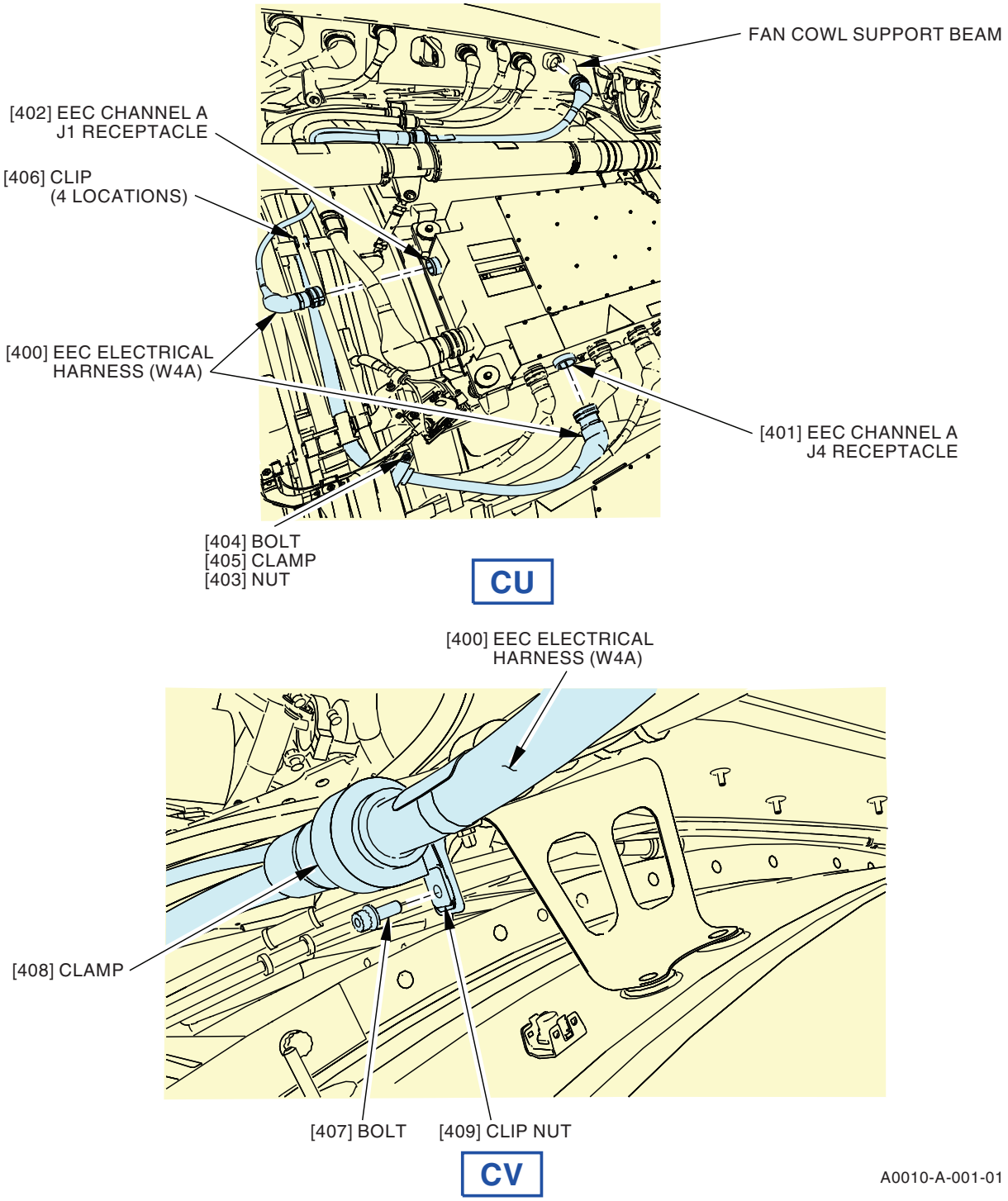


**EEC Electrical Harness (W4A) Installation**
**Figure 418/73-22-01-990-862-G00 (Sheet 2 of 2)**



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**TASK 73-22-01-400-810-G00**

## 21. Electronic Engine Control (EEC) Electrical Harness (W4A) Installation

(Figure 418)

### A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (W4A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-00-00-800-804-G00 | Power Plant Test Reference Table (P/B 501) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### E. Access Panels

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 424 | Right Fan Cowl, Engine 2 |

### F. Electronic Engine Control (EEC) Electrical Harness (W4A) Installation

**SUBTASK 73-22-01-420-025-G00**

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (W4A) [400] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.



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(c) Install the bolt [407], clip nut [409], and clamp [408] that attach the EEC electrical harness (W4A) [400] to the engine.

(d) Tighten the bolt [407] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(e) Install the EEC electrical harness (W4A) [400] in a correct position with the four clips [406] along its routing.

(f) Install the nut [403], bolt [404], and clamp [405] that attach the EEC electrical harness (W4A) [400] to the engine.

(g) Tighten the nut [403] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (W4A) [400] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (W4A) [400] connector to the fan cowl support beam.

2) Connect the EEC electrical harness (W4A) [400] connector to the EEC channel A J1 receptacle [402].

3) Connect the EEC electrical harness (W4A) [400] connector to the EEC channel A J4 receptacle [401].

(i) Make sure that the work area is clean of all tools and other items.

## G. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-22-01-410-034-G00**

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 424 | Right Fan Cowl, Engine 2 |

**SUBTASK 73-22-01-860-088-G00**

(2) Remove the safety tags and close these circuit breakers:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |



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**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-089-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |

SUBTASK 73-22-01-860-090-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-860-091-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**H. Electronic Engine Control Electrical (EEC) Electrical Harness (W4A) Installation Test**

SUBTASK 73-22-01-730-010-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

**——— END OF TASK ———**

**TASK 73-22-01-000-811-G00**

## 22. Electronic Engine Control (EEC) Electrical Harness (W4B) Removal

(Figure 419)

**A. General**

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (W4B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.



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**B. References**

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

**D. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**F. Prepare for the Removal**

SUBTASK 73-22-01-860-092-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-093-G00

(2) Open these applicable circuit breakers and install safety tags:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |



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**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-094-G00

(3) For the right engine, open this circuit breaker and install safety tag:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |

SUBTASK 73-22-01-860-095-G00

(4) For the left engine, open this circuit breaker and install safety tag:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-010-033-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).



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1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Electronic Engine Control (EEC) Electrical Harness (W4B) Removal

SUBTASK 73-22-01-020-026-G00

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (W4B) [420] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (W4B) [420] connector from the EEC channel B J4 receptacle [421].

(b) Disconnect the EEC electrical harness (W4B) [420] connector from the EEC channel B J1 receptacle [422].

(c) Disconnect the EEC electrical harness (W4B) [420] connector from the fan cowl support beam.

(d) Remove the bolt [423], clip nut [426], and clamp [424] that attach the EEC electrical harness (W4B) [420] to the engine.

(e) Remove the EEC electrical harness (W4B) [420] from the six clips [425] along its routing.

(f) Carefully remove the EEC electrical harness (W4B) [420] from the engine.

(g) Install the protective covers on all of the harness connectors and the component receptacles.

**END OF TASK**



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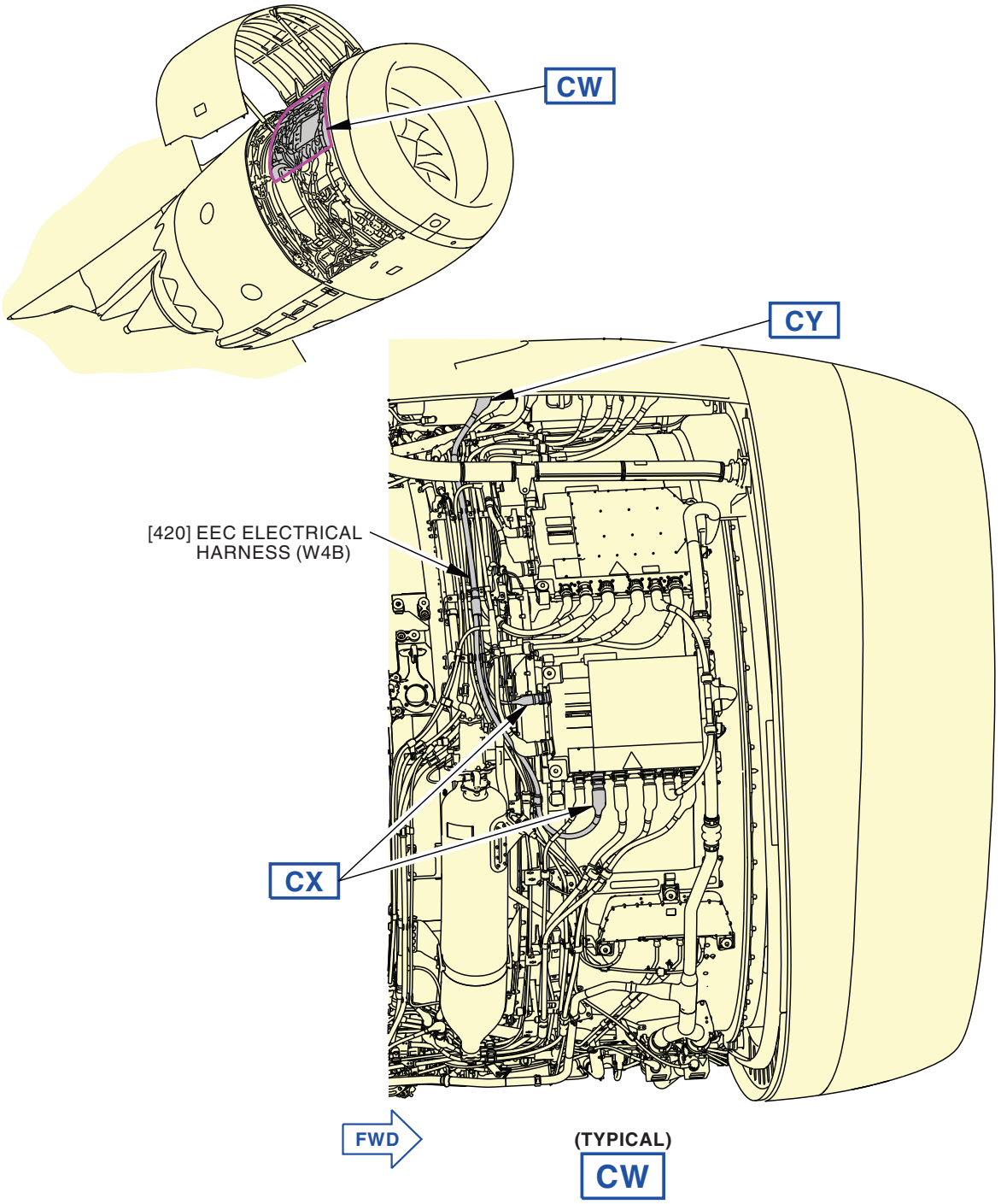


**EEC Electrical Harness (W4B) Installation**
**Figure 419/73-22-01-990-863-G00 (Sheet 1 of 2)**



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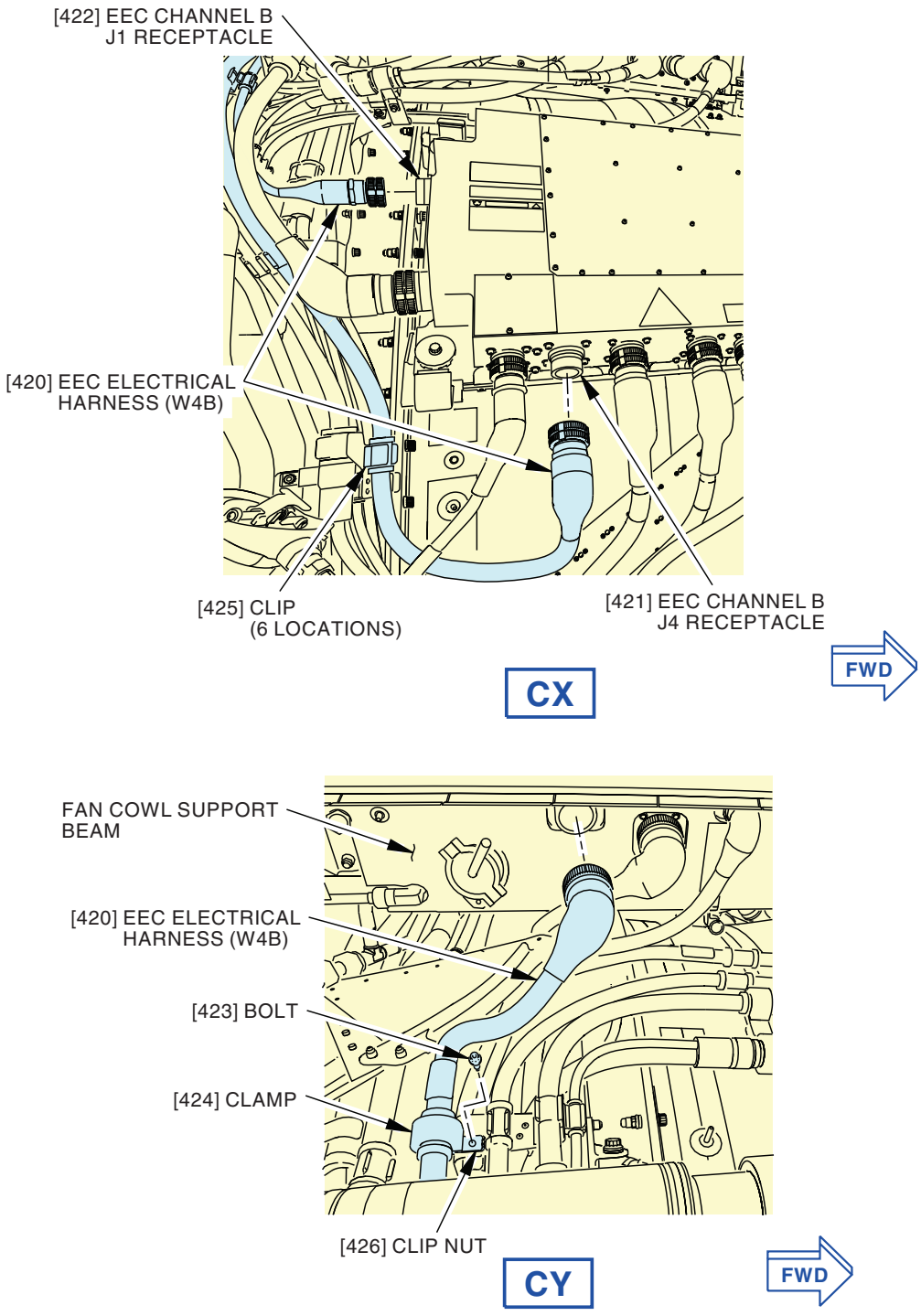


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**EEC Electrical Harness (W4B) Installation**
**Figure 419/73-22-01-990-863-G00 (Sheet 2 of 2)**



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**TASK 73-22-01-400-811-G00**

## 23. Electronic Engine Control (EEC) Electrical Harness (W4B) Installation

(Figure 419)

### A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (W4B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-00-00-800-804-G00 | Power Plant Test Reference Table (P/B 501) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

### F. Electronic Engine Control (EEC) Electrical Harness (W4B) Installation

**SUBTASK 73-22-01-420-026-G00**

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.



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**(CAUTION PRECEDES)**

**CAUTION** USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (W4B) [420] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the bolt [423], clip nut [426], and clamp [424] that attach the EEC electrical harness (W4B) [420] to the engine.

(d) Tighten the bolt [423] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(e) Install the EEC electrical harness (W4B) [420] in a correct position with the six clips [425] along its routing.

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION** USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(f) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (W4B) [420] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (W4B) [420] connector to the fan cowl support beam.

2) Connect the EEC electrical harness (W4B) [420] connector to the EEC channel B J1 receptacle [422].

3) Connect the EEC electrical harness (W4B) [420] connector to the EEC channel B J4 receptacle [421].

(g) Make sure that the work area is clean of all tools and other items.

## G. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-22-01-410-032-G00**

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |



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(Continued)

| Number | Name/Location |
|---|---|
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-096-G00

(2) Remove the safety tags and close these circuit breakers:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-097-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |

SUBTASK 73-22-01-860-098-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |



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SUBTASK 73-22-01-860-099-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**H. Electronic Engine Control Electrical (EEC) Electrical Harness (W4B) Installation Test**

SUBTASK 73-22-01-730-011-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

**——— END OF TASK ———**

**TASK 73-22-01-000-812-G00**

## 24. Electronic Engine Control (EEC) Electrical Harness (W2) Removal

(Figure 420 and Figure 421)

**A. General**

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harnesses (W2A and W2B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) Service Bulletin 737-CFM-LEAP-1B-75-0011 has two configurations:

- Configuration 1: deletion of blowers, air ducts and addition of W2 storage bracket
- Configuration 2: deletion of air ducts.

**B. References**

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

**D. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |



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(Continued)

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**F. Prepare for the Removal**

SUBTASK 73-22-01-860-100-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-101-G00

(2) Open these applicable circuit breakers and install safety tags:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-102-G00

(3) For the right engine, open this circuit breaker and install safety tag:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |



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SUBTASK 73-22-01-860-103-G00

(4) For the left engine, open this circuit breaker and install safety tag:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-010-034-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Electronic Engine Control (EEC) Electrical Harness (W2A and W2B) Removal

SUBTASK 73-22-01-020-027-G00

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.



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**(CAUTION PRECEDES)**

**CAUTION**

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (W2A) [430] connector from the EEC channel A J2 receptacle [432].

(b) Disconnect the EEC electrical harness (W2B) [431] connector from the EEC channel B J2 receptacle [433].

(c) Disconnect the EEC electrical harness (W2A) [430] connector from the fan cowl support beam.

(d) Disconnect the EEC electrical harness (W2B) [431] connector from the fan cowl support beam.

**PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-75-0010 AND PRE SB-CFM-LEAP-1B-75-0011 (CONFIG 1) AND PRE SB-CFM-LEAP-1B-72-0410**

(e) Disconnect the EEC electrical harness (W2A) [430] connector from the EEC cooling blower (FWD/channel A) [434].

(f) Disconnect the EEC electrical harness (W2B) [431] connector from the EEC cooling blower (AFT/channel B) [435].

**PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-75-0011 (CONFIG 1) AND PRE SB-CFM-LEAP-1B-72-0410**

(g) Disconnect the EEC electrical harness (W2A) [430] connector from the bracket assembly [516].

(h) Disconnect the EEC electrical harness (W2B) [431] connector from the bracket assembly [516].

**PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410**

(i) Disconnect the EEC electrical harness (W2B) [431] from the EEC electrical harness (HJ8A) [303] on the bracket [83].

**PTF ALL**

(j) Disconnect the EEC electrical harness (W2A) [430] connector from the ignition exciter 1 [436].

(k) Disconnect the EEC electrical harness (W2B) [431] connector from the ignition exciter 2 [437].

(l) Remove the nut [440], bolt [441], and clamps [442] that attach the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] to the engine.

(m) Remove the bolts [438], clip nuts [444], and clamps [439] that attach the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] to the engine.

(n) Remove the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] from the clips [443] along its routing.

(o) Remove blanket [190] from the strut 5 outer fitting [109] (View B, Figure 421).



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(p) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from the strut 5 outer fitting [109] as follows:

1) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].
2) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and strut 5 outer fitting [109].
3) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.
4) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].
5) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and around the harnesses.
6) Remove the harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].
7) Remove any remaining sealant residue.

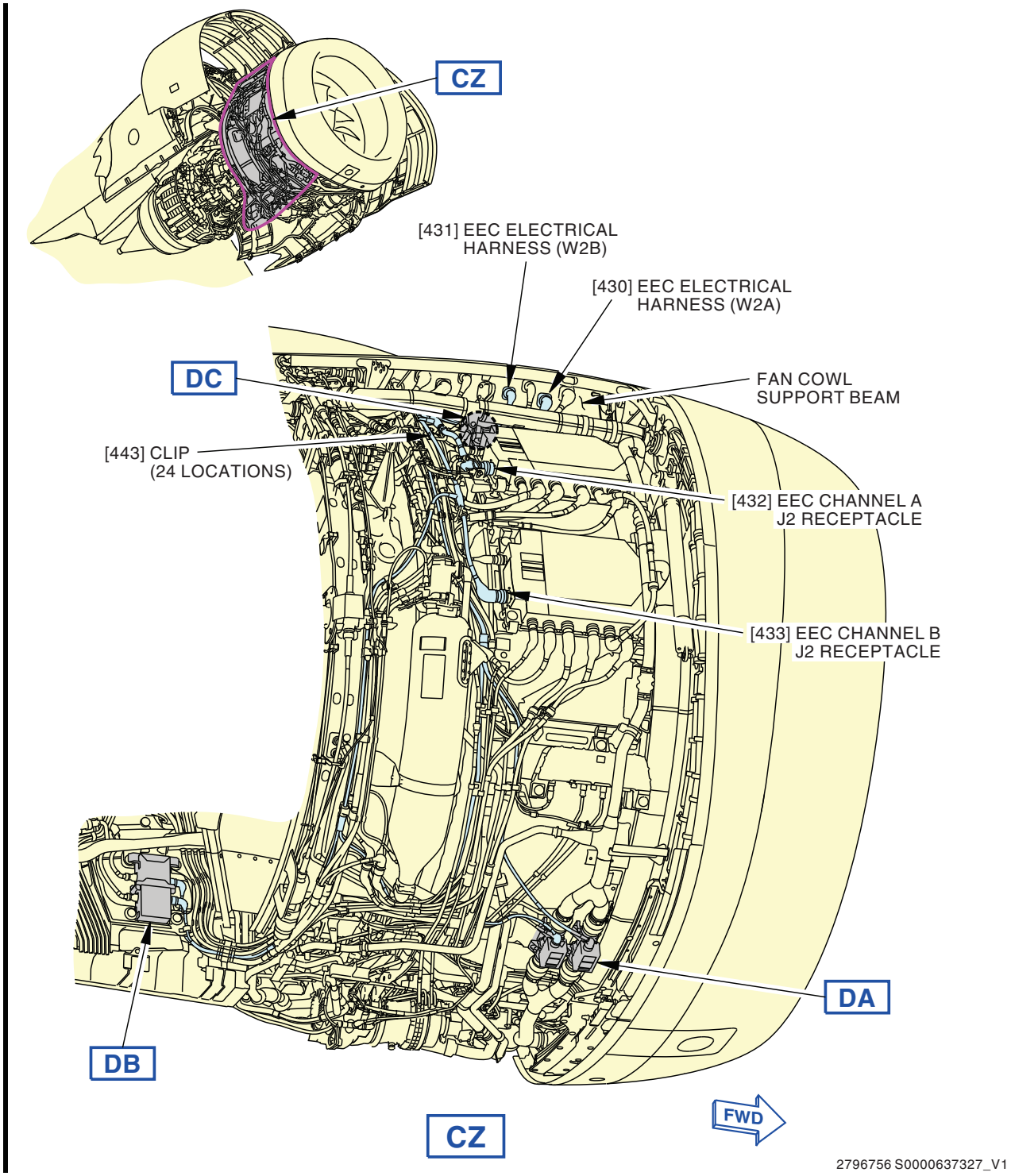
(q) Carefully remove the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] from the engine.

(r) Install the protective covers on all of the harness connectors and the component receptacles.

**——— END OF TASK ———**



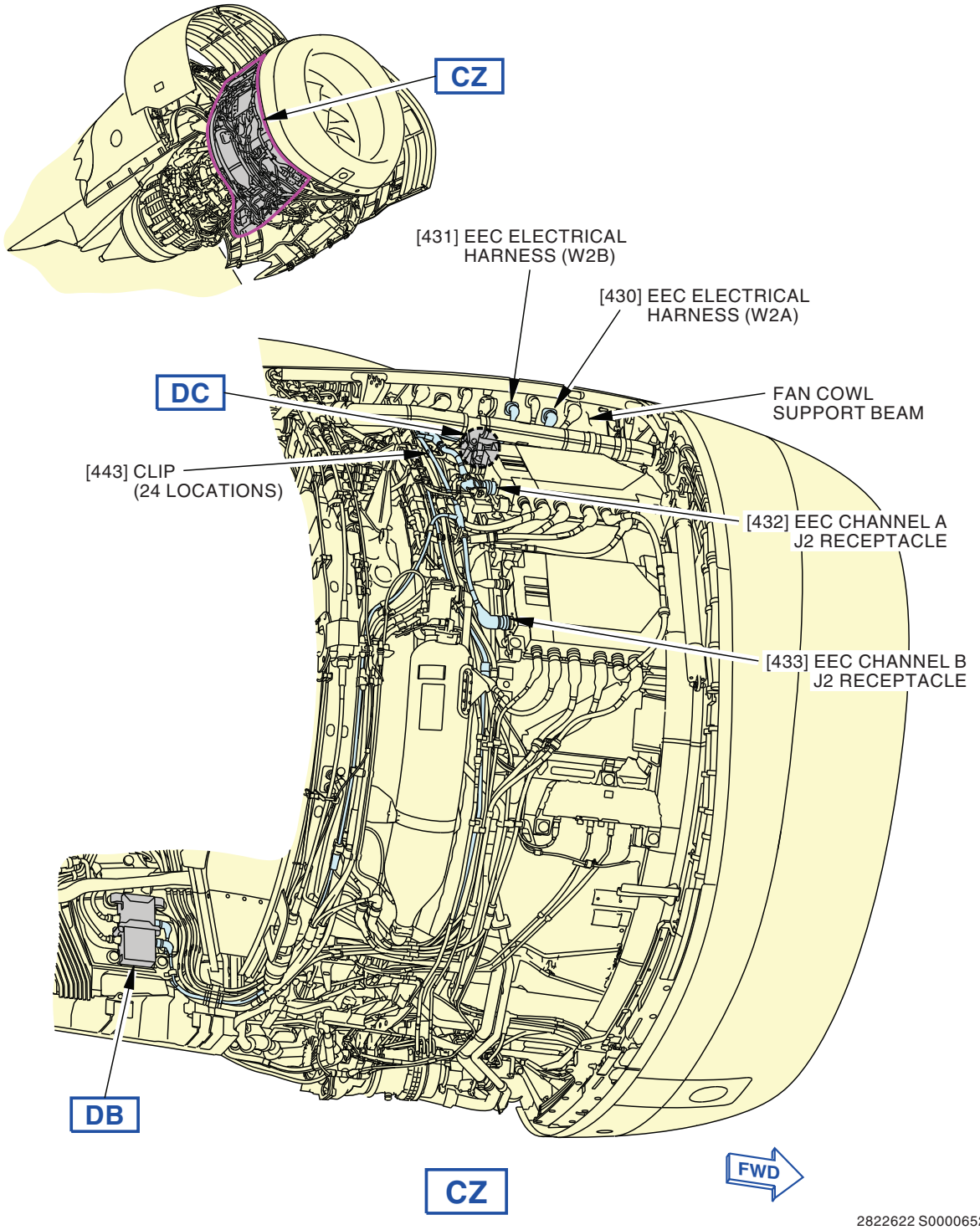
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 1 of 6)**



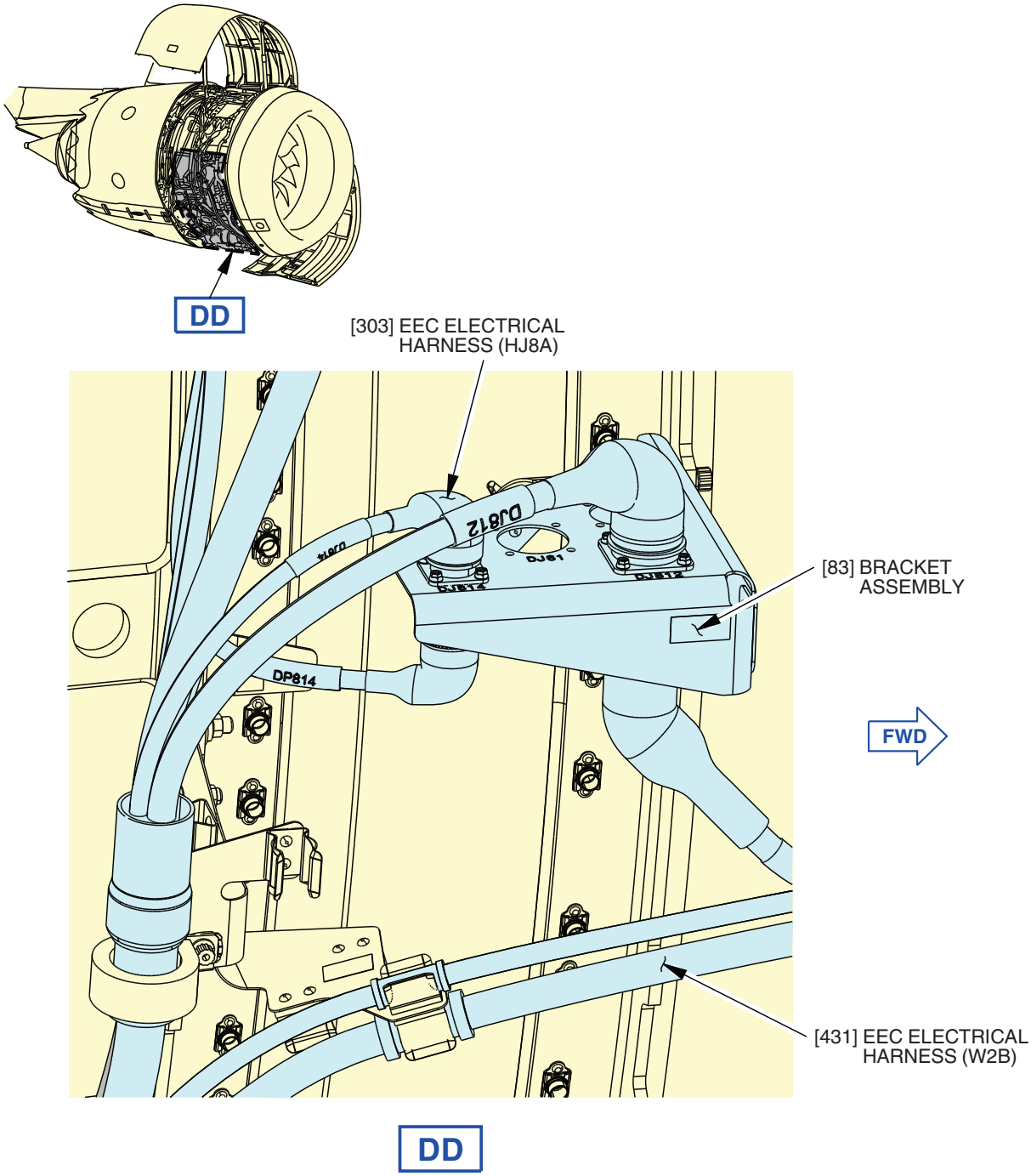
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 2 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

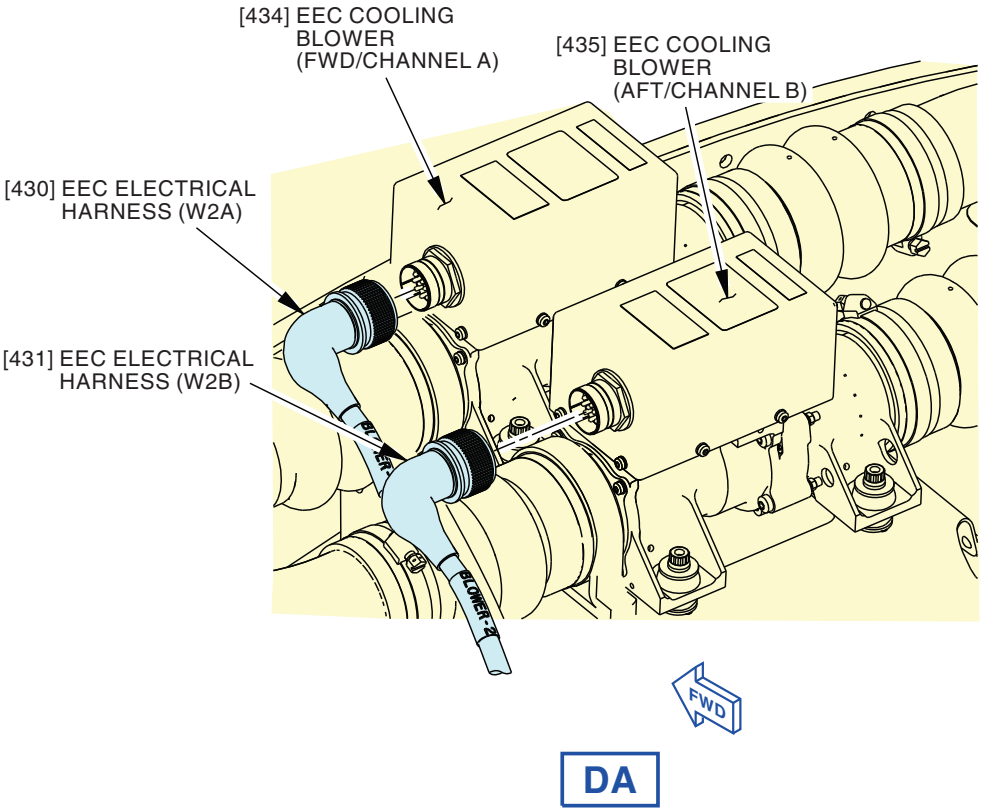
ICNLEAP1B732201D5882800159A00101

3132213 S0000862168_V1

**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 3 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

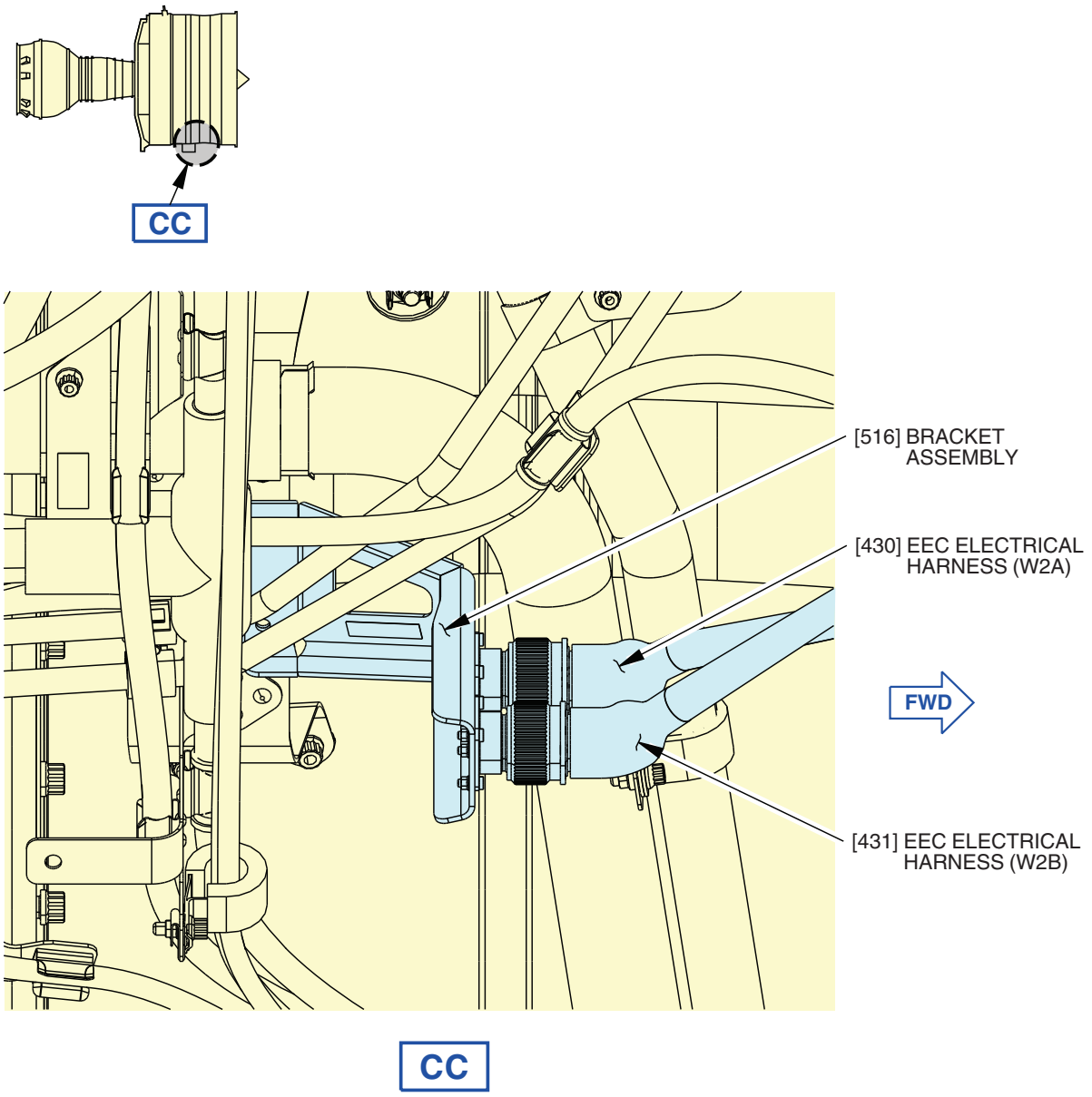



2797229 S0000637328_V1

**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 4 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

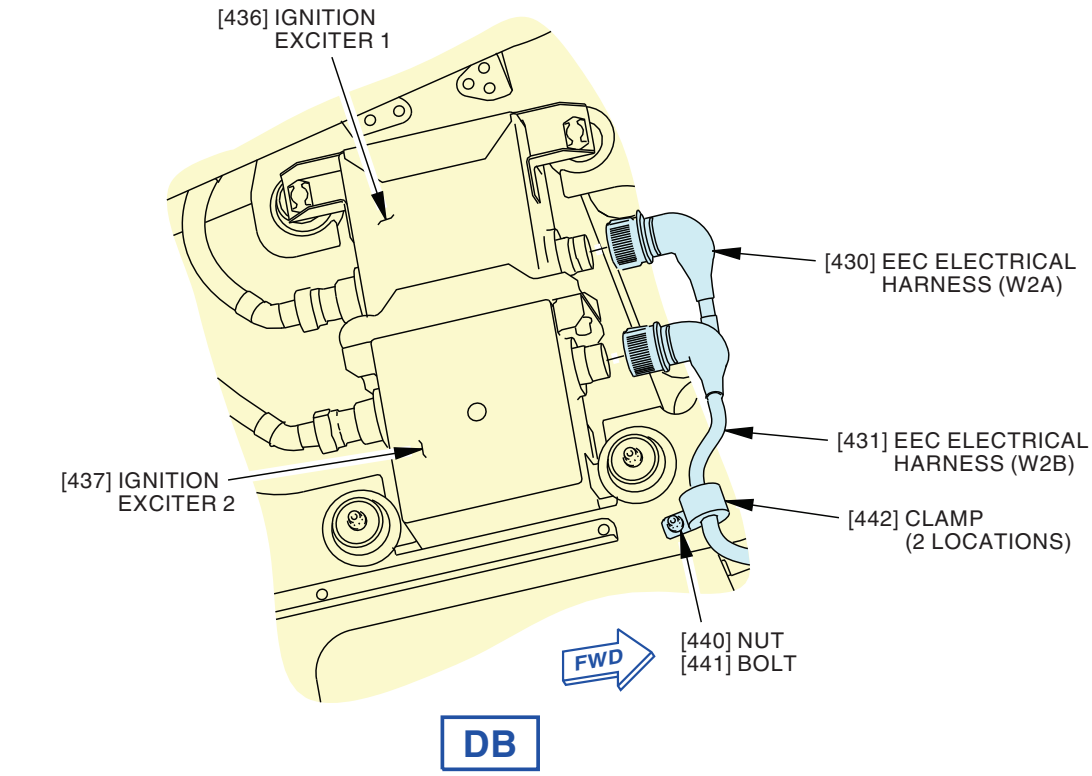
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

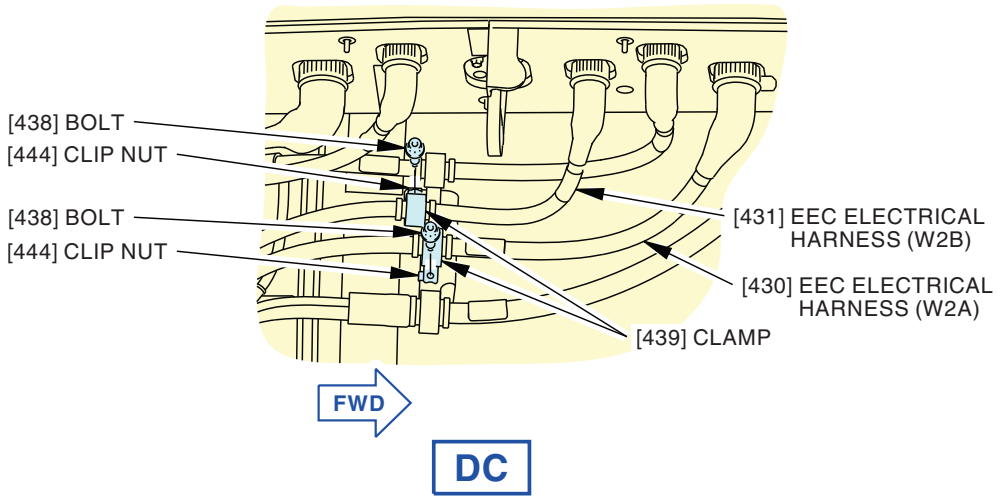
ICNLEAP1B732201D5882800166A00101
3132212 S0000862169_V1

**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 5 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



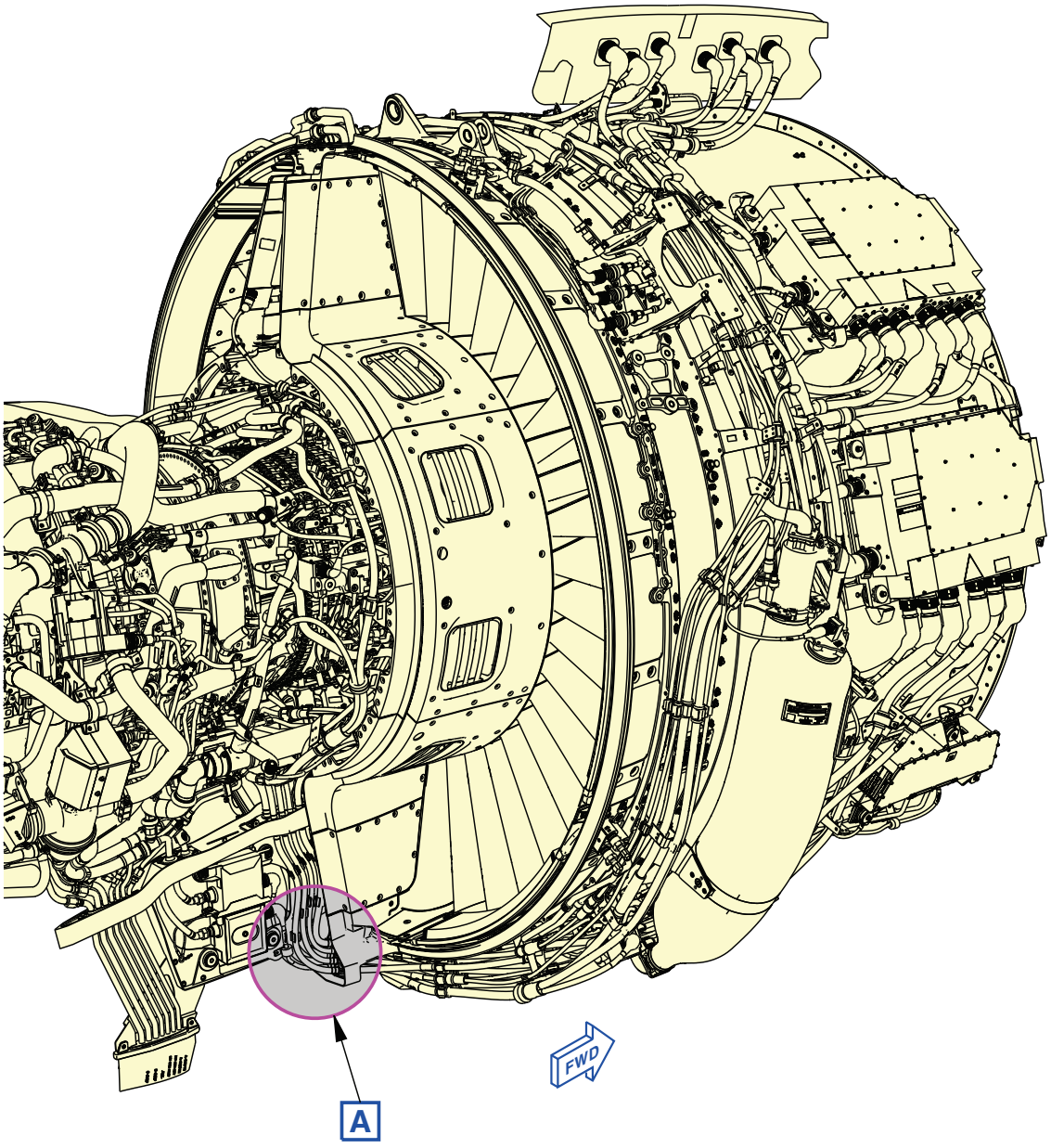




2797245 S0000637329_V2

**EEC Electrical Harness (W2A and W2B) Installation**
**Figure 420/73-22-01-990-864-G00 (Sheet 6 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

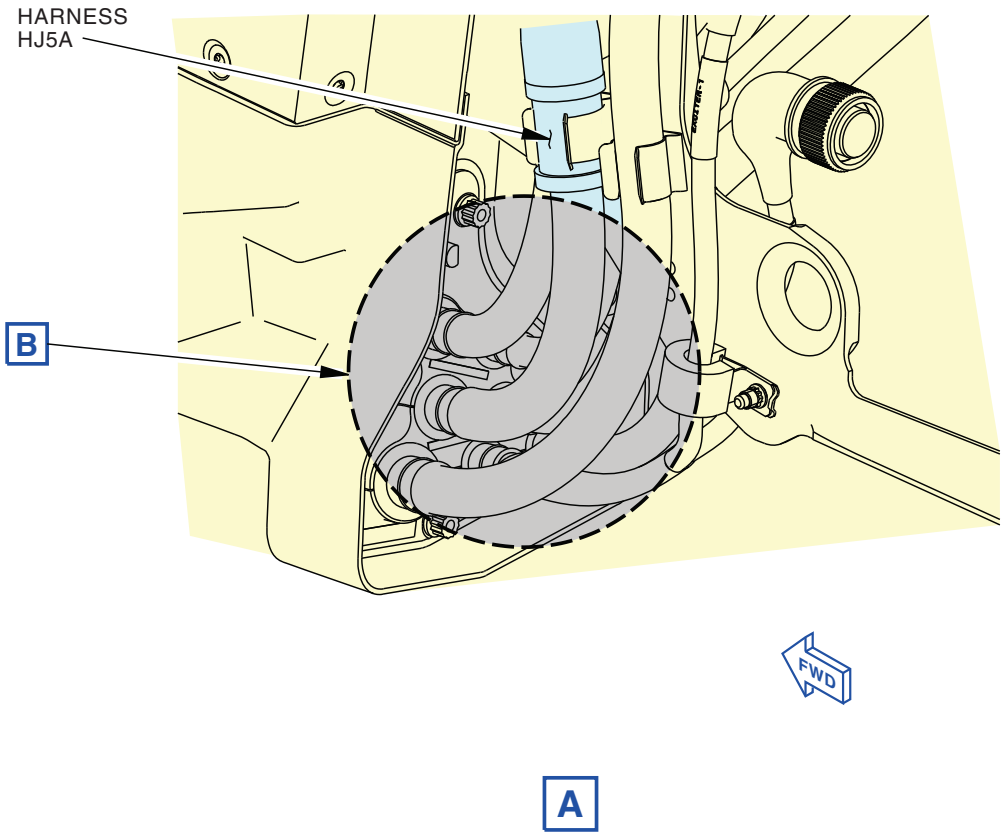



2937794 S0000713383_V3

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 1 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

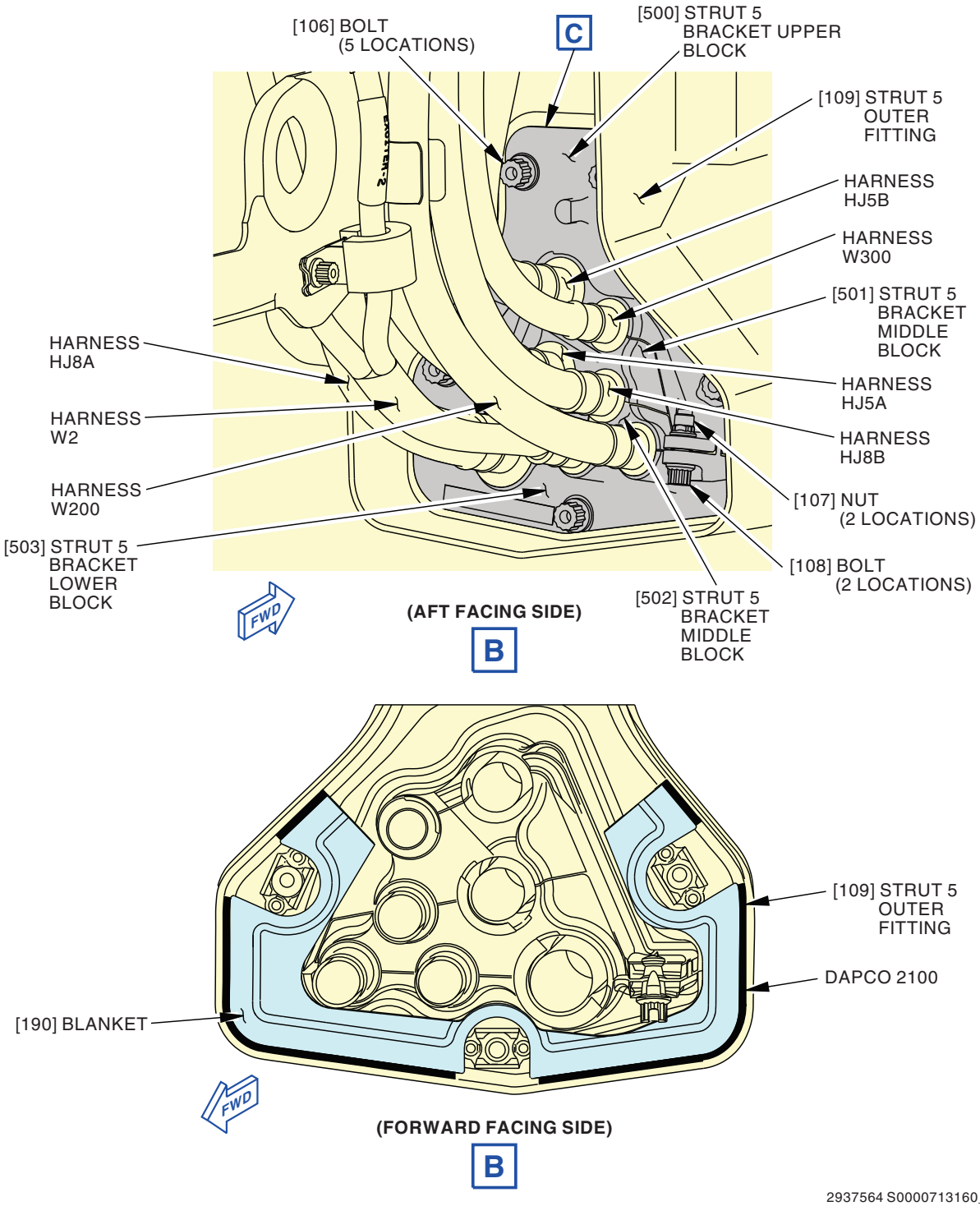



2937795 S0000713384_V2

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 2 of 9)**



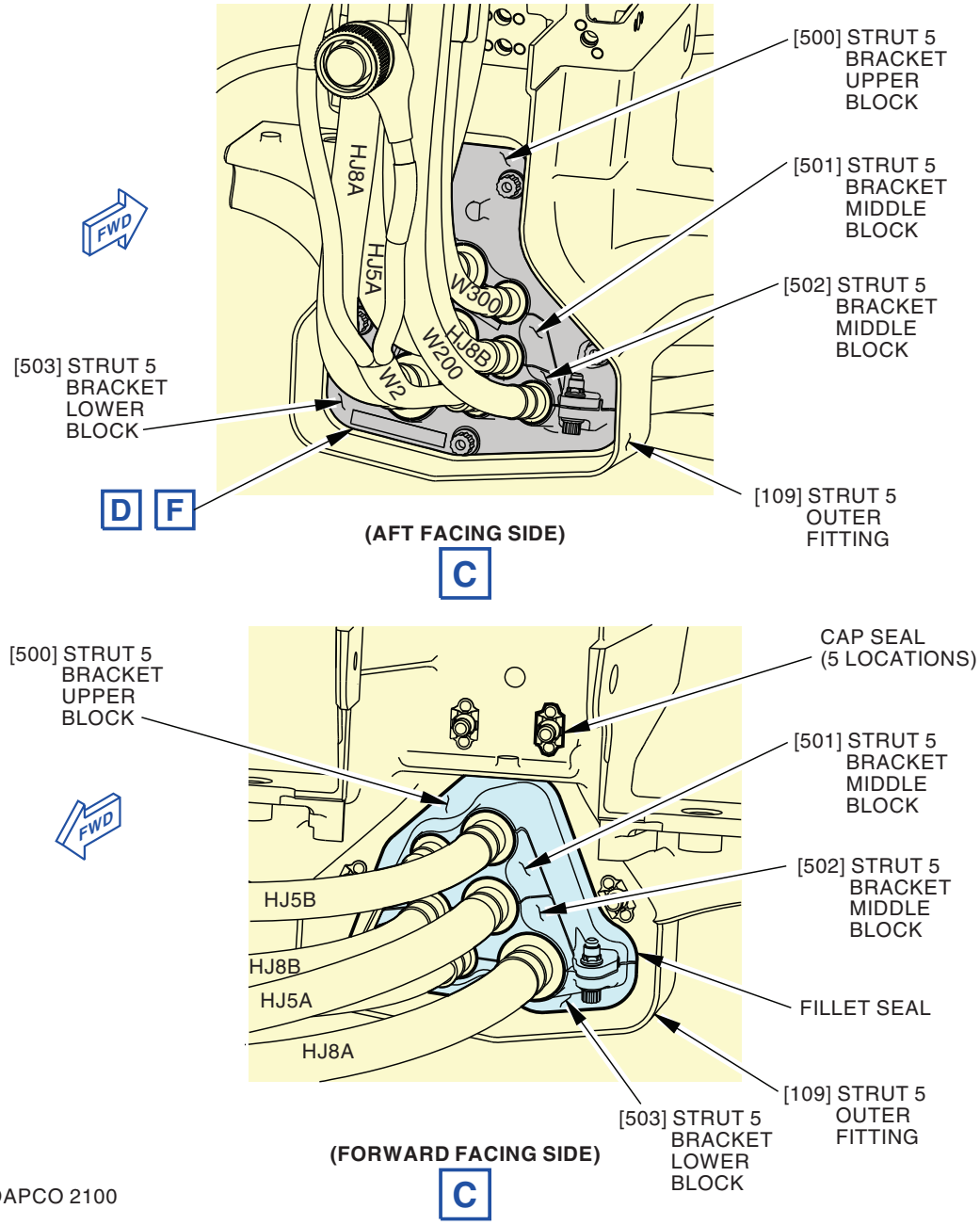
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 3 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

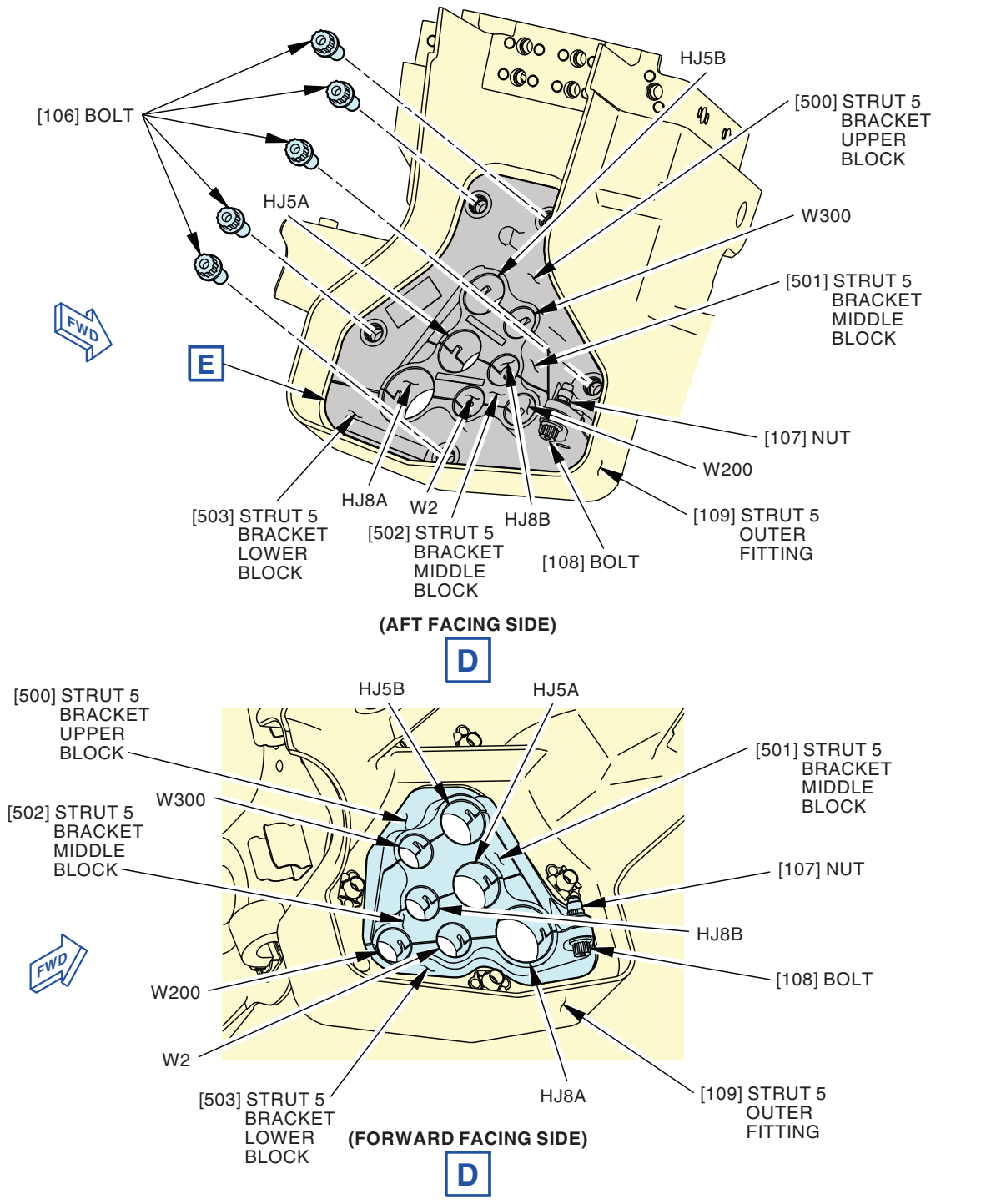
IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 4 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

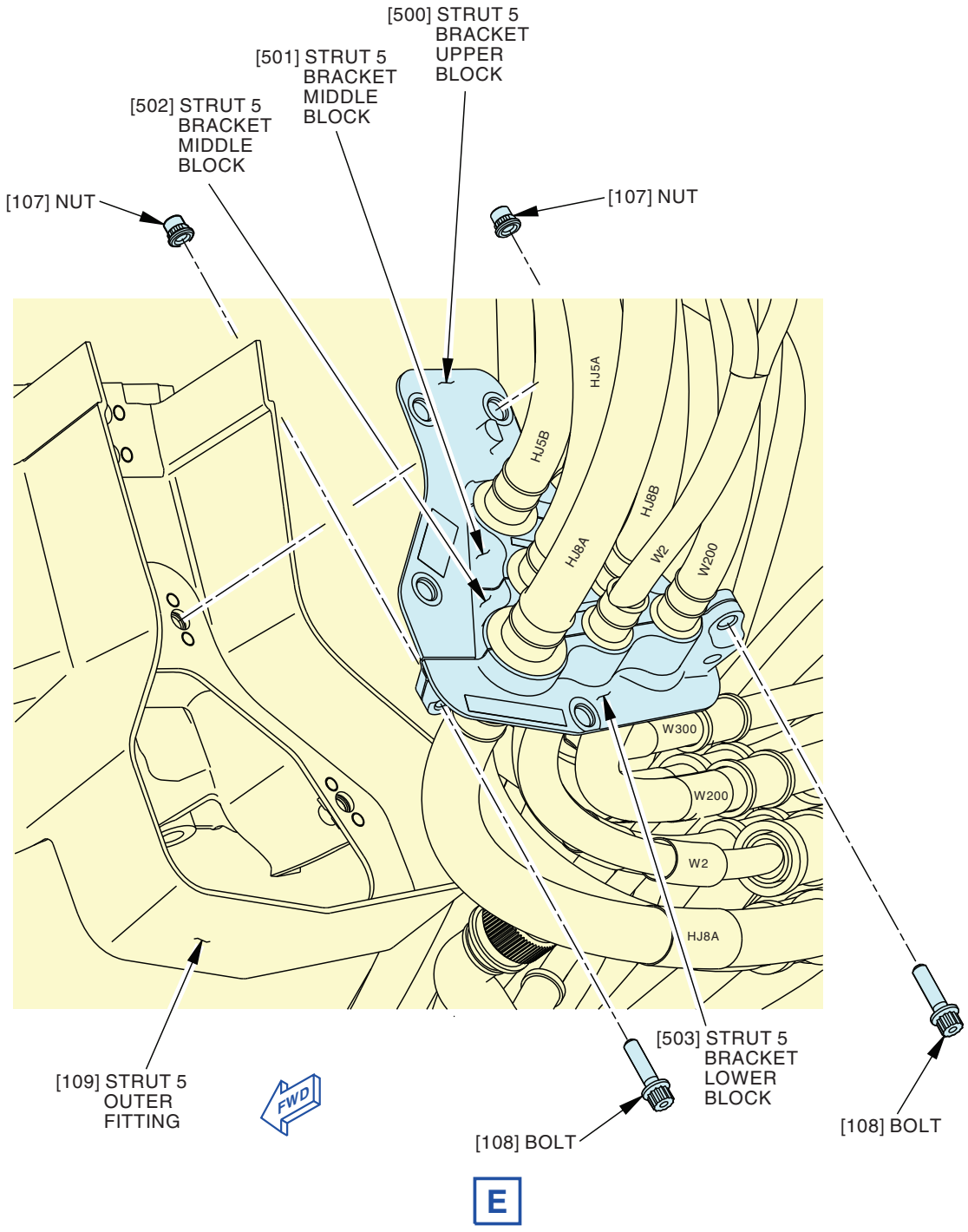
HARNESSES NOT SHOWN

2943980 S0000720369_V2

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 5 of 9)**



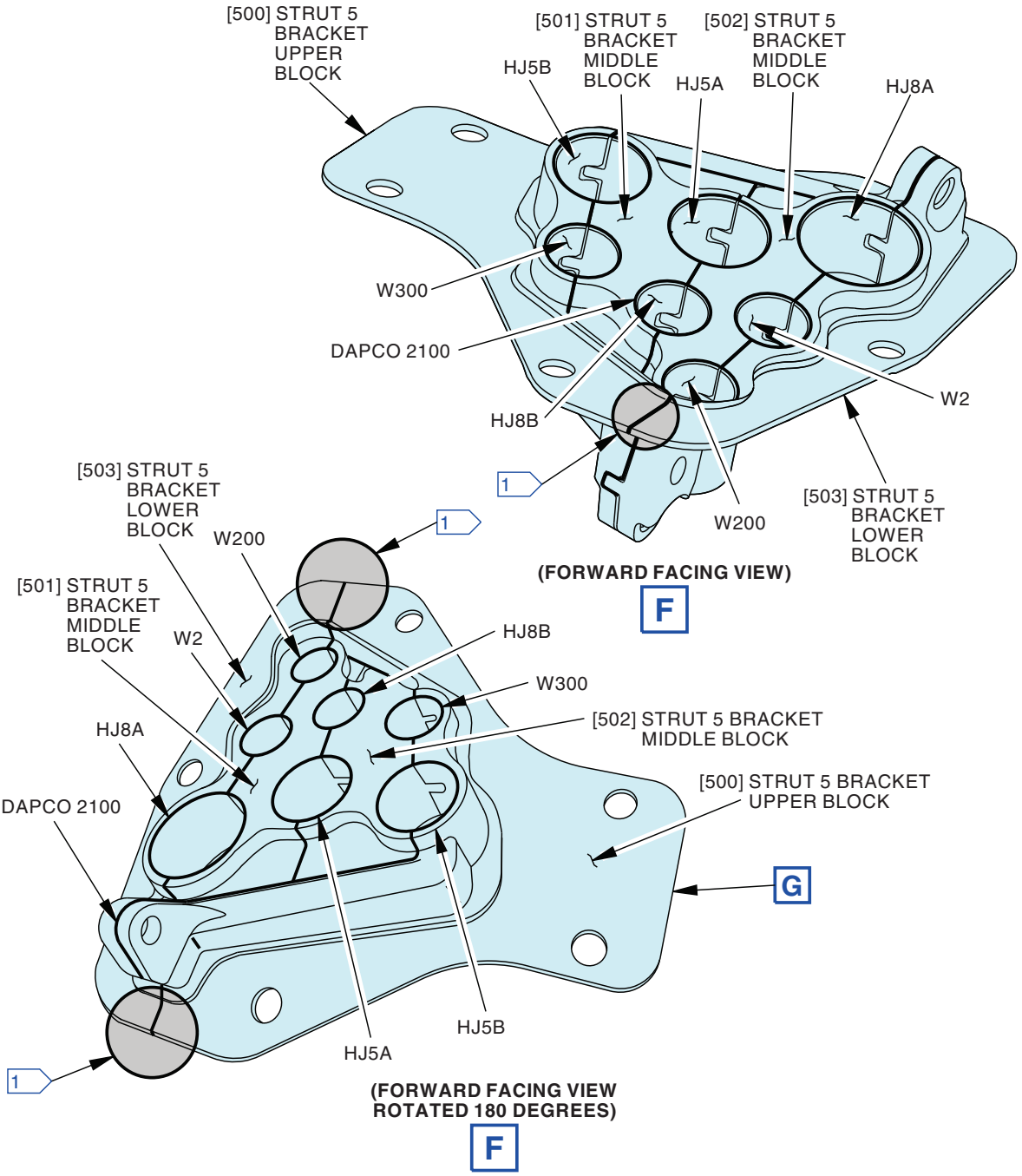
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 6 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

HARNESSES NOT SHOWN.

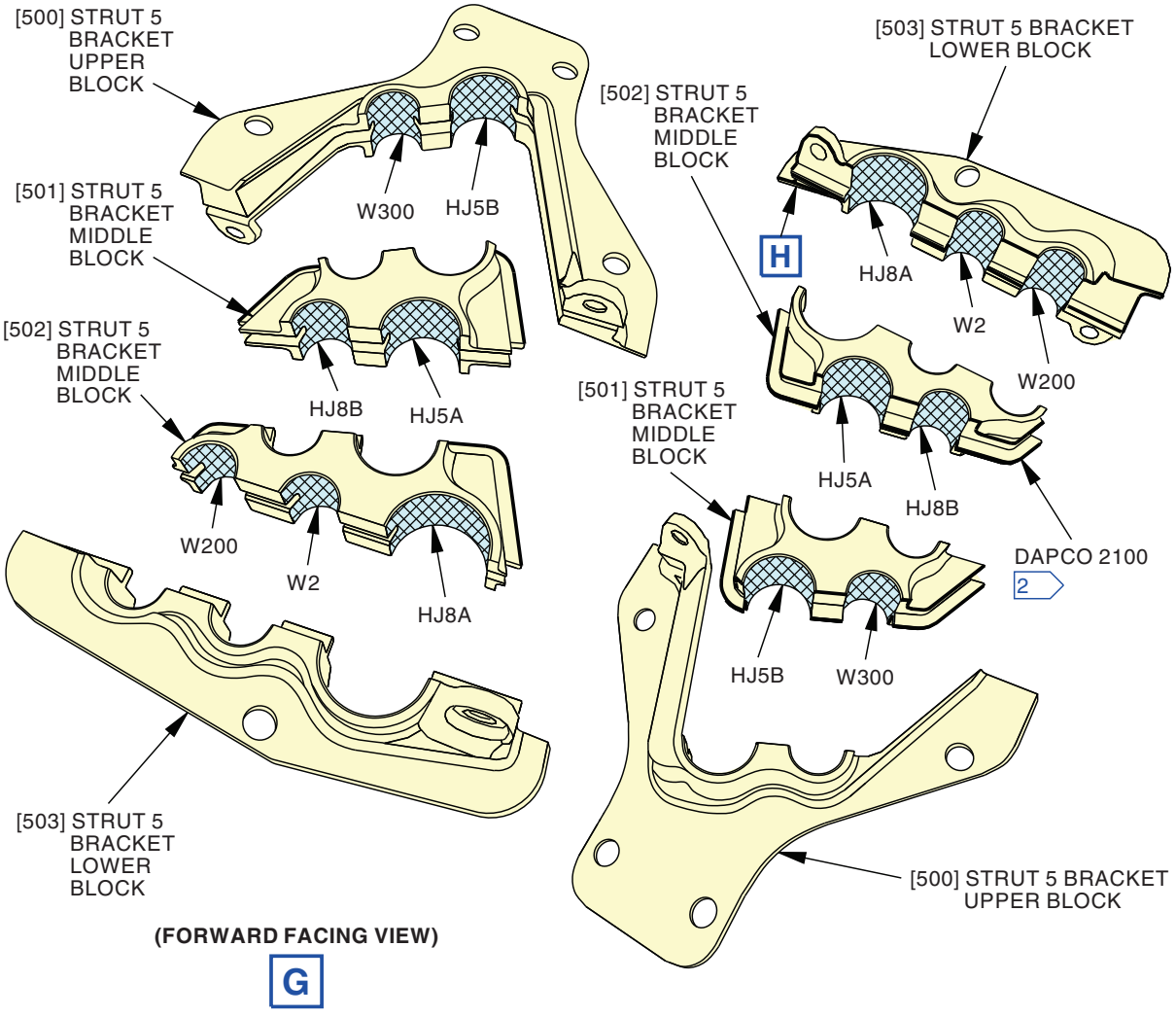
1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 7 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

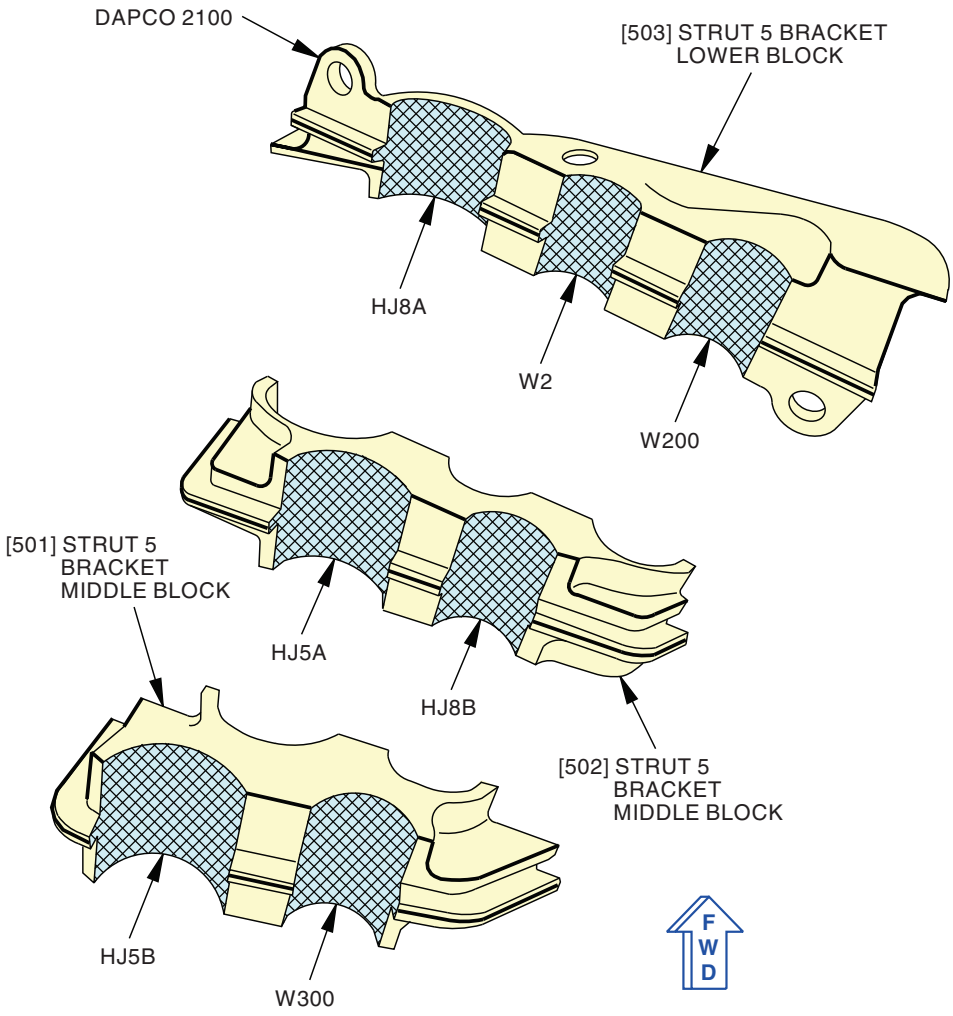
2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 8 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Strut Bracket Block Installation**
**Figure 421/73-22-01-990-880-G00 (Sheet 9 of 9)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-400-812-G00**

## 25. Electronic Engine Control (EEC) Electrical Harness (W2) Installation

(Figure 420 and Figure 421)

### A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (W2A and W2B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

(4) Service Bulletin 737-CFM-LEAP-1B-75-0011 has two configurations:

- Configuration 1: deletion of blowers, air ducts and addition of W2 storage bracket
- Configuration 2: deletion of air ducts.

### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-00-00-800-804-G00 | Power Plant Test Reference Table (P/B 501) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

### D. Consumable Materials

| Reference | Description | Specification |
|---|---|---|
| A50396 | Sealant - Dapco 2200 Primerless Silicone Firewall Sealant | BMS5-63 Type II Class B-1/2 |
| A50483 [CP2927] | Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant) | |
| D00601 [CP2101] | Vaseline - Graphite Mineral | |
| G51328 [CP2674] | Cloth - Lint-Free | |

### E. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### F. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**G. Electronic Engine Control (EEC) Electrical Harness (W2A and W2B) Installation**

SUBTASK 73-22-01-210-009-G00

(1) During the installation, make sure that EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] are not stressed or damaged.

SUBTASK 73-22-01-210-010-G00

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

(a) Make sure to apply silicone sealant during an assembly of harnesses and all related parts.

1) Apply sealant on edges that are the interfaces of parts.

SUBTASK 73-22-01-420-027-G00

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(3) Install the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the bolts [438], clip nuts [444], and clamps [439] that attach the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] to the engine.

(d) Tighten the bolt [438] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(e) Install the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] in a correct position with the clips [443] along its routing.

(f) Install the nut [440], bolt [441], and clamps [442] that attach the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] to the engine.

(g) Tighten the nut [440] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**(CAUTION PRECEDES)**

**CAUTION** USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] in fan area as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (W2A) [430] connector to the fan cowl support beam.

2) Connect the EEC electrical harness (W2B) [431] connector to the fan cowl support beam.

3) Connect the EEC electrical harness (W2A) [430] connector to the EEC channel A J2 receptacle [432].

4) Connect the EEC electrical harness (W2B) [431] connector to the EEC channel B J2 receptacle [433].

**PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-75-0010 AND PRE SB-CFM-LEAP-1B-75-0011 (CONFIG 1) AND PRE SB-CFM-LEAP-1B-72-0410**

5) Connect the EEC electrical harness (W2A) [430] connector to the EEC cooling blower (FWD/channel A) [434].

6) Connect the EEC electrical harness (W2B) [431] connector to the EEC cooling blower (AFT/channel B) [435].

**PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-75-0011 (CONFIG 1) AND PRE SB-CFM-LEAP-1B-72-0410**

7) Connect the EEC electrical harness (W2A) [430] connector to the bracket assembly [516].

8) Connect the EEC electrical harness (W2B) [431] connector to the bracket assembly [516].

**PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410**

9) Connect the EEC electrical harness (W2B) [431] to the EEC electrical harness (HJ8A) [303] on the bracket [83].

**PTF ALL**

(i) Install the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] on the strut 5 outer fitting [109] as follows:

1) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on thestrut 5 outer fitting [109] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the five bolts [106] and two nuts [107].



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b) Apply Dapco 2100 sealant, A50483 [CP2927], to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

c) Make sure that the EEC electrical harness (W2A) [430], EEC electrical harness (W2B) [431] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] (View E, Figure 421).

d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [107] with a lint-free cloth, G51328 [CP2674].

g) Make sure that the fillet seal of the squeeze-out is on the forward facing side.

**CAUTION**

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

h) Make the sealant squeeze-out flush with the surface on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 421).

i) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

j) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 outer fitting [109] (View C, Figure 421).

k) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109].

l) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

m) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the bolts [106] with a lint-free cloth, G51328 [CP2674].

**CAUTION**

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

n) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 421).



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o) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

**CAUTION**

MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

p) Clean the nut plate side of the bolts [106] and nut plates.

q) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

(j) Attach blanket [190] to the strut 5 outer fitting [109].

1) Apply Dapco 2100 sealant, A50483 [CP2927], between blanket and the strut 5 outer fitting [109] (View B, Figure 421).

NOTE: The Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

(k) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (W2A) [430] and EEC electrical harness (W2B) [431] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (W2A) [430] connector to the ignition exciter 1 [436].

2) Connect the EEC electrical harness (W2B) [431] connector to the ignition exciter 2 [437].

(l) Make sure that the work area is clean of all tools and other items.

## H. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-22-01-410-033-G00**

**WARNING**

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-104-G00

(2) Remove the safety tags and close these circuit breakers:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-105-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| C | 4 | C00154 | ENGINE 2 START VALVE |

SUBTASK 73-22-01-860-106-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 8 | C01103 | ENGINE 1 START VALVE |

SUBTASK 73-22-01-860-107-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

## I. Electronic Engine Control Electrical (EEC) Electrical Harness (W2A and W2B) Installation Test

SUBTASK 73-22-01-730-012-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

**——— END OF TASK ———**



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## ENGINE CONTROL ELECTRICAL HARNESSES - INSPECTION/CHECK

### 1. General

A. This procedure has two tasks:

(1) Engine control electrical harnesses inspection

(2) Engine control electrical harness connectors inspection.

**TASK 73-22-01-210-801-G00**

### 2. Engine Control Electrical Harnesses Inspection

(Figure 601, Figure 602, and Figure 603)

#### A. General

(1) This task gives the instructions to do the visual inspection of the engine control electrical harness.

#### B. References

| Reference | Title |
|---|---|
| 24-11-63-000-801 | W100 EDP and IDG Engine Wiring Harness Removal (P/B 401) |
| 24-11-63-400-801 | W100 EDP and IDG Engine Wiring Harness Installation (P/B 401) |
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01 P/B 401 | ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION |
| 73-22-01-300-801-G00 | Engine Wiring Harnesses W200 and W300 - BMS 1367 Cable at the Terminal Lugs Repair (P/B 801) |
| 73-22-01-300-802-G00 | Engine Wiring Harness - Harness Firewall Area Repair (P/B 801) |
| 73-22-01-300-803-G00 | Engine Wiring Harnesses W200 and W300 - Bobbin Replacement (P/B 801) |
| 73-22-01-900-801-G00 | Engine Wiring Harness Repair (Application of Silicone Tape Patch) (P/B 801) |
| 73-22-01-900-802-G00 | Engine Wiring Harness Repair (Application of Stainless Steel Braid) (P/B 801) |
| 73-22-01-900-803-G00 | Engine Wiring Harness Heat-Shrinkable Boot Repair (P/B 801) |
| 73-22-01-900-804-G00 | Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch) (P/B 801) |
| 73-22-01-900-805-G00 | Engine Wiring Harnesses W200 and W300 - Anti-Rotation Bobbin Repair (P/B 801) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |



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C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

F. Prepare for the Inspection

SUBTASK 73-22-01-860-073-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-074-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |



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**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Engine Control Electrical Harnesses Inspection

**SUBTASK 73-22-01-211-001-G00**

(1) Do the visual inspection of the engine control electrical harness (HJ3A) [1], engine control electrical harness (HJ3B) [2], engine control electrical harness (HJ5A) [3], engine control electrical harness (HJ5B) [4], engine control electrical harness (HJ6A) [5], engine control electrical harness (HJ6B) [6], engine control electrical harness (HJ7) [7], engine control electrical harness (HJ8A) [8], engine control electrical harness (HJ8B) [9], engine wiring harness (W100) [10], engine control electrical harness (W2) [11], engine fire detection harness (W200) [12], engine fire detection harness (W300) [13], engine control electrical harness (W4A) [14], and engine control electrical harness (W4B) [15] as follows:

(a) Examine the identification sleeve [17] for these conditions (Figure 602):

1) Chafing, burn, cut, removed characters

a) Permitted even if characters are unreadable.

b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Looseness

a) Permitted.

b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(b) Examine the identification plate [16] for these conditions (Figure 602):

1) Burn, cut

a) Permitted even if characters are unreadable.

b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Looseness

a) Permitted.



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b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

3) Removed characters

a) Permitted.

b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

4) Chafing

a) Permitted.

b) Replace the applicable engine harness at the next shop visit (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(c) Examine the heat-shrinkable bobbin [23] for these conditions (View B, Figure 602):

1) Burn, deformation, mark, chafing

a) Permitted, if it is not perforated.

b) If it is perforated, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Glue

a) Permitted.

3) Looseness

a) Not permitted.

b) If it is loose, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(d) Examine the heat-shrinkable transition [22] (Breakout) for these conditions (View B, Figure 602):

1) Burn, deformation, mark, chafing

a) Permitted, if not perforated.

b) If it is perforated, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Glue

a) Permitted.

3) Looseness

a) Not permitted.



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b) If it is loose, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(e) Examine the heat-shrinkable boot [18] for these conditions (View A, Figure 602):

1) Burn, deformation, chafing, mark, cut, nick, dent, scratches.

a) Permitted, if it is not perforated.

b) Repairable, if there is no finding (no cut strands, no braid deformation) on the shielding braid, if the hole in heat shrinkable boot is smaller than 0.39 in. (10 mm) x 0.19 in. (5 mm) and at least 0.39 in. (10 mm) from the end of the boot, connector side.

c) Repair method: refer to Engine Wiring Harness Heat-Shrinkable Boot Repair, TASK 73-22-01-900-803-G00.

d) If the shielding braid has finding, the hole in heat shrinkable boot is bigger than 0.39 in. (10 mm) x 0.19 in. (5 mm), or at least a part of the hole is within 0.39 in. (10 mm) from the end of the boot, replace the applicable engine control electrical harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801, TASK 24-11-63-400-801, TASK 26-11-02-000-801, and TASK 26-11-02-400-801).

2) Glue

a) Permitted.

3) Looseness

a) Not permitted.

b) If it is loose, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(f) Examine the connector [20] for these conditions (Figure 602):

1) Worn Master key

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Signs of short-circuit, carbonizing

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

3) Nicks, dents, scratches on connector locking ring

a) Permitted if the nicks, dents, scratches are not more than 0.08 in. (2 mm) long, 0.04 in. (1 mm) wide, 0.01 in. (0.3 mm) deep.

b) If the damage is more than the limits, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).



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4) Scratches on the connector and backshell parts

a) Permitted if below 0.01 in. (0.3 mm) deep.

b) If the damage is more than the limits, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

5) Nicks, dents on the connector thread

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

6) Nicks, dents on the connector backshell [19]

a) Permitted if the nicks, dents, scratches are not more than 0.08 in. (2 mm) long, 0.04 in. (1 mm) wide, 0.01 in. (0.3 mm) deep.

b) If the damage is more than the limits, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

7) Connector interfacial face deteriorated

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

8) Cut or looseness and deformated on connector RFI Ring

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

9) Bad connector assembly

a) Permitted if the color mark on the receptacle is not visible.

b) If the color mark is visible, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

10) Marking partially erased on connector and metallic parts

a) Permitted even if marking is unreadable.

b) Replace the component at the next shop visit.

(g) Examine the connector [20] and mechanical parts for these conditions (Figure 602):

1) Scratches on connectors and backshell

a) Permitted if the scratches are not more than 0.01 in. (0.3 mm).

b) If a scratch is more than the limit, replace the applicable engine control electrical component.

2) Nicks, dents on the connector backshell [19]



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a) Permitted if the nicks, dents are not more than 0.08 in. (2 mm) long, 0.04 in. (1 mm) wide, 0.01 in. (0.3 mm) deep.

b) If you find nicks or dents that are more the limits, replace the engine control electrical component.

3) Nicks, dents scratches on the connector mechanical parts (breakout type Y or T and transition)

a) Permitted if the nicks, dents, scratches are not more than 0.08 in. (2 mm) long, 0.04 in. (1 mm) wide, 0.01 in. (0.3 mm) deep.

b) If you find nicks, dents, scratches that are more the limits, replace the engine control electrical component.

4) Marking partially erased on connector and metallic parts

a) Permitted even if marking is unreadable.

b) Replace the component at the next shop visit.

5) Cut or looseness on mechanical parts (stainless steel clamps)

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(h) Examine the heat-shrinkable tubing [21] for these conditions (Figure 602 and Figure 603):

NOTE: Engine Wiring Harness Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-801-G00 is replaced by Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-804-G00 in this inspection task. There is no technical impact on harnesses repaired with Engine Wiring Harness Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-801-G00.

1) Burn, deformation, chafing

a) Permitted, if the external heat-shrinkable tubing [24] is not perforated.

b) To prevent perforation, repair task can be applied pro actively (TASK 73-22-01-900-801-G00).

c) Maximum repairable limit:

<1> In the limits of the applicable repair.

d) Do this task: Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-804-G00.

2) Mark

a) Permitted, if the external tubing is not perforated.

b) Maximum repairable limit:

<1> In the limits of the applicable repair.

c) Do this task: Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-804-G00.

d) If it is perforated, replace the applicable engine control electrical component (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801, TASK 24-11-63-400-801, TASK 26-11-02-000-801, and TASK 26-11-02-400-801).

3) Perforation, cut



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a) Not permitted.

b) Maximum repairable limit:

<1> In the limits of the applicable repair.

c) Do this task: Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-804-G00.

4) Perforation, cut on terminal lug (TFER, engine fire detection harness (W200) [12] and engine fire detection harness (W300) [13])

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(i) Examine the stainless outer copper braid [25] for these conditions (Figure 602):

1) Burn, cut, fretting (TFER)

a) Maximum repairable limit:

<1> 1.18 in. (30.0 mm) max on stainless outer copper braid [25]

<2> Max defective area angle is 90°.

b) Do this task: Engine Wiring Harness Repair (Application of Stainless Steel Braid), TASK 73-22-01-900-802-G00.

(j) Examine the outer copper braid (firewall area) at the 6:00 o'clock bifurcation for these conditions (Figure 602):

NOTE: This outer protection is present only on engine control electrical harness (HJ5A) and engine control electrical harness (W2).

1) Burn

a) Not permitted.

b) Only applicable to W200 and HJ5A harnesses, maximum repairable limit:

<1> Degradation of heat-shrinkable sheath under the copper braid is not permitted.

c) Only applicable to W200 and HJ5A harnesses, repair method: refer to Engine Wiring Harness - Harness Firewall Area Repair, TASK 73-22-01-300-802-G00.

d) In case of degradation of heat-shrinkable sheath under the copper braid, replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Cut

a) Permitted, if less than 100% of the strands of a carrier are cut.

b) Maximum repairable limit:

<1> No degradation of the heat-shrinkable sheath under the copper braid.

c) Do this task: Engine Wiring Harness - Harness Firewall Area Repair, TASK 73-22-01-300-802-G00.

d) If not repairable, replace the applicable engine harness (PAGEBLOCK 73-22-01/401).

3) Holes between strands, deformation (Area 1)



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a) Permitted, if the heat shrinkable sheath under the outer copper braid is not visible.

b) Maximum repairable limit:

<1> No degradation of the heat-shrinkable sheath under the copper braid.

c) Do this task: Engine Wiring Harness - Harness Firewall Area Repair, TASK 73-22-01-300-802-G00.

d) If not repairable, replace the applicable engine harness (PAGEBLOCK 73-22-01/401).

4) Holes between strands, deformation (Area 2)

a) Permitted, if the visible area of the heat shrinkable sheath under the outer copper braid is less than 0.031 $in^2$ (20 $mm^2$).

b) Maximum repairable limit:

<1> No degradation of the heat-shrinkable sheath under the copper braid.

c) Do this task: Engine Wiring Harness - Harness Firewall Area Repair, TASK 73-22-01-300-802-G00.

d) If not repairable, replace the applicable engine harness (PAGEBLOCK 73-22-01/401).

5) Looseness

a) Not permitted.

b) Maximum repairable limit:

<1> No degradation of the heat-shrinkable sheath under the copper braid.

c) Do this task: Engine Wiring Harness - Harness Firewall Area Repair, TASK 73-22-01-300-802-G00.

d) If not repairable, replace the applicable engine harness (PAGEBLOCK 73-22-01/401).

(k) Examine the wire [26] (engine fire detection harness (W200) [12] and engine fire detection harness (W300) [13]) for these conditions (Views X, Y, Z, and AA, Figure 601 and Figure 602):

1) Burn, hardening on specific cable for fire loop area (BMS1367)

a) Not permitted.

b) Replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Signs of short-circuit, carbonizing, tear, crushing on specific cable for fire loop area (BMS1367)

a) Not permitted.

b) Replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

3) Chafing, cut on insulation sheath on specific cable for fire loop area (BMS1367)

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: refer to Engine Wiring Harnesses W200 and W300 - BMS 1367 Cable at the Terminal Lugs Repair, TASK 73-22-01-300-801-G00.

d) If not repairable, replace the applicable Engine Control Electrical Harness.



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(l) Examine the contact [27] for these conditions (Figure 602):

1) Signs of wear, arcing or carbonizing

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Distortion

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

3) Missing or retracted

a) Not permitted.

b) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(m) Examine the receptacle connector mounting flange [28] for these conditions (Figure 602):

1) Scratches, dents, nicks

a) Permitted if the scratches, dent, nick is less than 0.012 in. (0.3 mm).

<1> Replace the component at the next shop visit.

b) If the condition is more than the limit, replace the engine electrical control component.

**PTF ALL PRE SB 737-CFM-LEAP-1B-72-0306**

(n) Examine the anti-rotation bobbin for these conditions (Figure 602):

1) Distortion, crack or tear on the anti-rotation lug

a) Not permitted.

b) Maximum repairable limit:

<1> In the limits of the applicable repair.

c) Do this task: Engine Wiring Harnesses W200 and W300 - Anti-Rotation Bobbin Repair, TASK 73-22-01-900-805-G00.

d) Replace the applicable engine harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801 and TASK 24-11-63-400-801, TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

**PTF ALL POST SB 737-CFM-LEAP-1B-72-0306**

(o) Only applicable to W200 and W300 harnesses, examine the split bobbin for these conditions (Figure 602 (Sheet 12)):

1) Crack, tear nicks, dents, scratches, distortion

a) Permitted, if less than 0.08 in. (2.0 mm) long, 0.04 in. (1.0 mm) wide and 0.012 in. (0.30 mm) depth.

b) Maximum repairable limit: any amount.

c) Repair method: refer to Engine Wiring Harnesses W200 and W300 - Bobbin Replacement, TASK 73-22-01-300-803-G00.



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**PTF ALL POST SB 737-CFM-LEAP-1B-72-0306 (Continued)**

d) If necessary, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

2) Marking

a) Permitted even if characters are unreadable.

b) Maximum repairable limit: any amount.

c) Repair method: refer to Engine Wiring Harnesses W200 and W300 - Bobbin Replacement, TASK 73-22-01-300-803-G00.

d) If necessary, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

3) Color change

a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method: refer to Engine Wiring Harnesses W200 and W300 - Bobbin Replacement, TASK 73-22-01-300-803-G00.

d) If necessary, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

4) Gap

a) Permitted if less than 0.10 in. (2.5 mm).

b) Maximum repairable limit: any amount.

c) Repair method: refer to Engine Wiring Harnesses W200 and W300 - Bobbin Replacement, TASK 73-22-01-300-803-G00.

d) If necessary, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(p) Only applicable to W200 and W300 harnesses, examine the External Heat-shrinkable TFER tubing (Figure 602 (Sheet 12)):

1) Burn, chafing, deformation

a) Permitted if external tubing is not perforated.

b) If external tubing is perforated, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

(q) Only applicable to W200 and W300 harnesses, examine the elastomer (Figure 602 (Sheet 13)):

1) Deformation

a) Permitted if the elastomer is always present between the cables and the metal part.

b) If the elastomer is missing, replace the applicable engine harness (TASK 26-11-02-000-801 and TASK 26-11-02-400-801).

**PTF ALL**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-075-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-076-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

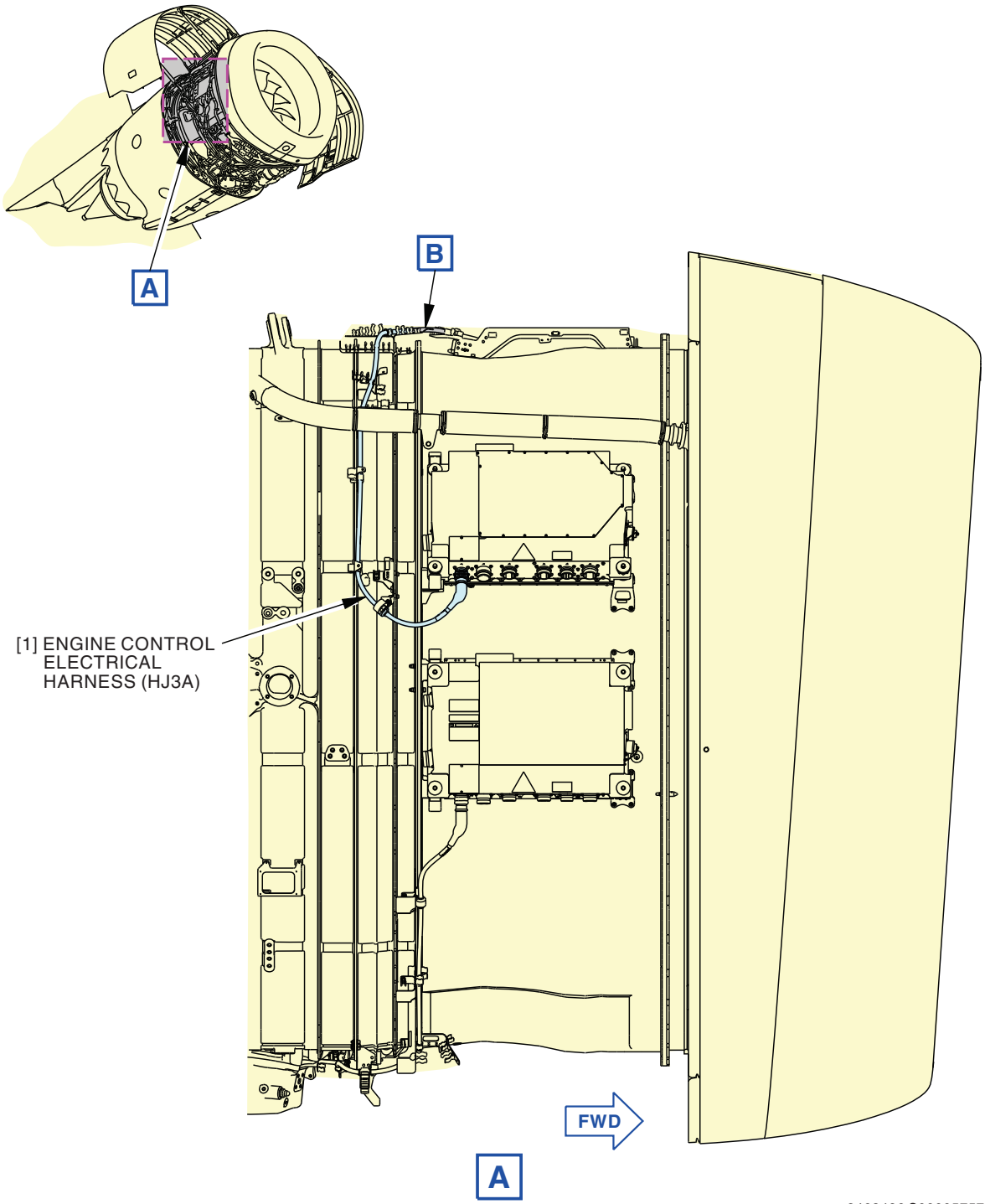
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



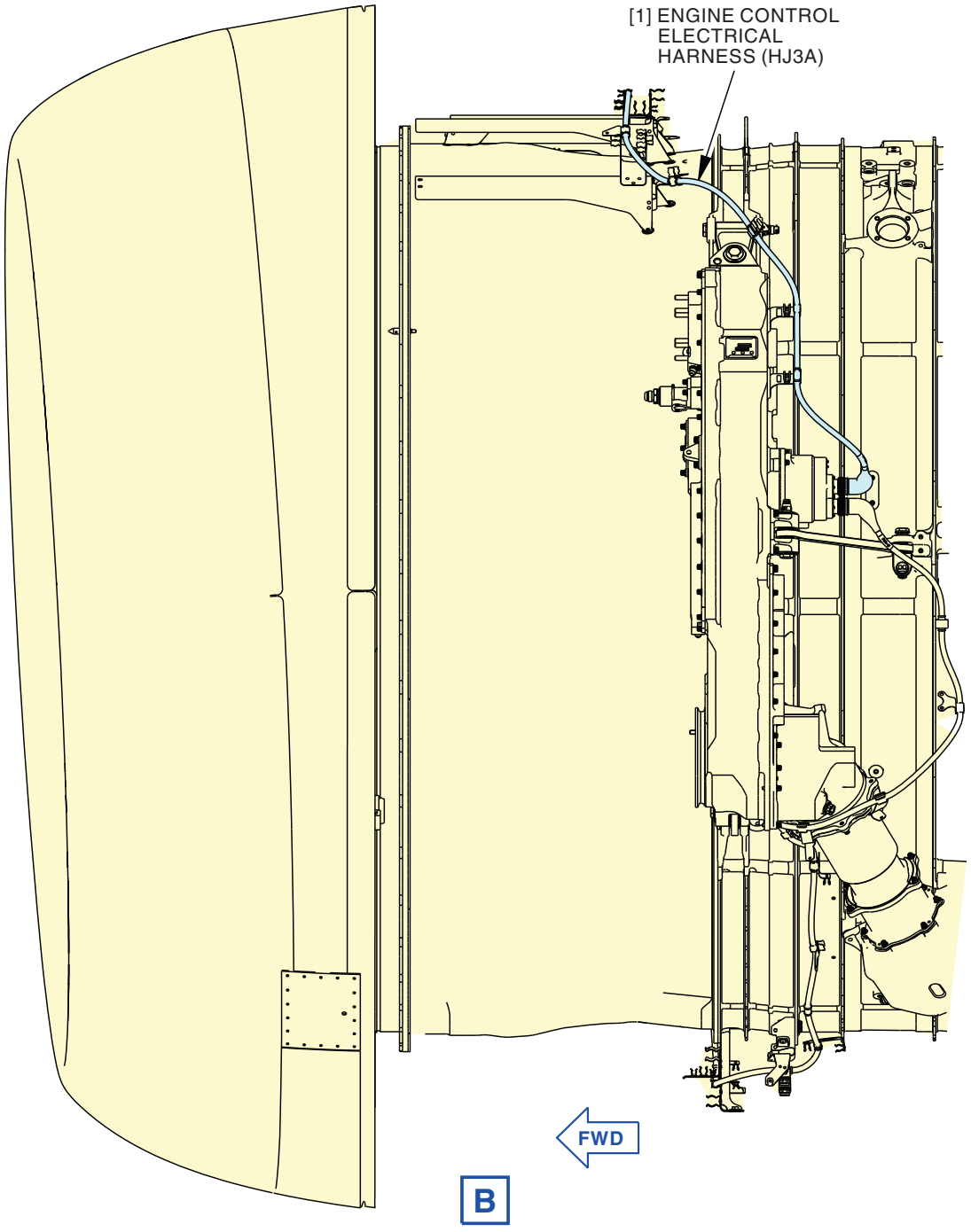
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 1 of 27)**



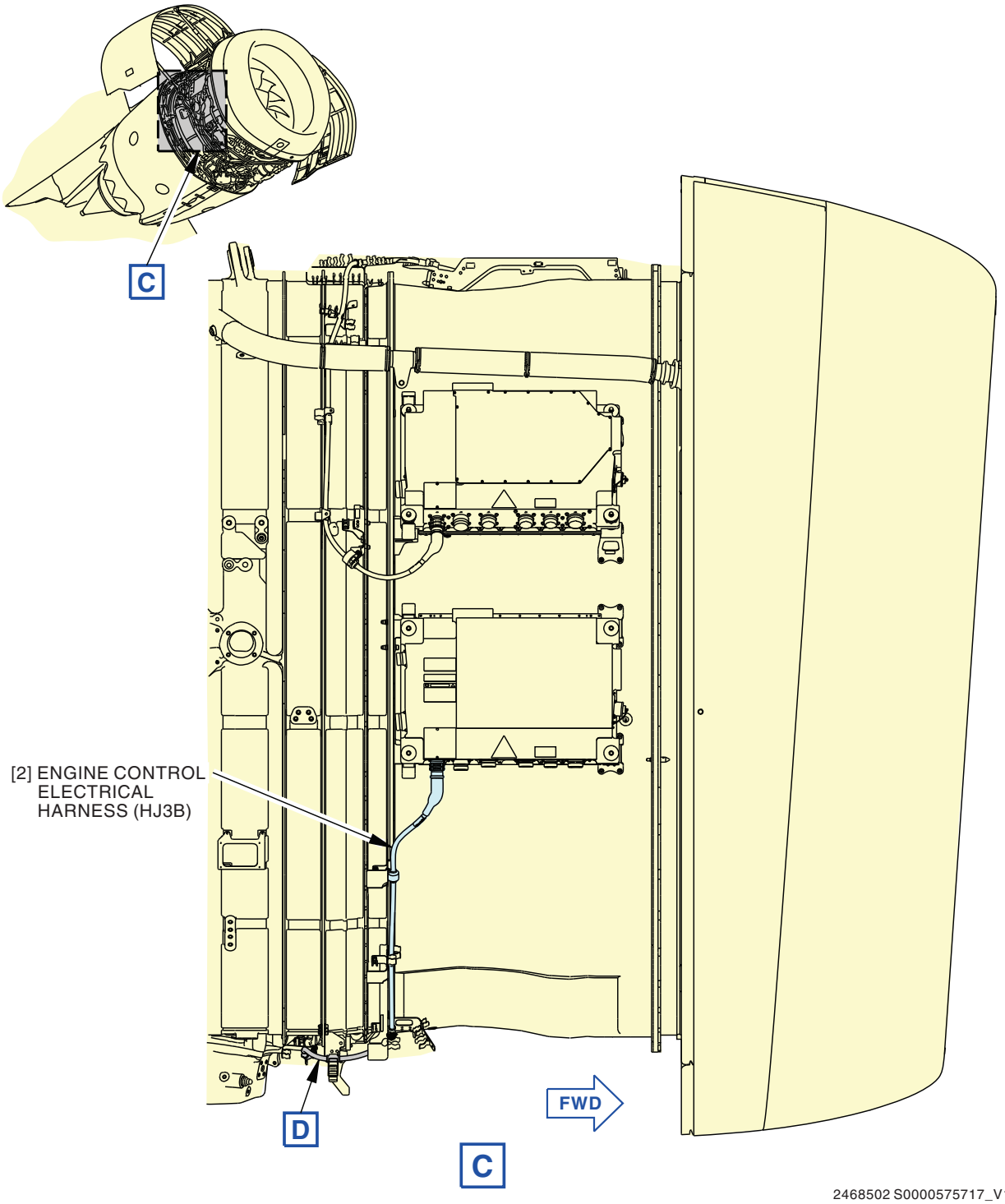
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 2 of 27)**



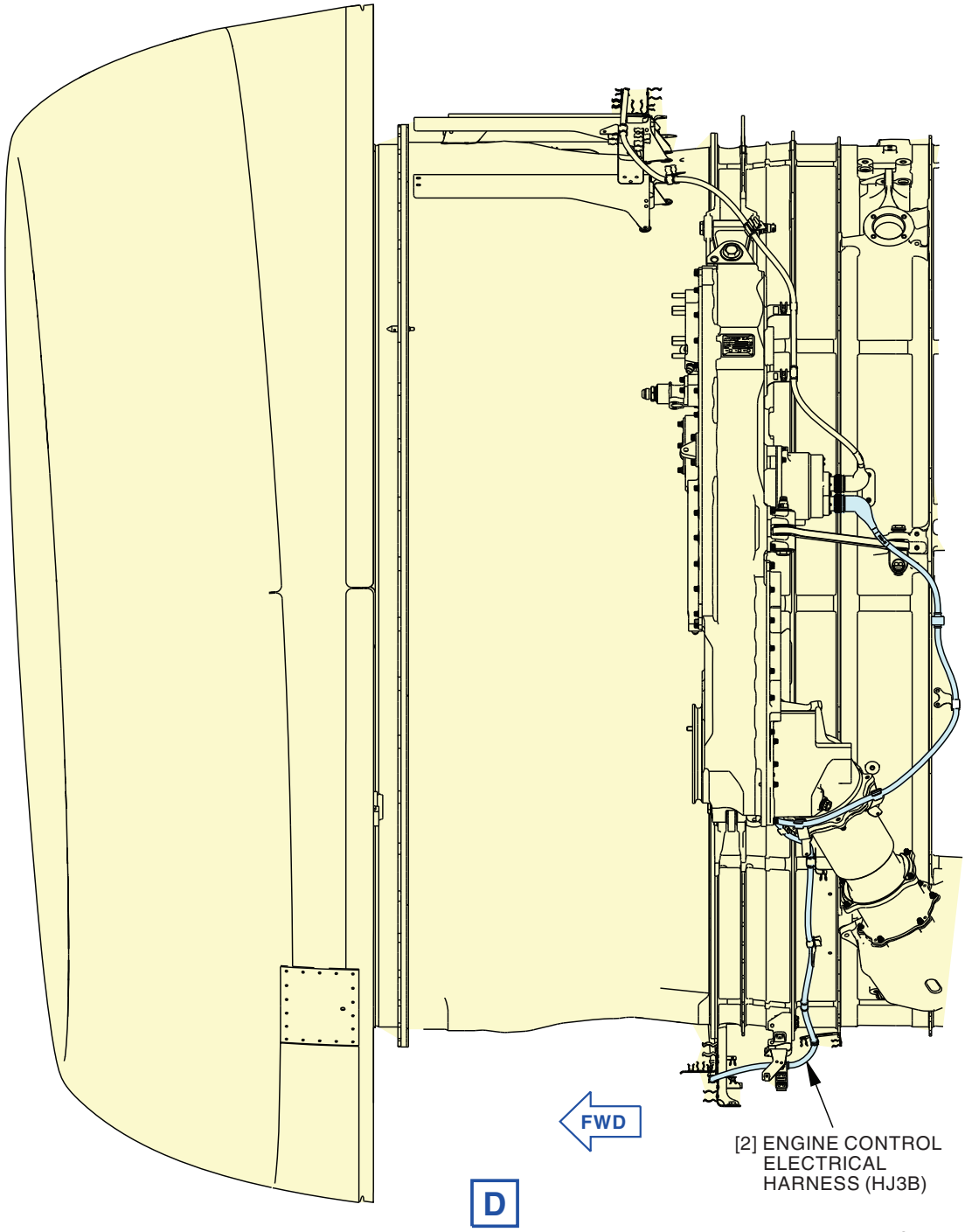
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 3 of 27)**



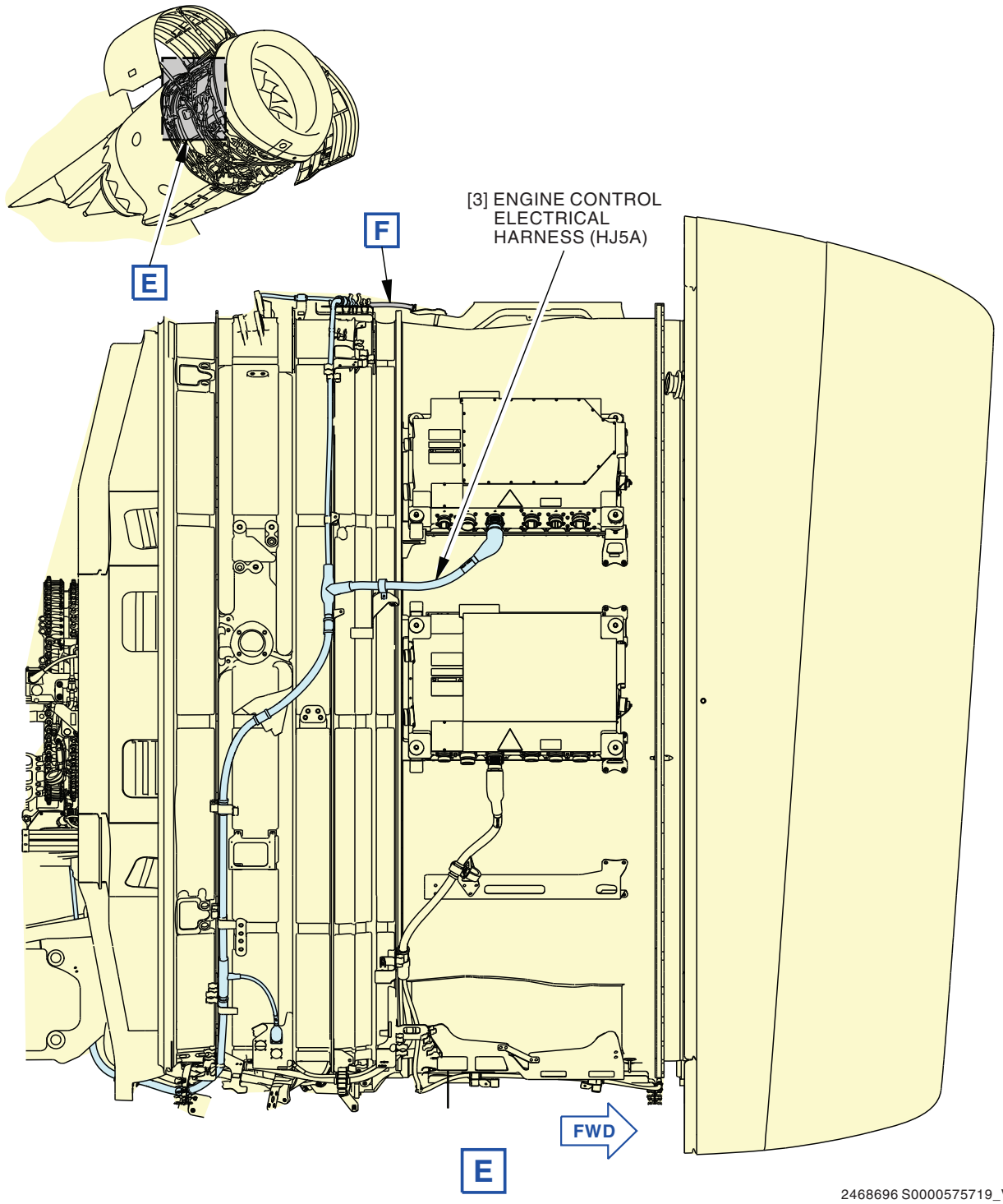
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 4 of 27)**



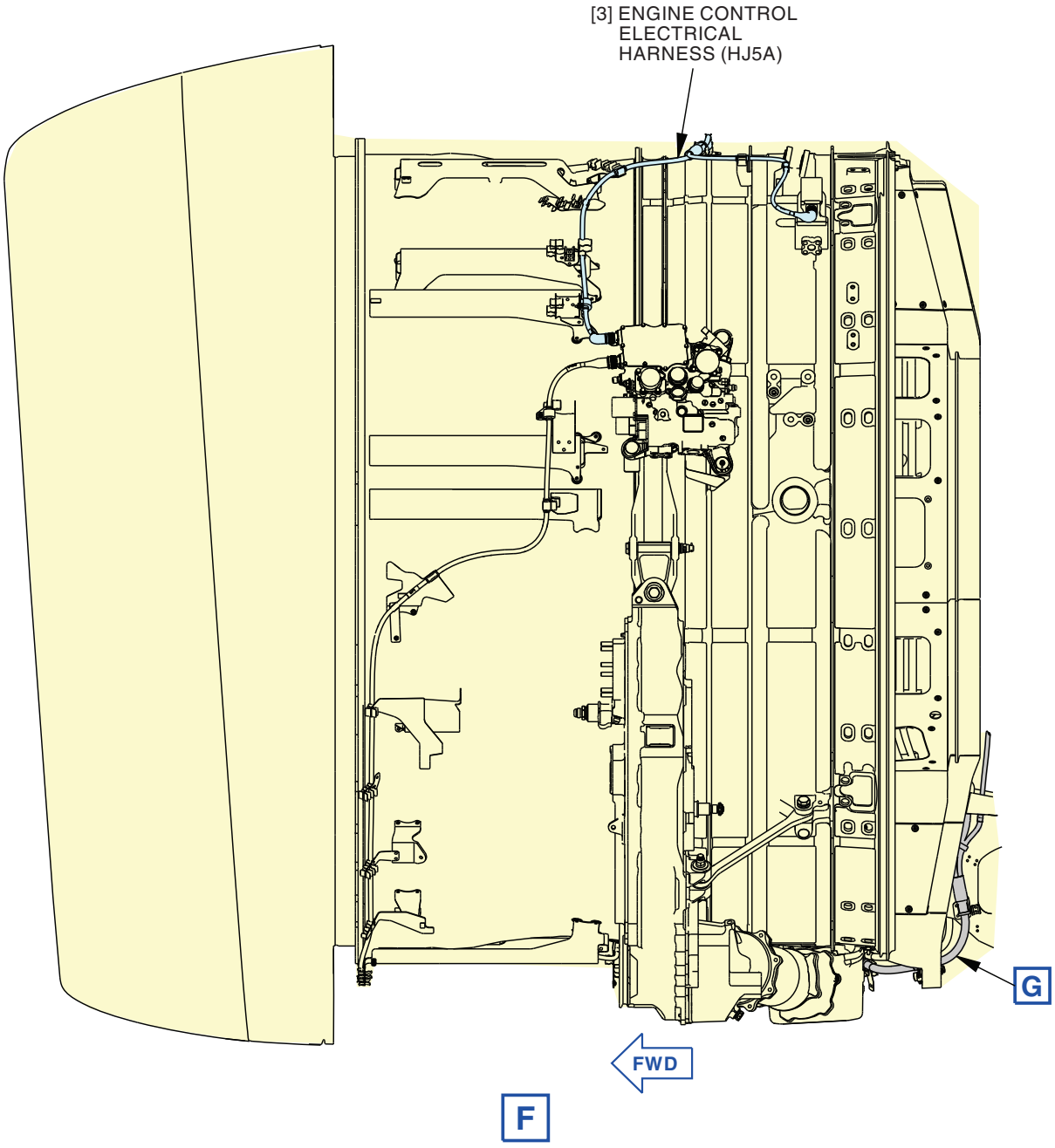
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 5 of 27)**



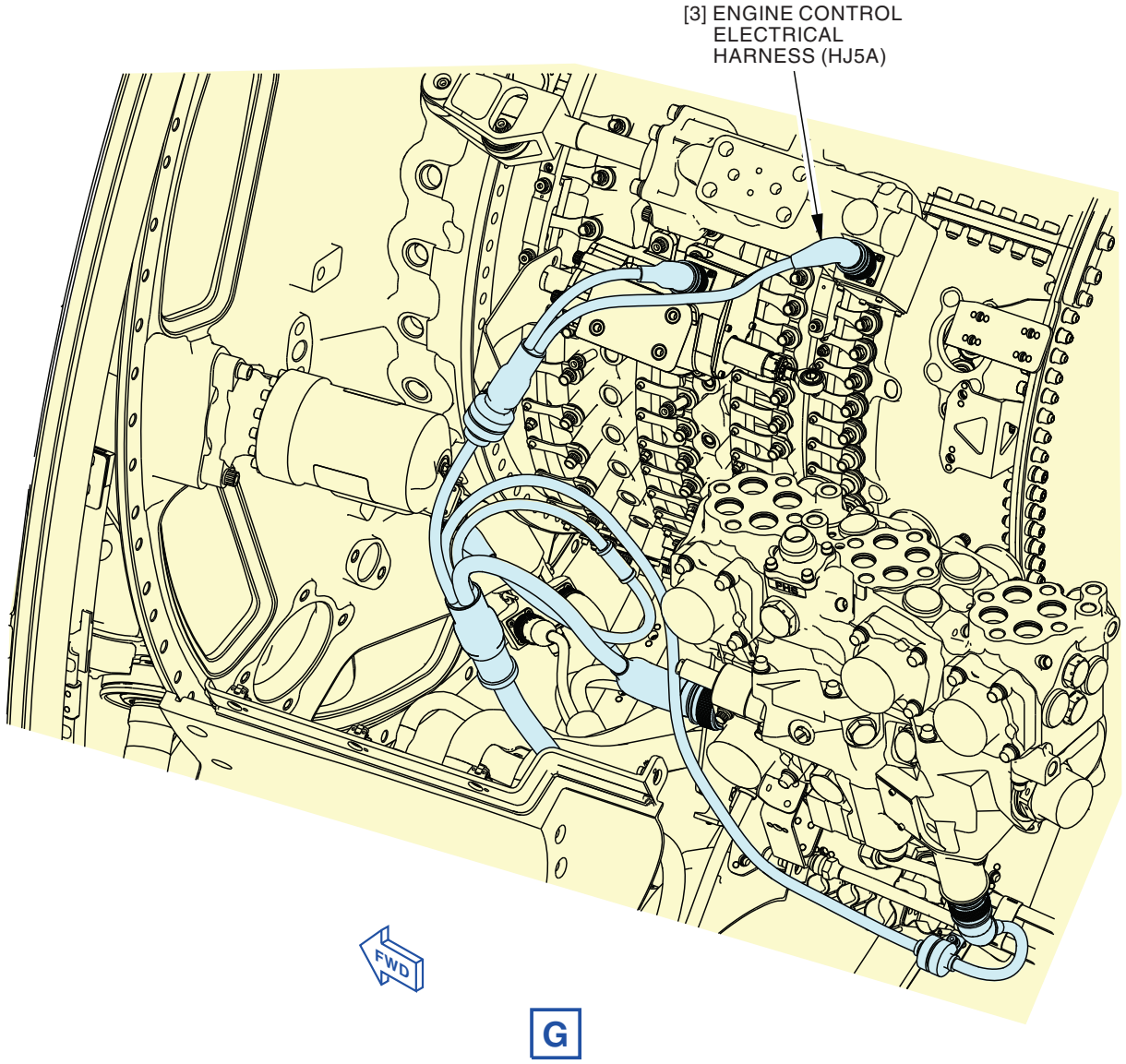
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 6 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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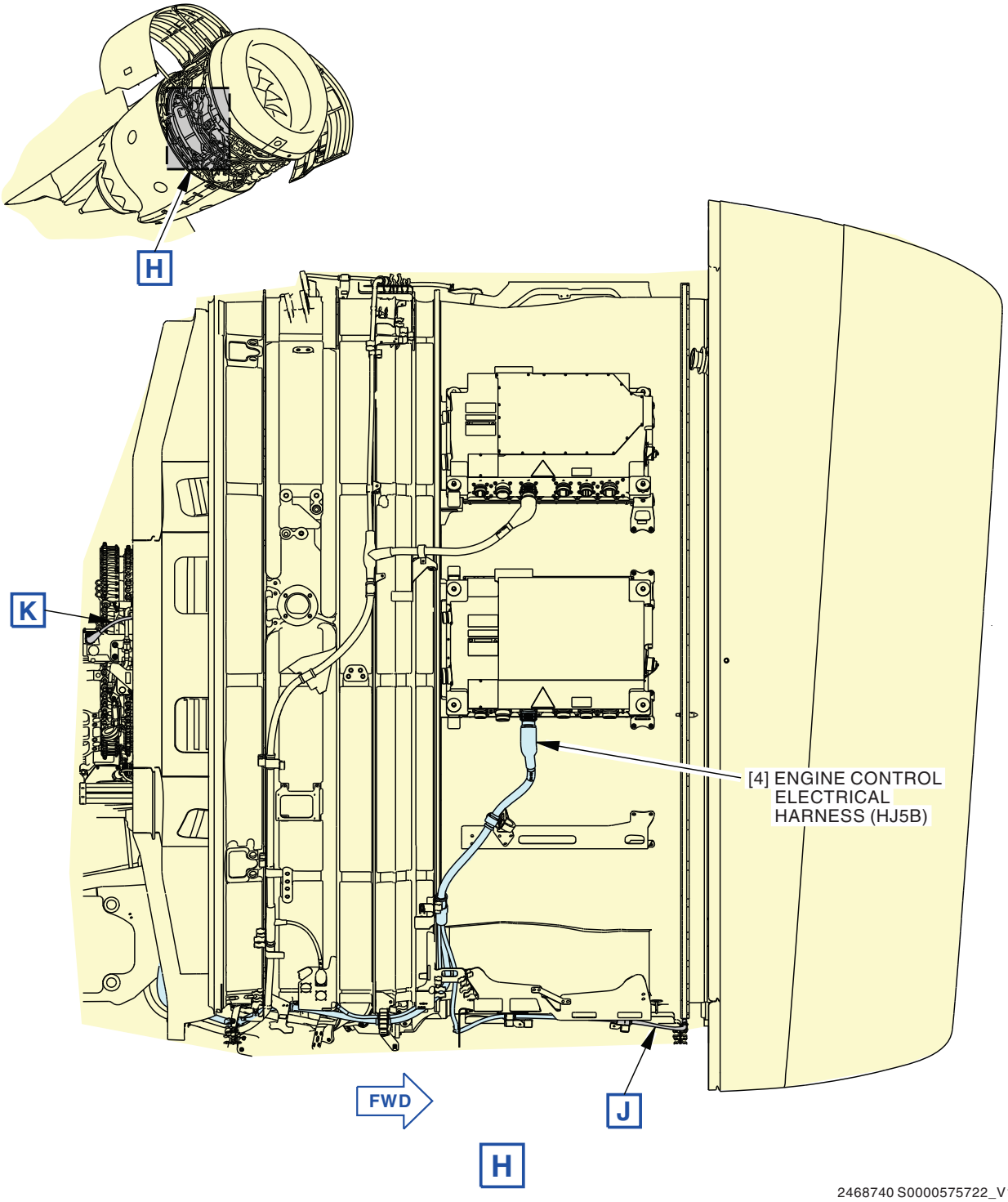
**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 7 of 27)**



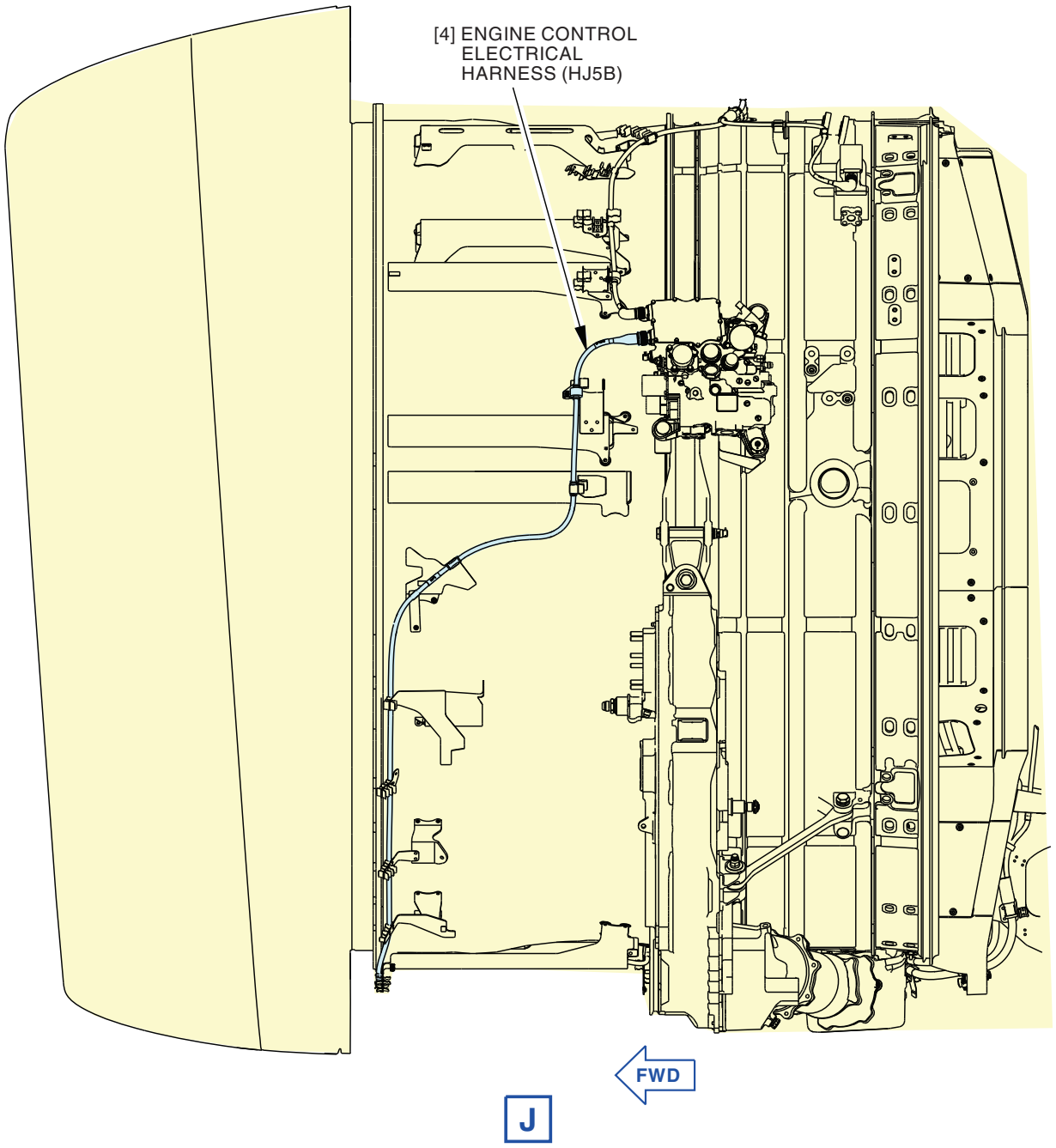
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 8 of 27)**



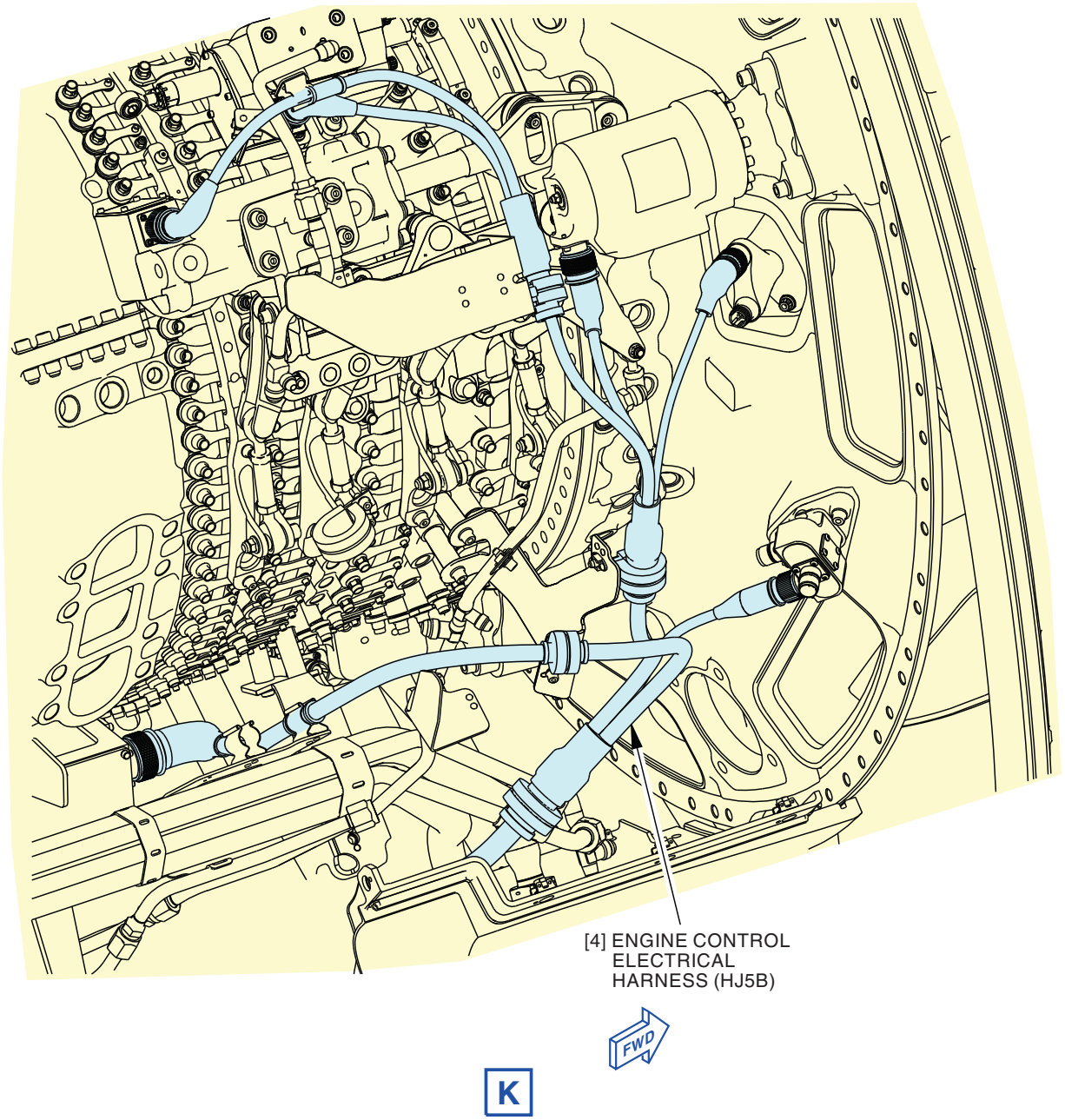
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 9 of 27)**



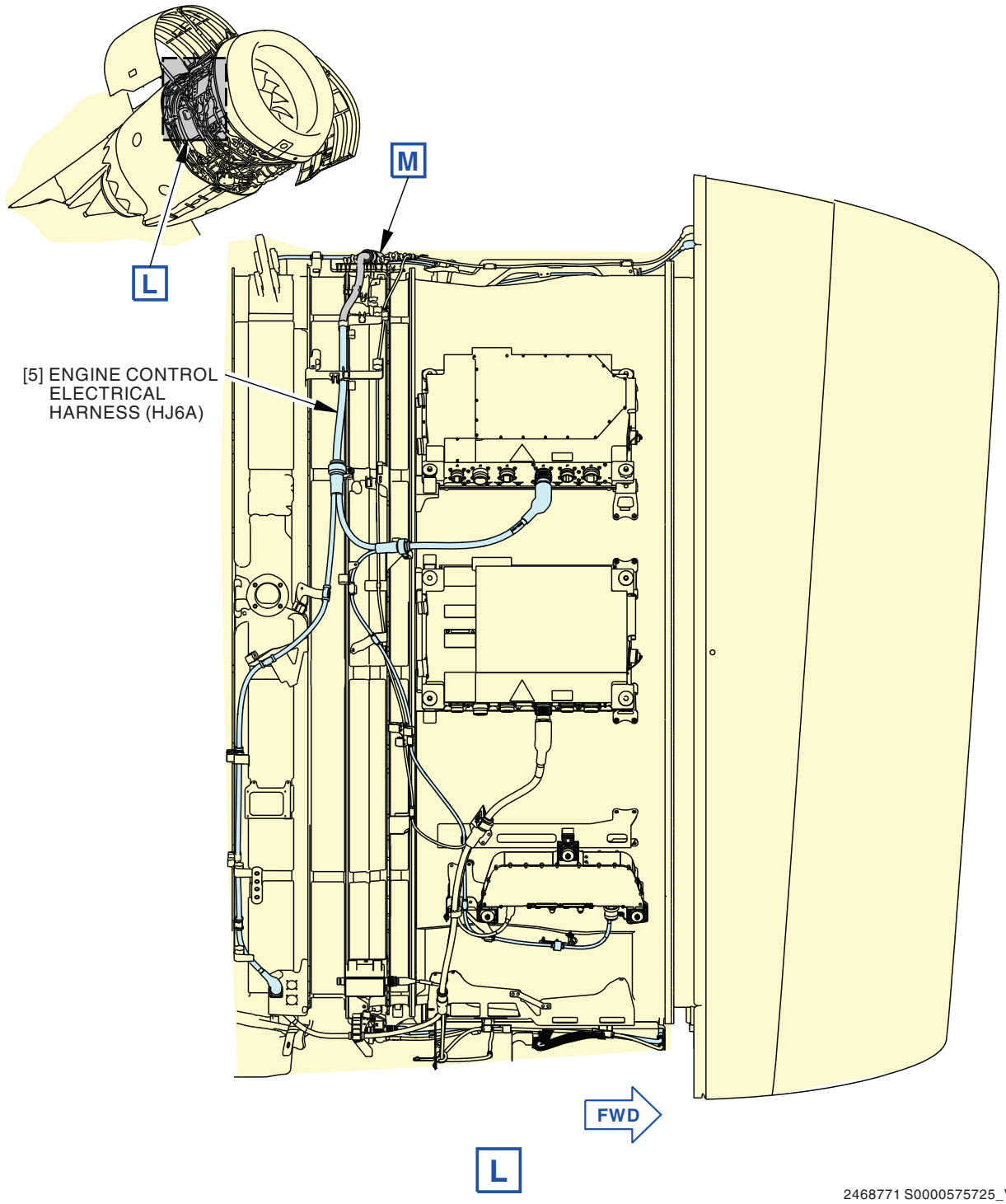
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 10 of 27)**



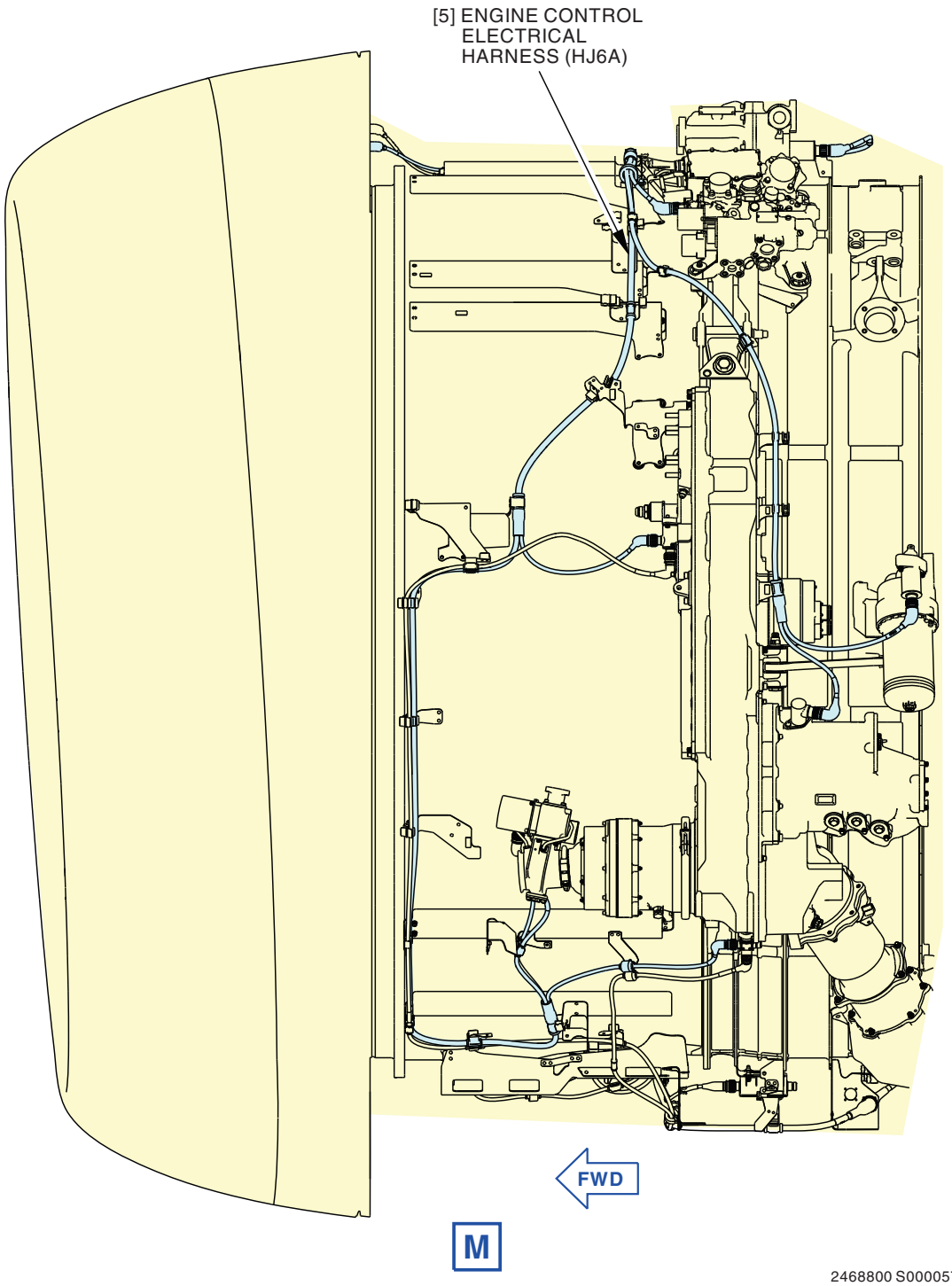
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 11 of 27)**



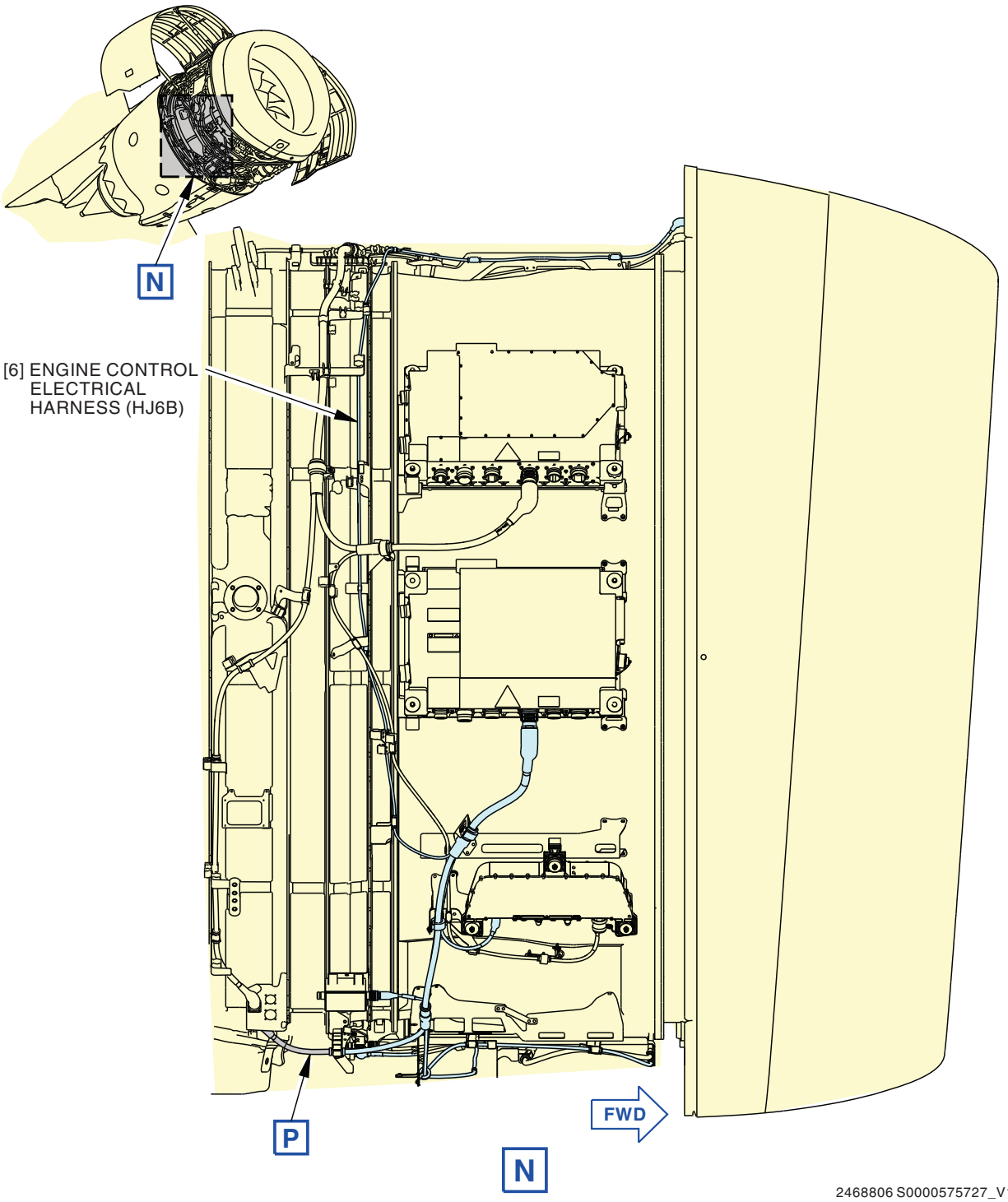
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 12 of 27)**



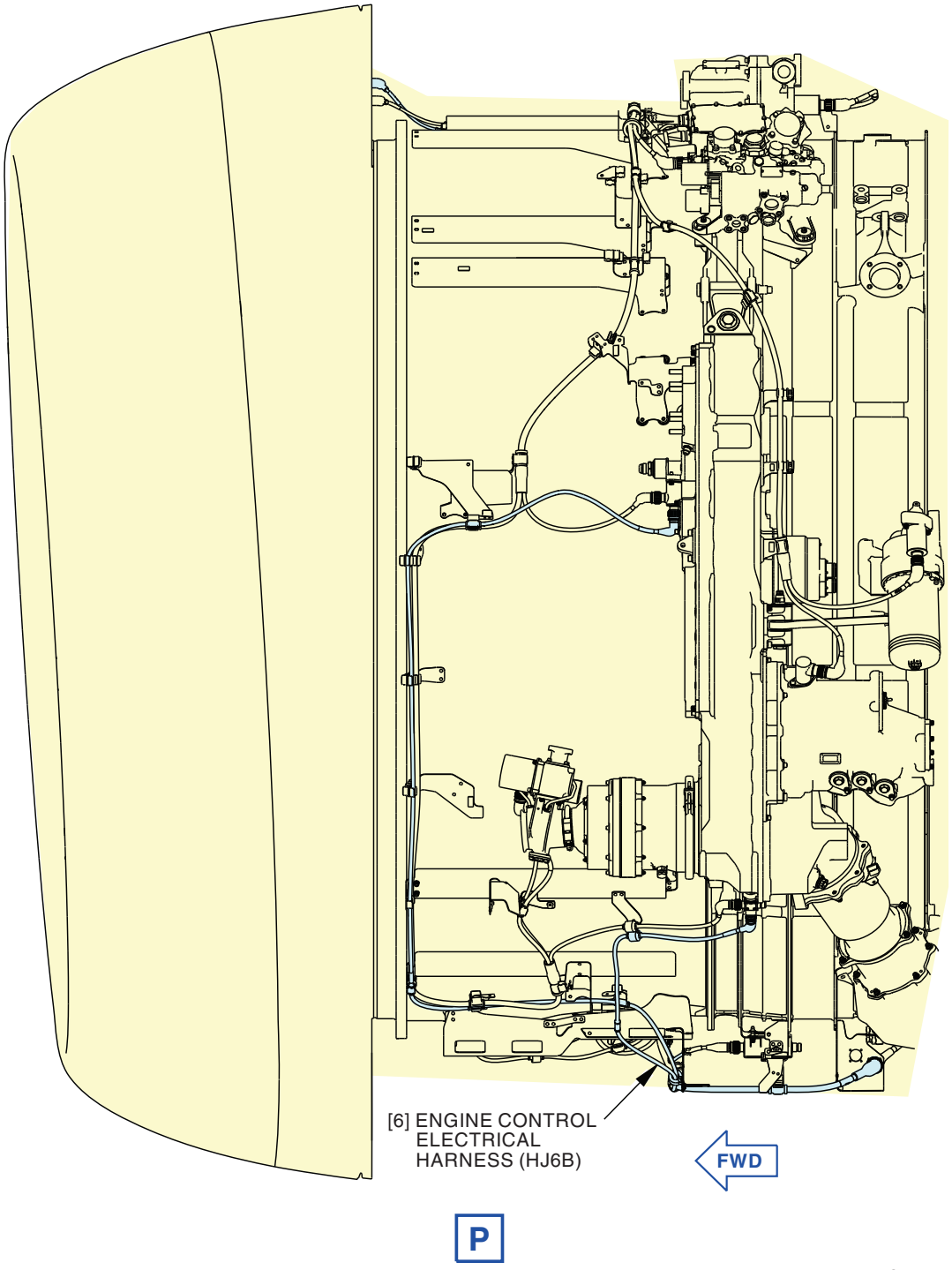
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 13 of 27)**



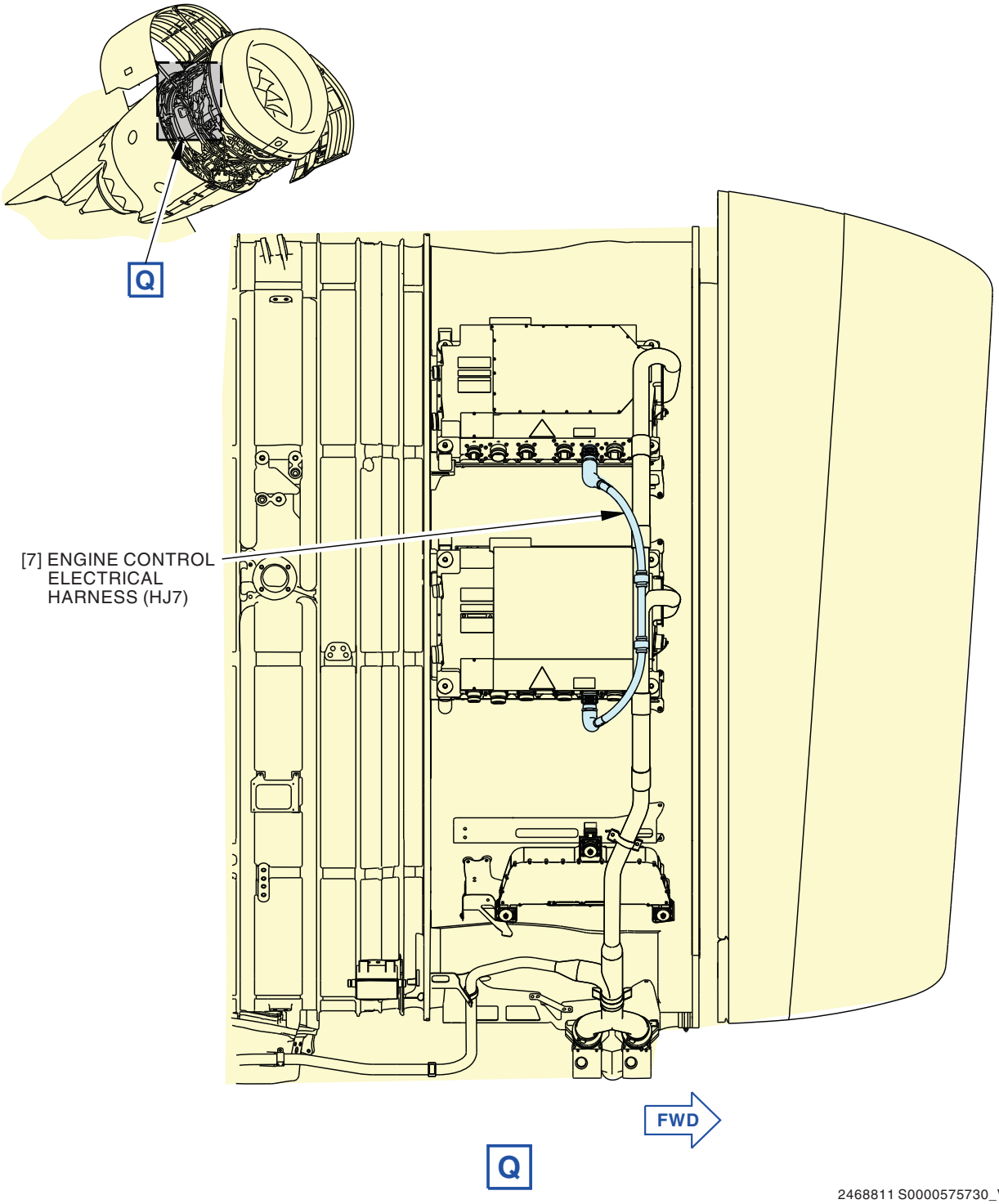
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 14 of 27)**



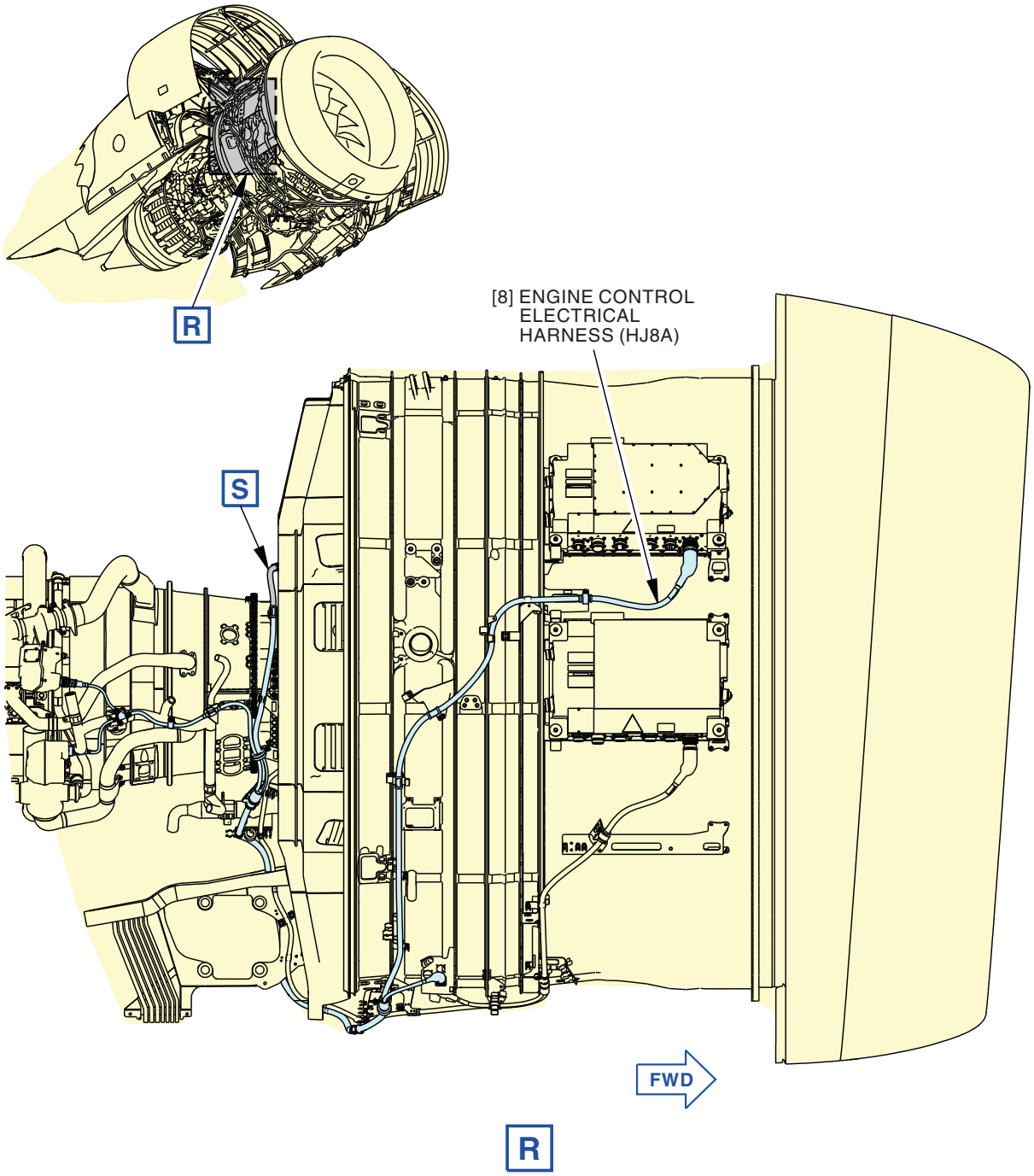
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 15 of 27)**



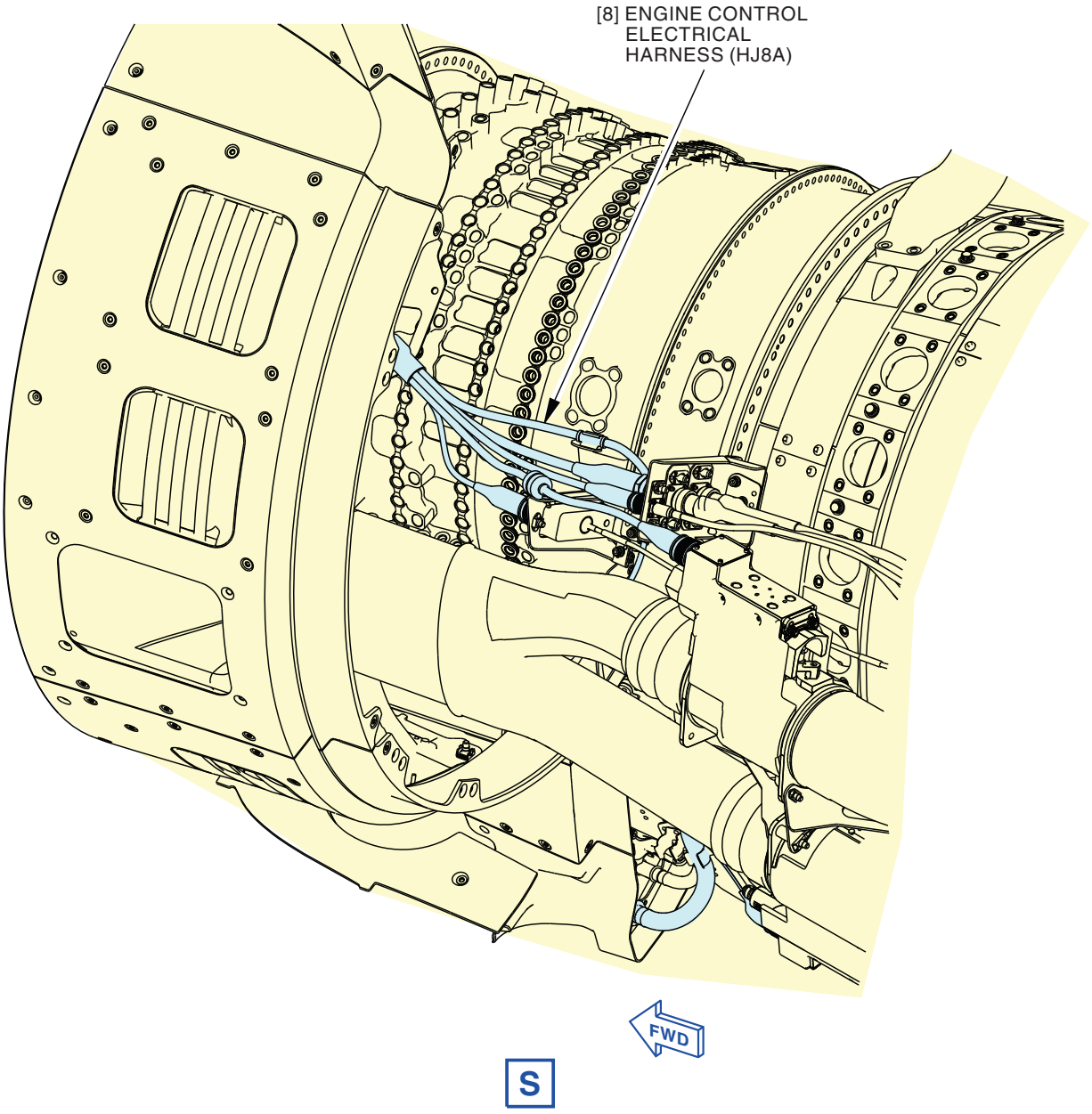
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 16 of 27)**



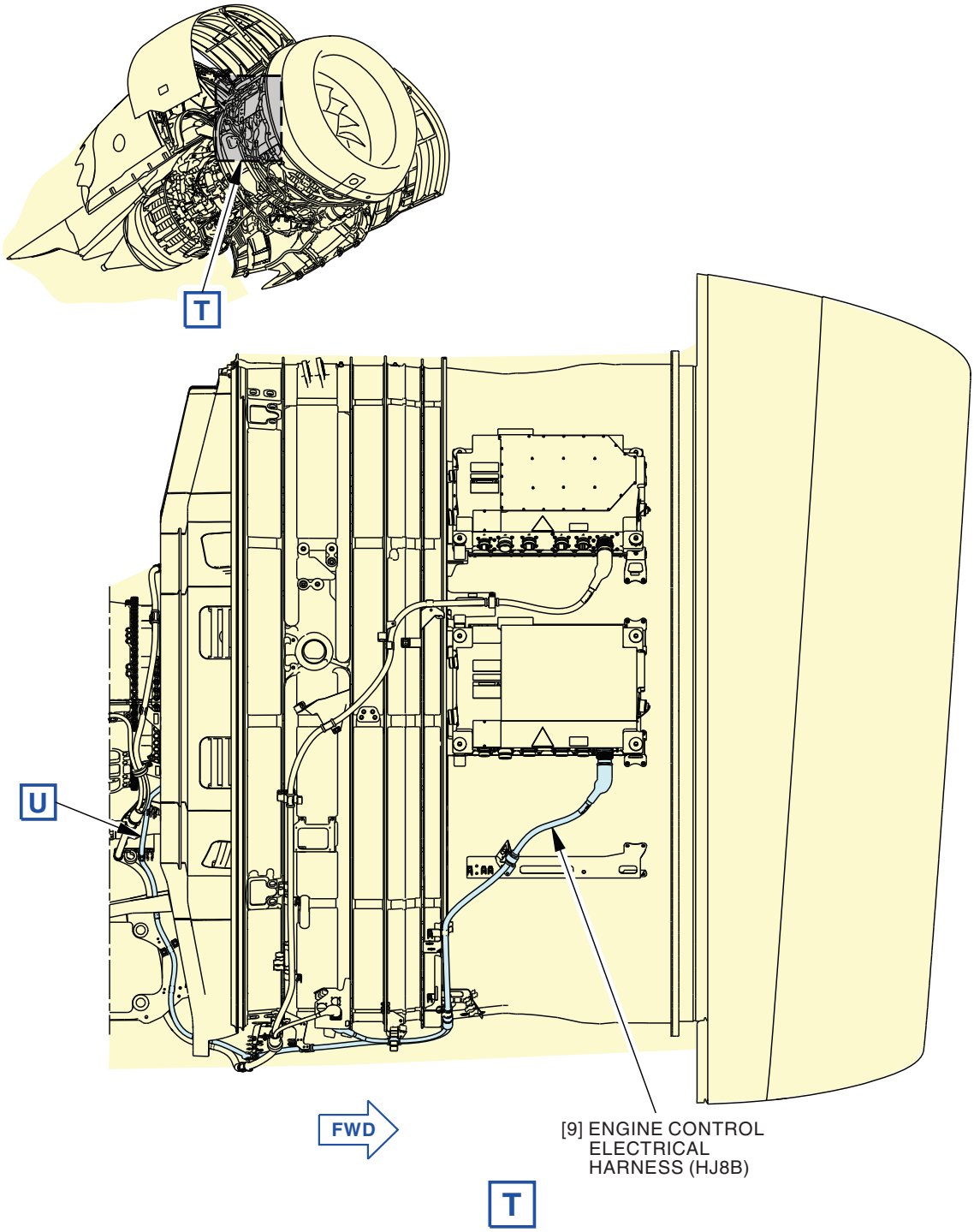
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 17 of 27)**



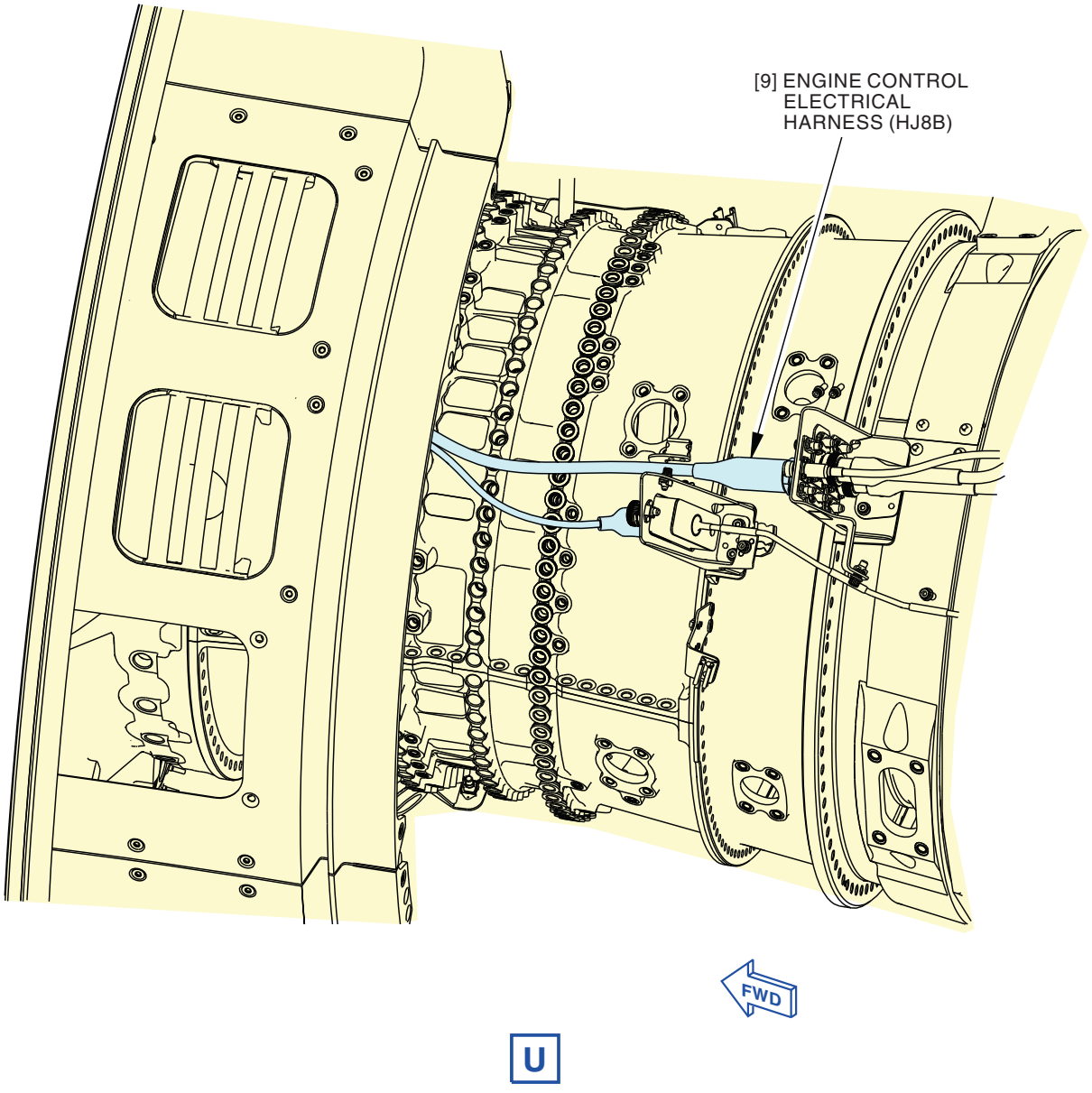
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 18 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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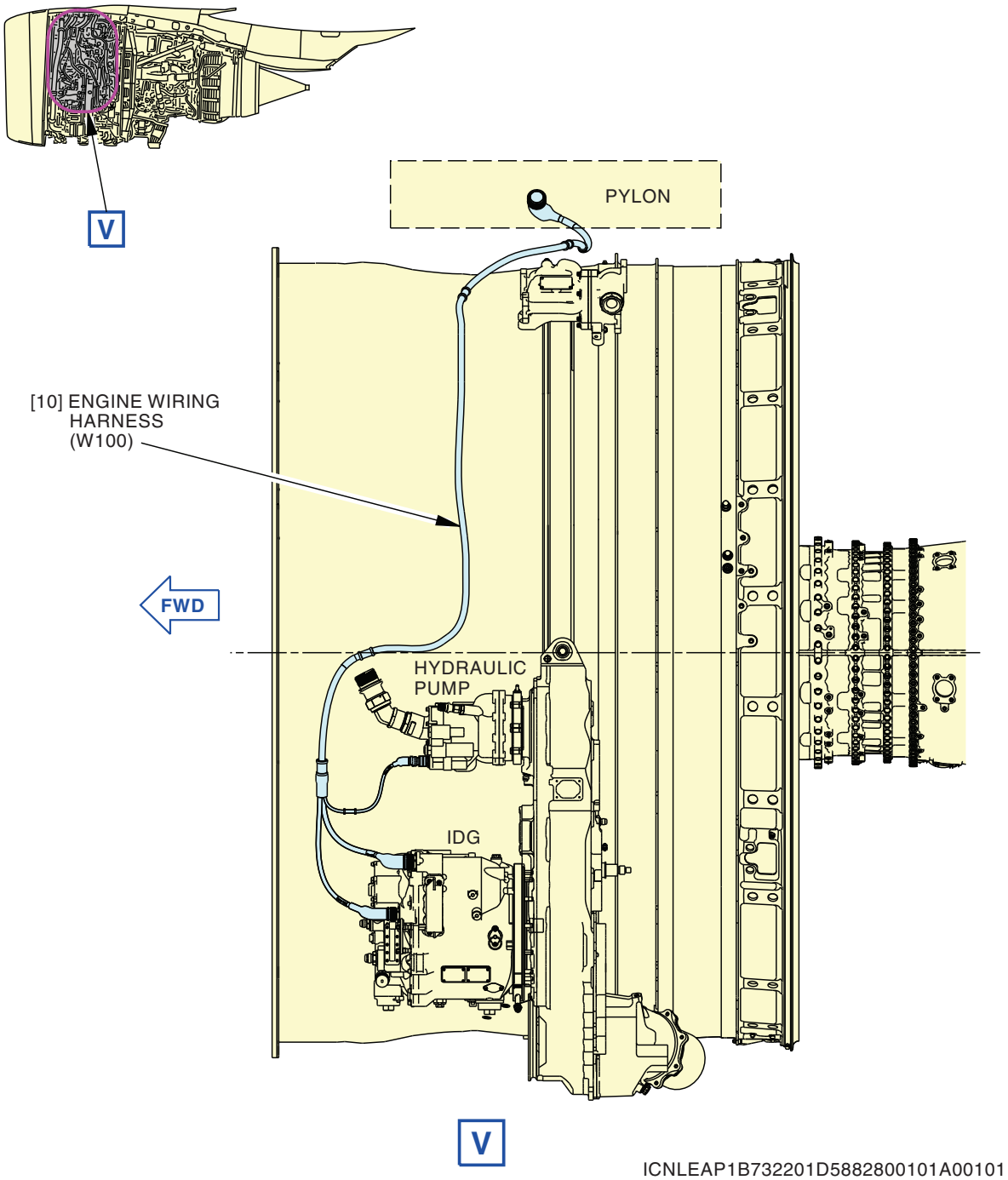
**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 19 of 27)**



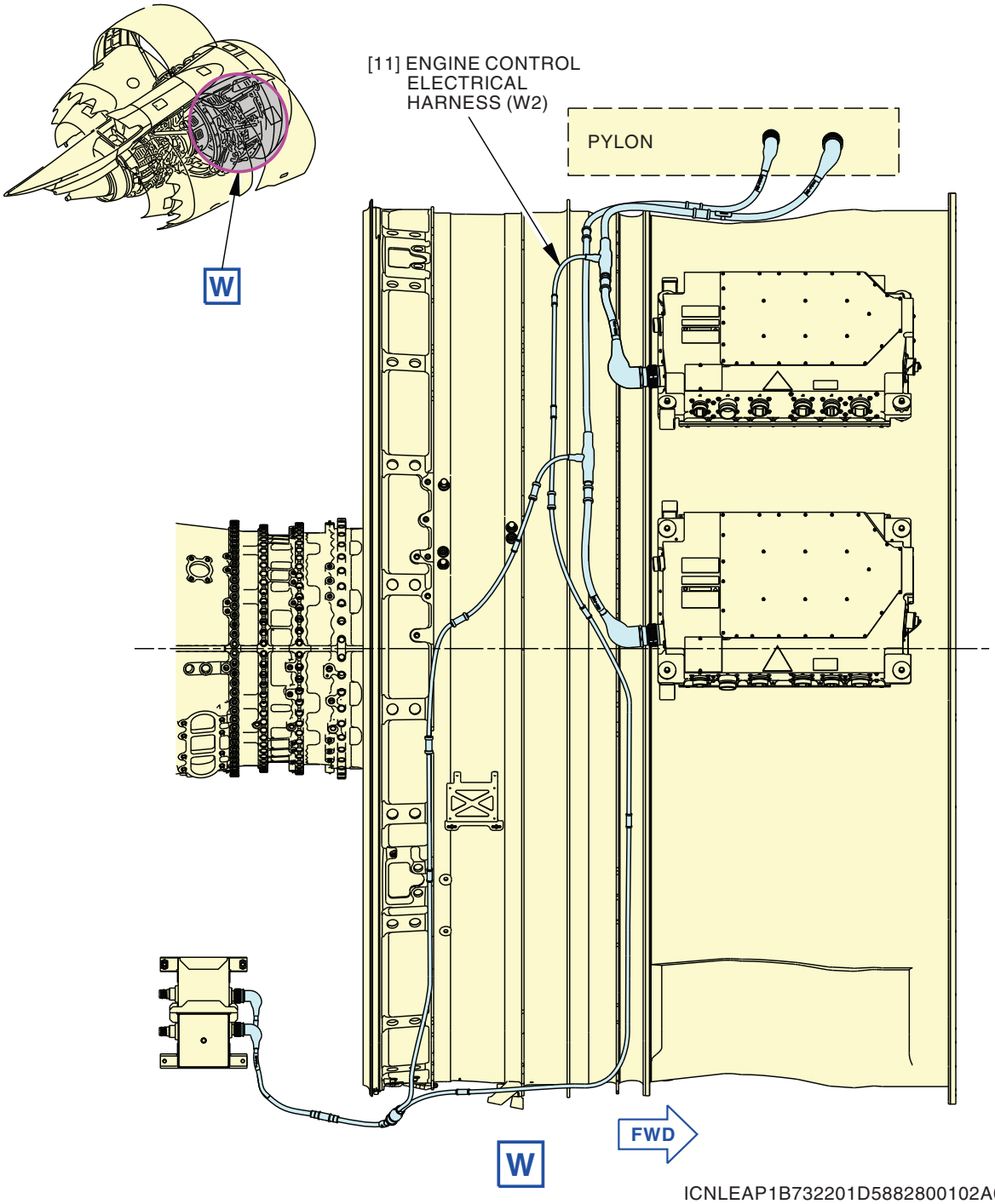
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 20 of 27)**



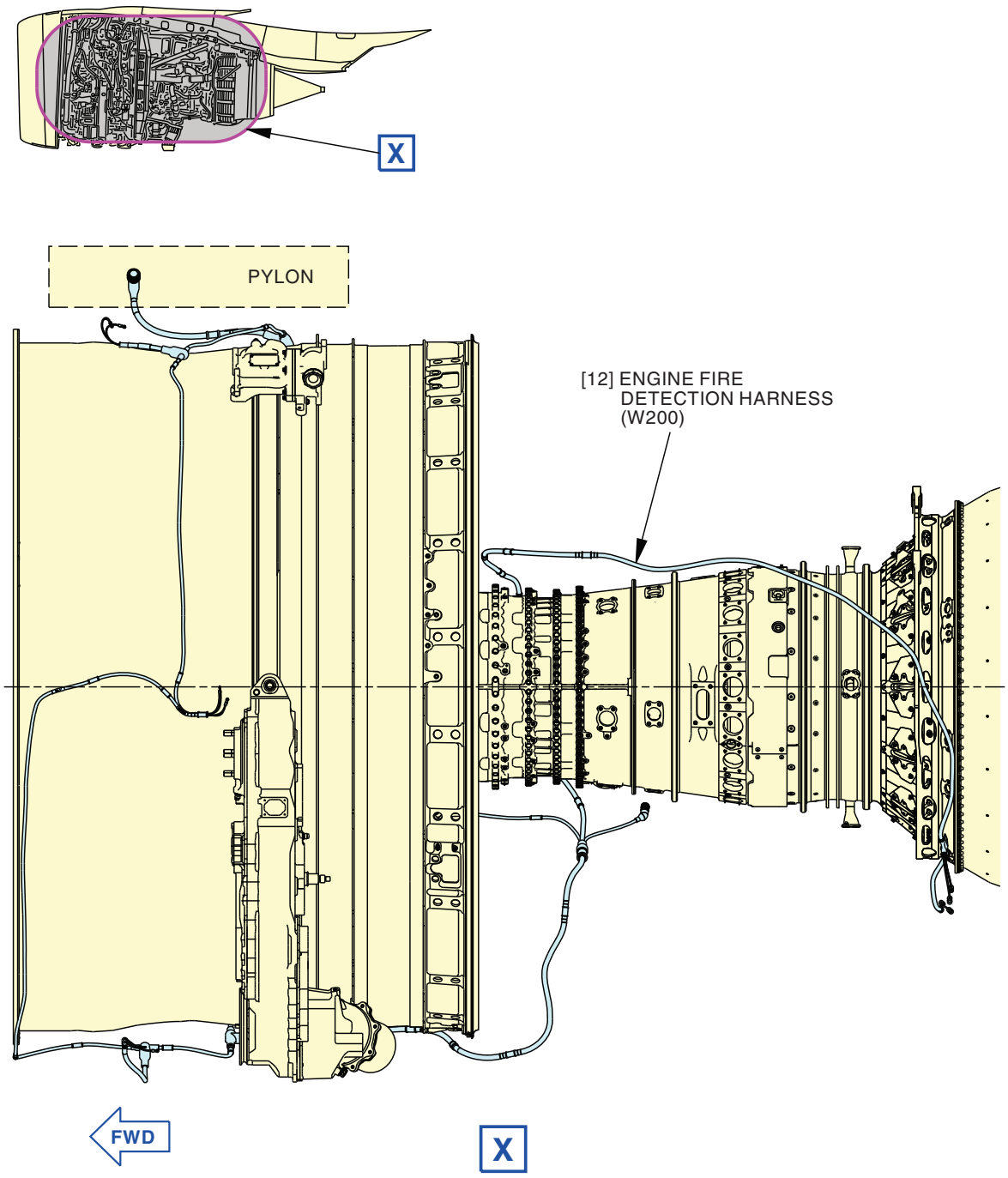
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 21 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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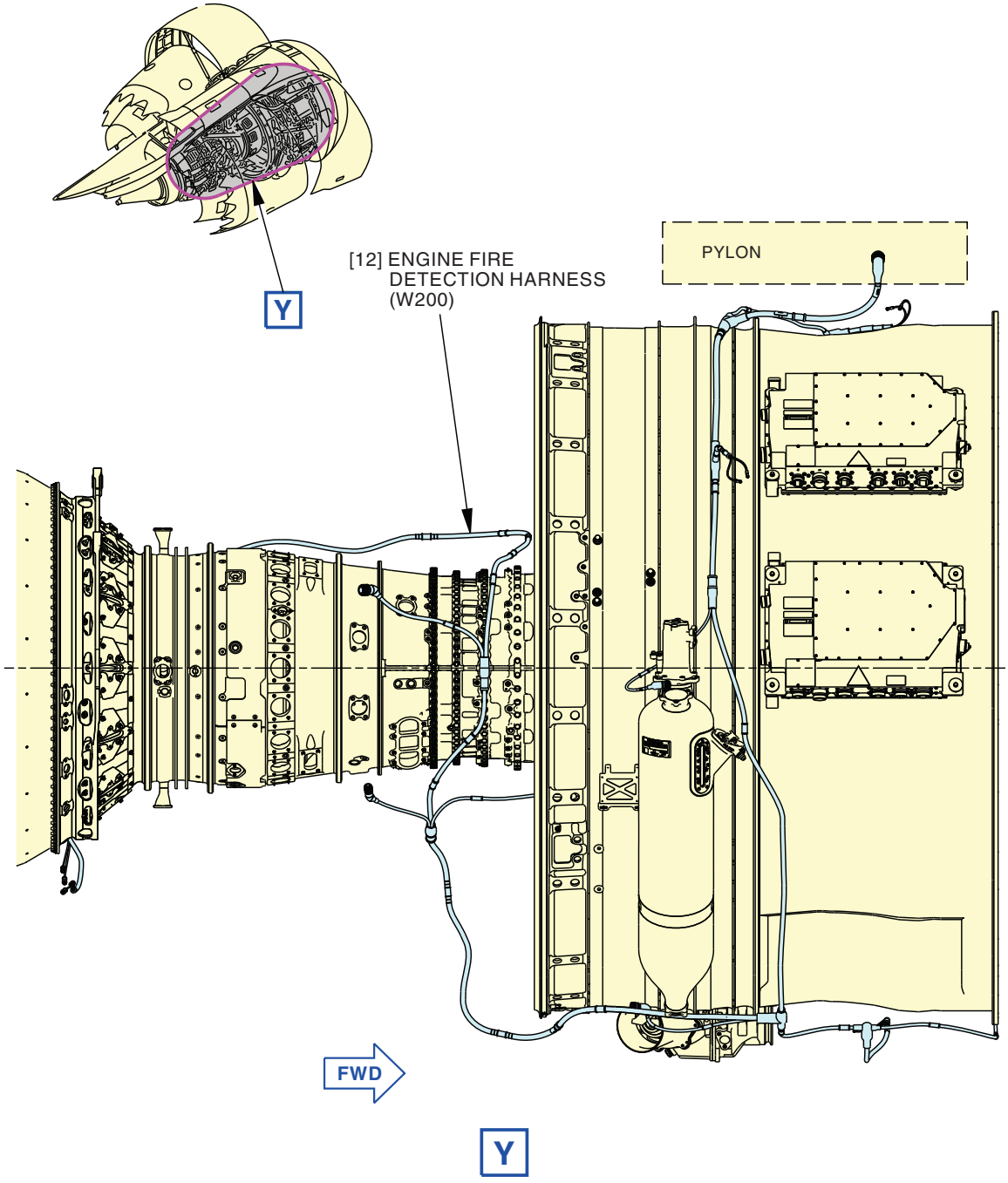
2966483 S0000745031_V1

**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 22 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



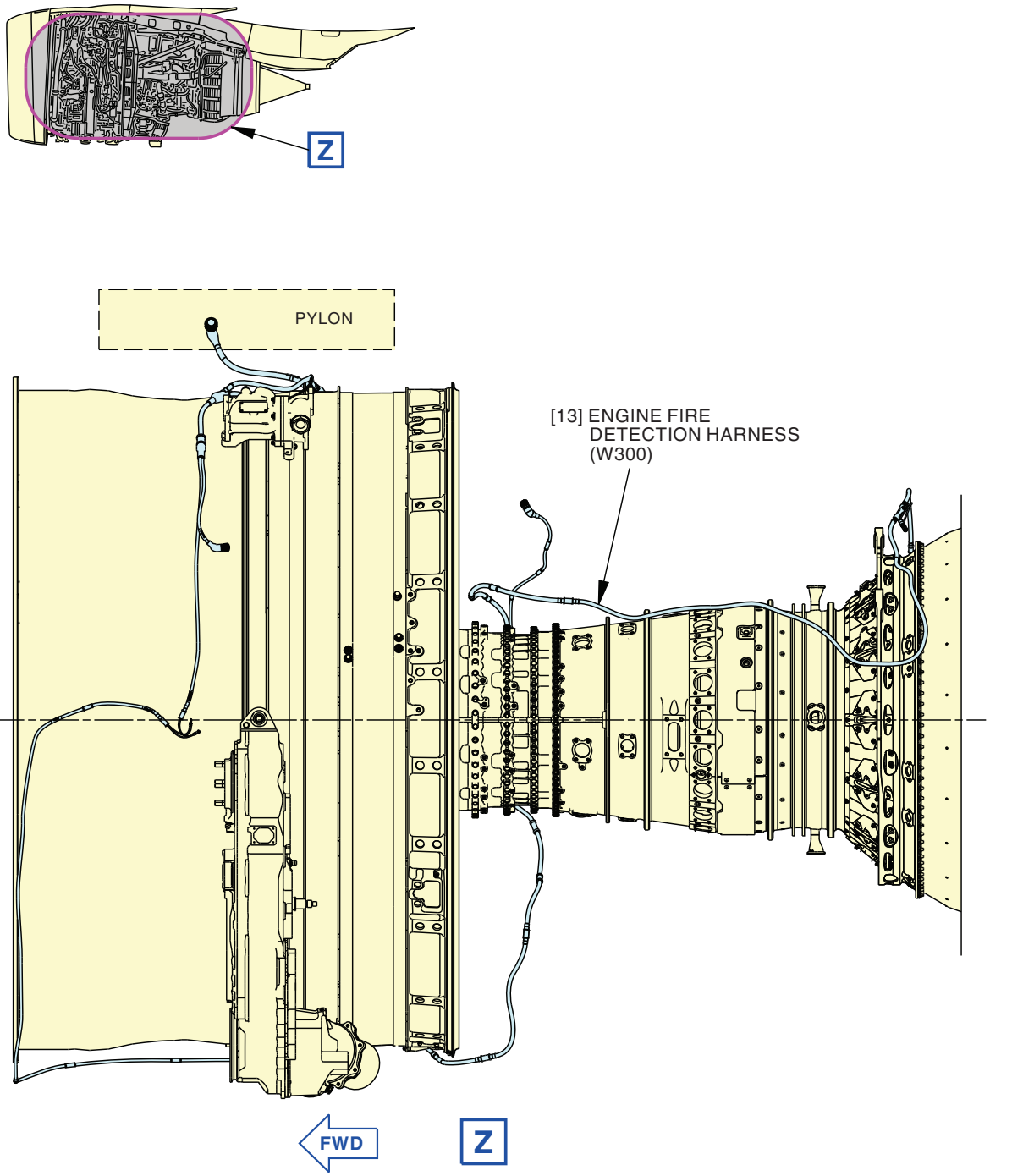

ICNLEAP1B732201D5882800104A00101

2966489 S0000745032_V1

**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 23 of 27)**



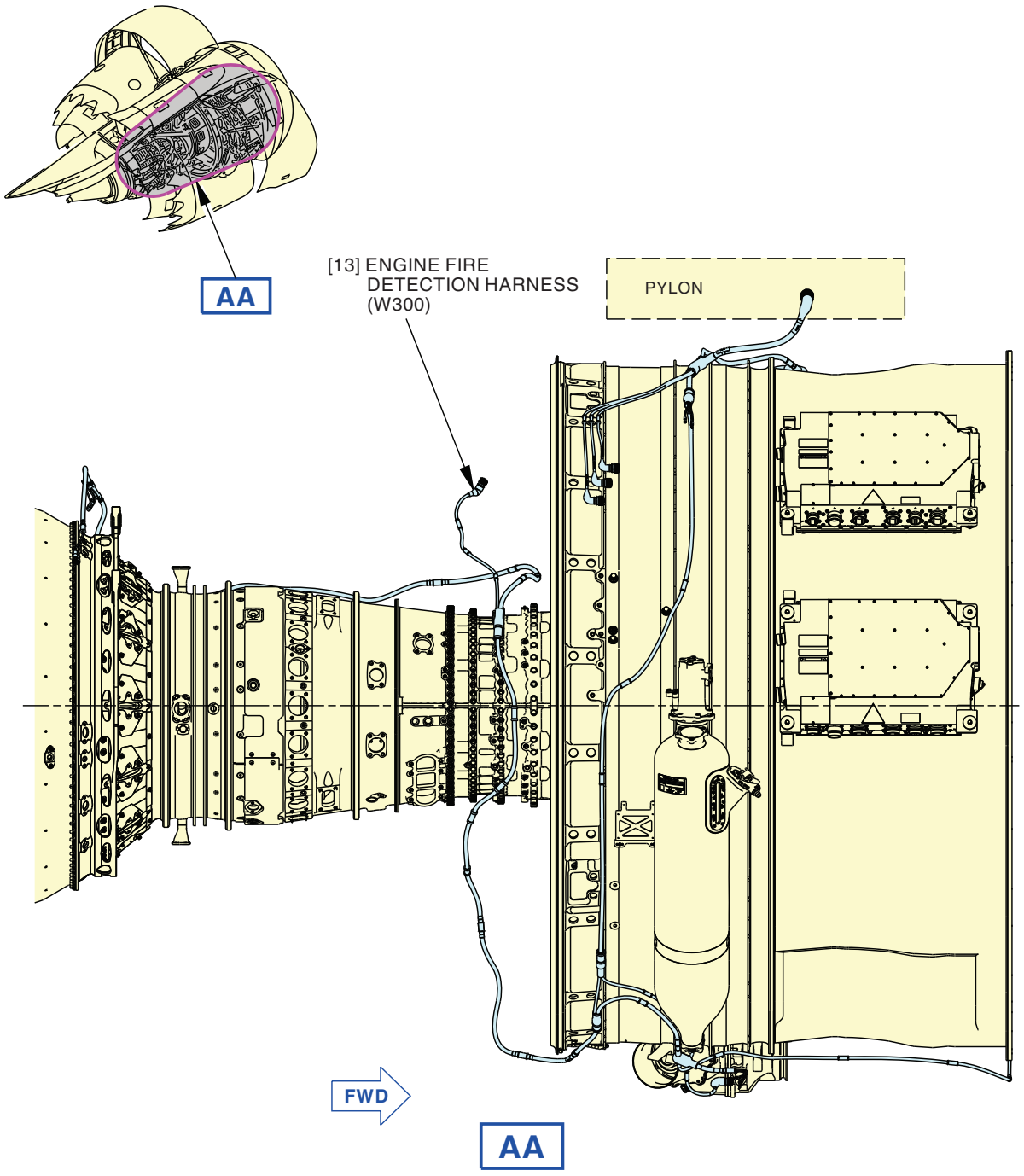
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 24 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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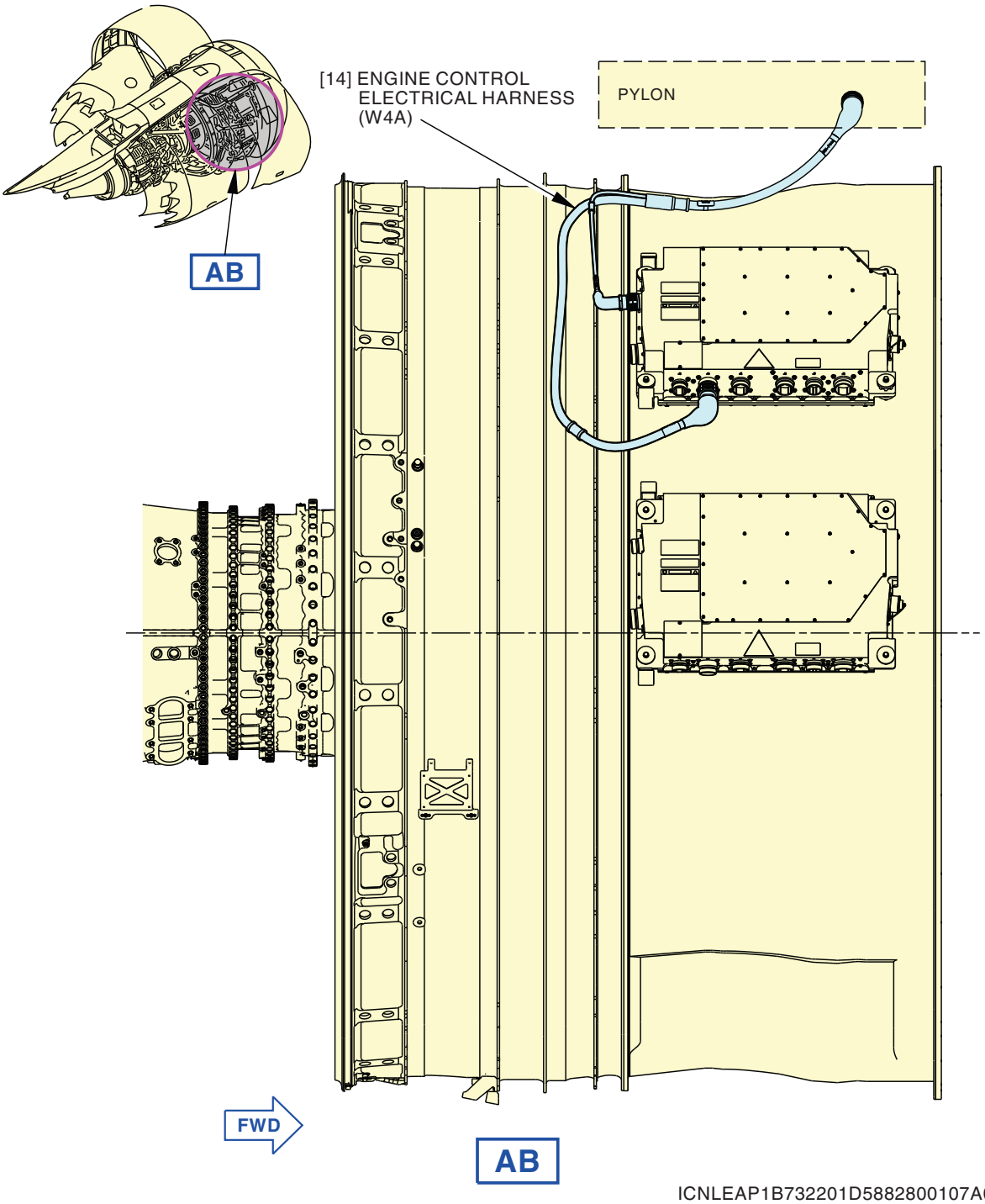
**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 25 of 27)**



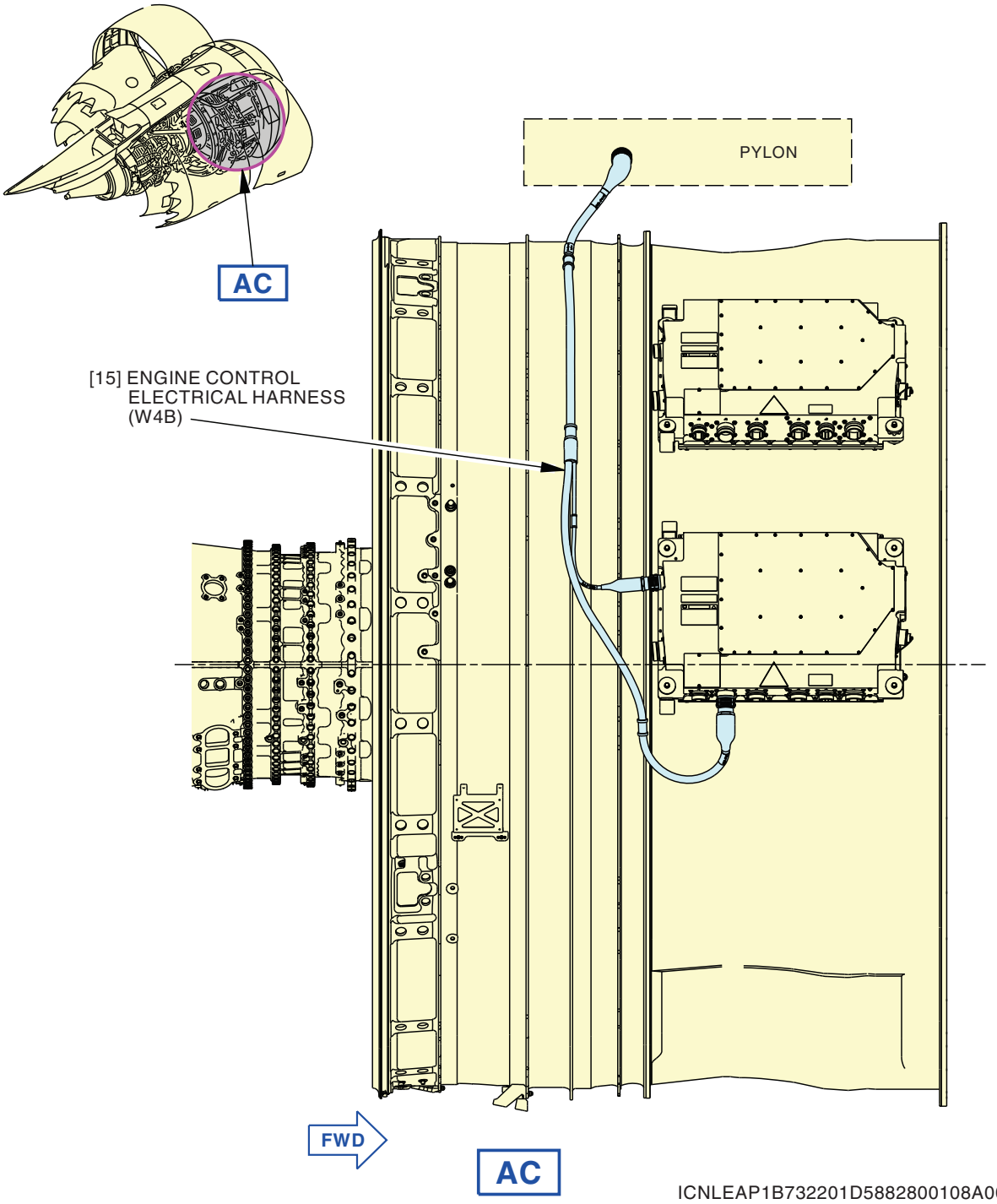
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 26 of 27)**



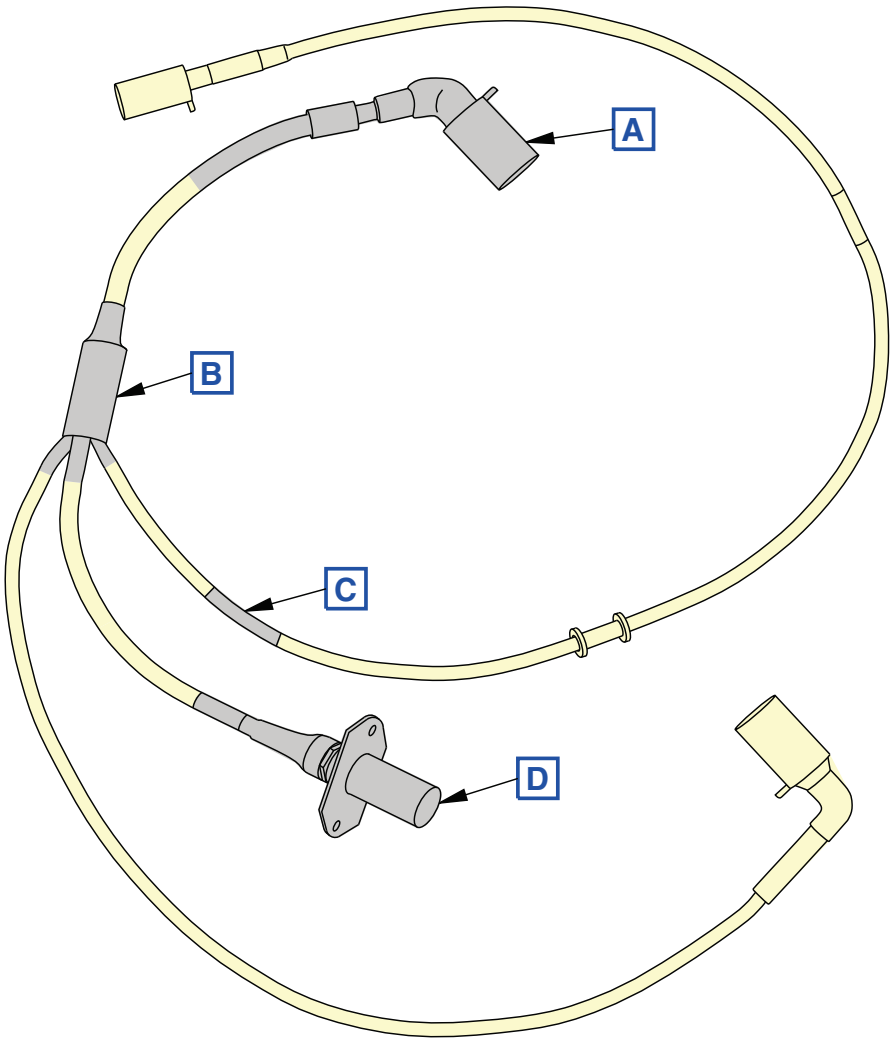
ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Harness Inspection**
**Figure 601/73-22-01-990-817-G00 (Sheet 27 of 27)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



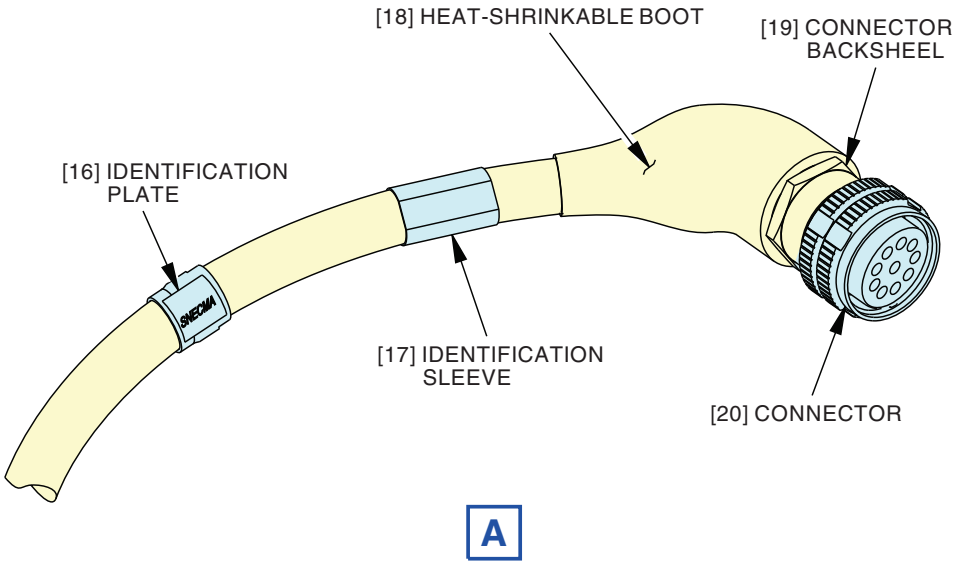

NOTE: THIS VIEW IS APPLICABLE TO THE HJ3A, HJ3B, HJ5A, HJ5B, HJ6A, HJ6B, HJ7, HJ8A AND HJ8B, W2, W4A, W4B, W100, W200, AND W300 ENGINE HARNESSES.

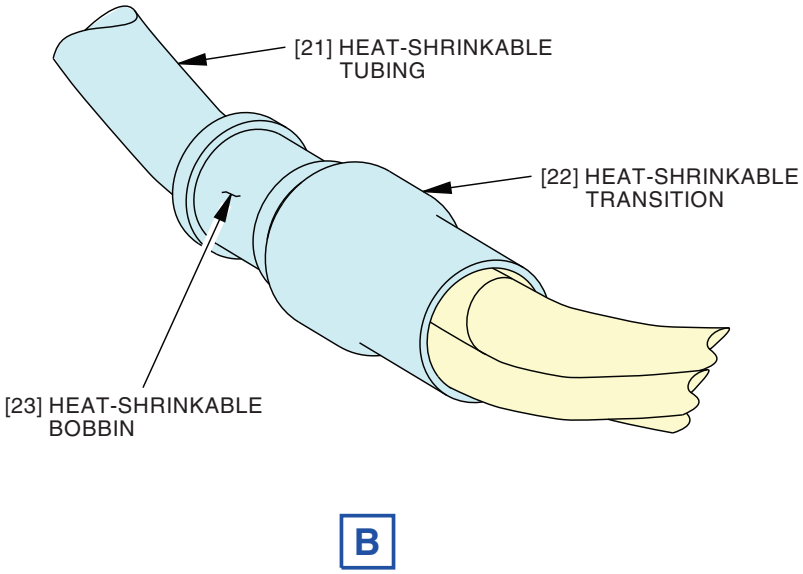
2468817 S0000575736_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 1 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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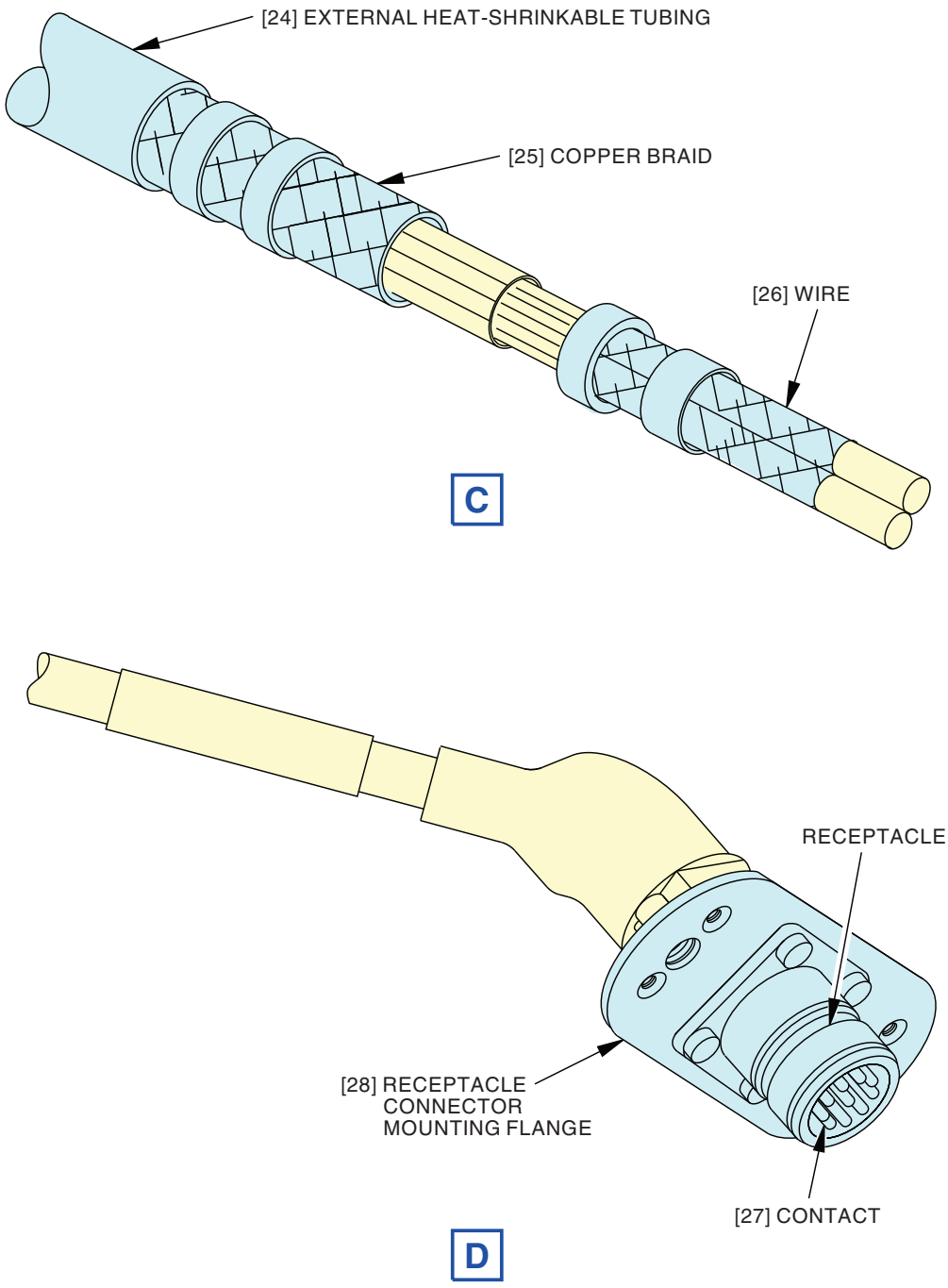
2468818 S0000575738_V3

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 2 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 3 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAFING, BURN, CUT, REMOVED CHARACTERS
IDENTIFICATION SLEEVE

BURN, CUT

CHAFING

IDENTIFICATION PLATE

BURN, DEFORMATION, CHAFING

LOOSENESS

GLUE

HEAT-SHRINKABLE PARTS

MARK, DEFORMATION
HEAT-SHRINKABLE PARTS, BOBBIN

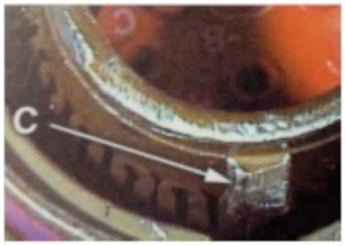
ICN-LEAP-1B-732201-B-F0301-A0087-A-001-01
2468820 S0000575741_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 4 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONNECTOR

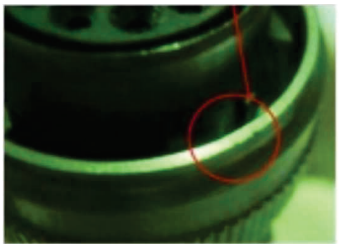



WORN MASTER KEY

CONNECTOR (THREAD)

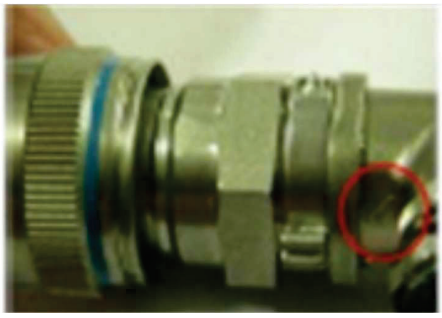
WORN THREADS

CONNECTOR (LOCKING RING)

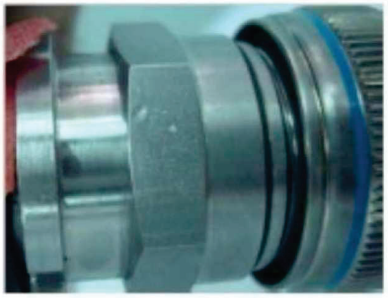


NICKS, DENTS, SCRATCHES

CONNECTOR AND BACKSHELL



SCRATCHES



NICKS, DENTS

ICN-LEAP-1B-732201-B-F0301-A0088-A-001-01

2468881 S0000575743_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 5 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONNECTOR AND METALLIC PARTS

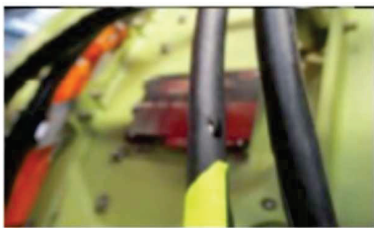
MARKING PARTIALLY ERASED

CONNECTOR

INTERFACIAL FACE DETERIORATED

EXTERNAL HEAT-SHRINKABLE TUBING

BURN, CHAFING, DEFORMATION

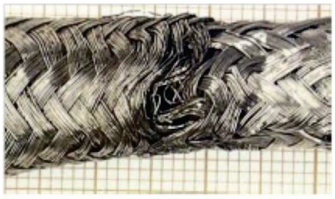


PERFORATION, CUT

INNER COPPER BRAID
(SHIELDING BRAID)

BURN, CUT, FRETTING

STAINLESS BRAID



BURN, CUT, FRETTING

RECEPTACLE CONNECTOR COUNTERPLATE

SCRATCHES, DENTS, NICKS

CONTACT

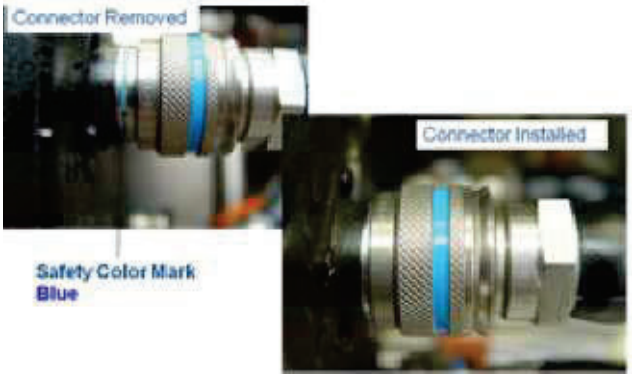
DISTORTION

ICN-LEAP-1B-732201-B-F0301-A0089-A-002-01
2468882 S0000575744_V3

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 6 of 13)**



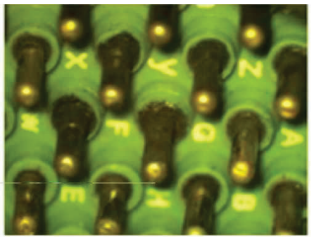
ECCN 9E991 BOEING PROPRIETARY - See title page for details



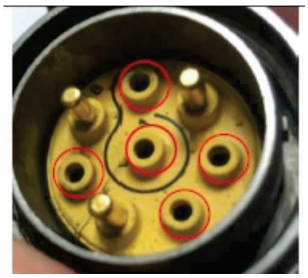

BAD ASSEMBLY CONNECTOR

SPECIFIC CABLES (BMS1367)
CHAFING

CONTACT



SIGNS OF WEAR,
ARCING OR CARBONIZING



MISSING OR RETRACTED

LUG
DISTORTION, CRACK OR TEAR

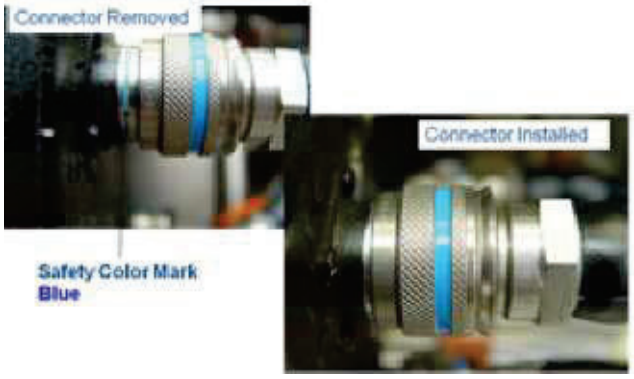
ANTI ROTATION LUG
CRACK OR TEAR

ICN-LEAP-1B-732201-B-F0301-A0090-A-001-01
2468883 S0000575745_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 7 of 13)**



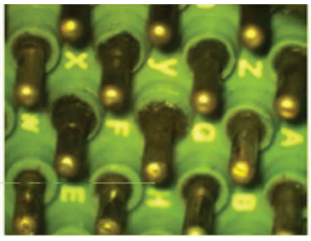
ECCN 9E991 BOEING PROPRIETARY - See title page for details



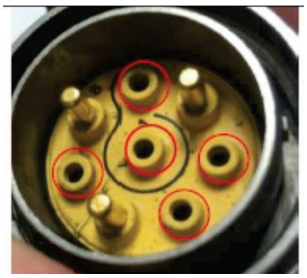

BAD ASSEMBLY CONNECTOR

SPECIFIC CABLES (BMS1367)
CHAFING

CONTACT



SIGNS OF WEAR,
ARCING OR CARBONIZING



MISSING OR RETRACTED

LUG
DISTORTION, CRACK OR TEAR

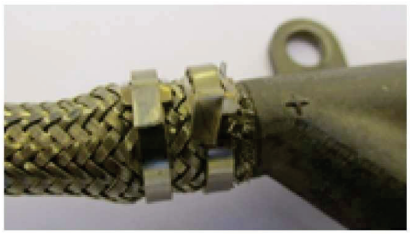
ICN-LEAP-1B-732201-B-F0301-A0090-A-001-01
3050363 S0000814540_V1

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 8 of 13)**



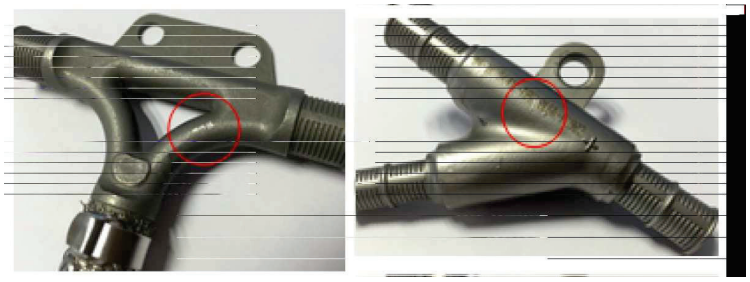
ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONNECTOR RFI RING
CUT LOOSENESS AND DEFORMATED



STAINLESS STEEL CLAMPS
CUT OR LOOSENESS

EXTERNAL HEATSHRINKABLE
TUBING FOR FIRE LOOP (TFER)
PERFORATION, CUT



MECHANICAL PARTS (BREAKOUT TYPE Y OR T AND TRANSITION)
NICKS, DENTS, SCRATCHES

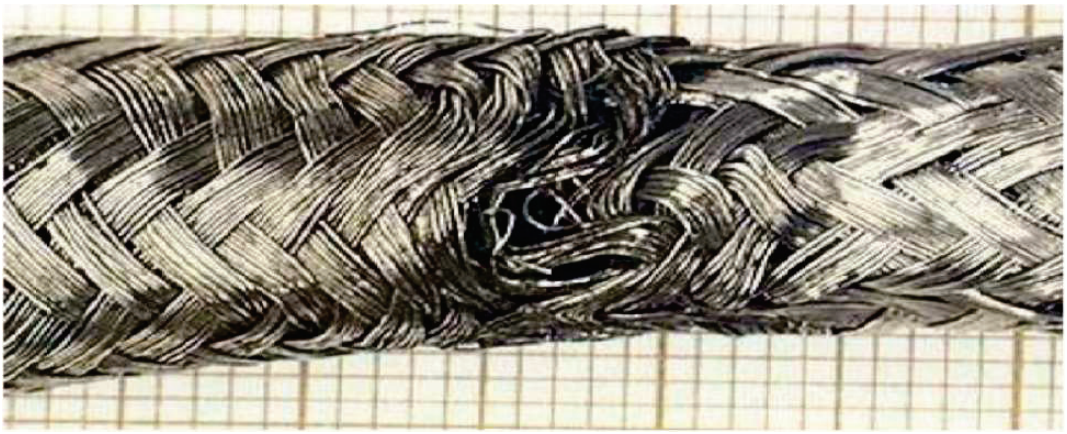
ICN-LEAP-1B-732201-B-F0301-A0094-A-001-01
2549135 S0000606883_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 9 of 13)**

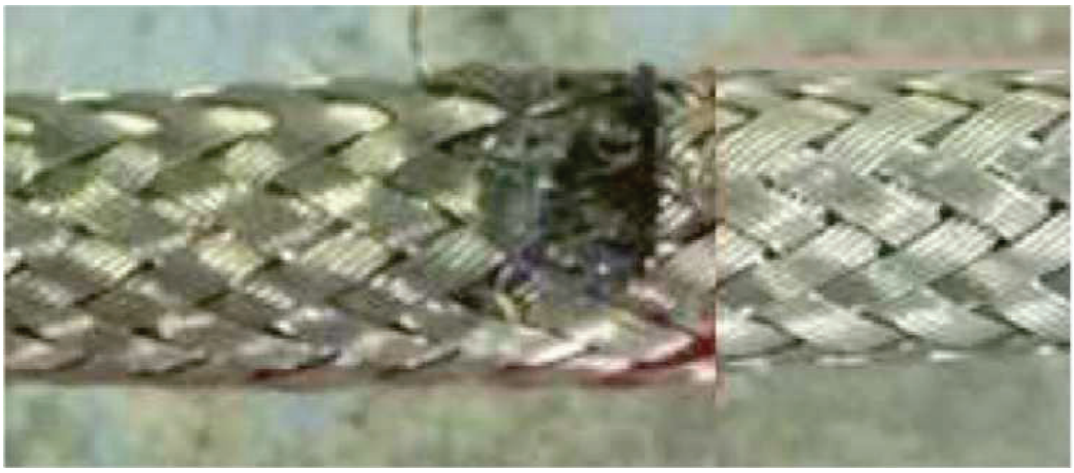


ECCN 9E991 BOEING PROPRIETARY - See title page for details

STAINLESS OUTER COPPER BRAID



BURN, CUT



FRETTING

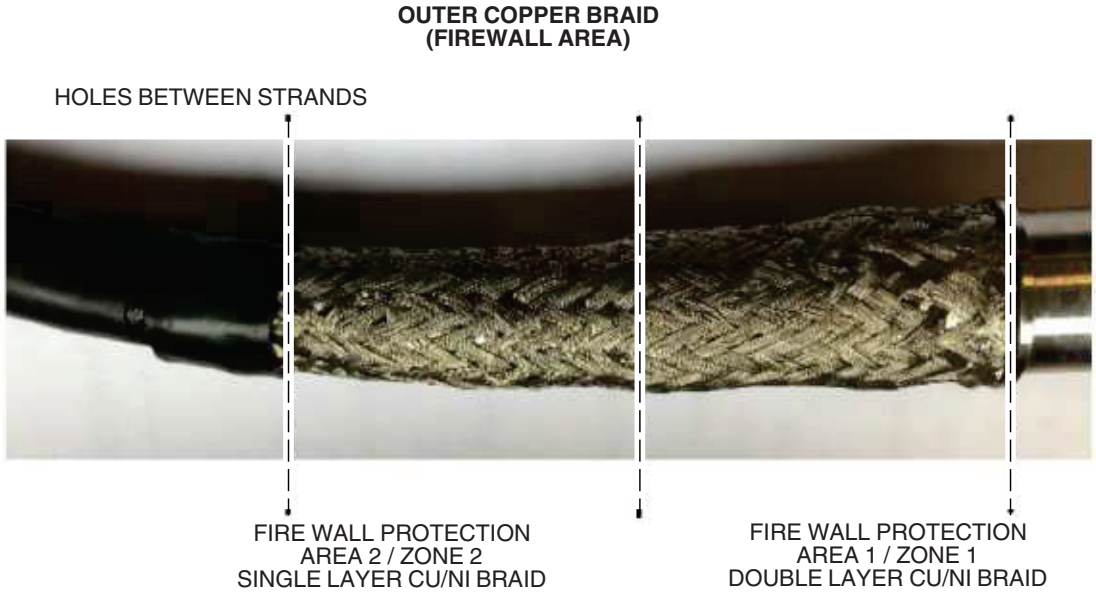
ICNLEAP1B732201D5882800109A00101

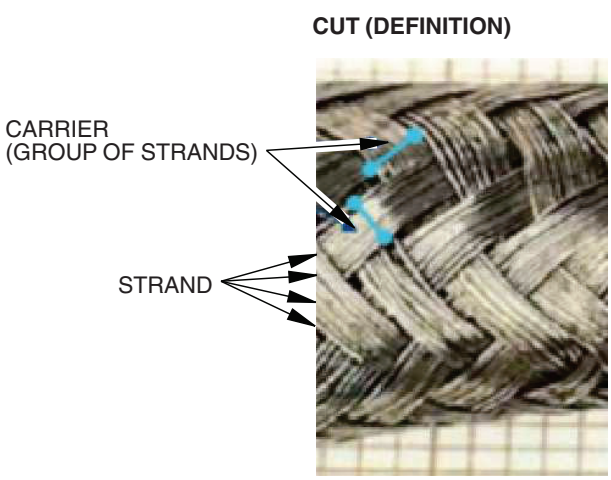
2967003 S0000746064_V1

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 10 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



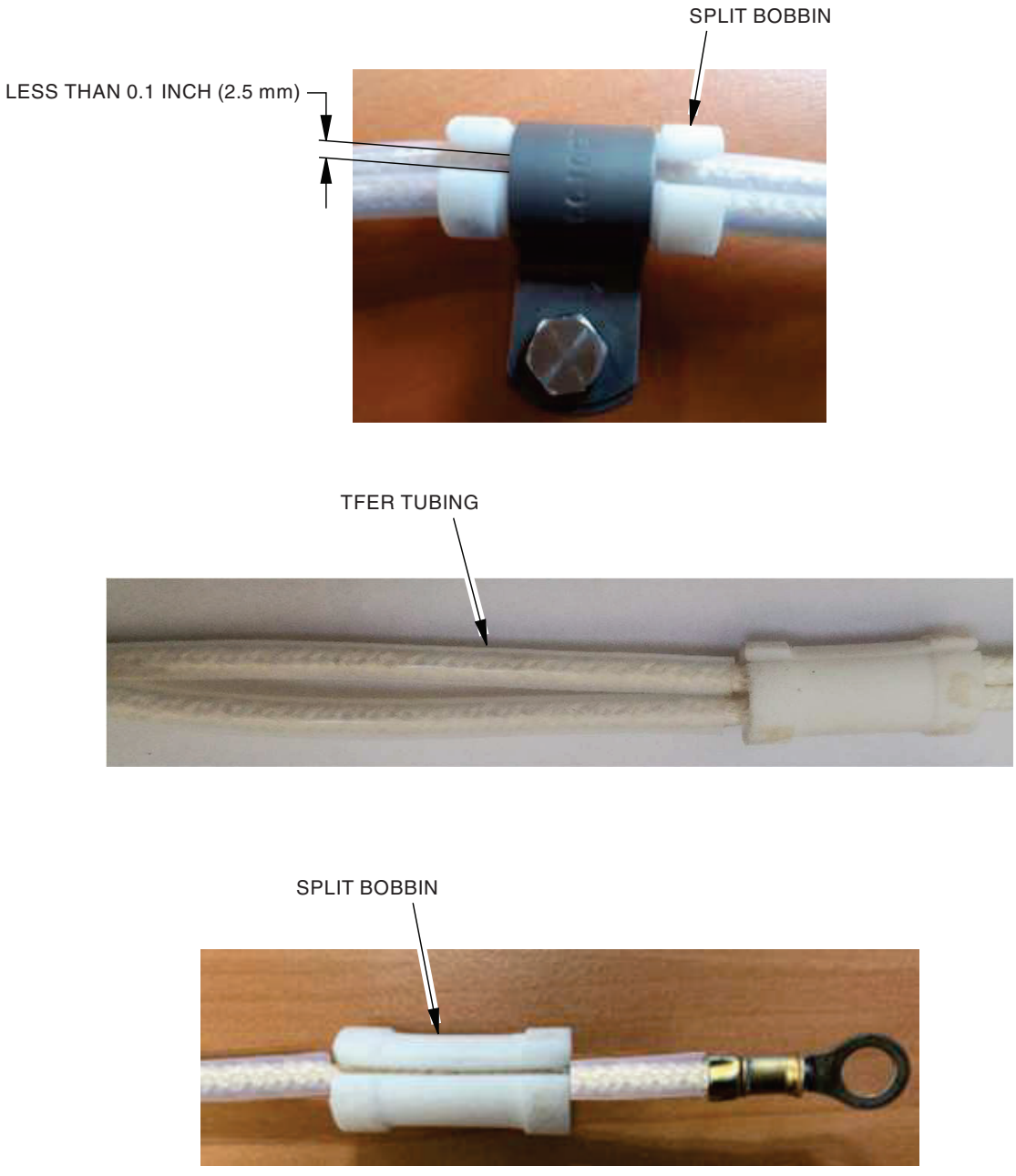


ICN-LEAP-1B-732201-D-5882800124-A-001-01
3050364 S0000814452_V2

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 11 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



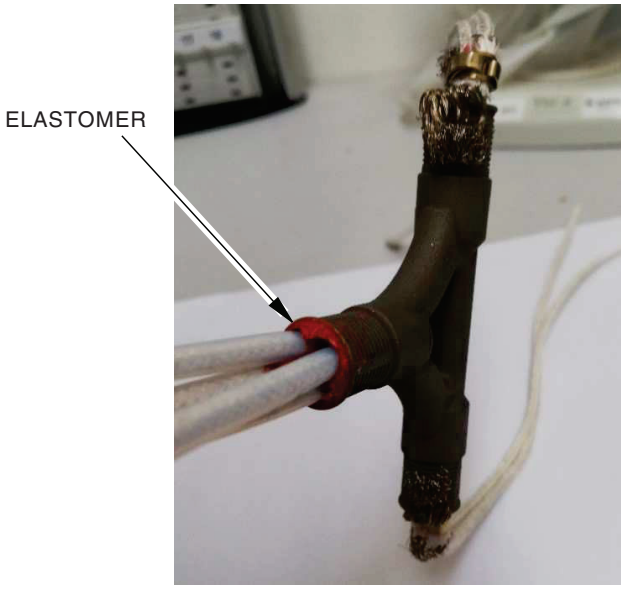

ICN-LEAP-1B-732201-D-5882800116-A-001-01
ICN-LEAP-1B-732201-D-5882800117-A-001-01
ICN-LEAP-1B-732201-D-5882800118-A-001-01

3050365 S0000814541_V1

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 12 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

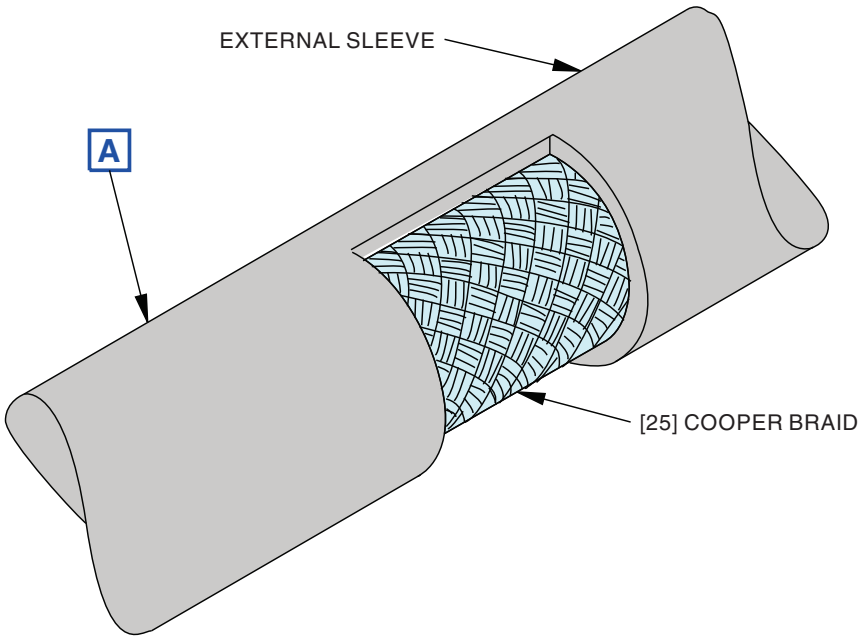


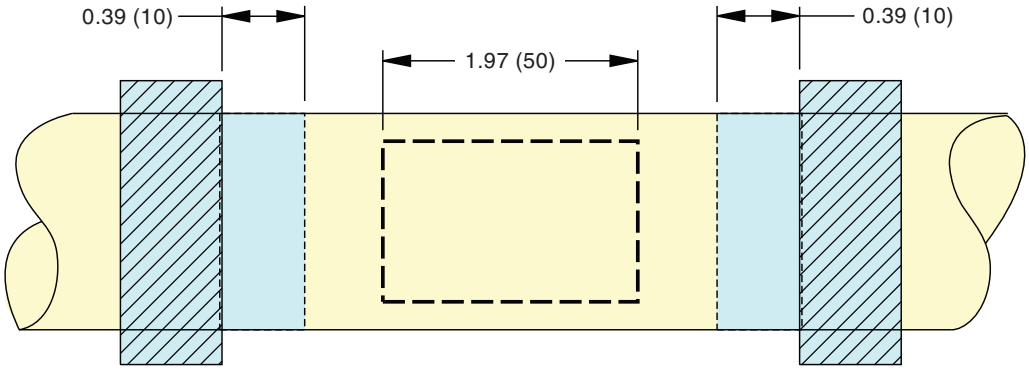

ICN-LEAP-1B-732201-D-5882800119-A-001-01
3050366 S0000814542_V1

**Engine Control Electrical Harness Inspection**
**Figure 602/73-22-01-990-883-G00 (Sheet 13 of 13)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**NOTE:**

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

MOUNTING POINT (OMEGA CLIP OR P CLAMP)

NO REPAIR AREA

ICNLEAP1B732201D5882800100A00101

2966522 S0000745037_V1

**Heat-Shrinkable Tubing Inspection**
**Figure 603/73-22-01-990-884-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-210-802-G00**

## 3. Engine Control Electrical Harness Connectors Inspection

(Figure 604)

### A. General

(1) This task gives the instructions to do the visual inspection of the engine control electrical harness connectors.

### B. References

| Reference | Title |
|---|---|
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01-000-801-G00 | Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal (P/B 401) |
| 73-22-01-400-801-G00 | Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation (P/B 401) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

### F. Prepare for the Inspection

**SUBTASK 73-22-01-860-077-G00**

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-078-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Engine Control Electrical Harness Connectors Inspection

SUBTASK 73-22-01-211-014-G00

(1) Do a visual inspection of the connector [1] for these conditions:

(a) Worn master key

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(b) Signs of short-circuit, carbonizing

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(c) Nicks, dents, scratches on the connector locking ring


ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Permitted, if the nicks, dents and scratches are not more than 0.1 in. (2.0 mm) long, 0.04 in. (1.0 mm) wide and 0.01 in. (0.3 mm) deep.

2) If findings arer more than the limits, replace the applicable engine control electrical component.

(d) Scratches on connector and backshell parts

1) Permitted, if scratches are not more than 0.01 in. (0.3 mm) in depth.

2) If scratches are more than the limits, replace the applicable engine control electrical component.

(e) Nicks, Dents on connector thread

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(f) Nicks and dents on connector backshell

1) Permitted, if the nicks and dents are not more than 0.1 in. (2.0 mm) long, 0.04 in. (1.0 mm) wide and 0.01 in. (0.3 mm) deep.

2) If findings are more than the limits, replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(g) Connector interfacial face deteriorated

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(h) Cut or looseness and deformated on connector RFI Ring

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(i) Bad connector assembly

1) Not permitted if color mark on the receptacle is visible

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(j) Marking partially erased on connector and metallic parts

1) Permitted even if marking is not readable.

2) Replace the component at the next shop visit.

**SUBTASK 73-22-01-211-015-G00**

(2) Do a visual inspection of the connectors and mechanical parts for this conditions:

(a) Scratches on connector and backshell parts

1) Permitted, if scratches are not more than 0.01 in. (0.3 mm) deep.

2) If scratches are more than the limit, replace the applicable engine control electrical component.

(b) Nicks and dents on connector backshell

1) Permitted, if the nicks and dents are not more than 0.1 in. (2.0 mm) long, 0.04 in. (1.0 mm) wide and 0.01 in. (0.3 mm) deep.



ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If findings are more than the limits, replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(c) Nicks and dents on the connector mechanical parts (breakout type Y or T)

1) Permitted, if the nicks and dents are not more than 0.1 in. (2.0 mm) long, 0.04 in. (1.0 mm) wide and 0.01 in. (0.3 mm) deep.

2) If findings are more than the limits, replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(d) Marking partially erased on connector and metallic parts

1) Permitted even if marking is not readable.

2) Replace the component at the next shop visit.

(e) Cut or looseness on the mechanical parts (stainless steel clamps)

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

**SUBTASK 73-22-01-211-016-G00**

(3) Do the visual inspection of the heat-shrinkable boot [18] for these conditions:

(a) Signs of wear, arcing or carbonizing

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(b) Distortion

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

(c) Missing or retracted

1) Not permitted.

2) Replace the applicable engine control electrical harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

## H. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-22-01-860-079-G00**

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |


ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

**SUBTASK 73-22-01-860-080-G00**

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

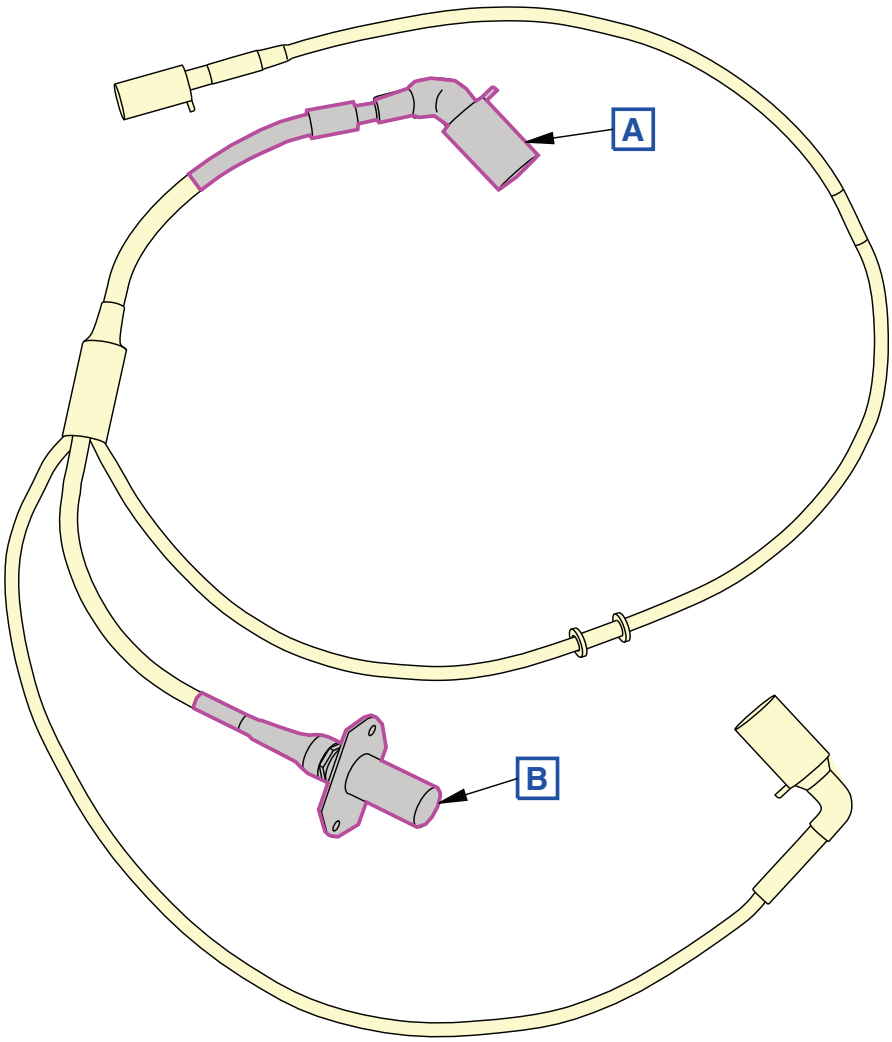
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**——— END OF TASK ———**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



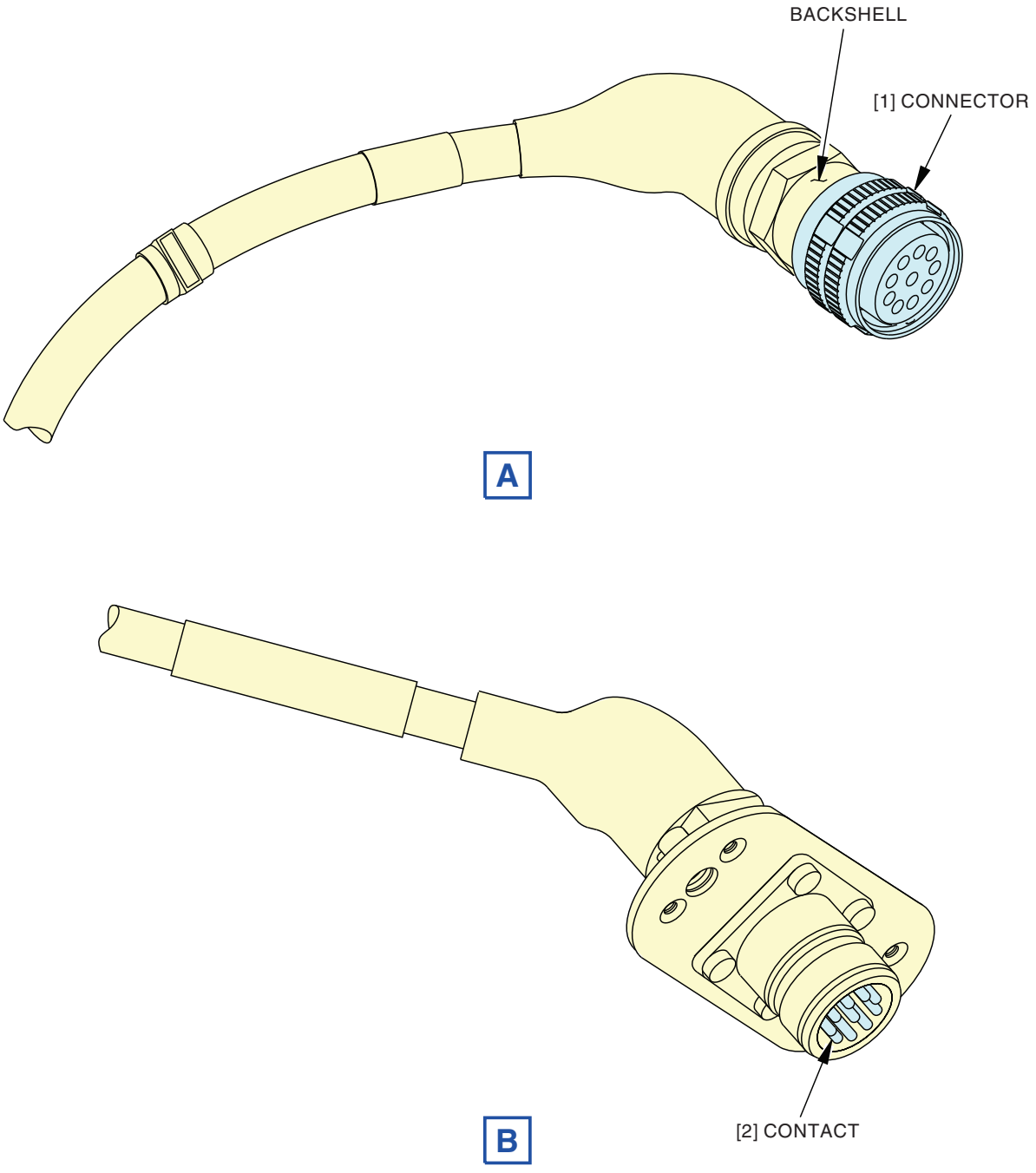

NOTE: THIS VIEW IS APPLICABLE TO THE HJ3A, HJ3B, HJ5A, HJ5B, HJ6A, HJ6B, HJ7, HJ8A AND HJ8B ENGINE CONTROL ELECTRICAL HARNESS.

2470325 S0000575954_V1

**Engine Control Electrical Harness Connectors Inspection**
**Figure 604/73-22-01-990-834-G00 (Sheet 1 of 4)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Control Electrical Harness Connectors Inspection**
**Figure 604/73-22-01-990-834-G00 (Sheet 2 of 4)**

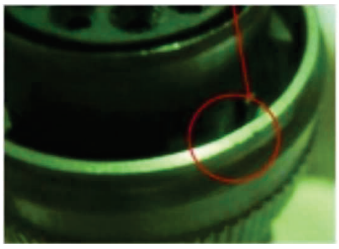


ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONNECTOR

WORN MASTER KEY ON CONNECTOR

WORN THREADS ON CONNECTOR AND BACKSHELL



NICKS, DENTS, SCRATCHES ON CONNECTOR LOCKING RING

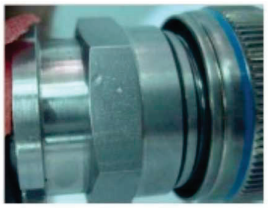
FAILURE OF MACHINING ON BACKSHELLS

2470330 S0000575959_V2

**Engine Control Electrical Harness Connectors Inspection**
**Figure 604/73-22-01-990-834-G00 (Sheet 3 of 4)**

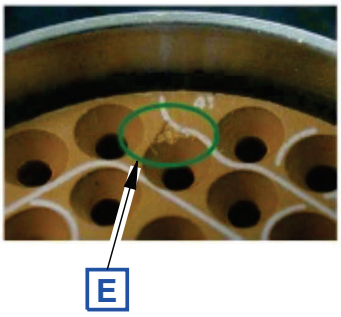


ECCN 9E991 BOEING PROPRIETARY - See title page for details

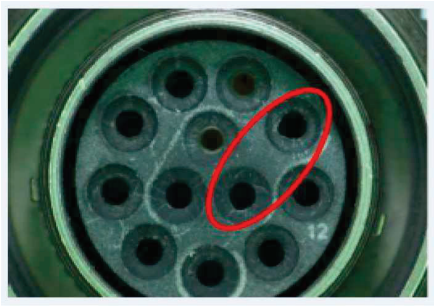


NICKS, DENTS ON BACKSHELLS

MARKING PARTIALLY ERASED ON BACKSHELL AND CONNECTOR




GROMMET DETERIORATED - PERMITTED ON CONNECTOR



INTERFACIAL FACE DETERIORATED

CONTACT DISTORTION

2470332 S0000575961_V2

**Engine Control Electrical Harness Connectors Inspection**
**Figure 604/73-22-01-990-834-G00 (Sheet 4 of 4)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## ENGINE CONTROL ELECTRICAL HARNESSES - CLEANING/PAINTING

### 1. General

A. This procedure contains one task:

(1) Engine control electrical harnesses cleaning.

**TASK 73-22-01-100-801-G00**

### 2. Engine Control Electrical Harness Cleaning

#### A. General

(1) This task gives the instructions to clean the engine control electrical harnesses.

#### B. References

| Reference | Title |
|---|---|
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |

#### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |

#### D. Consumable Materials

| Reference | Description | Specification |
|---|---|---|
| B00679 [C04-035] | Alcohol - Isopropyl | |
| C50233 [C04-258] | Cleaner - Electrical Contact Cleaner / Lubricant - DeoxIT GOLD Gx100L or Gx5S-6N | |
| G50031 [C10-108] | Brush - Disposable, Natural Bristle - No. 7237T4 or 7798T2 | |
| G50938 [C10-228] | Brushes - Nylon Miniature | |
| G50939 [C10-207] | Swab - Lint Free Rayon Tip | |

#### E. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

#### F. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

#### G. Prepare for the Cleaning

**SUBTASK 73-22-01-860-081-G00**

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |



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(Continued)

| Number | Name/Location |
|---|---|
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

SUBTASK 73-22-01-020-023-G00

(2) Open these applicable circuit breakers and install safety tags:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-082-G00

(3) Make sure that the EEC Maint switch, on the P461 panel, is in the NORM position.

## H. Engine Control Electrical Harness Cleaning

SUBTASK 73-22-01-020-024-G00

(1) Disconnect the electrical connector on the applicable EEC from the component.

SUBTASK 73-22-01-200-001-G00

(2) Examine the socket contacts and pins of the electrical connector for damage.

(a) If socket contacts or pins are heavily damaged, oxidized or blackened, replace the electrical connector and mated component or sensor.

SUBTASK 73-22-01-160-001-G00

**WARNING**

DO NOT GET ISOPROPYL ALCOHOL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE ISOPROPYL ALCOHOL. KEEP THE ISOPROPYL ALCOHOL AWAY FROM SPARKS, FLAME, AND HEAT. ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

**CAUTION**

USE ONLY ISOPROPYL ALCOHOL WHEN YOU CLEAN THE ELECTRICAL CONNECTORS. IF YOU USE SOME OTHER SOLVENT, YOU CAN CAUSE DAMAGE TO THE CONNECTOR OR CAUSE THE FAILURE OF A COMPONENT.

(3) Clean the electrical connector pins and connector face as follows:

(a) Flush the electrical connector with alcohol, B00679 [C04-035] to remove contamination and other unwanted materials.



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(b) Use a No. 7237T4 or 7798T2 brush, G50031 [C10-108] and clean the electrical connector pins and connector face with alcohol, B00679 [C04-035].

(c) Flush the electrical connector again with alcohol, B00679 [C04-035] to remove the contamination that remains and other unwanted materials.

NOTE: Do the steps again as necessary to remove all contamination.

(d) Remove the alcohol, B00679 [C04-035] that remains from the connector.

**WARNING**

WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

(e) Let the electrical connector dry fully.

NOTE: If it is necessary, blow the electrical connector dry with clean, filtered air at a maximum pressure of 30.0 PSI (2.07 bar).

**CAUTION**

DO NOT APPLY THE CLEANER/LUBRICANT TO THE INSIDE SURFACE OF THE CONNECTOR RECEPTACLE OR TO THE FACE OF THE INTERFACIAL SEAL. APPLY THE CLEANER/LUBRICANT ONLY TO THE TIP AND SIDES OF THE CONNECTOR PINS. USE A NATURAL BRISTLE BRUSH AND APPLY THE CLEANER/LUBRICANT WITH LIGHT PRESSURE SO YOU DO NOT DAMAGE THE CONNECTOR PINS.

(f) Apply a layer of electrical contact cleaner, C50233 [C04-258] to the electrical connector pins.

1) Use the liquid version of the electrical contact cleaner lubricant.

2) Use a No. 7237T4 or 7798T2 brush, G50031 [C10-108] and lightly apply the cleaner lubricant on the tips and sides of the connector pins.

NOTE: Keep the cleaner lubricant from the inside surface of the receptacle and interfacial seal, as much as possible.

(g) Clean unwanted electrical contact cleaner lubricant from the inside surface of the connector receptacle and from the face of the interfacial seal with clean, lint free swabs, G50939 [C10-207].

(h) Let the electrical connector pins and face air dry for 10 - 15 seconds.

**SUBTASK 73-22-01-160-002-G00**

**WARNING**

DO NOT GET ISOPROPYL ALCOHOL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE ISOPROPYL ALCOHOL. KEEP THE ISOPROPYL ALCOHOL AWAY FROM SPARKS, FLAME, AND HEAT. ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Clean the electrical connector socket contacts and connector face as follows:

(a) Flush the electrical connector with alcohol, B00679 [C04-035] to remove contamination and other unwanted materials from the socket contacts and connector face.



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**CAUTION**

MAKE SURE YOU USE THE CORRECT SIZE MINIATURE NYLON BRUSH TO CLEAN THE ELECTRICAL CONNECTOR SOCKET CONTACTS. USE A MINIATURE NYLON BRUSH THAT IS SIZED FOR THE SPECIFIC SOCKET CONTACT. OTHERWISE, DAMAGE TO THE SOCKET CONTACT CAN OCCUR OR THE SOCKET CONTACTS WILL NOT BE SUFFICIENTLY CLEANED.

(b) Use a nylon miniature brushes, G50938 [C10-228] and clean the electrical connector socket contacts with alcohol, B00679 [C04-035] as follows:

1) Use a 0.031 inch (0.79 mm) size nylon brush for the 20 gauge socket contact, or

2) Use a 0.062 inch (1.57 mm) brush for the 16 gauge socket contact.

(c) Use a No. 7237T4 or 7798T2 brush, G50031 [C10-108] and clean the electrical connector face with alcohol, B00679 [C04-035].

(d) Flush the electrical connector again with alcohol, B00679 [C04-035] to remove the contamination that remains and other unwanted materials.

NOTE: Do the steps again as necessary to remove all contamination.

(e) Remove the remaining alcohol, B00679 [C04-035] from the connector.

**WARNING**

WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

1) Let the electrical connector dry fully.

NOTE: If it is necessary, blow the electrical connector dry with clean, filtered air at a maximum pressure of 30.0 PSI (2.07 bar).

(f) Apply a layer of electrical contact cleaner, C50233 [C04-258] to the electrical connector socket contacts.

1) If you use the spray version of the electrical contact cleaner, C50233 [C04-258], apply the cleaner lubricant with a short burst of spray directly into the socket contacts.

2) If you use a liquid version of electrical contact cleaner, C50233 [C04-258], apply the cleaner lubricant with nylon miniature brushes, G50938 [C10-228] as follows:

**CAUTION**

MAKE SURE YOU USE THE CORRECT SIZE MINIATURE NYLON BRUSH TO CLEAN THE ELECTRICAL CONNECTOR SOCKET CONTACTS. USE A MINIATURE NYLON BRUSH THAT IS SIZED FOR THE SPECIFIC SOCKET CONTACT. OTHERWISE, DAMAGE TO THE SOCKET CONTACT CAN OCCUR OR THE SOCKET CONTACTS WILL NOT BE SUFFICIENTLY CLEANED.

a) Use a miniature nylon brush that is sized for the specific socket contact.

<1> Use a 0.031 inch (0.79 mm) brush for 20 gauge socket contacts, or

<2> Use a 0.062 inch (1.57 mm) brush for 16 gauge socket contacts.

(g) Remove the electrical contact cleaner from the face of the electrical connector.

(h) Let the electrical connector socket contacts air dry for 10 - 15 seconds.



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SUBTASK 73-22-01-420-023-G00

(5) Connect the electrical connector on the applicable EEC to the component.

(a) Tighten by hand then use teflon-jawed pliers, STD-664 to tighten the connector until the pliers slip.

NOTE: This will be approximately 1/8 turn more than hand-tight.

## I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-420-024-G00

(1) Remove the safety tags and close these circuit breakers:

**CAPT Electrical System Panel, P18-2**

| Row | Col | Number | Name |
|---|---|---|---|
| A | 4 | C04008 | ENGINE 1 ALTN PWR CHAN B |
| A | 5 | C04007 | ENGINE 1 ALTN PWR CHAN A |
| B | 3 | C01312 | ENGINE 1 RUN/PWR |

**F/O Electrical System Panel, P6-2**

| Row | Col | Number | Name |
|---|---|---|---|
| B | 5 | C01313 | ENGINE 2 RUN/PWR |
| D | 7 | C04010 | ENGINE 2 ALTN PWR CHAN B |
| D | 8 | C04009 | ENGINE 2 ALTN PWR CHAN A |

**F/O Electrical System Panel, P6-5**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C04018 | ENG 1 STBY ALTN PWR RELAY CHAN A |
| E | 4 | C04019 | ENG 1 STBY ALTN PWR CHAN A |

SUBTASK 73-22-01-860-083-G00

(2) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**END OF TASK**



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## ENGINE WIRING HARNESS - REPAIRS

### 1. General

A. This procedure has these tasks:

(1) An engine wiring harness repair (application of silicone tape patch)

(2) An engine wiring harness repair (application of stainless steel braid)

(3) An engine wiring harness heat-shrinkable boot repair

(4) An engine wiring harness heat-shrinkable tubing repair (application of silicone tape patch)

(5) An engine wiring harnesses W200 and W300 - BMS 1367 cable at the terminal lugs repair

(6) An engine wiring harnesses W200 and W300 bobbin replacement

(7) An engine wiring harnesses W200 and W300 anti-rotation bobbin repair

(8) An engine wiring harnesses firewall area repair.

**TASK 73-22-01-900-801-G00**

### 2. Engine Wiring Harness Repair (Application of Silicone Tape Patch)

(Figure 801)

#### A. General

(1) This repair is superseded by Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch), TASK 73-22-01-900-804-G00.

(2) This task gives instructions for the application of silicon rubber tape for interim protection of harness outer sheath.

(3) The Table 801 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 801/73-22-01-993-802-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-083-111-0 | 72-0001 | Harness HJ3A |
| 364-083-211-0 | 72-0001 | Harness HJ3B |
| 364-083-311-0 | 72-0013 | Harness HJ5A |
| 364-083-312-0 | 72-0001 | |
| 364-083-313-0 | 73-0019 | |
| 364-083-411-0 | 72-0013 | Harness HJ5B |
| 364-083-412-0 | 72-0001 | |
| 364-083-413-0 | 73-0019 | |
| 364-083-511-0 | 72-0013 | Harness HJ6A |
| 364-083-512-0 | 72-0001 | |
| 364-083-513-0 | 73-0019 | |
| 364-083-514-0 | 73-0031 | |



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**Table 801/73-22-01-993-802-G00 (Continued)**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-083-611-0 | 72-0013 | Harness HJ6B |
| 364-083-612-0 | 72-0001 | |
| 364-083-613-0 | 73-0019 | |
| 364-083-614-0 | 73-0031 | |
| 364-083-710-0 | 72-0001 | Harness HJ7 |
| 364-083-711-0 | 75-0010 | |
| 364-083-811-0 | 72-0001 | Harness HJ8A |
| 364-083-812-0 | 73-0021 | |
| 364-083-911-0 | 72-0001 | Harness HJ8B |
| 364-055-211-0 | 72-0001 | Harness W2 |
| 364-055-212-0 | 75-0010 | |
| 364-055-311-0 | 72-0001 | Harness W4A |
| 364-055-411-0 | 72-0001 | Harness W4B |
| 364-056-511-0 | 72-0001 | Harness W100 |
| 364-056-612-0 | 72-0001 | Harness W200 |
| 364-056-613-0 | 72-0132 | |
| 364-056-614-0 | 72-0306 | |
| 364-056-711-0 | 72-0001 | Harness W300 |
| 364-056-712-0 | 72-0132 | |
| 364-056-713-0 | 72-0214 | |
| 364-056-714-0 | 72-0214 | |
| 364-056-715-0 | 72-0306 | |

**B. References**

| Reference | Title |
|---|---|
| 70-21-23-100-801-G00 | Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G51335 [CP2674] | Lint-Free Cloth | |
| G51695 [CP2940] | Tape - Silicone, Black - 69N | |
| G51696 [CP2939] | Tape - Silicone, Red/Brown - 68N | |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**F. Prepare for the Repairs**

SUBTASK 73-22-01-860-112-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-113-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |



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**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Engine Wiring Harness Repairs

**WARNING** REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

SUBTASK 73-22-01-020-032-G00

(1) Loosen the clamps as required until the area to repair is clear.

SUBTASK 73-22-01-220-001-G00

(2) Do a visual inspection of the area to repair (Figure 801).

NOTE: This repair only applies on external heat shrinkable tubing.

NOTE: Repairs are allowed between attachments points only.

NOTE: Repairs under attachment points are not permitted.

SUBTASK 73-22-01-220-002-G00

(3) Do a dimensional inspection of the area to repair (View A, Figure 801).

(a) Do not use this repair if there is one of the following conditions:

1) The length of the area to repair is more than 1.97 in. (50 mm) (DIM. A), or

2) The distance between the area to repair and nearest repaired area is less than 1.38 in. (35 mm) (DIM. B + DIM. C), or

3) The distance between the area to repair and nearest attachment point is less than 1.38 in. (35 mm) (DIM. B + DIM. C).

SUBTASK 73-22-01-100-002-G00

(4) Clean the external heat shrinkable tubing in and around the area to repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 73-22-01-220-003-G00

(5) Examine the area to repair as follows:

(a) Remove all material on the area to repair (Figure 801).

NOTE: Degradation of copper braid is not permitted.

SUBTASK 73-22-01-350-001-G00

(6) Wind silicone tape 68N, G51696 [CP2939], or silicone tape 69N, G51695 [CP2940], around the harness branch as follows (View A, Figure 801):

(a) Extend the ribbon 0.99 in. (25 mm) (DIM. B) from the area to repair limits.



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(b) Do an 50% overlap while winding.

(c) Continue the winding in the opposite direction with the same overlap.

1) Do a straight turn to complete the winding.

(d) Finish the winding with one complete turn.

SUBTASK 73-22-01-350-002-G00

(7) Do a visual inspection of the repaired area.

(a) Make sure that the tape is fully attached.

NOTE: Nicks and perforations are not permitted.

SUBTASK 73-22-01-410-035-G00

(8) Put the harness back on its supports.

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-114-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-115-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



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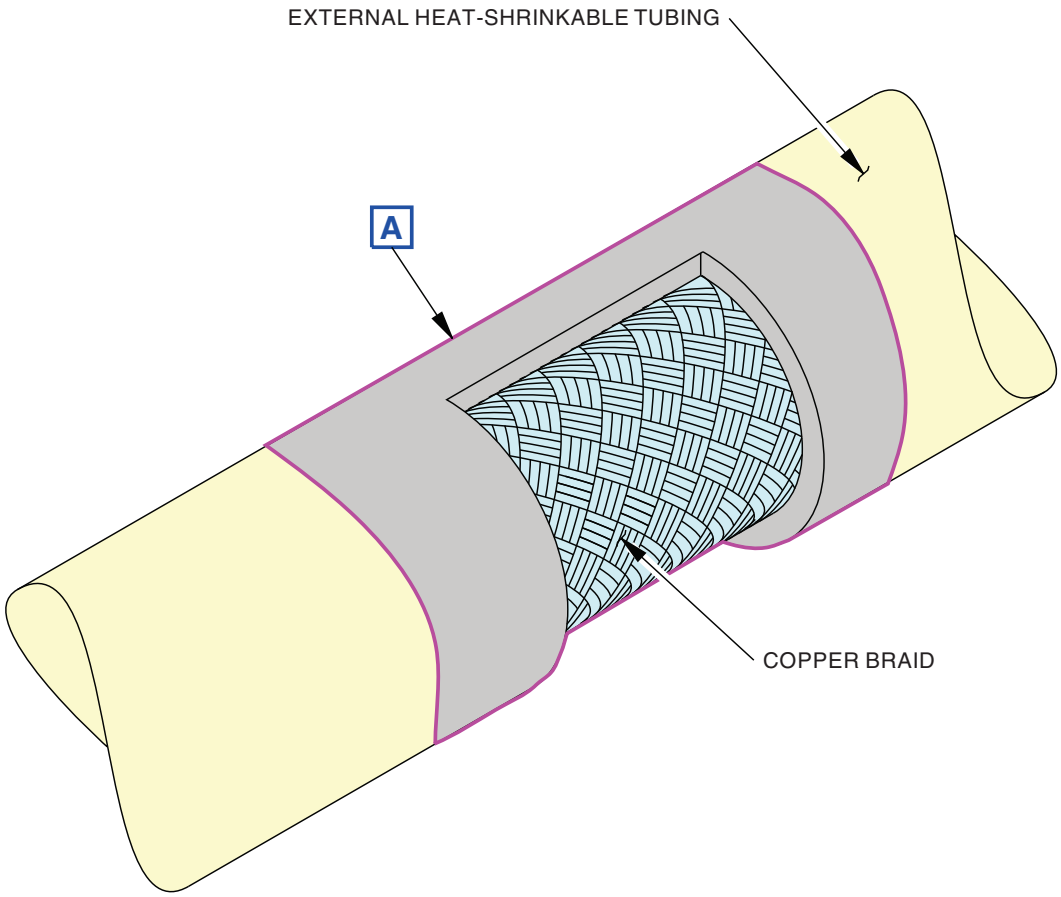


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**Engine Wiring Harness Repair**
**Figure 801/73-22-01-990-882-G00 (Sheet 1 of 2)**



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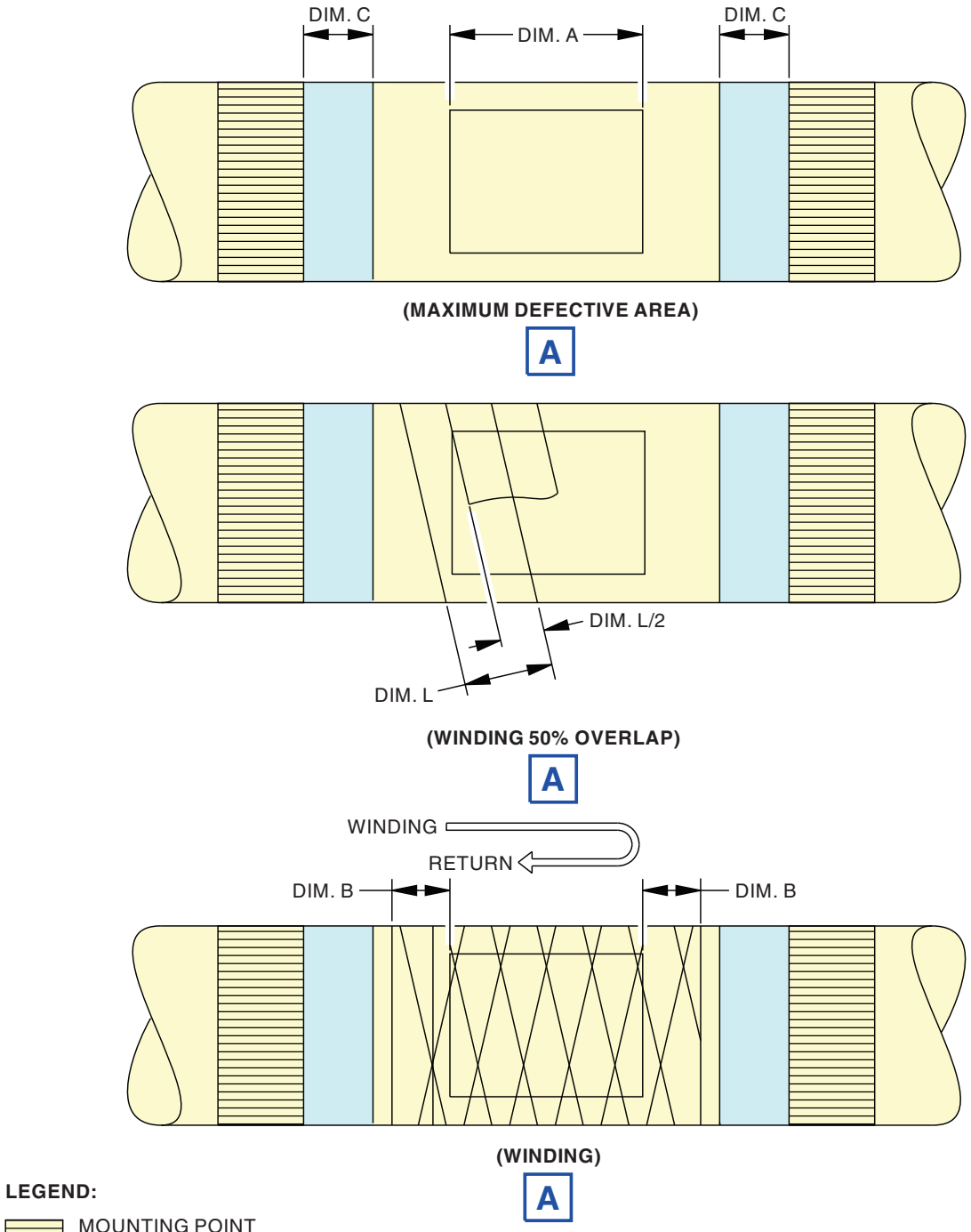


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**Engine Wiring Harness Repair**
**Figure 801/73-22-01-990-882-G00 (Sheet 2 of 2)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-900-802-G00**

**3. Engine Wiring Harness Repair (Application of Stainless Steel Braid)**

(Figure 802)

**A. General**

(1) This tasks gives instructions for the application of stainless steel wire for interim protection of harness outer stainless steel braid.

(2) The Table 802 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 802/73-22-01-993-803-G00**

| Part Number | Service Bulletin | Designation | Engine |
|---|---|---|---|
| 364-083-811-0 | 72-0001 | Harness HJ8A | LEAP 1B |
| 364-056-612-0 | 72-0001 | Harness W200 | |
| 364-056-711-0 | 72-0001 | Harness W300 | |

**B. References**

| Reference | Title |
|---|---|
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G50142 | Lockwire - MS20995C15, Corrosion Resistant Steel - 0.015 Inch (0.381 mm) Diameter | NASM20995 |
| G50145 [CP2628] | Tape - High Temperature Aluminum Foil with Glass Cloth, Silicone Adhesive - 3M 363 | |
| G51335 [CP2674] | Lint-Free Cloth | |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 426 | Right Thrust Reverser, Engine 2 |

## F. Prepare for the Repairs

SUBTASK 73-22-01-860-116-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-117-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## G. Engine Wiring Harness Repairs

**WARNING** REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

SUBTASK 73-22-01-020-033-G00

(1) Loosen the clamps as required until the area to repair is clear.

SUBTASK 73-22-01-220-004-G00

(2) Do a visual inspection of the area to repair.

NOTE: This repair only applies on harness branches in stainless steel.

SUBTASK 73-22-01-220-005-G00

(3) Do a dimensional inspection of the area to repair (Figure 802).

(a) Do not use this repair if there is one of the following conditions:

1) The length of the area to repair is more than 1.18 in. (30.0 mm) (DIM. A) (View A, Figure 802)

2) The distance between the area to repair and nearest repaired area is less than 1.97 in. (50.0 mm)

3) The defective area angle is more than 90° (DIM. D) (View B, Figure 802).

SUBTASK 73-22-01-100-003-G00

(4) Clean the external sleeve in and around the area to repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041].

SUBTASK 73-22-01-220-006-G00

(5) Examine the area to repair as follows:

(a) Remove all material on the area to repair.

(b) Degradation of copper braid is not permitted.

SUBTASK 73-22-01-350-003-G00

(6) Wind 3M 363 tape, G50145 [CP2628] (wide 2 in. (51 mm)), as follows (View C, Figure 802):

(a) Make two turns minimum with 3M 363 tape, G50145 [CP2628] (DIM. B).

(b) Center 3M 363 tape, G50145 [CP2628], with the defective area.

SUBTASK 73-22-01-350-004-G00

(7) Apply MS20995C15 lockwire, G50142, as follows (View C, Figure 802).

(a) Hold MS20995C15 lockwire, G50142, along the length of the worn area beginning from the left.

(b) Start at the opposite end and apply MS20995C15 lockwire, G50142, tightly and equally over the outer sheath to cover the repair area (DIM. X).

(c) Keep the short end of the loop and make sure that you can see it, when approximately two thirds of the wound lockwire length is completed.

(d) Continue to wind MS20995C15 lockwire, G50142, to complete the remaining distance DIM. X/3.

(e) Put MS20995C15 lockwire, G50142, “1” into the loop after the final turn.

(f) Pull the loop end “3” until MS20995C15 lockwire, G50142, “1” goes under the wound lockwire.



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(g) Cut MS20995C15 lockwire, G50142, end “2” and “3” flush.

SUBTASK 73-22-01-350-005-G00

(8) Do a visual inspection of the repaired area.

(a) A bad wire wind up is not permitted, do the repair again.

SUBTASK 73-22-01-410-036-G00

(9) Put the harness back on its supports.

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-118-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-119-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



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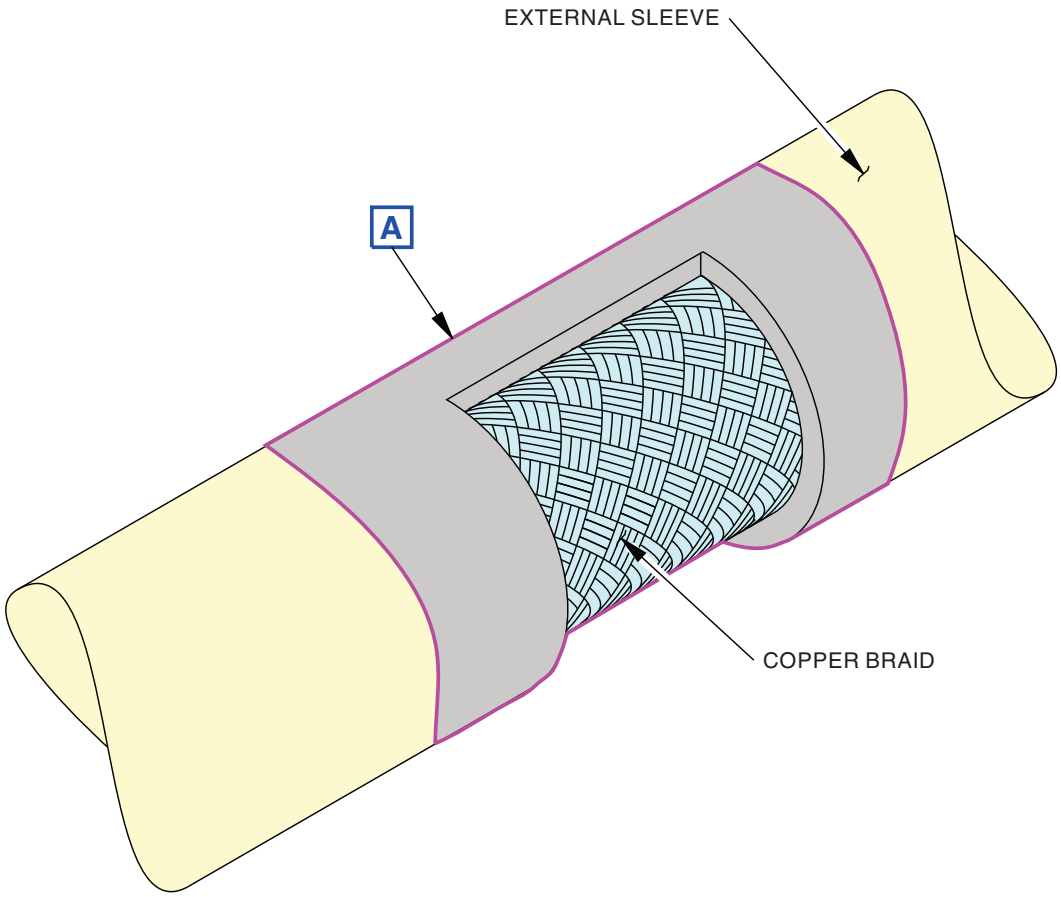


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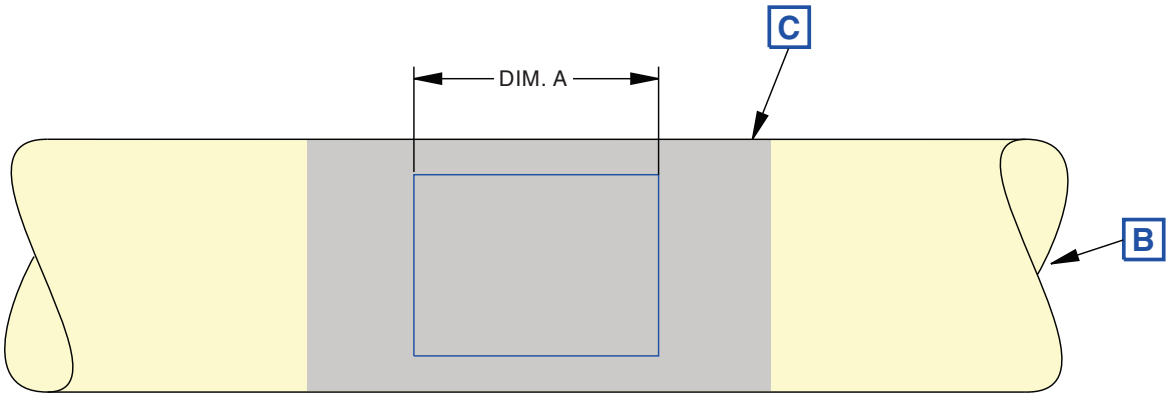
**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 1 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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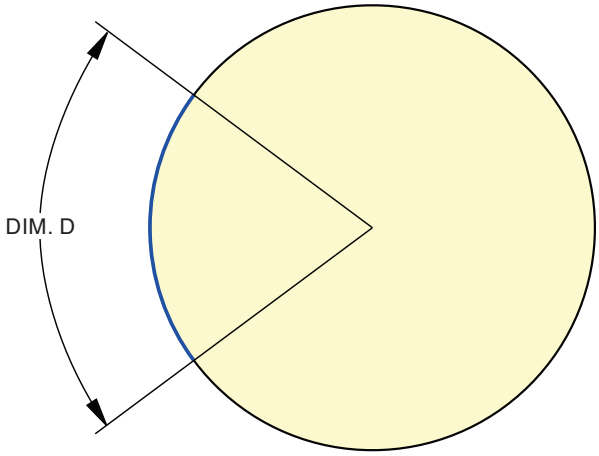
**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 2 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




(MAXIMUM DEFECTIVE AREA ANGLE)

B

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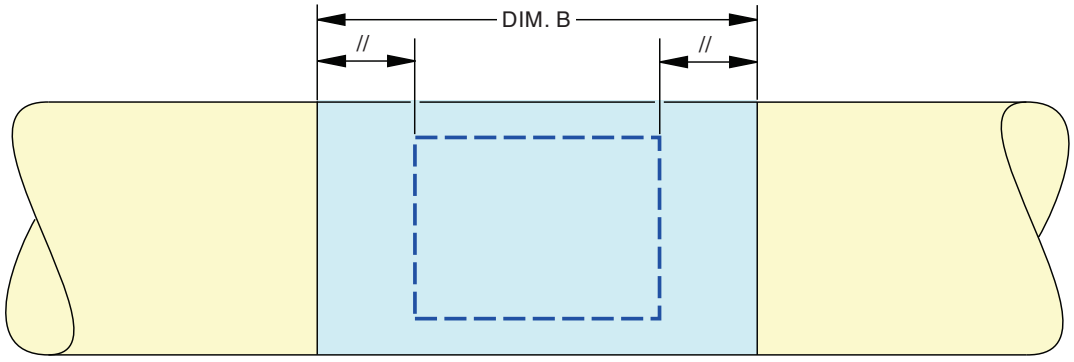
**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 3 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




(ADHESIVE TAPE)

C

DOUBLE TURN MINIMUM

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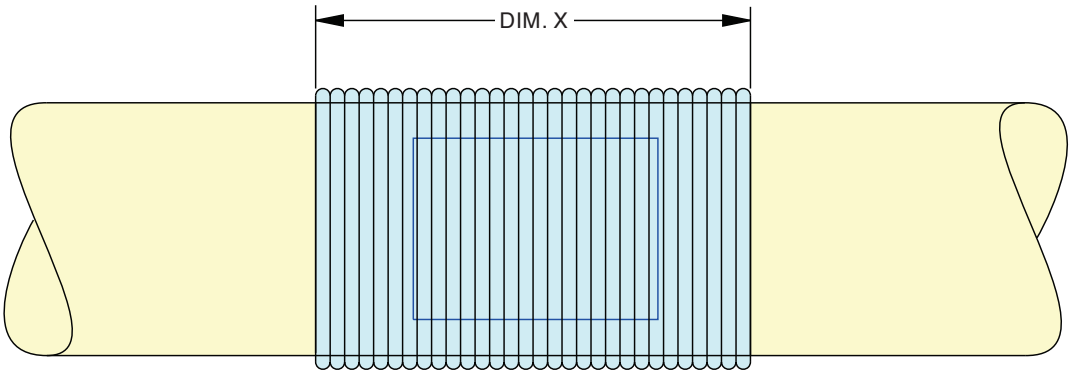
**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 4 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




(STAINLESS STEEL WIRE)

C

NOTE: STAINLESS STEEL WIRE OVER TAPE.

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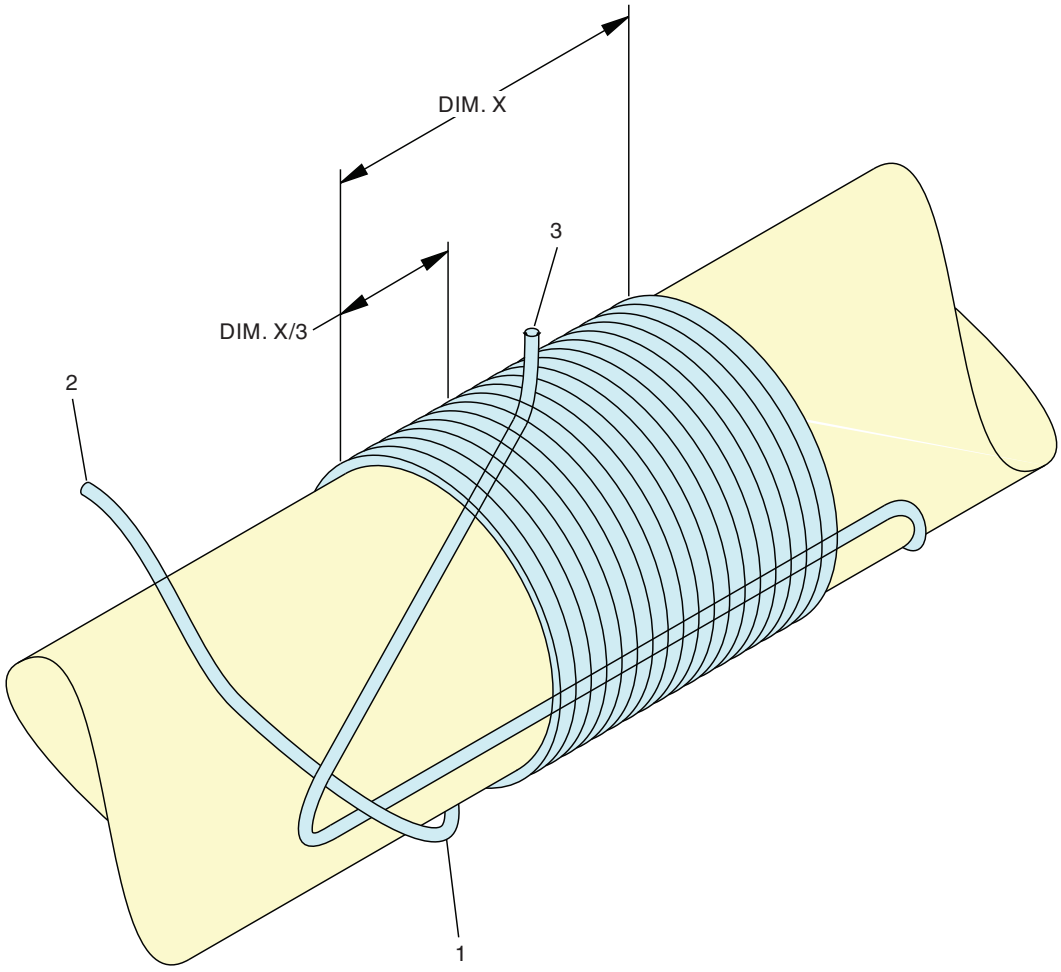
**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 5 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




(KNOT)

C

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**Engine Wiring Harness Repair (Application of Stainless Steel Braid)**
**Figure 802/73-22-01-990-885-G00 (Sheet 6 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-900-803-G00**

**4. Engine Wiring Harness Heat-Shrinkable Boot Repair**

(Figure 803)

**A. General**

(1) This task gives instructions on how to repair harness heat-shrinkable boots with silicone tape and applies to the harness heat-shrinkable boots only.

(2) There are several geometries of heat-shrinkable boots. This repair is applicable to all boots geometries.

(3) The Table 803 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 803/73-22-01-993-804-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-083-111-0 | 72-0001 | Harness HJ3A |
| 364-083-211-0 | 72-0001 | Harness HJ3B |
| 364-083-311-0 | 72-0013 | Harness HJ5A |
| 364-083-313-0 | 73-0019 | |
| 364-083-411-0 | 72-0013 | Harness HJ5B |
| 364-083-413-0 | 73-0019 | |
| 364-083-511-0 | 72-0013 | Harness HJ6A |
| 364-083-513-0 | 73-0019 | |
| 364-083-514-0 | 73-0031 | |
| 364-083-611-0 | 72-0013 | Harness HJ6B |
| 364-083-613-0 | 73-0019 | |
| 364-083-614-0 | 73-0031 | |
| 364-083-710-0 | 72-0001 | Harness HJ7 |
| 364-083-711-0 | 75-0010 | |
| 364-083-811-0 | 72-0001 | Harness HJ8A |
| 364-083-812-0 | 73-0021 | |
| 364-083-911-0 | 72-0001 | Harness HJ8B |
| 364-055-211-0 | 72-0001 | Harness W2 |
| 364-055-212-0 | 75-0010 | |
| 364-055-311-0 | 72-0001 | Harness W4A |
| 364-055-411-0 | 72-0001 | Harness W4B |
| 364-056-511-0 | 72-0001 | Harness W100 |



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**Table 803/73-22-01-993-804-G00 (Continued)**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-056-612-0 | 72-0001 | Harness W200 |
| 364-056-613-0 | 72-0132 | |
| 364-056-614-0 | 72-0306 | |
| 364-056-711-0 | 72-0001 | Harness W300 |
| 364-056-712-0 | 72-0132 | |
| 364-056-713-0 | 72-0214 | |
| 364-056-714-0 | 72-0214 | |
| 364-056-715-0 | 72-0306 | |

**B. References**

| Reference | Title |
|---|---|
| 24-11-63-000-801 | W100 EDP and IDG Engine Wiring Harness Removal (P/B 401) |
| 24-11-63-400-801 | W100 EDP and IDG Engine Wiring Harness Installation (P/B 401) |
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 70-21-23-100-801-G00 | Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01 P/B 401 | ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G51335 [CP2674] | Lint-Free Cloth | |
| G51741 [CP2990] | Tape - Silicone - 66N | |

**E. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## F. Prepare for the Repairs

SUBTASK 73-22-01-860-120-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-121-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## G. Engine Wiring Harness Heat-Shrinkable Boot Repair

SUBTASK 73-22-01-020-034-G00

(1) Loosen the connectors and clamps as required until the area to repair is clear.

SUBTASK 73-22-01-210-011-G00

(2) Do a visual inspection of the harness and heat-shrinkable boots.

(a) If the harness or heat-shrinkable boots have been repaired and the silicone tape is damaged, replace the harness (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801, TASK 24-11-63-400-801, TASK 26-11-02-000-801, and TASK 26-11-02-400-801).

SUBTASK 73-22-01-220-007-G00

(3) Do a dimensional inspection of the area to repair.

(a) Do not use this repair if there is one of the following conditions:

1) There is a damage of the heat-shrinkable boot over a 0.39 in. (10 mm) area on the connector side end (DIM. C).

2) There is a damage of the heat-shrinkable boot with dimension more than 0.39 in. (10 mm) x 0.19 in. (5 mm).

3) There is a damage of the shielding braid.

a) Cut strands and braid deformation are not permitted.

SUBTASK 73-22-01-100-004-G00

(4) Clean the area to repair as follows:

**WARNING** REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

**WARNING** ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(a) Clean the area with lint-free cloth, G51335 [CP2674], soaked with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(b) Remove all material on the area to repair.

SUBTASK 73-22-01-350-006-G00

(5) Wind a silicone tapes 66N, G51741 [CP2990], around the boot as follows:

**Dimensional Information**

| Description | Minimum In-Process Dimension | Maximum In-Process Dimension | Minimum Finish Dimension | Maximum Finish Dimension | Reference Dimension |
|---|---|---|---|---|---|
| DIM. A | 0.40 in. (10 mm)*[1]*[2] | - | - | - | -*[3] |
| DIM. B | - | - | 0 in. (0 mm)*[1]*[2] | 0.12 in. (3 mm)*[1]*[2] | -*[3] |



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*[1] The dimensional data identified within this task was designed using millimeters as the base unit of measurement for dimensional values. The conversion to inches is rounded to two additional decimal digits than millimeters. For minimum dimensions, converted values are rounded up. For maximum dimensions, converted values are rounded down. To avoid conversion errors, do not re-convert from inches to millimeters.

*[2] Dimensions are in inches with millimeters in parentheses.

*[3] Reference dimensions do not require inspection.

(a) Start the winding on identification sleeve side with a complete turn of the tape on itself with the respect of DIM. A.

1) An overlap on the identification sleeve is permitted if reading the identification is possible.

(b) Wind tightly the silicone tape on the heat-shrinkable boot with an overlap.

(c) At DIM. B, continue the winding in the opposite direction with overlap.

(d) Finish the winding with one complete turn of the silicone tape on itself, perpendicularly to the harness branch, going over the first layer.

**SUBTASK 73-22-01-350-007-G00**

(6) Do a visual inspection of the part.

(a) Make sure that the part meet the following criteria:

1) No detachment of the silicone tape is permitted.

2) No nick and no perforation of the silicone tape are permitted.

3) Make sure that silicone tape has a complete overlap of heat-shrinkable boot respecting DIM. B.

(b) If one of the criteria is not eligible, remove the tape applied and do the repair again.

1) Start the repair at the step to clean the area.

**SUBTASK 73-22-01-410-037-G00**

(7) Put the harness back on its supports.

## H. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-22-01-860-122-G00**

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

**SUBTASK 73-22-01-860-123-G00**

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

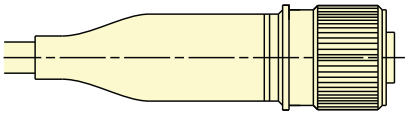
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

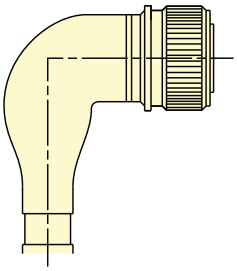
**——— END OF TASK ———**



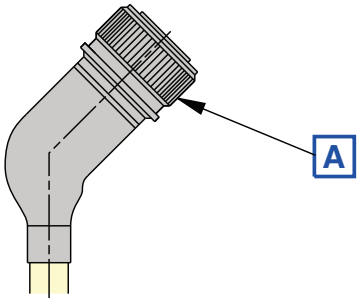
ECCN 9E991 BOEING PROPRIETARY - See title page for details



STRAIGHT BOOT



ELBOW BOOT




ELBOW BOOT

POSSIBLE GEOMETRIES OF
HEAT-SHRINKABLE BOOT

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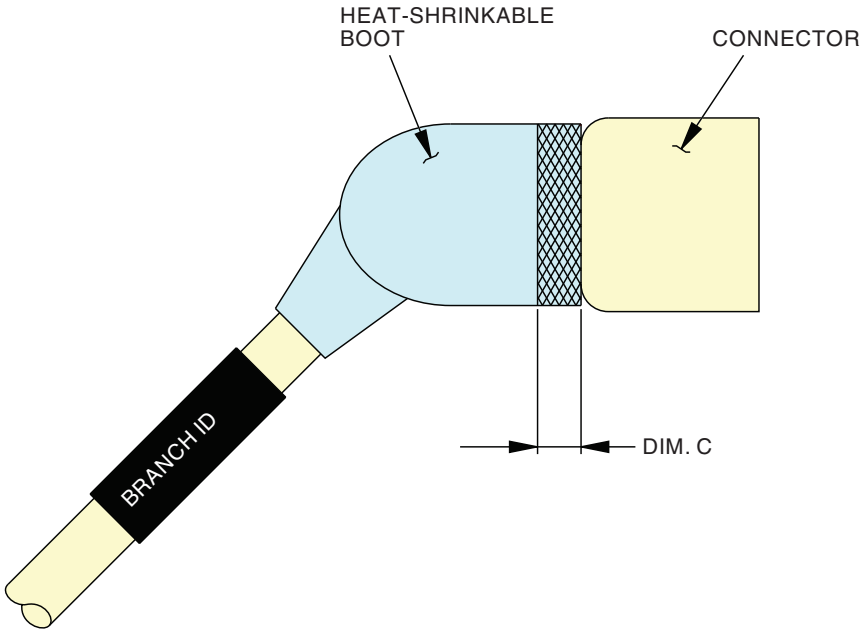
**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 1 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(ELBOW BOOT IS SHOWN, STRAIGHT BOOT IS EQUIVALENT)**

A

AREA WHERE DAMAGES ARE REPAIRABLE

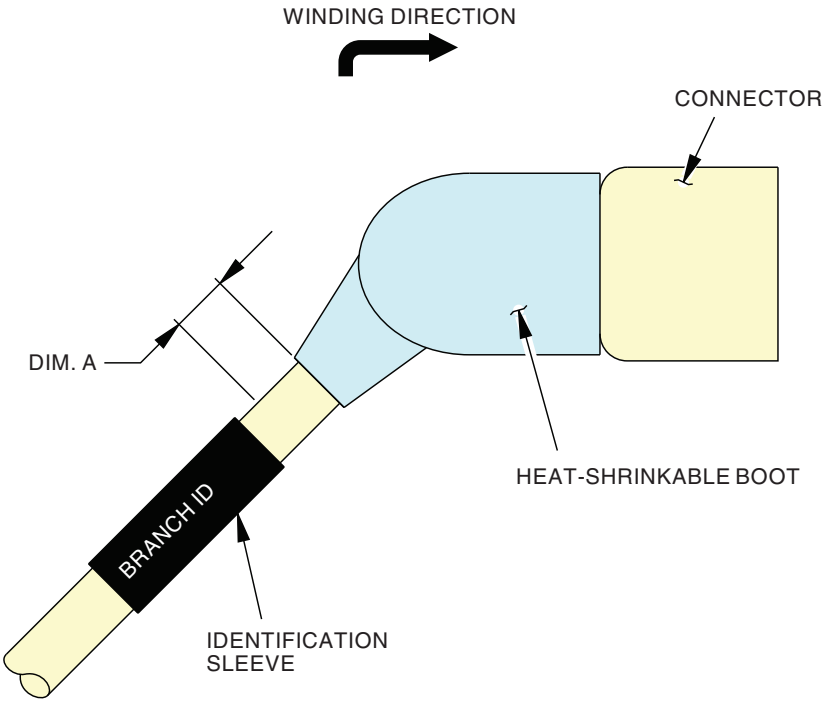
AREA WHERE DAMAGES ARE NOT REPAIRABLE

2979875 S0000758017_V1

**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 2 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(WINDING START)**
**(EXAMPLE)**

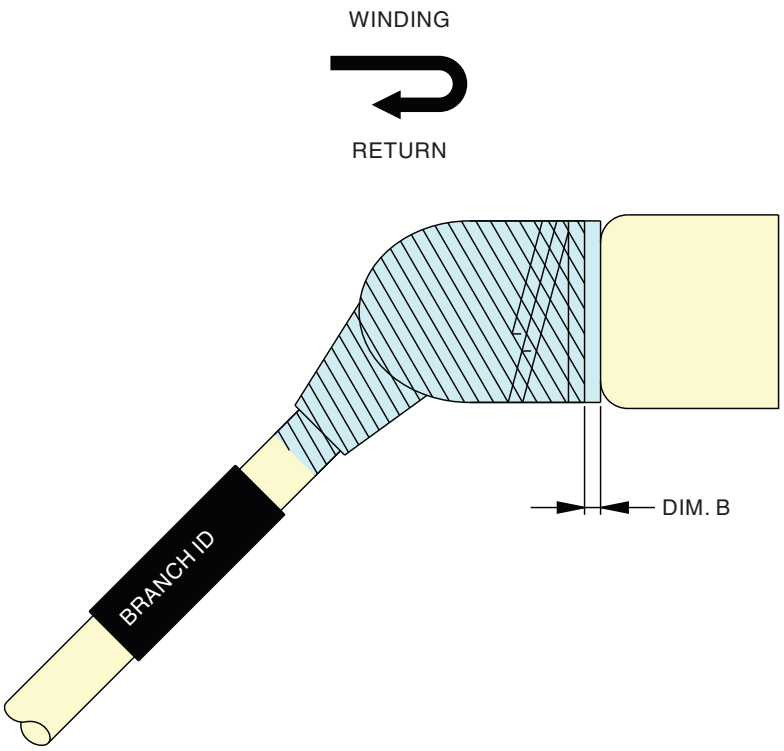
A

2979876 S0000758018_V1

**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 3 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(WINDING HALF TURN)**
**(EXAMPLE)**

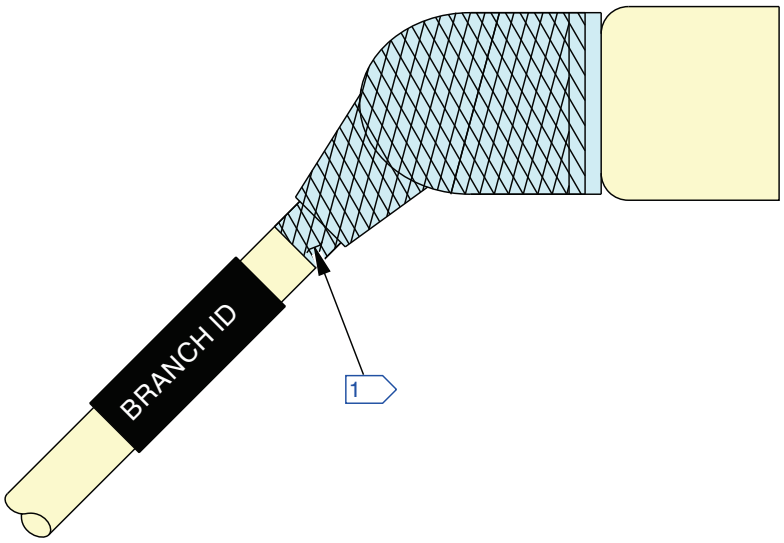
**A**

2979878 S0000758019_V1

**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 4 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**(WINDING FINISH)**
**(EXAMPLE)**

**A**

1 FINISH THE WINDING WITH ONE COMPLETE TURN OF THE SILICONE TAPE ON ITSELF, PERPENDICULARLY TO THE HARNESS BRANCH, GOING OVER THE FIRST LAYER.

2979881 S0000758020_V1

**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 5 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**DAMAGE**
**(EXAMPLE)**

2979882 S0000758021_V1

**Engine Wiring Harness Heat-Shrinkable Boot Repair**
**Figure 803/73-22-01-990-886-G00 (Sheet 6 of 6)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-900-804-G00**

**5. Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch)**

(Figure 804, Figure 805, and Figure 806)

**A. General**

(1) This task gives instructions for the application of silicone tape to repair the external heat-shrinkable tubing.

(2) The Table 804 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 804/73-22-01-993-808-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-083-111-0 | 72-0001 | Harness HJ3A |
| 364-083-211-0 | 72-0001 | Harness HJ3B |
| 364-083-311-0 | 72-0013 | Harness HJ5A |
| 364-083-313-0 | 73-0019 | |
| 364-083-411-0 | 72-0013 | Harness HJ5B |
| 364-083-413-0 | 73-0019 | |
| 364-083-511-0 | 72-0013 | Harness HJ6A |
| 364-083-513-0 | 73-0019 | |
| 364-083-514-0 | 73-0031 | |
| 364-083-611-0 | 72-0013 | Harness HJ6B |
| 364-083-613-0 | 73-0019 | |
| 364-083-614-0 | 73-0031 | |
| 364-083-710-0 | 72-0001 | Harness HJ7 |
| 364-083-711-0 | 75-0010; 75-0011 | |
| 364-083-811-0 | 72-0001 | Harness HJ8A |
| 364-083-812-0 | 73-0021 | |
| 364-083-911-0 | 72-0001 | Harness HJ8B |
| 364-055-211-0 | 72-0001 | Harness W2 |
| 364-055-212-0 | 75-0010 | |
| 364-055-311-0 | 72-0001 | Harness W4A |
| 364-055-411-0 | 72-0001 | Harness W4B |
| 364-056-511-0 | 72-0001 | Harness W100 |
| 364-056-612-0 | 72-0001 | Harness W200 |
| 364-056-613-0 | 72-0132 | |
| 364-056-614-0 | 72-0306 | |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**Table 804/73-22-01-993-808-G00 (Continued)**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-056-711-0 | 72-0001 | Harness W300 |
| 364-056-712-0 | 72-0132 | |
| 364-056-713-0 | 72-0214 | |
| 364-056-714-0 | 72-0214 | |
| 364-056-715-0 | 72-0306 | |

**B. References**

| Reference | Title |
|---|---|
| 24-11-63-000-801 | W100 EDP and IDG Engine Wiring Harness Removal (P/B 401) |
| 24-11-63-400-801 | W100 EDP and IDG Engine Wiring Harness Installation (P/B 401) |
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 70-21-23-100-801-G00 | Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01 P/B 401 | ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION |
| 73-22-01-210-801-G00 | Engine Control Electrical Harnesses Inspection (P/B 601) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-762 | Scissors - Electrical |
| STD-858 | Tag - DO NOT OPERATE |

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G51335 [CP2674] | Lint-Free Cloth | |
| G51695 [CP2940] | Tape - Silicone, Black - 69N | |
| G51696 [CP2939] | Tape - Silicone, Red/Brown - 68N | |
| G51936 [CP2822] | Tape - Silicone, 58N, Red/Brown | |
| G51937 [CP2823] | Tape - Silicone, 59N, Black | |

**E. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**F. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**G. Prepare for the Repair**

SUBTASK 73-22-01-860-154-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-155-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).


ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## H. Engine Wiring Harness Heat-Shrinkable Tubing Repair

**WARNING** REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

SUBTASK 73-22-01-020-036-G00

(1) If it is necessary, loosen the clamps as required until the area to repair is clear (PAGEBLOCK 73-22-01/401, TASK 24-11-63-000-801, and TASK 26-11-02-000-801).

**WARNING** BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

(a) If there is a silicone tape already in place that needs replacement, remove it manually or with the electrical scissors, STD-762, and do the repair procedure again.

1) Make sure that you do not damage the heat-shrinkable tubing.

**WARNING** BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

(b) If there is a repaired area next to the area to repair, you can remove the tape manually or with the electrical scissors, STD-762, and do the repair procedure for the two indications.

1) Make sure that you do not damage the heat-shrinkable tubing.

SUBTASK 73-22-01-220-012-G00

(2) Do a dimensional inspection of the area to repair (Figure 805).

(a) The length of the area to repair must be less than 2.00 in. (50.8 mm) (DIM A.).

NOTE: The unserviceable area is acceptable over a 360° circumference.

(b) The distance between the area to repair and the nearest unserviceable area must be more than 0.99 in. (25 mm) (DIM B).

(c) The length of the heat-shrinkable tubing must be more than DIM. L + 0.20 in. (5 mm) on the both sides of the area to repair.

- 0.71 in. (18 mm) (DIM. L + 0.20 in. (5 mm)) for silicone tape, G51936 [CP2822], or silicone tape, G51937 [CP2823]
- 0.95 in. (24 mm) (DIM. L + 0.20 in. (5 mm)) for silicone tape 68N, G51696 [CP2939], or silicone tape 69N, G51695 [CP2940].

SUBTASK 73-22-01-210-015-G00

(3) Do a visual inspection of the area to repair (TASK 73-22-01-210-801-G00).



ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-100-006-G00

**WARNING** ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(4) Clean the external heat-shrinkable tubing in and around the area to repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 73-22-01-210-016-G00

(5) Do a visual inspection of the shielding braid (TASK 73-22-01-210-801-G00) (Figure 804).

(a) Remove all material on the area to repair.

SUBTASK 73-22-01-350-011-G00

(6) Wind silicone tape, G51936 [CP2822], or silicone tape, G51937 [CP2823], or silicone tape 68N, G51696 [CP2939], or silicone tape 69N, G51695 [CP2940], around the harness branch as follows (Figure 806):

(a) If the diameter of the branch to be repaired is less than 0.47 in. (12 mm), use silicone tape, G51936 [CP2822], or silicone tape, G51937 [CP2823],

(b) Cut the edge of silicone tape at approximately 45°.

(c) Start to wind tightly silicone tape near the unserviceable area.

1) Make sure that you stretch the silicone tape longitudinally.

(d) Wind away from the unserviceable area with one full turn of tape on itself to cover the extremity of the tape.

- 0.71 in. (18 mm) (DIM. L + 0.20 in. (5 mm)) for silicone tape, G51936 [CP2822], or silicone tape, G51937 [CP2823]
- 0.95 in. (24 mm) (DIM. L + 0.20 in. (5 mm)) for silicone tape 68N, G51696 [CP2939], or silicone tape 69N, G51695 [CP2940].

(e) Wind silicone tape to cover the unserviceable area with a 50% overlap.

(f) Extend tape outside the unserviceable area.

1) Make sure that more than a complete width of silicone tape outside the unserviceable area (DIM. L + 0.20 in. (5 mm)).

(g) Finish the winding with one full turn of tape on itself.

**WARNING** BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

(h) Cut the tape with the electrical scissors, STD-762.

(i) Apply an additional pressure on the tape with fingers.

SUBTASK 73-22-01-210-017-G00

(7) Do a visual inspection of the repaired area.

(a) If the tape is not fully attached, do the repair again.

(b) If there are holes or nicks, do the repair again.

SUBTASK 73-22-01-420-034-G00

(8) Put the harness back on its supports (PAGEBLOCK 73-22-01/401, TASK 24-11-63-400-801, and TASK 26-11-02-400-801).



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-156-G00

**WARNING**

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-157-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

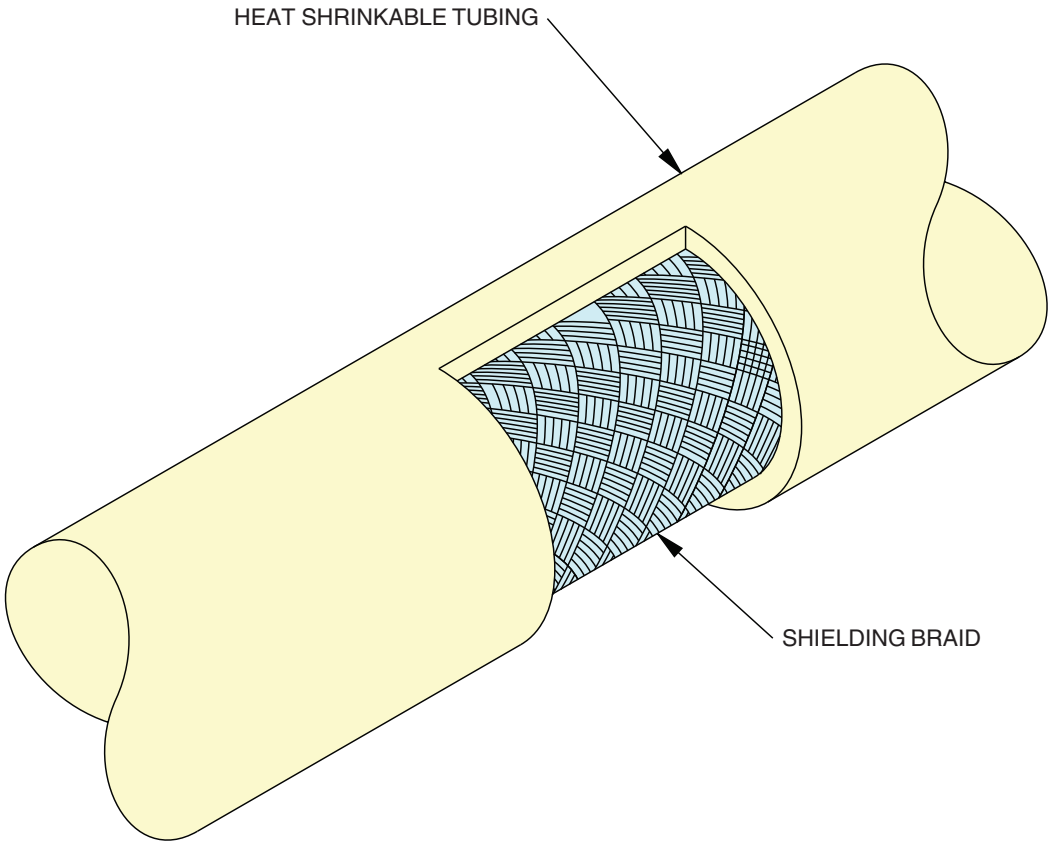
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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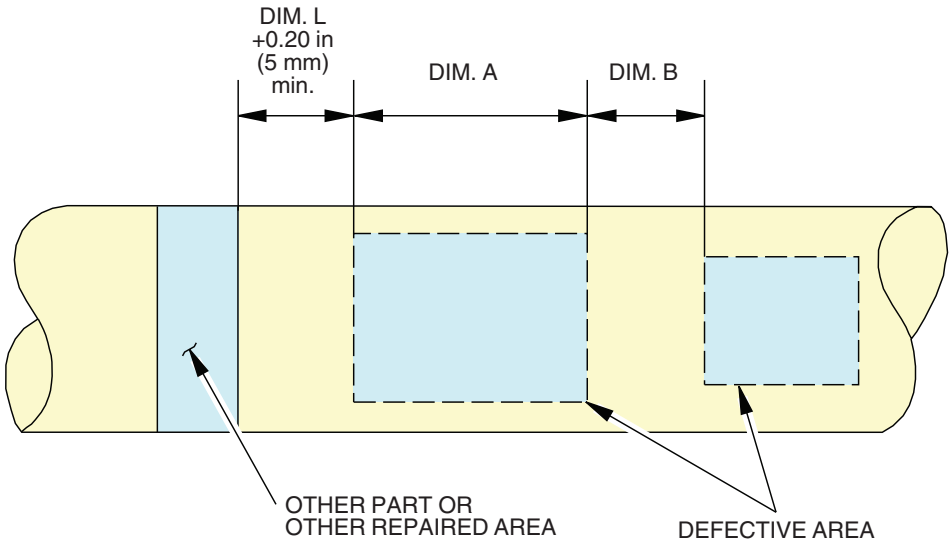
**Shielding Braid**
**Figure 804/73-22-01-990-890-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




DIM L: WIDTH OF THE SILICONE TAPE

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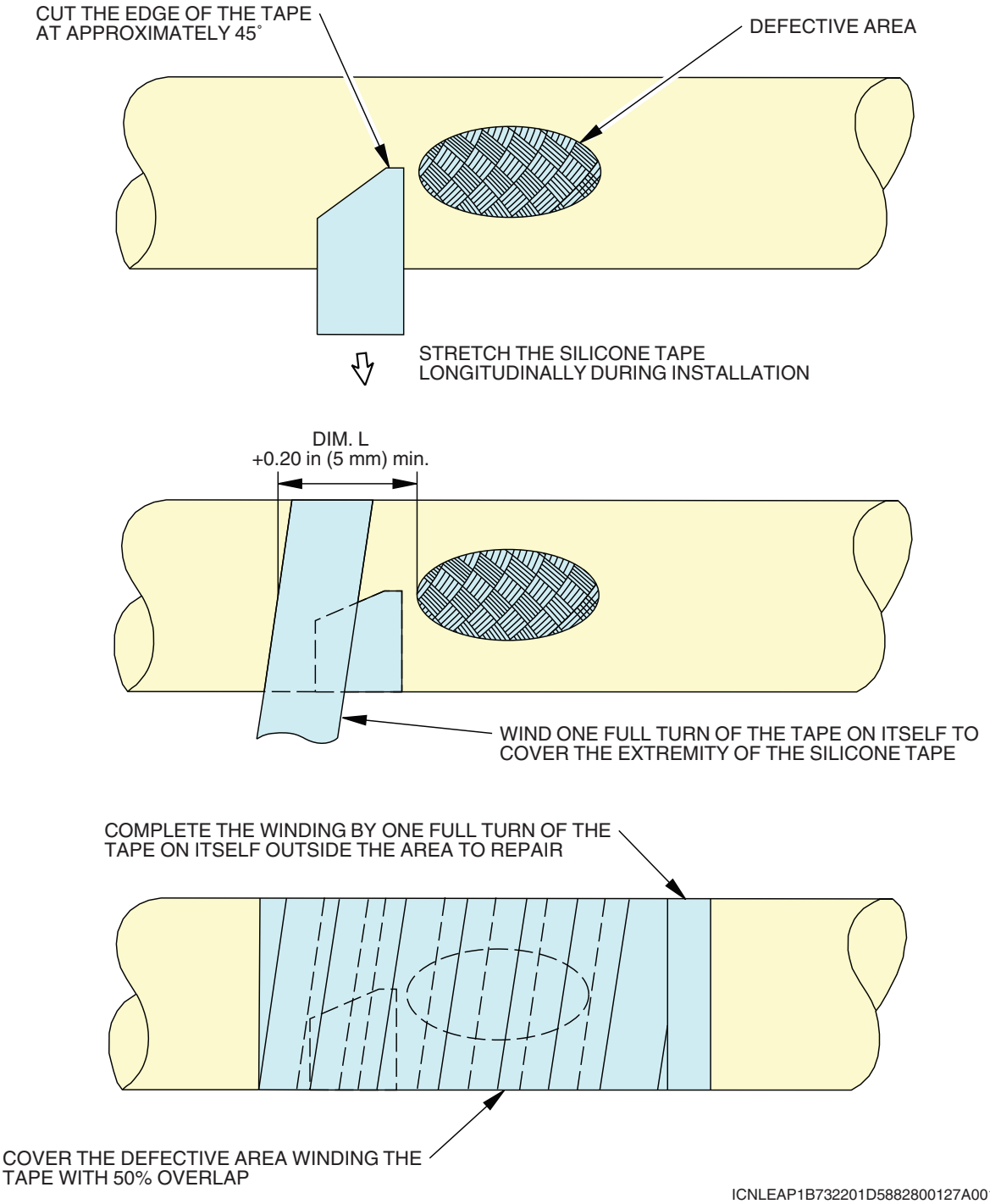
3079154 S0000835604_V1

**Maximum Unserviceable Area**
**Figure 805/73-22-01-990-891-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**Engine Wiring Harness Heat-Shrinkable Tubing Repair**
**Figure 806/73-22-01-990-892-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-300-801-G00**

## 6. Engine Wiring Harnesses W200 and W300 - BMS 1367 Cable at the Terminal Lugs Repair

(Figure 807)

### A. General

(1) This task gives instructions to repair the BMS 1367 cable on the “FD CORE RIGHT and FD CORE LEFT” branches of the harnesses.

(2) The Table 801 gives a list of the part numbeat are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 805/73-22-01-993-805-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-056-612-0 | 72-0001 | Harness W200 |
| 364-056-613-0 | 72-0132 | Harness W200 |
| 364-056-711-0 | 72-0001 | Harness W300 |
| 364-056-712-0 | 72-0132 | Harness W300 |
| 364-056-713-0 | 72-0214 | Harness W300 |
| 364-056-714-0 | 72-0214 | Harness W300 |

### B. References

| Reference | Title |
|---|---|
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

### D. Consumable Materials

| Reference | Description | Specification |
|---|---|---|
| G51328 [CP2674] | Cloth - Lint-Free | |
| G51864 [CP9004] | Tape - Lacing, Glass Fiber | A-A-52083-D-4 |

### E. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### F. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Prepare for the Repairs

SUBTASK 73-22-01-860-124-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-128-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

| Number | Name/Location |
|---|---|
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**H. Cable at the Terminal Lugs Repairs**

SUBTASK 73-22-01-020-035-G00

(1) Remove the nuts, lugs, and P-clamps (View B, Figure 807) (TASK 26-11-02-000-801).

SUBTASK 73-22-01-220-008-G00

(2) Do a visual inspection of the area to repair:

(a) Damage maximum length is 1.97 in. (50 mm) (Dim. B) (View E, Figure 807).

(b) If the white internal layer is damaged, replace the harness (View D, Figure 807).

(c) The maximum circumferential length of the damage is 180 degrees (Angle C) (View A-A, Figure 807).

SUBTASK 73-22-01-100-005-G00

(3) Clean the area to be repair with lint-free cloth, G51328 [CP2674].

(a) Remove all material on the area to repair.

SUBTASK 73-22-01-220-009-G00

(4) Locate the damaged area (View D, Figure 807).

(a) If the harness has been repaired and protection is damaged, replace it.

SUBTASK 73-22-01-960-006-G00

(5) Cut lacing tape, G51864 [CP9004], on approximately 40 in. (1016 mm).

SUBTASK 73-22-01-350-008-G00

(6) Lace the cord on the damaged area as follows (Figure 807 (Sheet 4)):

(a) Start a minimum of 0.20 in. (5 mm) (Dim. A) from the damaged area and finish a minimum of 0.20 in. (5 mm) from the damaged area (View E, Figure 807).

(b) Make the loop and then wrap around the wire, then go through the loop (Figure 807 (Sheet 5)).

(c) Pull on the end to move the loop under the lacing cord (Figure 807 (Sheet 5)).

NOTE: Minimum Loop Limit lacing of the cord (Figure 807 (Sheet 5)).

(d) Cut both ends.

(e) Make sure to collect all debris to prevent Foreign Object Debris (FOD).

SUBTASK 73-22-01-210-014-G00

(7) Do a visual inspection of the repaired area.

(a) Make sure that the lacing cord is fully attached.

(b) Make sure that the lacing cord has a complete overlap of the wire (BMS 1367 cable).

NOTE: If one criteria is not correct, remove the lacing cord and start again step (4).

SUBTASK 73-22-01-420-030-G00

(8) Install the nuts, lugs, and P-clamps (TASK 26-11-02-400-801).



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-129-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-133-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

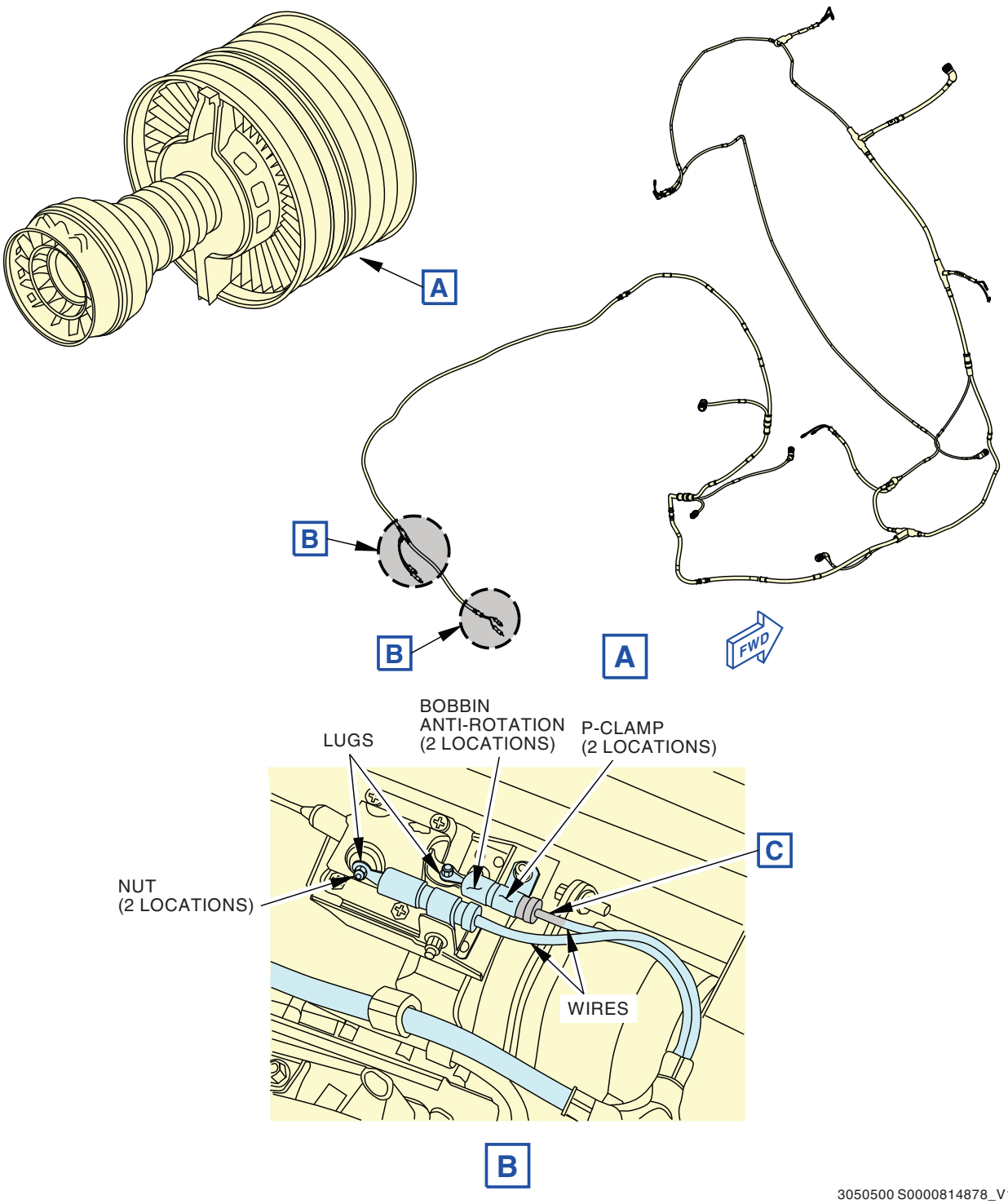
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



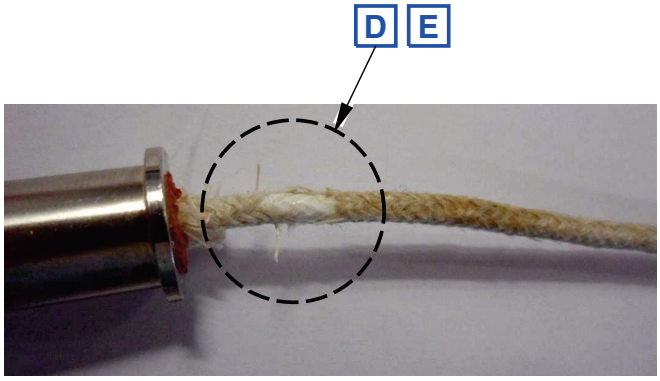
ECCN 9E991 BOEING PROPRIETARY - See title page for details

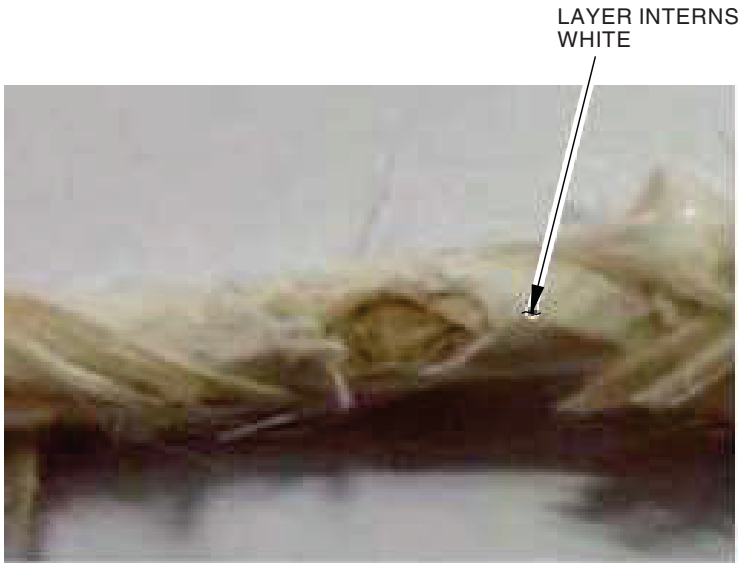



**Engine Wiring Cable Repair**
**Figure 807/73-22-01-990-889-G00 (Sheet 1 of 5)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



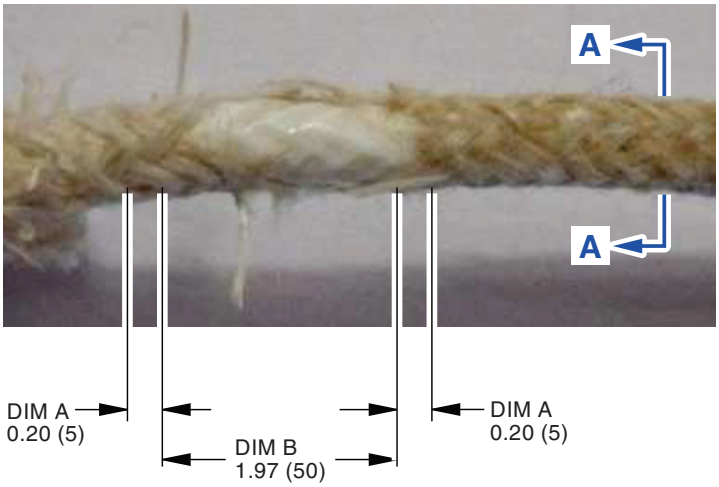




3050527 S0000814879_V1

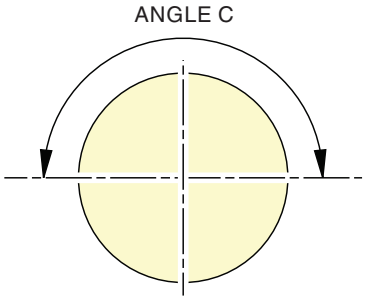
**Engine Wiring Cable Repair**
**Figure 807/73-22-01-990-889-G00 (Sheet 2 of 5)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




E




A-A

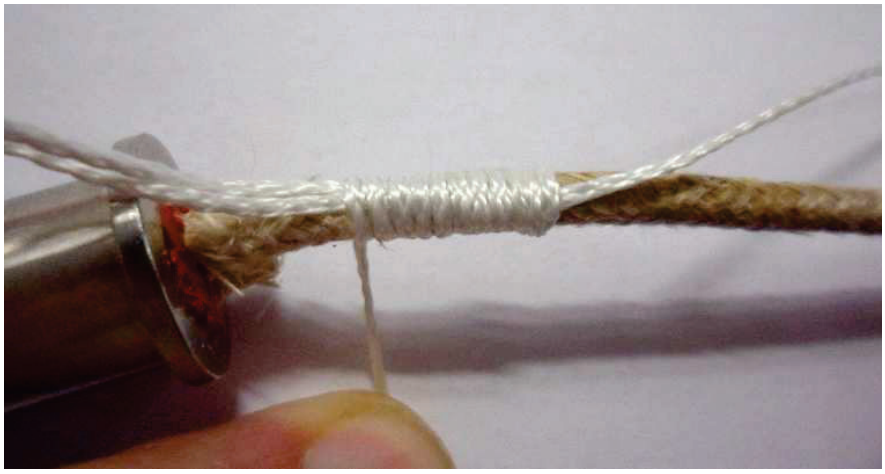
**NOTE:**
ALL DIMENSIONS ARE IN INCHES (mm).

3050532 S0000814880_V1

**Engine Wiring Cable Repair**
**Figure 807/73-22-01-990-889-G00 (Sheet 3 of 5)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



LACING THE CORD

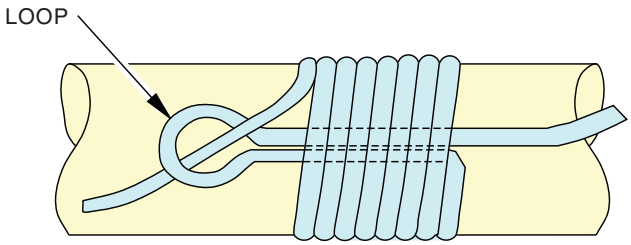
FULLY ATTACHED THE LACING CORD

3050534 S0000815234_V1

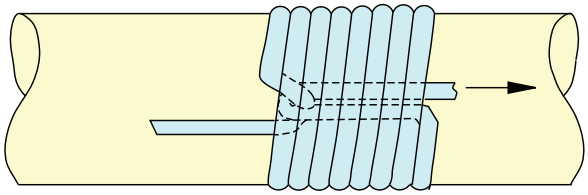
**Engine Wiring Cable Repair**
**Figure 807/73-22-01-990-889-G00 (Sheet 4 of 5)**



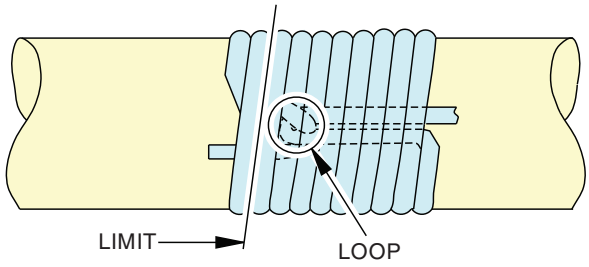
ECCN 9E991 BOEING PROPRIETARY - See title page for details



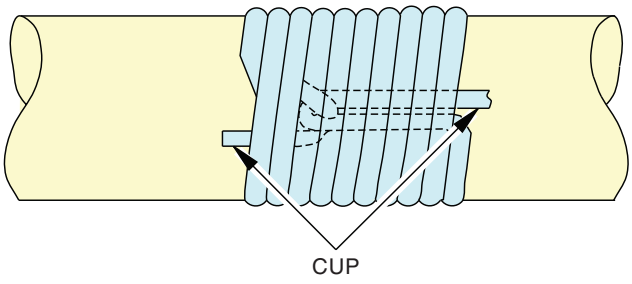

MAKING A LOOP



PULLING ON THE LACING CORD




MINIMUM LOOP LIMIT




CUTTING BOTH ENDS

3050535 S0000814881_V1

**Engine Wiring Cable Repair**
**Figure 807/73-22-01-990-889-G00 (Sheet 5 of 5)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-300-803-G00**

## 7. Engine Wiring Harnesses W200 and W300 - Bobbin Replacement

(Figure 808)

### A. General

(1) This task gives instructions to replace the bobbin on the harness.

(2) The Table 806 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 806/73-22-01-993-806-G00**

| Part number | Designation |
|---|---|
| 364-056-614-0 | Harness W200 |
| 364-056-715-0 | Harness W300 |

### B. References

| Reference | Title |
|---|---|
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## F. Prepare for the Repairs

SUBTASK 73-22-01-860-134-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-138-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## G. Bobbin Replacement

SUBTASK 73-22-01-960-001-G00

(1) Remove the cable lug [3] and p-clamp [2] from the equipment (TASK 26-11-02-000-801).

SUBTASK 73-22-01-960-002-G00

(2) Remove the clamp that holds the bobbin [1].



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SUBTASK 73-22-01-960-003-G00

(3) Open the bobbin [1] manually to remove it from the wire (View C, Figure 808).

(a) Discard the bobbin [1].

SUBTASK 73-22-01-960-004-G00

(4) Take a new bobbin [1] according to the table below:

| Harness | Number of wire | Supply Number | Marking |
|---|---|---|---|
| W200 | 1 | 1019657790 | Loop A/1 |
| W300 | 1 | 1019657792 | Loop B/2 |
| W200 | 2 | 1019657794 | Loop A/1 |
| W300 | 2 | 1019657796 | Loop B/2 |

SUBTASK 73-22-01-960-005-G00

(5) Open the bobbin [1] manually to install it on the wire or two wires (View E, Figure 808).

SUBTASK 73-22-01-420-031-G00

(6) Install the clamp that holds the bobbin [1].

SUBTASK 73-22-01-420-033-G00

(7) Install the cable lug [3] and the p-clamp [2] to the equipment (TASK 26-11-02-400-801).

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-139-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-143-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:



ECCN 9E991 BOEING PROPRIETARY - See title page for details

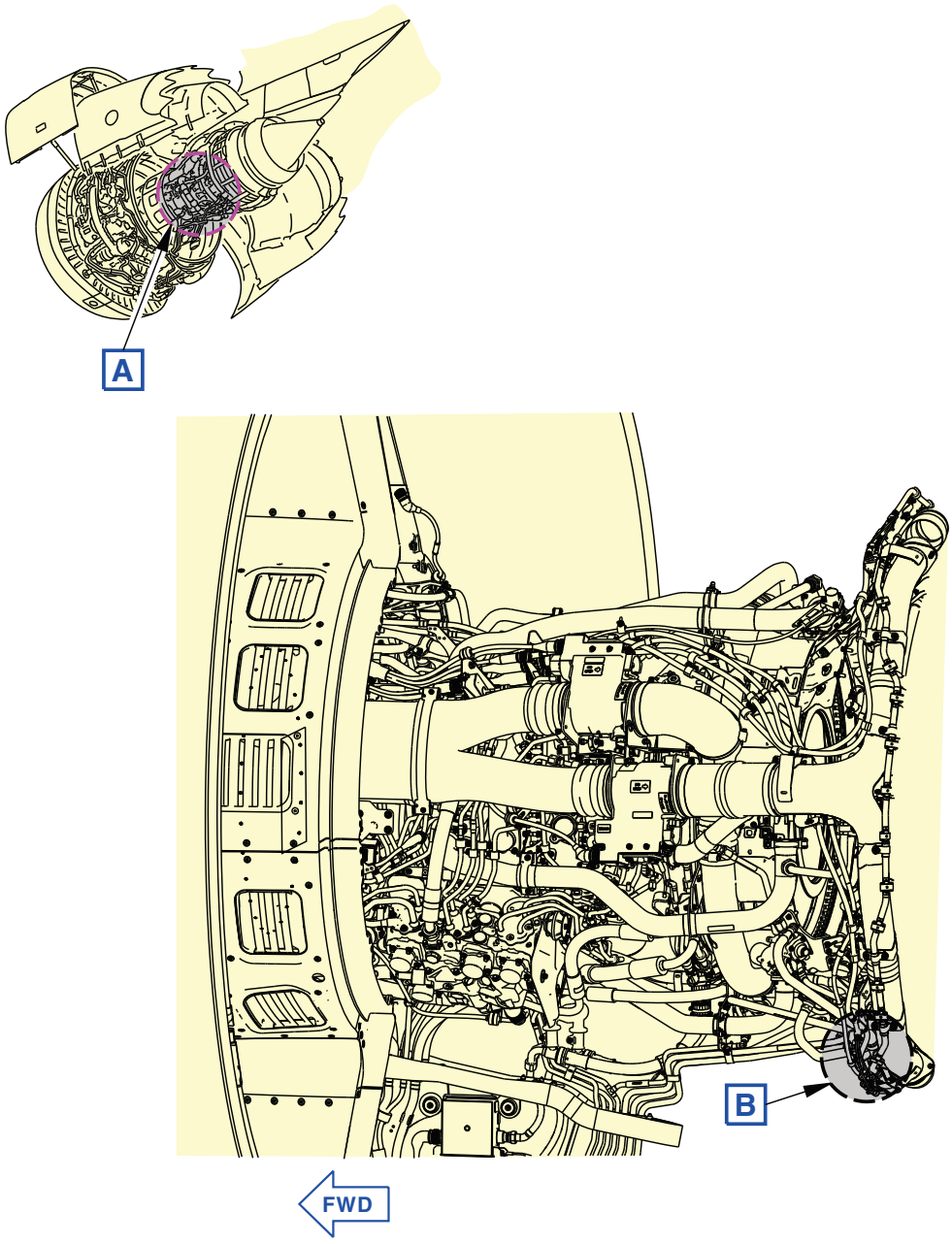
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**——— END OF TASK ———**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



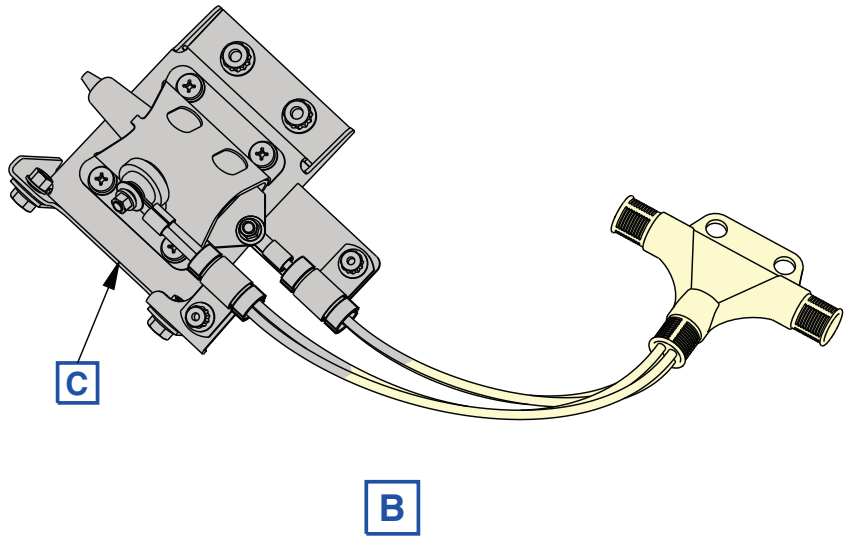

ICNLEAP1B732201D5882800001A00101

3050331 S0000814882_V1

**Bobbin replacement**
**Figure 808/73-22-01-990-887-G00 (Sheet 1 of 4)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

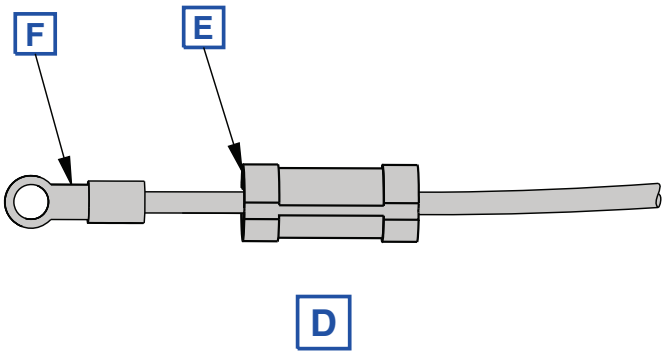


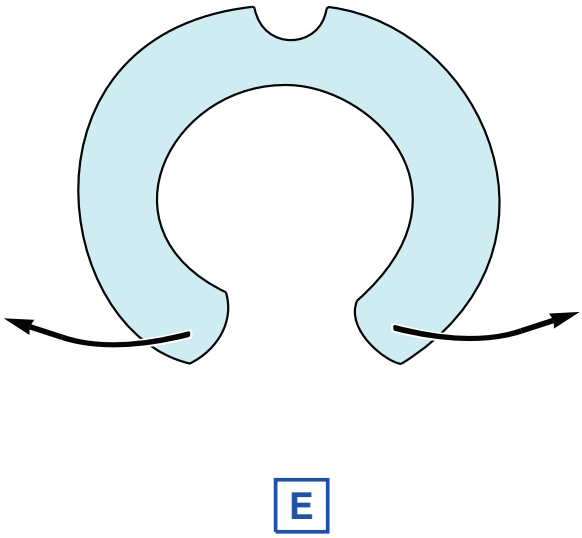

ICNLEAP1B732201D5882800002A00101
3050334 S0000814883_V1

**Bobbin replacement**
**Figure 808/73-22-01-990-887-G00 (Sheet 2 of 4)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details



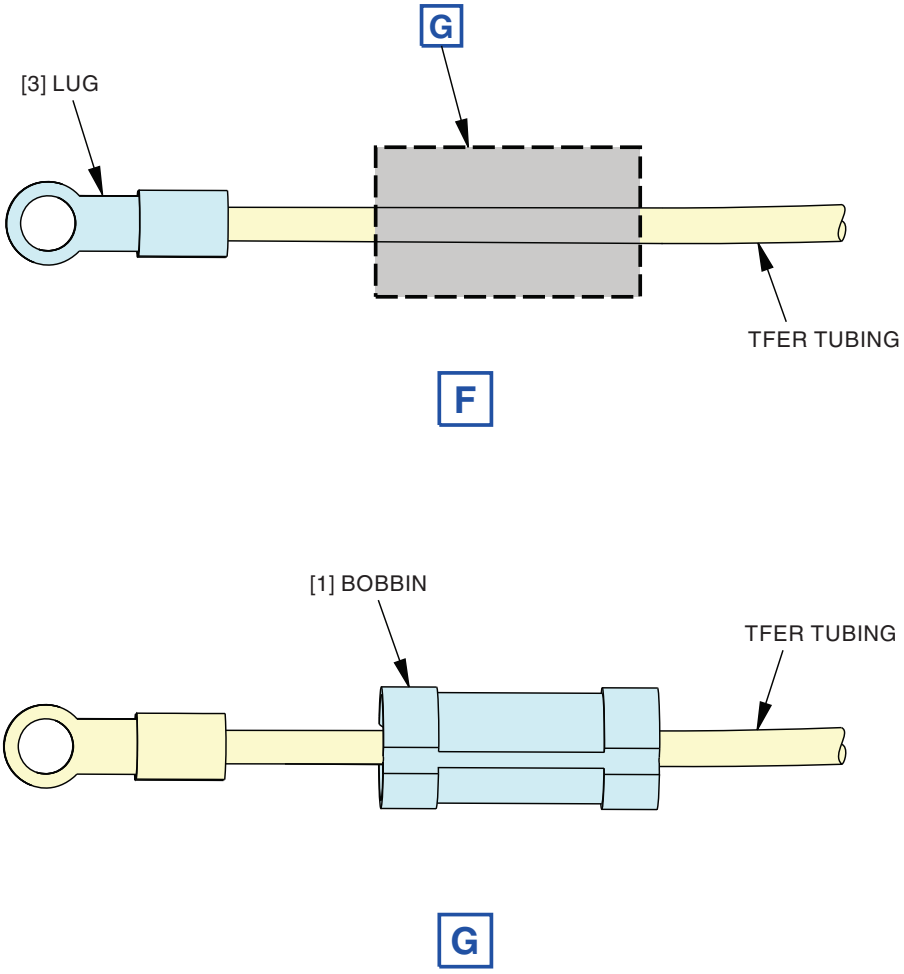




ICNLEAP1B732201D5882800003A00101
3050340 S0000814884_V1

**Bobbin replacement**
**Figure 808/73-22-01-990-887-G00 (Sheet 3 of 4)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




ICNLEAP1B732201D5882800004A00101

3050362 S0000814885_V1

**Bobbin replacement**
**Figure 808/73-22-01-990-887-G00 (Sheet 4 of 4)**



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**TASK 73-22-01-900-805-G00**

## 8. Engine Wiring Harnesses W200 and W300 - Anti-Rotation Bobbin Repair

(Figure 809, Figure 810, Figure 811)

### A. General

(1) This task gives the instructions to repair anti-rotation bobbins on the “FD CORE RIGHT” and “FD CORE LEFT” branches of the W200 and W300 engine harnesses.

(a) There are two possible configurations:

- FD CORE RIGHT branch: one BMS 1367 cable in the anti-rotation bobbin
- FD CORE LEFT branch: two BMS 1367 cables in the anti-rotation bobbin.

(2) This repair task is not applicable if some lacing tape is already present on the BMS 1367 cable (TASK 73-22-01-300-801-G00).

(3) The Table 807 gives a list of the part numbers are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 807/73-22-01-993-809-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-056-612-0 | 72-0001 | Harness W200 |
| 364-056-613-0 | 72-0132 | |
| 364-056-711-0 | 72-0001 | Harness W300 |
| 364-056-712-0 | 72-0132 | |
| 364-056-713-0 | 72-0214 | |
| 364-056-714-0 | 72-0214 | |

(4) Dimensional Information.

| Description | Minimum In-Process Dimension | Maximum In-Process Dimension | Minimum Finish Dimension | Maximum Finish Dimension | Reference Dimension |
|---|---|---|---|---|---|
| DIM. A | 1.97 in. (50 mm)*[1]*[2] | - | - | - | -*[3] |
| DIM. B | - | - | 0.40 in. (10 mm)*[1]*[2] | 0.78 in. (20 mm)*[1]*[2] | -*[3] |
| DIM. C | 1.58 in. (40 mm)*[1]*[2] | - | - | - | -*[3] |
| DIM. D | - | - | 0.00 in. (0 mm)*[1]*[2] | 0.11 in. (3 mm)*[1]*[2] | -*[3] |
| DIM. E | 12.60 in. (320 mm)*[1]*[2] | - | - | - | -*[3] |

*[1] The dimensional data identified within this task was designed using millimeters as the base unit of measurement for dimensional values. The conversion to inches is rounded to two additional decimal digits than millimeters. For minimum dimensions, converted values are rounded up. For maximum dimensions, converted values are rounded down. To avoid conversion errors, do not re-convert from inches to millimeters.

*[2] Dimensions are in inches with millimeters in parentheses.

*[3] Reference dimensions do not require inspection.



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**B. References**

| Reference | Title |
|---|---|
| 26-11-02-000-801 | Engine Fire Detection Harness Removal (P/B 401) |
| 26-11-02-400-801 | Engine Fire Detection Harness Installation (P/B 401) |
| 70-21-23-100-801-G00 | Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01 P/B 401 | ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION |
| 73-22-01-210-801-G00 | Engine Control Electrical Harnesses Inspection (P/B 601) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-762 | Scissors - Electrical |
| STD-809 | Spatula |
| STD-858 | Tag - DO NOT OPERATE |
| STD-6554 | Gun - Heat |
| STD-10670 | Pliers |

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| A50618 [CP2267] | Adhesive - Sealant, Silicone Rubber, RTV 106 | |
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G51335 [CP2674] | Lint-Free Cloth | |
| G51938 [CP2999] | Tape - Adhesive, 3M 218 | |
| G51939 [C1306] | Tubing - Heat Shrinkable, DERAY-PTFE 4:1, 1/2, Clear | |
| G51940 [F0286] | Tubing - Heat Shrinkable, TFE-R 1/2-X, Clear | |

**E. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

**F. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |



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(Continued)

| Number | Name/Location |
|---|---|
| 426 | Right Thrust Reverser, Engine 2 |

## G. Prepare for the Repair

SUBTASK 73-22-01-860-158-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-159-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## H. Engine Wiring Harnesses W200 and W300 Anti-Rotation Bobbin Repair

SUBTASK 73-22-01-020-037-G00

(1) Loosen the connectors and clamps as required until the area to repair is clear (PAGEBLOCK 73-22-01/401 and TASK 26-11-02-000-801).

SUBTASK 73-22-01-210-018-G00

(2) Do a visual inspection of the area to repair (TASK 73-22-01-210-801-G00).

SUBTASK 73-22-01-350-012-G00

**WARNING** REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(3) Remove the anti-rotation bobbin and resin (Figure 810).

(a) Install a length (DIM. A) of 3M 218 adhesive tape, G51938 [CP2999], on the BMS 1367 cable.

(b) Move the anti-rotation bobbin above the BMS 1367 cable protected area.

1) Make sure that the insulation sheath of the BMS 1367 cable is not damaged.

(c) Apply multiple pressures with the pliers, STD-10670, all around the anti-rotation bobbin to break it.

1) Remove minimum material when you break the bobbin.

(d) Remove and discard the remaining particles of the anti-rotation bobbin.

(e) Remove and discard the 3M 218 adhesive tape, G51938 [CP2999], from the BMS 1367 cable.

1) Make sure that the insulation sheath of the BMS 1367 cable is not damaged.

(f) Remove a depth (DIM. B) of resin in the mechanical part (transition or breakout) with an adapted spatula, STD-809.

SUBTASK 73-22-01-350-013-G00

(4) Install the heat-shrinkable sheath, resin, and split bobbin (Figure 811).

(a) Cut a piece (DIM. E) of heat shrink tubing, G51939 [C1306], or heat shrink tubing, G51940 [F0286], and install it above each cable.

**WARNING** USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT PARTS. IF YOU DO NOT OBEY, HOT PARTS CAN CAUSE INJURY.

**CAUTION** DO NOT POINT THE HOT AIR GUN AT THE ENGINE COMPONENTS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Blow hot air with a heat gun, STD-6554 (maximum 446°F (230°C)), to shrink a length (DIM. C) of the heat-shrinkable sheath on the mechanical part side.

(c) Move the heat-shrinkable sheath into the mechanical part (DIM. B).



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**WARNING** USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT PARTS. IF YOU DO NOT OBEY, HOT PARTS CAN CAUSE INJURY.

**CAUTION** DO NOT POINT THE HOT AIR GUN AT THE ENGINE COMPONENTS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Blow hot air with the heat gun, STD-6554 (maximum 446°F (230°C)), to continue to shrink the heat-shrinkable sheath to the end of the cable.

**WARNING** BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

(e) Cut the extremity of the heat-shrinkable sheath along the lug (DIM. C) with the cutting pliers or electrical scissors, STD-762.

NOTE: The heat-shrinkable sheath must not overlap the lug.

1) Make sure that the BMS 1367 cable is not damaged.

**WARNING** DO NOT GET ADHESIVE IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM ADHESIVE. PUT ON GOGGLES, AND GLOVES WHEN YOU USE ADHESIVE. ADHESIVE IS POISONOUS AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(f) Fill the mechanical part with resin RTV 106 adhesive, A50618 [CP2267].

**WARNING** DO NOT GET THE SOLVENT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE IN THE FUMES FROM THE SOLVENT. PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME, AND HEAT. THE SOLVENT IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(g) Clean the remaining resin with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(h) Let the resin polymerize for approximately 24 hours at ambient air (minimum 68°F (20°C)).

(i) Get a new split bobbin according to the table below:

| Harness | Branch | Number of cables in the bobbin | Marking | Supply Number |
|---|---|---|---|---|
| W200 | FD CORE LEFT | 2 | LOOP A/1 | 1019657794 |
| W200 | FD CORE RIGHT | 1 | LOOP A/1 | 1019657790 |
| W300 | FD CORE LEFT | 2 | LOOP B/2 | 1019657796 |
| W300 | FD CORE RIGHT | 1 | LOOP B/2 | 1019657792 |

(j) Install the split bobbin on the one or two cables.



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SUBTASK 73-22-01-210-019-G00

(5) Do a visual inspection of the area (TASK 73-22-01-210-801-G00).

SUBTASK 73-22-01-420-035-G00

(6) Put the harness back on its supports (PAGEBLOCK 73-22-01/401 and TASK 26-11-02-400-801).

## I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-160-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-161-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

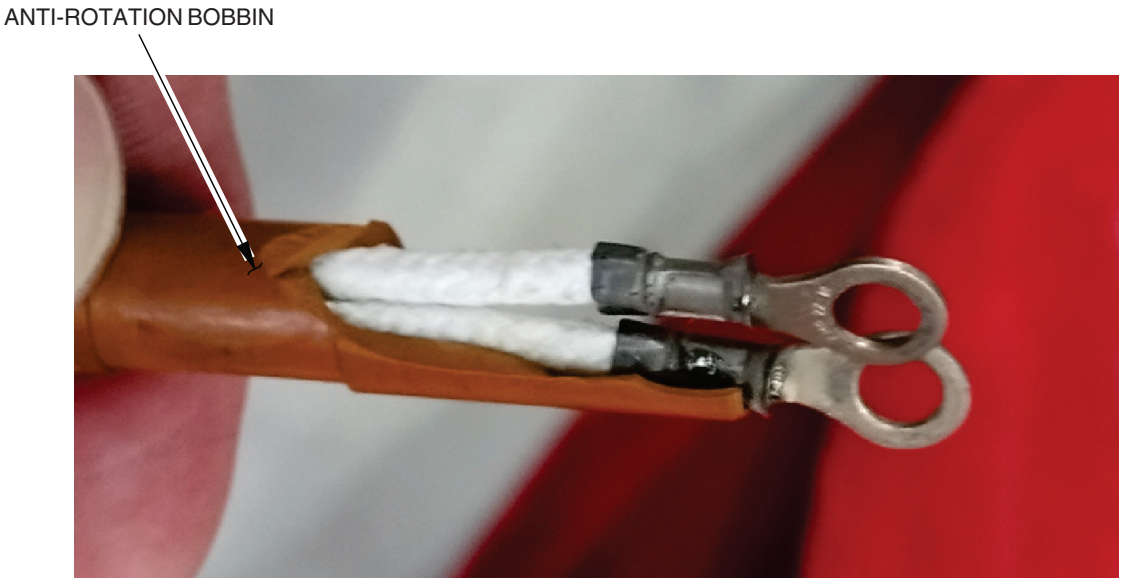
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

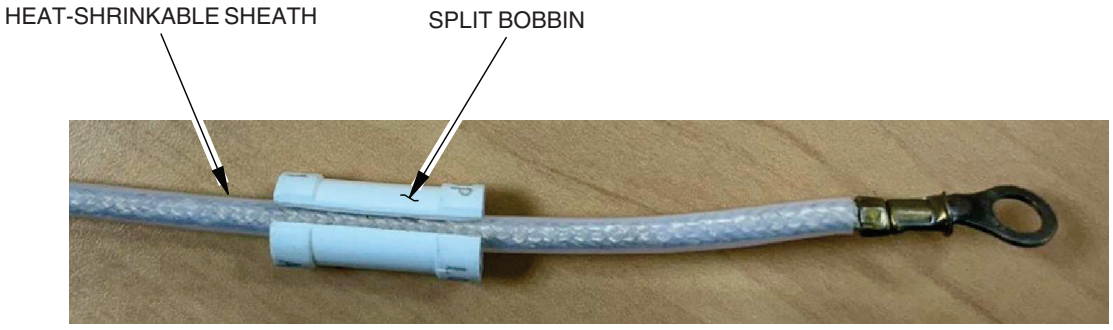
**END OF TASK**



ECCN 9E991 BOEING PROPRIETARY - See title page for details




**BEFORE REPAIR**




**AFTER REPAIR**

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**Bobbin**
**Figure 809/73-22-01-990-893-G00**



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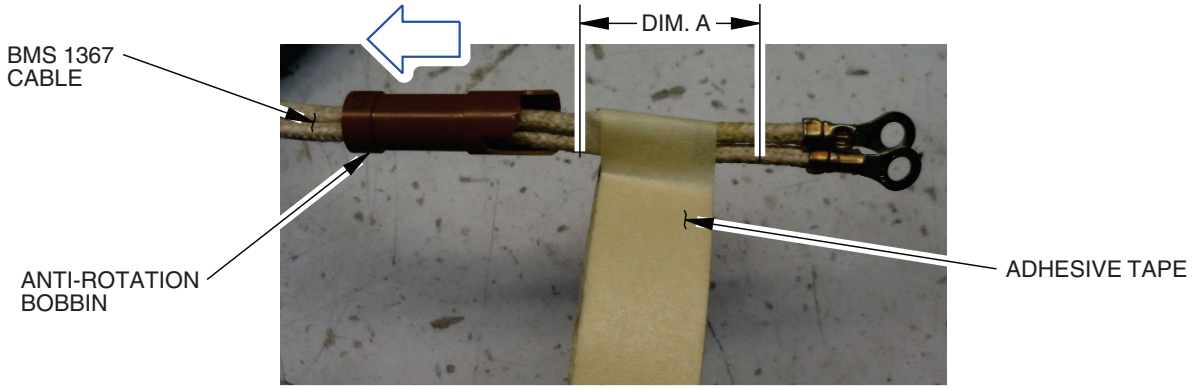
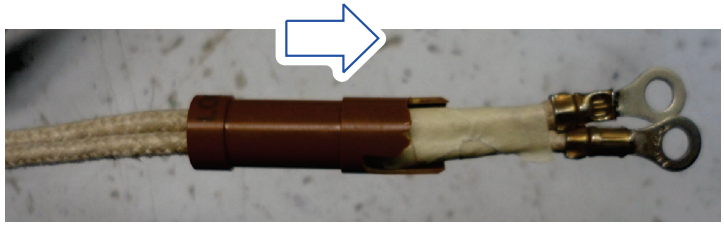
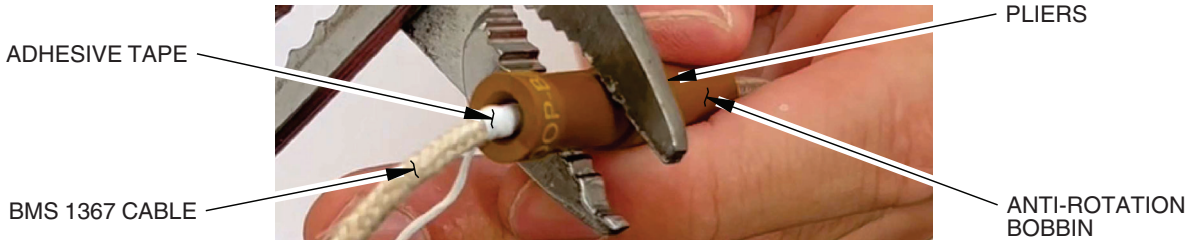


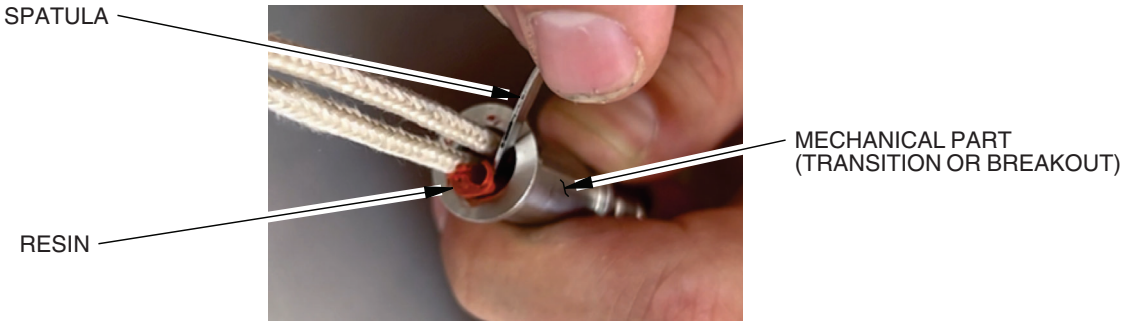

**BMS 1367 CABLE PROTECTION**



**MOVE THE ANTI-ROTATION BOBBIN ABOVE THE PROTECTED AREA**




**BREAK THE ANTI-ROTATION BOBBIN**




**RESIN REMOVAL**

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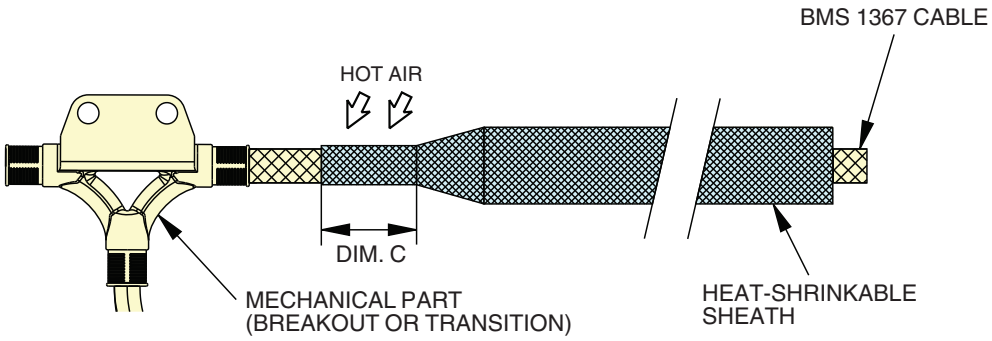
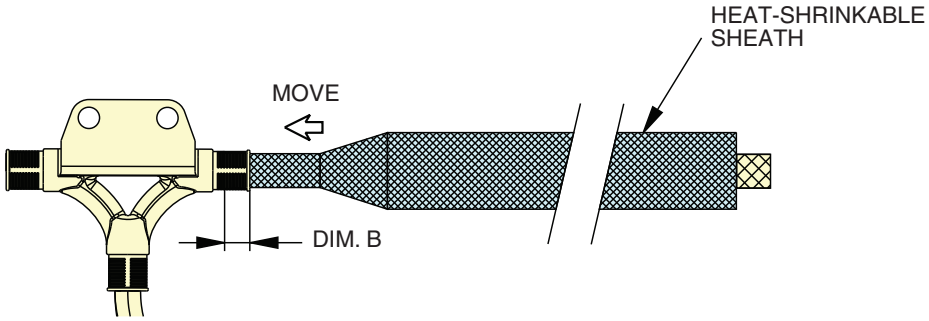
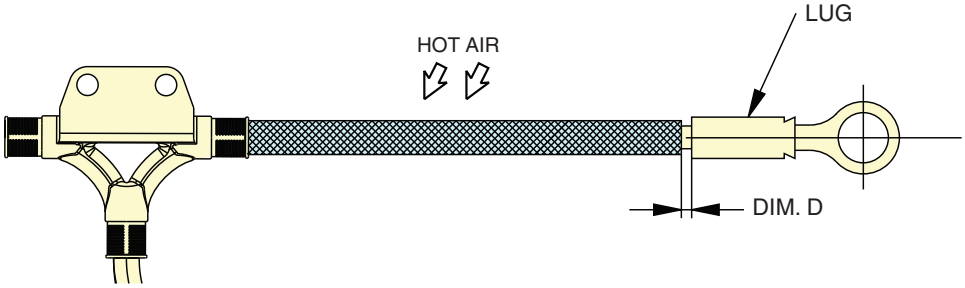
**Anti-Rotation Bobbin and Resin**
**Figure 810/73-22-01-990-894-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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**Heat-Shrinkable Sheath Installation**
**Figure 811/73-22-01-990-895-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**TASK 73-22-01-300-802-G00**

**9. Engine Wiring Harness - Harness Firewall Area Repair**

(Figure 812)

**A. General**

(1) This task gives instructions to repair the copper braid of the harness installed in the firewall area.

(2) The Table 808 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

**Table 808/73-22-01-993-807-G00**

| Part Number | Service Bulletin | Designation |
|---|---|---|
| 364-083-311-0 | 72-0013 | Harness HJ5A |
| 364-083-313-0 | 73-0019 | Harness HJ5A |
| 364-055-211-0 | 72-0001 | Harness W2 |
| 364-055-212-0 | 75-0010 | Harness W2 |

**B. References**

| Reference | Title |
|---|---|
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-22-01-210-801-G00 | Engine Control Electrical Harnesses Inspection (P/B 601) |
| 78-31-00-010-801-G00 | Open the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-010-802-G00 | Close the Thrust Reverser (Selection) (P/B 201) |
| 78-31-00-040-801-G00 | Thrust Reverser Deactivation For Ground Maintenance (P/B 201) |
| 78-31-00-440-801-G00 | Thrust Reverser Activation After Ground Maintenance (P/B 201) |

**C. Tools/Equipment**

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

**D. Consumable Materials**

| Reference | Description | Specification |
|---|---|---|
| B00676 [CP1041] | Alcohol - Isopropyl | |
| G50145 [CP2628] | Tape - High Temperature Aluminum Foil with Glass Cloth, Silicone Adhesive - 3M 363 | |
| G51328 [CP2674] | Cloth - Lint-Free | |
| G51411 [CP8005] | Cable, Safety, Stainless Steel, 0.024 inch (0.61 mm) Diameter | AMS 5689 |
| G51957 [CP8008] | Wire - Safety, 0.020 Inch (0.5 mm) Diameter | AMS 5689 |
| G51958 [CP8026] | Wire - Safety, 0.020 Inch (0.5 mm) Diameter, MS20995C20 | NASM20995 |
| G51959 [CP8027] | Wire - Safety, 0.025 Inch (0.6 mm) Diameter, MS20995C25 | NASM20995 |



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**E. Location Zones**

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

**F. Access Panels**

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

**G. Prepare for the Repairs**

SUBTASK 73-22-01-860-144-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-148-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

**WARNING** DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |



ECCN 9E991 BOEING PROPRIETARY - See title page for details

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

## H. Harness Firewall Area Repair

SUBTASK 73-22-01-210-012-G00

(1) If necessary, loose the clamps as required until the area to repair is clear.

SUBTASK 73-22-01-220-010-G00

(2) Do a visual inspection of the area to repair (TASK 73-22-01-210-801-G00).

NOTE: This repair only applies on harness branches in firewall braiding.

(a) Maximum finish defective area (DIM. A) is 3.15 in. (80 mm).

(b) Maximum finish defective area (Angle D) is 360 degrees.

SUBTASK 73-22-01-110-001-G00

**WARNING** MAKE SURE THAT WHEN YOU USE SOLVENT, TANK COVERS AND EXTINGUISHERS ARE AVAILABLE IF FIRE OCCURS. SOLVENT IS FLAMMABLE. IF YOU DO NOT OBEY, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

**WARNING** ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(3) Clean the external copper braid around the area to repair with a lint-free cloth, G51328 [CP2674], moist with alcohol, B00676 [CP1041].

SUBTASK 73-22-01-210-013-G00

(4) Prepare the area to repair as follows:

(a) Remove all material from the area to repair.

**WARNING** USE EYE PROTECTION WHEN YOU CUT MATERIALS OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

**WARNING** USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(b) Cut the strands of the damaged copper braid with the cutting pliers.



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(c) Make sure that the heat-shrinkable sleeve (black) under the copper braid is not worn.

SUBTASK 73-22-01-350-009-G00

(5) Wind 3M 363 tape, G50145 [CP2628], as follows (View C, Figure 812):

(a) Start to apply the first piece of 3M 363 tape, G50145 [CP2628], from the end of the metallic transition.

(b) Make two turns minimum and three turns maximum with the first 3M 363 tape, G50145 [CP2628].

1) Minimum finish 3M 363 tape, G50145 [CP2628], width (DIM. B) is 1.97 in. (50 mm).

(c) Apply the second 3M 363 tape, G50145 [CP2628], with an overlap minimum 0.20 in. (5 mm) (DIM. C) on the first 3M 363 tape, G50145 [CP2628], and a complete overlap of the braiding.

NOTE: An overlap on the heat-shrinkable sleeve is permitted.

(d) Make two turns minimum and three turns maximum with the second 3M 363 tape, G50145 [CP2628].

1) Minimum finish 3M 363 tape, G50145 [CP2628], width (DIM. B) is 1.97 in. (50 mm).

SUBTASK 73-22-01-350-010-G00

(6) Apply cable, G51411 [CP8005], or safety wire, G51957 [CP8008], or safety wire, G51958 [CP8026], or safety wire, G51959 [CP8027], as follows (View D, Figure 812):

(a) Start to apply cable, G51411 [CP8005], or safety wire, G51957 [CP8008], or safety wire, G51958 [CP8026], or safety wire, G51959 [CP8027], from the metallic transition side as follows:

**WARNING** USE EYE PROTECTION WHEN YOU CUT MATERIALS OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

**WARNING** USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

1) Cut a piece of cable, G51411 [CP8005], or safety wire, G51957 [CP8008], or safety wire, G51958 [CP8026], or safety wire, G51959 [CP8027], approximately 19.70 in. (500 mm) long.

2) Hold the two ends of the piece of cable, G51411 [CP8005], or safety wire, G51957 [CP8008], or safety wire, G51958 [CP8026], or safety wire, G51959 [CP8027], with the hands and put its middle part approximately 0.20 in. (5 mm) from the end of the copper braid.

3) Make two turns around the first 3M 363 tape, G50145 [CP2628].

(b) Twist the two strands approximately three turns with pliers, to brake on.

(c) Cross the strands three times around 3M 363 tape, G50145 [CP2628], homogeneous and equidistant way.

(d) Make two turns approximately 0.20 in. (5 mm) from the edge of the second 3M 363 tape, G50145 [CP2628].

(e) Twist the two strands approximately five to seven turns with pliers, to brake on again.



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**WARNING** USE EYE PROTECTION WHEN YOU CUT MATERIALS OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

**WARNING** USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(f) Cut the cable with cutting pliers and fold it on itself with pliers.

SUBTASK 73-22-01-220-011-G00

(7) Do a visual inspection of the repaired area.

(a) If the aluminum adhesive tape does not cover completely the copper braid, do the repair again.

(b) If the stainless steel wire has not three successive crossings, do the repair again.

SUBTASK 73-22-01-410-040-G00

(8) Put the harness back on its supports.

(a) If loosen, tighten the clamps.

## I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-860-149-G00

**WARNING** OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 415 | Left Thrust Reverser, Engine 1 |
| 416 | Right Thrust Reverser, Engine 1 |
| 425 | Left Thrust Reverser, Engine 2 |
| 426 | Right Thrust Reverser, Engine 2 |

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 414 | Right Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |
| 424 | Right Fan Cowl, Engine 2 |

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-153-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:



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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———



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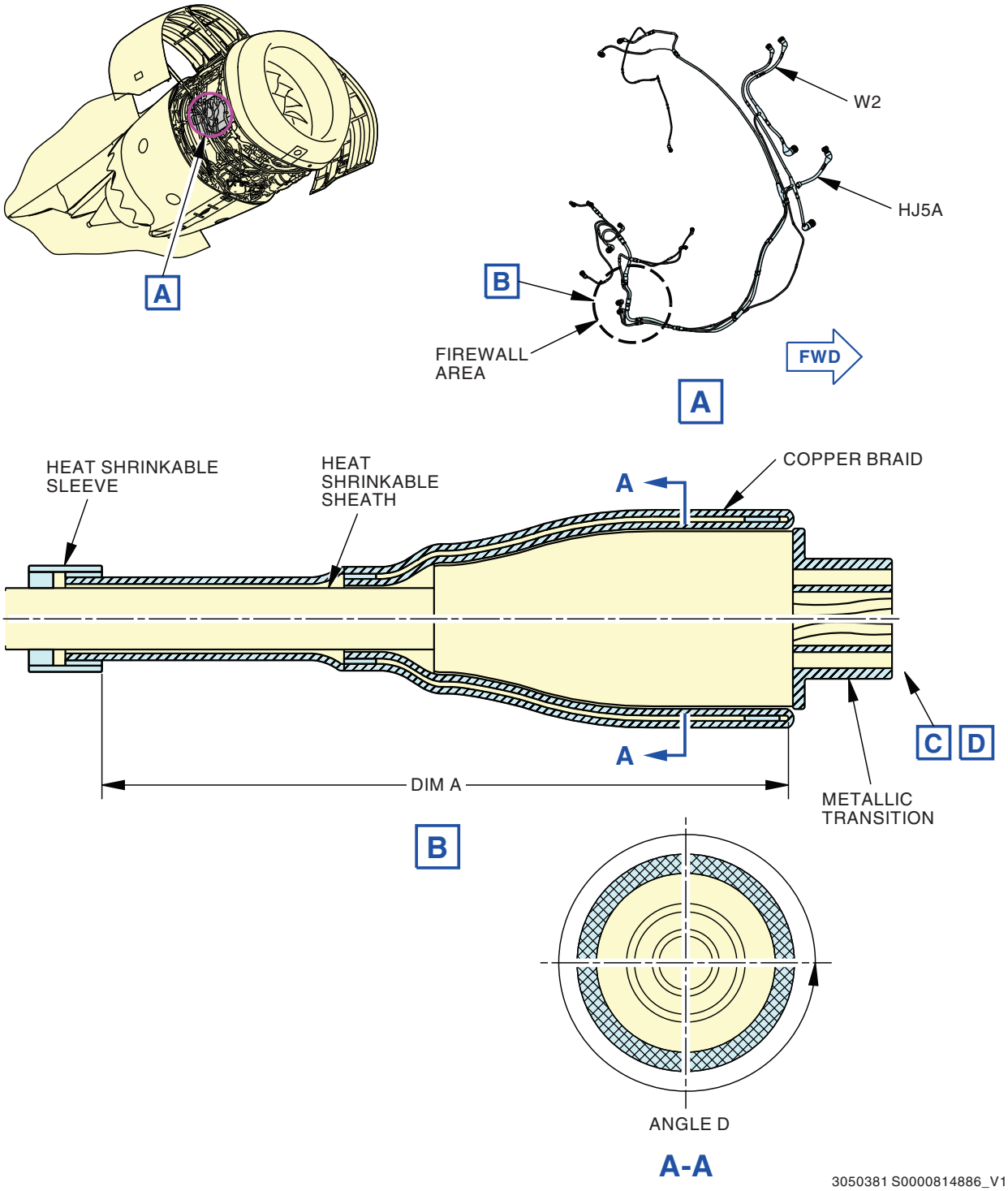
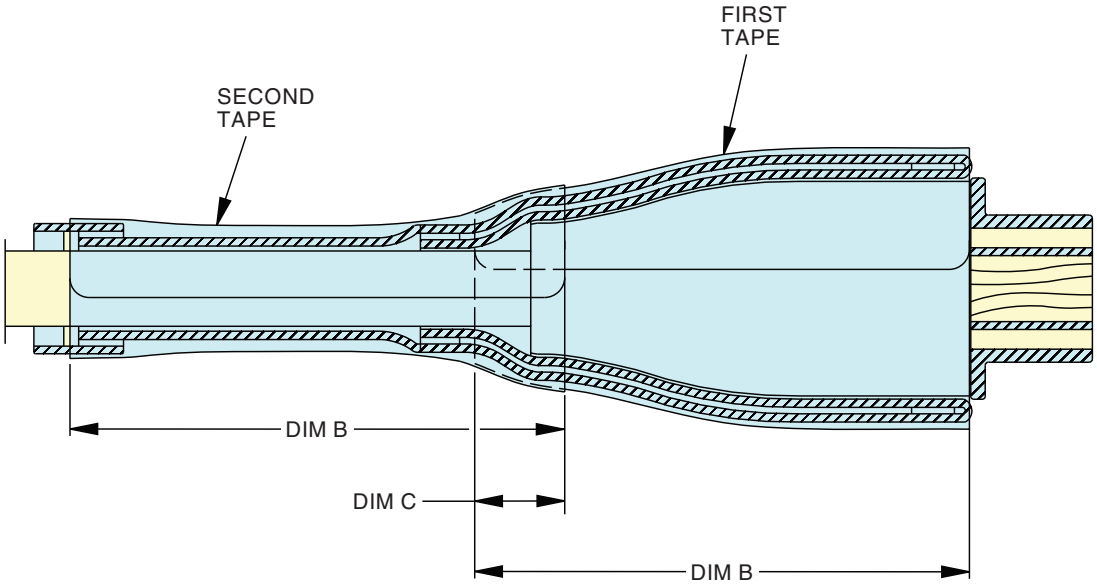



**Engine Wiring Harness Firewall Repair**
**Figure 812/73-22-01-990-888-G00 (Sheet 1 of 3)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

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**Engine Wiring Harness Firewall Repair**
**Figure 812/73-22-01-990-888-G00 (Sheet 2 of 3)**



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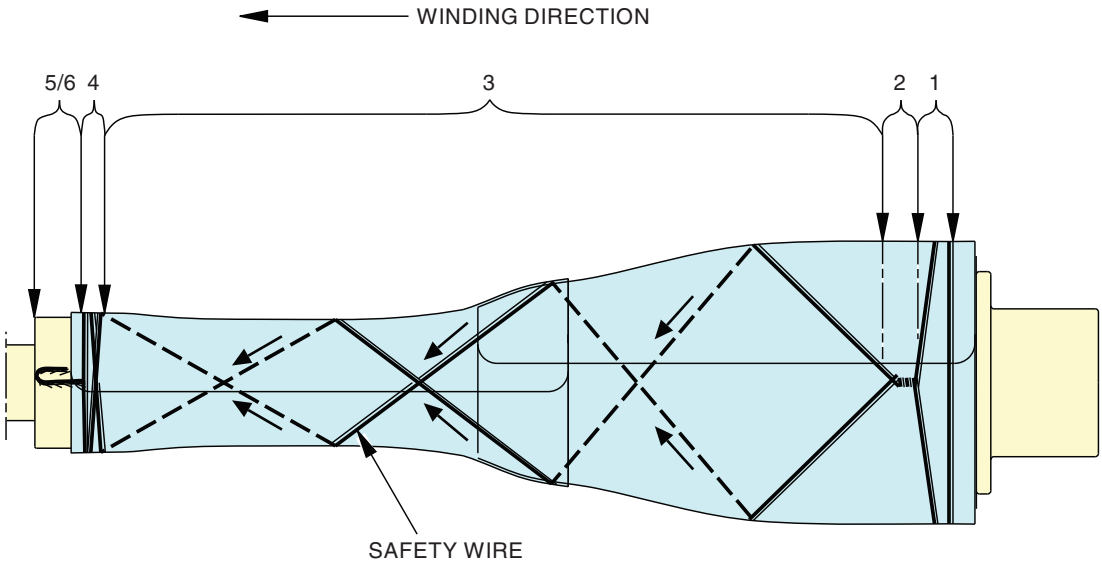


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**Engine Wiring Harness Firewall Repair**
**Figure 812/73-22-01-990-888-G00 (Sheet 3 of 3)**



ECCN 9E991 BOEING PROPRIETARY - See title page for details

## FUEL FLOW INDICATING SYSTEM - INSPECTION/CHECK

### 1. General

A. This procedure has one task:

(1) Fuel flow indicating system inspection.

**TASK 73-31-00-200-801-G00**

### 2. Fuel Flow Indicating System Inspection

#### A. General

(1) This task provides the instructions on how to examine the fuel flow indicating system.

#### B. References

| Reference | Title |
|---|---|
| 27-81-00-040-801 | Leading Edge Flaps and Slats - Deactivation (P/B 201) |
| 27-81-00-440-801 | Leading Edge Flaps and Slats - Activation (P/B 201) |
| 27-81-00-860-804 | Leading Edge Flaps and Slats Retraction (P/B 201) |
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-31-01-000-801-G00 | Fuel Flow Transmitter Removal (P/B 401) |
| 73-31-01-400-801-G00 | Fuel Flow Transmitter Installation (P/B 401) |

#### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-858 | Tag - DO NOT OPERATE |

#### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

#### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

#### F. Prepare for the Inspection

**SUBTASK 73-31-00-860-001-G00**

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

**SUBTASK 73-31-00-860-002-G00**

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.



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SUBTASK 73-31-00-040-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-31-00-010-001-G00

(4) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

## G. Fuel Flow Indicating System Inspection

SUBTASK 73-31-00-212-001-G00

(1) Examine the fuel flow transmitter as follows:

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, replace the fuel flow transmitter gasket seals (TASK 73-31-01-000-801-G00 and TASK 73-31-01-400-801-G00).

(b) Make sure that the fuel flow transmitter is not loose.

1) If the fuel flow transmitter is loose, tighten the connection (TASK 73-31-01-400-801-G00).

a) Tighten the installation bolts.

(c) Look for dents or cracks in the fuel flow transmitter.

1) If you see dents or cracks, replace the fuel flow transmitter (TASK 73-31-01-000-801-G00 and TASK 73-31-01-400-801-G00).

(d) Make sure that the electrical connector is not loose.

1) If the electrical connector is loose, disconnect and connect the electrical connector on the fuel flow transmitter (TASK 70-00-01-910-803-G00).

## H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-31-00-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

SUBTASK 73-31-00-440-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 73-31-00-860-007-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



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## FUEL FLOW TRANSMITTER- REMOVAL/INSTALLATION

### 1. General

A. This procedure has two tasks:

(1) Fuel flow transmitter removal

(2) Fuel flow transmitter installation.

**TASK 73-31-01-000-801-G00**

### 2. Fuel Flow Transmitter Removal

(Figure 401)

#### A. General

(1) This task provides the instructions on how to remove the Fuel Flow Transmitter (FFT).

#### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 73-11-01-000-801-G00 | Main Fuel Pump Removal (P/B 401) |
| 73-11-02-000-801-G00 | Fuel Filter Cartridge Removal (P/B 401) |
| 73-11-04-000-801-G00 | Fuel Metering Unit Strainer Removal (P/B 401) |

#### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-201 | Container - Fuel Resistant, 5-Gallon (19-Liter) |
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

#### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

#### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

#### F. Prepare for the Removal

**SUBTASK 73-31-01-860-001-G00**

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.



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SUBTASK 73-31-01-860-002-G00

(2) For the left engine, open these applicable circuit breakers and install safety tags:

**F/O Electrical System Panel, P6-3**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 5 | C01320 | ENGINE FUEL ENGINE 1 HPSOV CONT |
| E | 6 | C01395 | ENGINE FUEL ENGINE 1 HPSOV IND |

SUBTASK 73-31-01-860-003-G00

(3) For the right engine, open these applicable circuit breakers and install safety tags:

**F/O Electrical System Panel, P6-3**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C01321 | ENGINE FUEL ENGINE 2 HPSOV CONT |
| E | 4 | C01396 | ENGINE FUEL ENGINE 2 HPSOV IND |

SUBTASK 73-31-01-010-001-G00

(4) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

## G. Fuel Flow Transmitter Removal

SUBTASK 73-31-01-010-002-G00

**WARNING** DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the fuel flow transmitter [1] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the fuel flow transmitter [1].

(b) Drain the fuel system as follows:

1) Drain the main fuel pump (TASK 73-11-01-000-801-G00).

2) Drain the fuel metering unit strainer (TASK 73-11-04-000-801-G00).

3) Drain the main fuel filter housing (TASK 73-11-02-000-801-G00).

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION** USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(c) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector from the fuel flow transmitter [1] as follows (TASK 70-00-01-910-803-G00):


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1) Disconnect the electrical connector [2] from the fuel flow transmitter [1].
2) Put the protective covers on the electrical connector [2] and the electrical receptacle of the fuel flow transmitter [1].

(d) Disconnect the fuel tube [9] from the fuel flow transmitter [1] as follows:
1) Remove the four bolts [10] that attach the fuel tube [9] to the fuel flow transmitter [1].
2) Remove and discard the gasket seal [11].
3) Install the protective covers on the fuel tube [9].

(e) Disconnect the fuel tube [3] from the fuel flow transmitter [1] as follows:
1) Loosen the two nuts [4] and the two bolts [5] that attach the fuel tube [3] to the bracket [6].
2) Remove the four bolts [7] that attach the fuel tube [3] to the fuel flow transmitter [1].
3) Remove and discard the gasket seal [8].
4) Install the protective covers on the fuel tube [3].

(f) Remove the fuel flow transmitter [1] from the engine.

(g) Install the protective covers on the fuel flow transmitter [1].

**——— END OF TASK ———**



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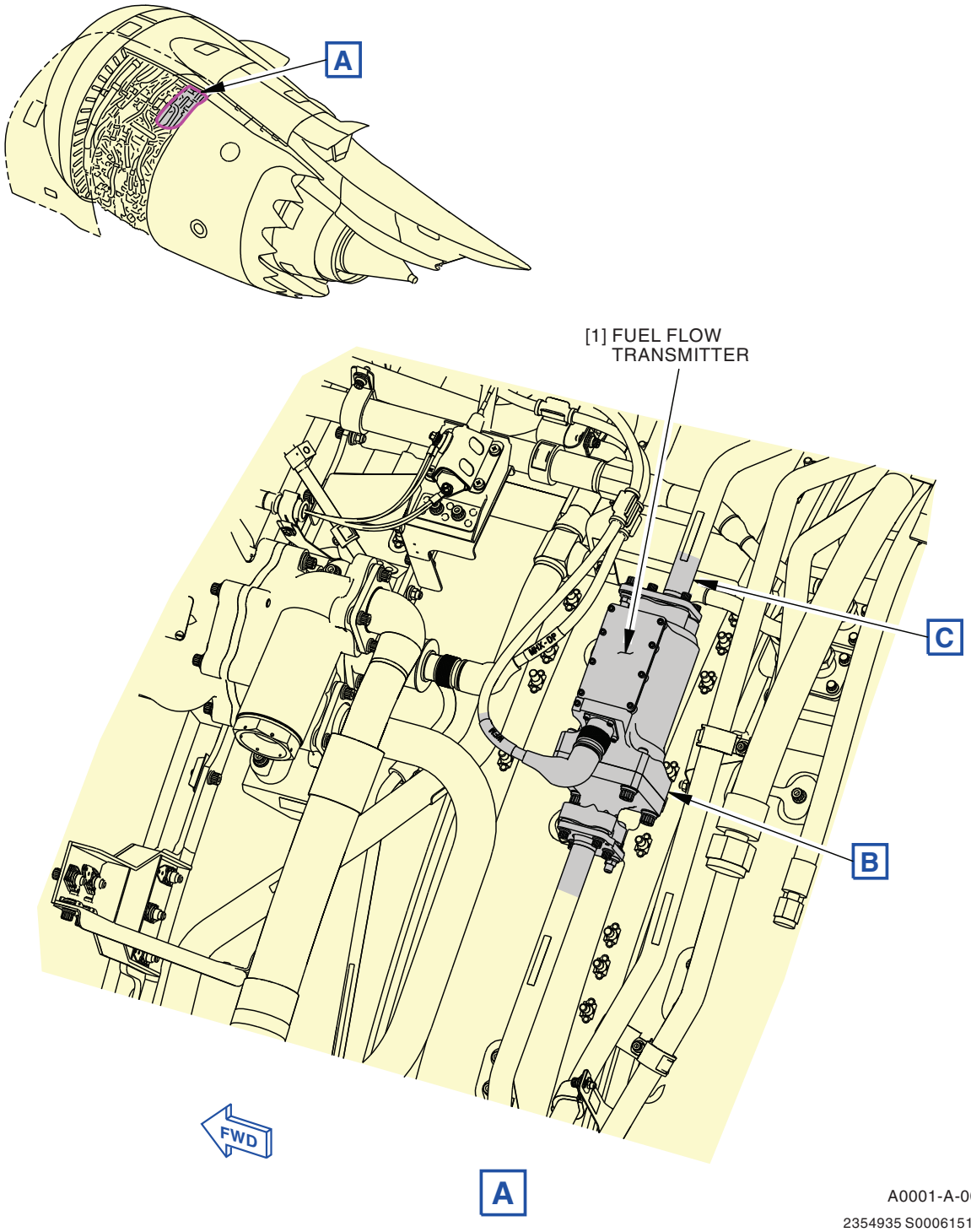


**Fuel Flow Transmitter Installation**
**Figure 401/73-31-01-990-801-G00 (Sheet 1 of 2)**



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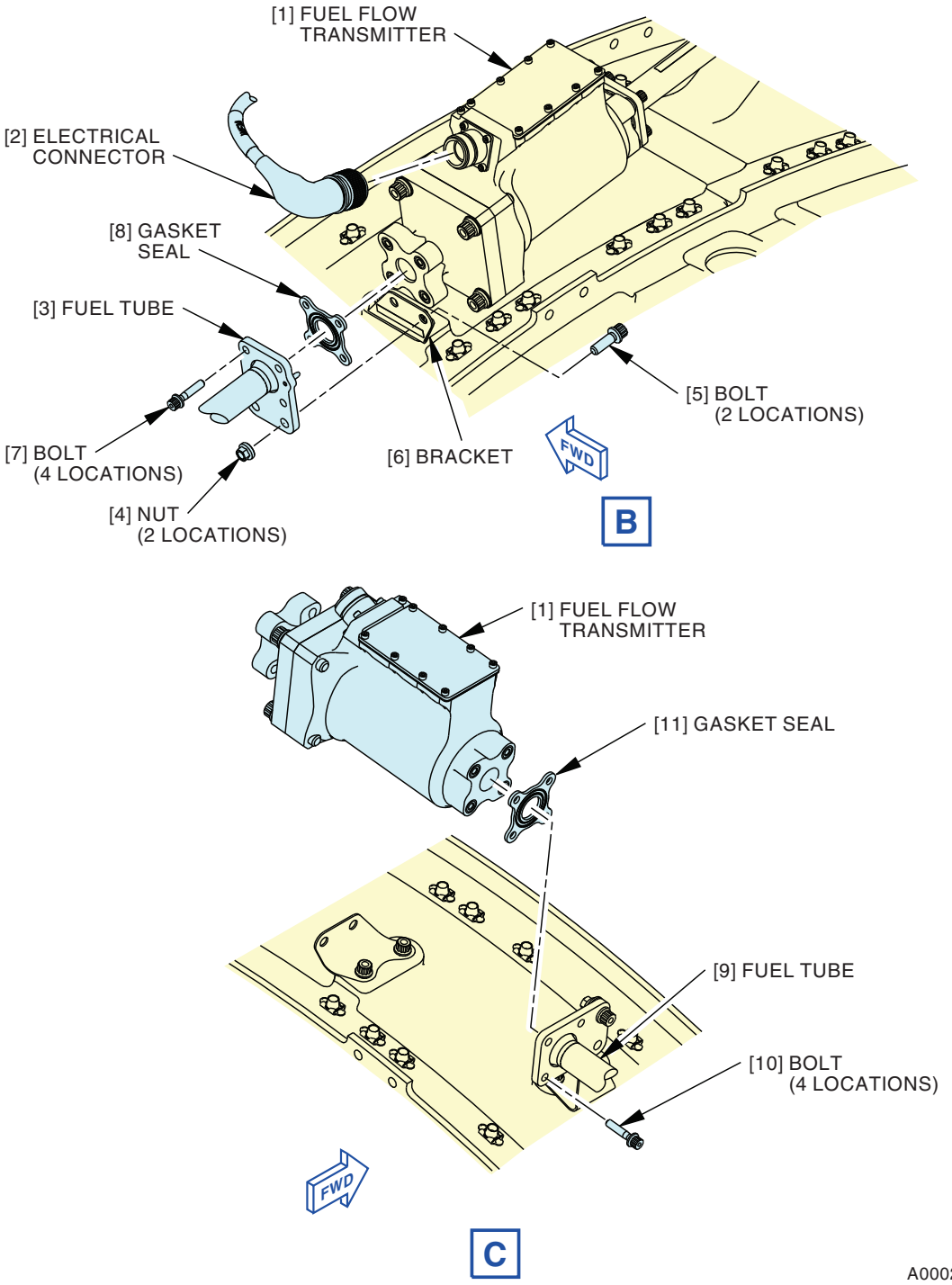


**Fuel Flow Transmitter Installation**
**Figure 401/73-31-01-990-801-G00 (Sheet 2 of 2)**



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**TASK 73-31-01-400-801-G00**

## 3. Fuel Flow Transmitter Installation

(Figure 401)

### A. General

(1) This task provides the instructions on how to install the Fuel Flow Transmitter (FFT).

### B. References

| Reference | Title |
|---|---|
| 70-00-01-910-803-G00 | Electrical Connector Disconnection and Connection (P/B 201) |
| 71-00-00-800-804-G00 | Power Plant Test Reference Table (P/B 501) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |

### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

### D. Consumable Materials

| Reference | Description | Specification |
|---|---|---|
| D00599 [CP2442] | Oil - Engine | |
| D00601 [CP2101] | Vaseline - Graphite Mineral | |
| D00672 [CP5070] | Vaseline - Pure Mineral | V V-P-236 |

### E. Expendables/Parts

| AMM Item | Description | AIPC Reference | AIPC Effectivity |
|---|---|---|---|
| 1 | Fuel flow transmitter | 73-31-01-01-010 | PTF ALL |
| 8 | Gasket seal | 73-31-01-01-050 | PTF ALL |
| 11 | Gasket seal | 73-31-01-01-030 | PTF ALL |

### F. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

### G. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

### H. Fuel Flow Transmitter Installation

**SUBTASK 73-31-01-410-001-G00**

**WARNING** DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Install the fuel flow transmitter [1] as follows:

(a) Make sure that the mating surfaces are clean and clear of unwanted materials.

(b) Remove the protective covers from the fuel flow transmitter [1].



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(c) Put the fuel flow transmitter [1] on the engine.

(d) Connect the fuel tube [3] to the fuel flow transmitter [1] as follows:

1) Remove the protective covers from the fuel tube [3].
2) Lubricate the new gasket seal [8] with vaseline, D00672 [CP5070].
3) Install the new gasket seal [8] and the fuel tube [3] on the fuel flow transmitter [1].
4) Lubricate the threads of the four bolts [7] with graphite mineral vaseline, D00601 [CP2101].

**CAUTION** MAKE SURE THAT YOU INSTALL THE CORRECT BOLT PART NUMBERS THAT ATTACH THE FUEL TUBE TO THE FUEL FLOW TRANSMITTER. IF NOT, FUEL LEAKAGE CAN OCCUR.

5) Install the four bolts [7] to attach the fuel tube [3] to the fuel flow transmitter [1].
6) Tighten the bolts [7] in a crisscross pattern to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).
7) Verify the torque of the bolts [7] in a crisscross pattern.
8) Lubricate the threads of the two bolts [5] and the two nuts [4] with engine oil, D00599 [CP2442].
9) Install the two bolts [5] and the two nuts [4] to attach the fuel tube [3] to the bracket [6].
10) Tighten the nuts [4] to 109.2 in-lb (12.34 N·m) - 120.7 in-lb (13.64 N·m).

(e) Connect the fuel tube [9] to the fuel flow transmitter [1] as follows:

1) Remove the protective covers from the fuel tube [9].
2) Lubricate the new gasket seal [11] with vaseline, D00672 [CP5070].
3) Install the new gasket seal [11] and the fuel tube [9] on the fuel flow transmitter [1].
4) Lubricate the threads of the four bolts [10] with graphite mineral vaseline, D00601 [CP2101].

**CAUTION** MAKE SURE THAT YOU INSTALL THE CORRECT BOLT PART NUMBERS THAT ATTACH THE FUEL TUBE TO THE FUEL FLOW TRANSMITTER. IF NOT, FUEL LEAKAGE CAN OCCUR.

5) Install the four bolts [10] that attach the fuel tube [9] to the fuel flow transmitter [1].
6) Tighten the bolts [10] in a crisscross pattern to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).
7) Verify the torque of the bolts [10] in a crisscross pattern.

**CAUTION** MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.


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**(CAUTION PRECEDES)**

**CAUTION**

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(f) Use the teflon-jawed pliers, STD-664, to connect the electrical connector to the fuel flow transmitter [1] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the electrical connector [2] and the electrical receptacle of the fuel flow transmitter [1].

2) Connect the electrical connector [2] to the fuel flow transmitter [1].

(g) Clean the work area and clear the tools and other items.

## I. Put the Airplane Back to Its Usual Condition

**SUBTASK 73-31-01-410-002-G00**

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

**SUBTASK 73-31-01-860-004-G00**

(2) For the left engine, remove the safety tags and close these applicable circuit breakers:

**F/O Electrical System Panel, P6-3**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 5 | C01320 | ENGINE FUEL ENGINE 1 HPSOV CONT |
| E | 6 | C01395 | ENGINE FUEL ENGINE 1 HPSOV IND |

**SUBTASK 73-31-01-860-005-G00**

(3) For the right engine, remove the safety tags and close these applicable circuit breakers:

**F/O Electrical System Panel, P6-3**

| Row | Col | Number | Name |
|---|---|---|---|
| E | 3 | C01321 | ENGINE FUEL ENGINE 2 HPSOV CONT |
| E | 4 | C01396 | ENGINE FUEL ENGINE 2 HPSOV IND |

**SUBTASK 73-31-01-860-006-G00**

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

## J. Fuel Flow Transmitter Installation Test

**SUBTASK 73-31-01-710-001-G00**

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

**END OF TASK**



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## FUEL FLOW TRANSMITTER - INSPECTION/CHECK

### 1. General

A. This procedure has one task:

(1) Fuel Flow Transmitter Inspection

**TASK 73-31-01-210-801-G00**

### 2. Fuel Flow Transmitter Inspection

(Figure 601)

#### A. General

(1) This task provides the instructions to inspect the Fuel Flow Transmitter.

(2) The Fuel Flow Transmitter is located on the left side of the fan case.

#### B. References

| Reference | Title |
|---|---|
| 71-11-04-010-801-G00 | Open the Fan Cowl Panels (Selection) (P/B 201) |
| 71-11-04-410-801-G00 | Close the Fan Cowl Panels (Selection) (P/B 201) |
| 73-31-01-000-801-G00 | Fuel Flow Transmitter Removal (P/B 401) |
| 73-31-01-400-801-G00 | Fuel Flow Transmitter Installation (P/B 401) |

#### C. Tools/Equipment

| Reference | Description |
|---|---|
| STD-664 | Pliers - Teflon-jawed (or Equivalent Soft-Jawed) |
| STD-858 | Tag - DO NOT OPERATE |

#### D. Location Zones

| Zone | Area |
|---|---|
| 411 | Engine 1 - Engine |
| 421 | Engine 2 - Engine |

#### E. Access Panels

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

#### F. Prepare for the Inspection

**SUBTASK 73-31-01-860-007-G00**

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

**SUBTASK 73-31-01-010-003-G00**

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).



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(a) Open these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

### G. Fuel Flow Transmitter Inspection

SUBTASK 73-31-01-210-001-G00

(1) Inspect the fuel flow transmitter.

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, do these tasks to replace the transmitter preformed packings (Fuel Flow Transmitter Removal, TASK 73-31-01-000-801-G00 and Fuel Flow Transmitter Installation, TASK 73-31-01-400-801-G00).

(b) Make sure that the fuel flow transmitter is not loose.

1) If the fuel flow transmitter is loose, tighten the bolts that attach the fuel flow transmitter to the support brackets (Fuel Flow Transmitter Installation, TASK 73-31-01-400-801-G00).

(c) Make sure that the electrical connector is not loose.

**CAUTION**

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

**CAUTION**

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If the electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

### H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-31-01-410-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

| Number | Name/Location |
|---|---|
| 413 | Left Fan Cowl, Engine 1 |
| 423 | Left Fan Cowl, Engine 2 |

SUBTASK 73-31-01-860-008-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

**END OF TASK**



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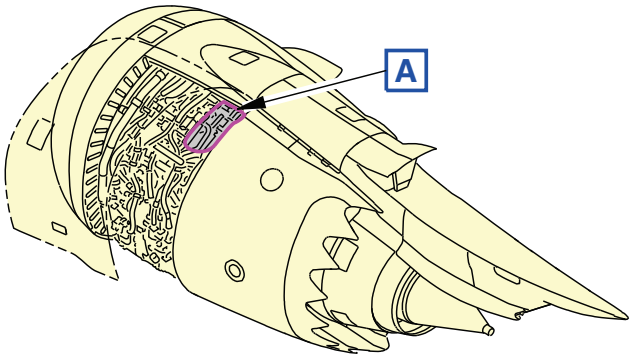


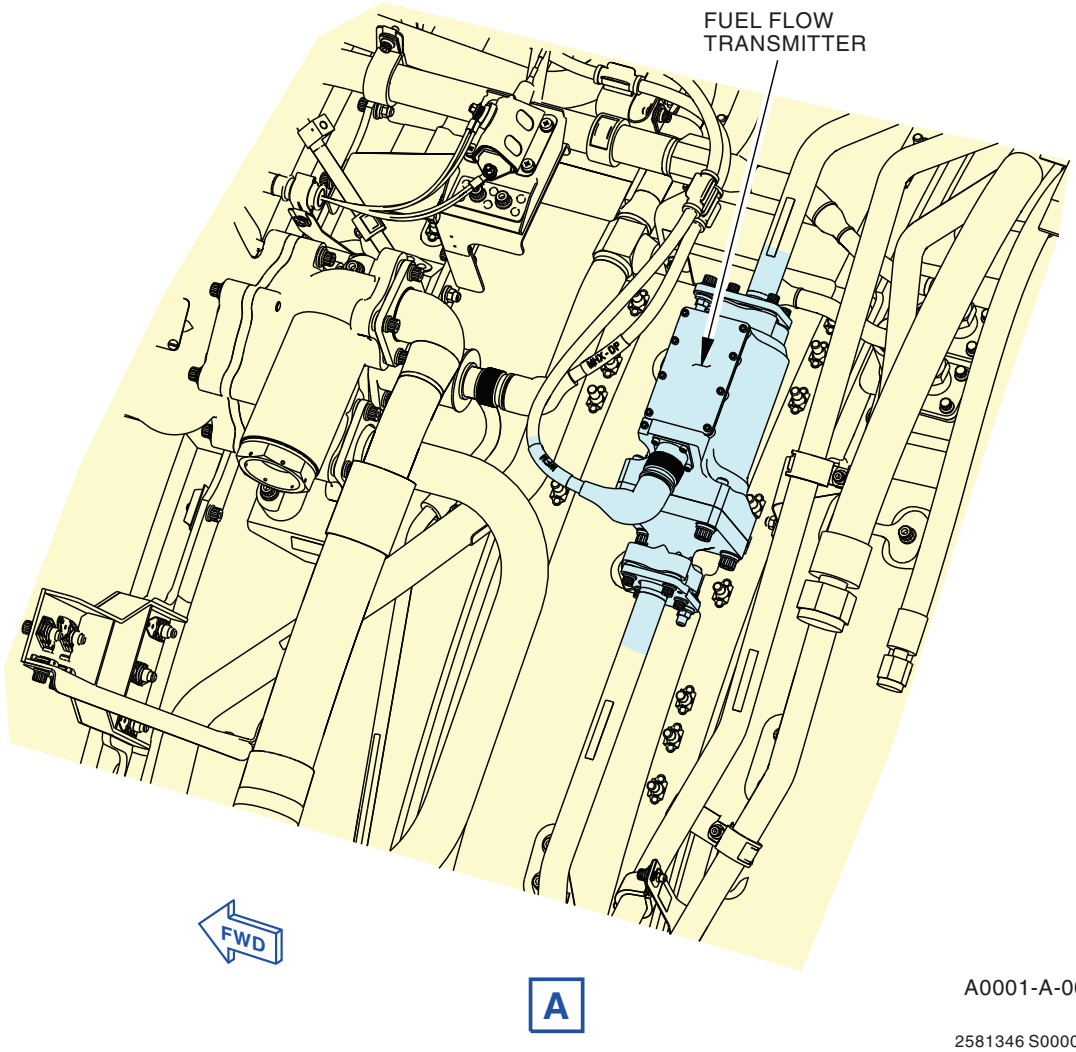


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**Fuel Flow Transmitter Inspection**
**Figure 601/73-31-01-990-802-G00**



ECCN 9E991 BOEING PROPRIETARY - See title page for details